

CYP2D6 DDGI Qualification

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This qualification report is filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>

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1 Introduction

1.1 Objective

This qualification report evaluates the developed physiologically based pharmacokinetic (PBPK) drug-drug-gene interaction (DDGI) network for the ability to perform simulations with the intended purpose to predict cytochrome P450 2D6 (**CYP2D6**)-mediated drug-drug interactions (DDIs) and DDGIs.

To demonstrate the level of confidence, the predictive performance is assessed using a network of PBPK models for selected CYP2D6 perpetrators, victims, and genotype-dependent interaction scenarios together with clinical DD(G)I data from published studies. The models are whole-body PBPK models and allow dynamic interaction simulations in tissues expressing the relevant enzymes or transporters.

The respective *qualification plan* to produce this *qualification report* is transparently documented and provided open-source (<https://github.com/Open-Systems-Pharmacology/DDGI-CYP2D6>). The same applies for all presented PBPK models including *evaluation reports* on model building and evaluation of each model (<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library>).

Evaluation reports including descriptions on model building and detailed evaluations of the included models are documented separately (see [Section 1.2](#)).

Please refer to the [Appendix](#) to learn more details:

- An overview over the Open Systems Pharmacology (OSP) Suite is given in chapter [Section 5.1](#)
- [Section 5.2](#) shows the implementation of the underlying mathematical equations for DDIs in the OSP suite.
- A detailed general description of the performed qualification workflow (*qualification plan*, *qualification report*, etc.) can be found in chapter [Section 5.3](#).

2 DDI Evaluations

The following section shows the correlations between observed and model-predicted AUC_{last} and C_{max} ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and **C_{max} ratio (CmaxR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
- the solid lines denote the limits suggested by [Guest et al. 2011](#),
- the bold solid line denotes the unity line,
- each color represents one combination of perpetrator and victim (parent drug or metabolite)

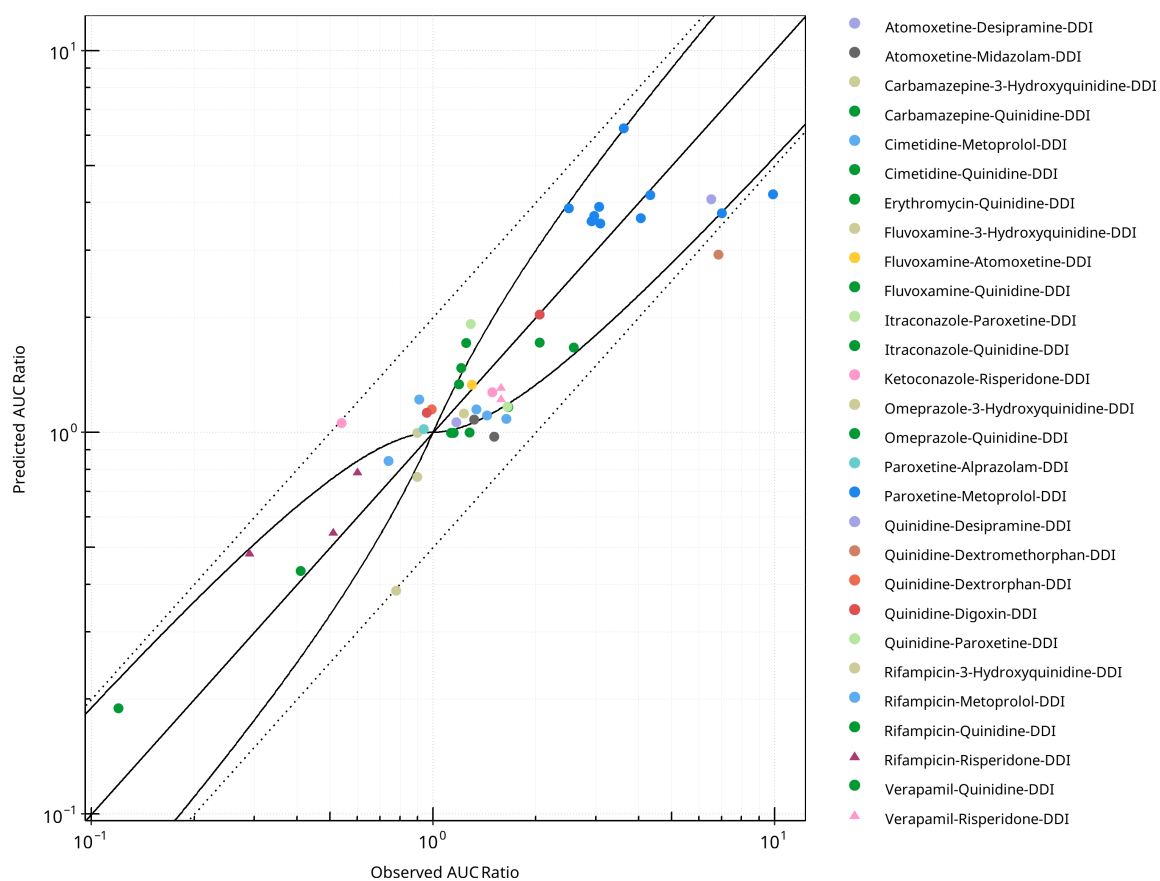


Figure 2-1: CYP2D6 DDIs. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

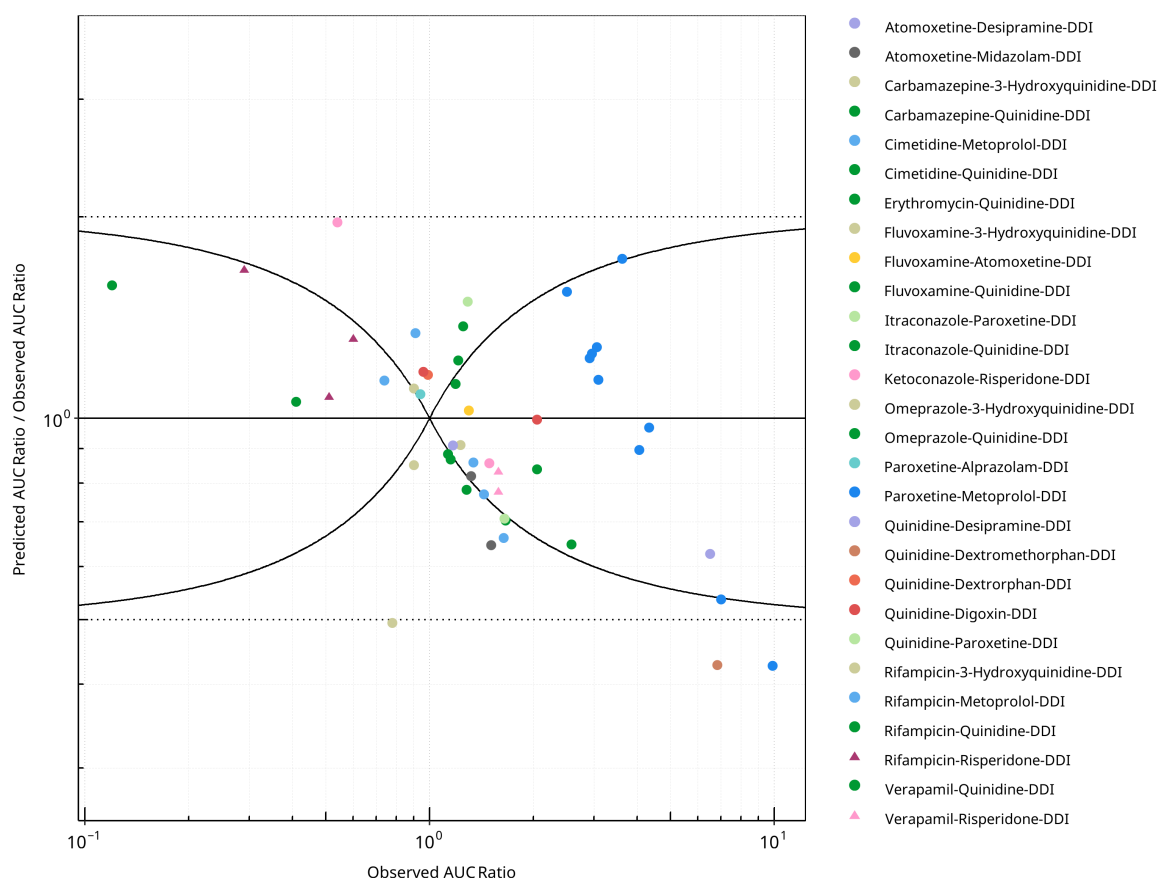


Figure 2-2: CYP2D6 DDIs. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

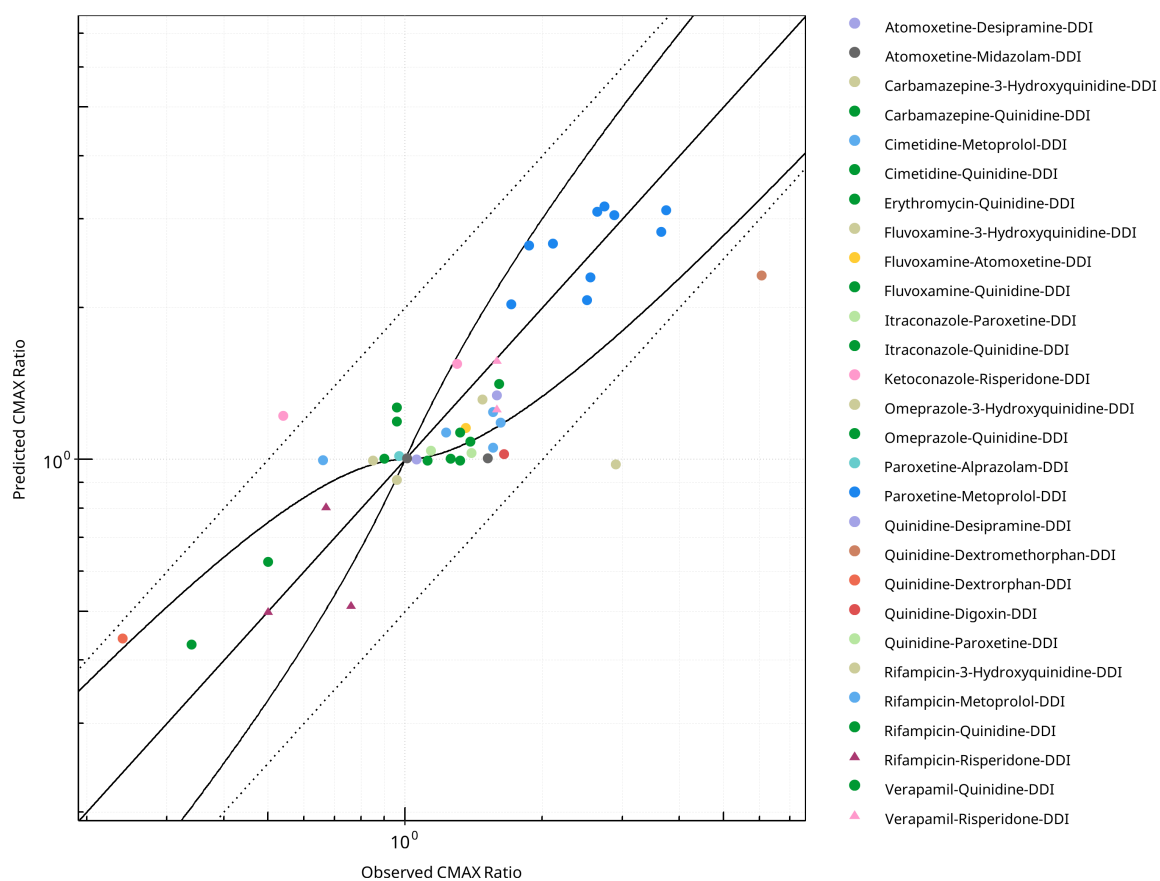


Figure 2-3: CYP2D6 DDIs. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

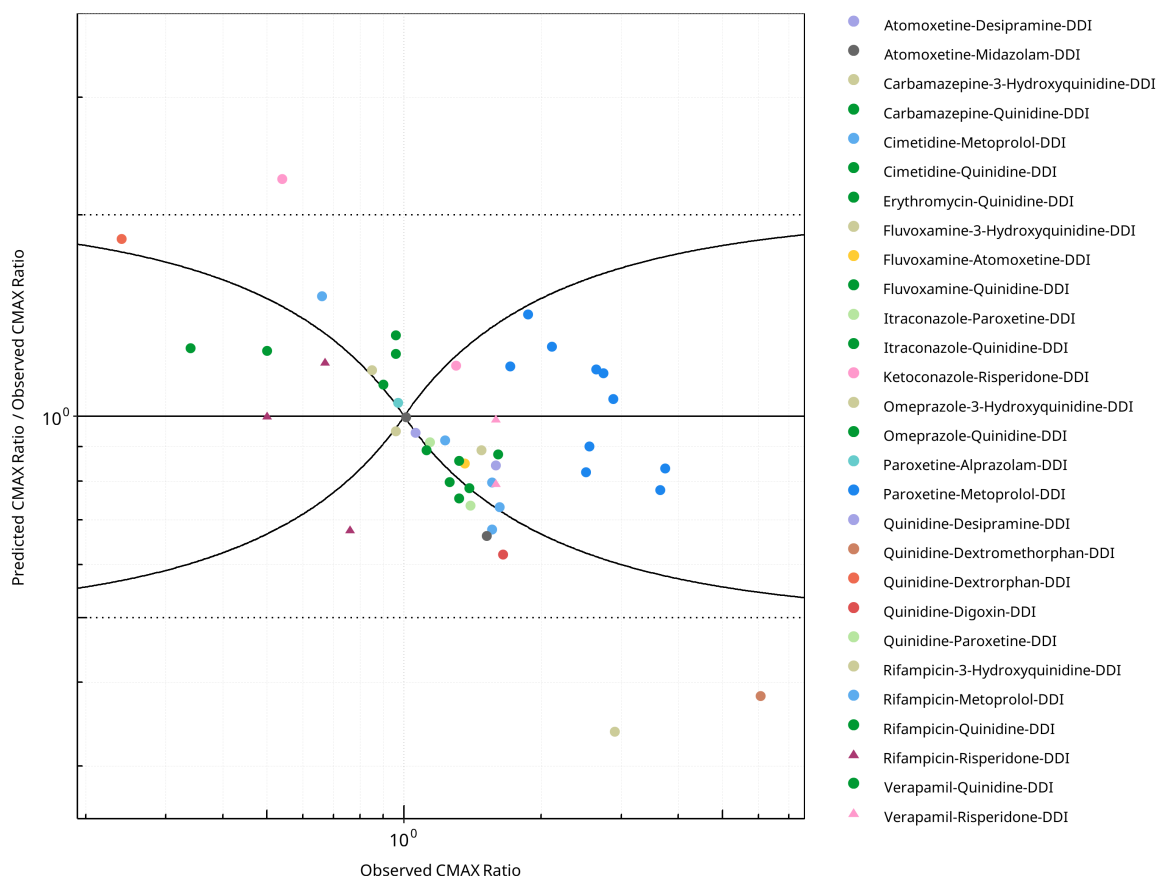


Figure 2-4: CYP2D6 DDIs. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 2-1: GMFE for CYP2D6 DDIs Ratio

PK parameter	GMFE
AUC	1.32
CMAX	1.28

Table 2-2: Summary table for CYP2D6 DDIs - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	49	-
Points within Guest <i>et al.</i>	27	55.10
Points within 2 fold	46	93.88

Table 2-3: Summary table for CYP2D6 DDIs - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

C _{MAX}	Number	Ratio [%]
Points total	48	-
Points within Guest <i>et al.</i>	28	58.33
Points within 2 fold	45	93.75

Table 2-4: Summary table for CYP2D6 DDIs

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16940	Quinidine, 50 mg, PO, SD: 1h before DEX	Dextromethorphan, PO	2.93	6.84	0.43	2.32	6.07	0.38	Capon 1996
16942	Quinidine, 50 mg, PO, SD: 1h before DEX	Total dextrophan, PO	1.15	0.99	1.16	0.44	0.24	1.84	Capon 1996
16946	Quinidine, 200 mg, PO, BID	Digoxin, IV	1.13	0.96	1.18	0.68	-	-	Steiness 1980
16948	Quinidine, 200 mg, PO, QID	Digoxin, IV	2.04	2.05	1.00	1.03	1.65	0.62	Ochs 1981
16954	Quinidine, 30 mg, PO, BID	Paroxetine, PO	1.93	1.29	1.50	1.04	1.14	0.92	Schoedel 2012
16956	Carbamazepine, 200/400 mg, PO, BID	Quinidine, PO	0.43	0.41	1.06	0.63	0.50	1.26	Andreassen 2007
16958	Carbamazepine, 200/400 mg, PO, BID	3-Hydroxyquinidine, PO	0.77	0.90	0.85	1.32	1.48	0.89	Andreassen 2007
16960	Cimetidine, 300 mg, PO, QD	Quinidine, PO	1.00	1.13	0.89	1.01	0.90	1.12	Kolb 1984
16962	Cimetidine, 300 mg, PO, QID	Quinidine, PO	1.00	1.28	0.78	1.01	1.26	0.80	Hardy 1983
16964	Fluvoxamine, 100 mg, PO, QD	Quinidine, PO	1.17	1.66	0.70	1.00	1.32	0.76	Damkier 1999a
16966	Fluvoxamine, 100 mg, PO, QD	3-Hydroxyquinidine, PO	1.12	1.23	0.91	0.91	0.96	0.95	Damkier 1999a
16968	Omeprazole, 40 mg, PO, QD	Quinidine, PO	1.00	1.15	0.87	1.00	1.12	0.89	Ching 1991
16970	Omeprazole, 40 mg, PO, QD	3-Hydroxyquinidine, PO	1.00	0.90	1.11	1.00	0.85	1.17	Ching 1991
16972	Rifampicin, 600 mg, PO, QD	Quinidine, PO	0.19	0.12	1.58	0.43	0.34	1.27	Damkier 1999
16974	Rifampicin, 600 mg, PO, QD	3-Hydroxyquinidine, PO	0.39	0.78	0.50	0.98	2.90	0.34	Damkier 1999
16976	Verapamil, 80 mg, PO, TID	Quinidine, PO	1.48	1.21	1.22	1.19	0.96	1.24	Edwards 1987
16978	Verapamil, 120 mg, PO, TID	Quinidine, PO	1.72	1.25	1.38	1.27	0.96	1.32	Edwards 1987
16980	Paroxetine, 20 mg, PO, QD	Alprazolam, PO	1.02	0.94	1.09	1.02	0.97	1.05	Calvo 2004
16986	Fluvoxamine, 50/100 mg, PO, QD/QD	Atomoxetine, PO	1.34	1.30	1.03	1.16	1.36	0.85	Todor 2017

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17078	Atomoxetine, 60 mg, PO, BID	Desipramine, PO	1.07	1.17	0.91	1.00	1.06	0.95	Sauer 2004
17096	Quinidine, 200 mg, PO, QD	Desipramine, PO	4.10	6.52	0.63	1.34	1.59	0.85	Brøsen 1989
17108	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.22	0.91	1.34	1.24	1.56	0.80	Chellingworth 1988
17110	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.15	1.34	0.86	1.19	1.62	0.73	Chellingworth 1988
17112	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.11	1.44	0.77	1.13	1.23	0.92	Toon 1988
17114	Cimetidine, 200/200/200/400 mg, PO, QD	Metoprolol, PO	1.09	1.64	0.66	1.06	1.56	0.68	Kirch 1982
17116	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	4.20	4.33	0.97	2.07	2.51	0.83	Stout 2011
17118	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	6.28	3.62	1.73	2.84	3.65	0.78	Stout 2011
17120	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.70	2.96	1.25	3.18	2.74	1.16	Parker 2011
17122	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.59	2.91	1.23	3.11	2.64	1.18	Parker 2011
17124	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.92	3.06	1.28	2.69	2.11	1.27	Parker 2011
17126	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.87	2.50	1.55	2.66	1.87	1.42	Parker 2011
17128	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.65	4.06	0.90	3.13	3.74	0.84	Parker 2011
17130	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.54	3.09	1.14	3.06	2.88	1.06	Parker 2011
17132	Paroxetine, 10 mg, PO, BID	R-Metoprolol, PO	4.22	9.88	0.43	2.30	2.55	0.90	Hemeryck 2000
17134	Paroxetine, 10 mg, PO, BID	S-Metoprolol, PO	3.77	7.01	0.54	2.04	1.71	1.19	Hemeryck 2000
17152	Rifampicin, 600 mg, PO, QD	Metoprolol, PO	0.84	0.74	1.14	1.00	0.66	1.52	Bennett 1982
17154	Atomoxetine, 60 mg, PO, BID	Midazolam, PO	1.08	1.32	0.82	1.01	1.01	1.00	Sauer 2004
17156	Atomoxetine, 60 mg, PO, BID	Midazolam, PO	0.98	1.51	0.65	1.01	1.52	0.66	Sauer 2004

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17158	Paroxetine, 100 mg, PO, BID	Paroxetine, PO	1.17	1.65	0.71	1.03	1.40	0.74	Yasui-Furukori 2007
17160	Erythromycin, 250/250/250/250 mg, PO, QID	Quinidine, PO	1.34	1.19	1.13	1.09	1.39	0.78	Damkier 1999b
17162	Itraconazole, 200 mg, PO, QD	Quinidine, PO	1.72	2.05	0.84	1.42	1.61	0.88	Kaukonen 1997
17164	Itraconazole, 100 mg, PO, QD	Quinidine, PO	1.67	2.58	0.65	1.13	1.32	0.86	Damkier 1999b
17166	Ketoconazole, 200 mg, PO, QD	Risperidone, PO	1.28	1.49	0.86	1.55	1.30	1.19	Mahatthanatrakul 2012
17168	Ketoconazole, 200 mg, PO, QD	9-Hydroxyrisperidone, PO	1.06	0.54	1.97	1.22	0.54	2.27	Mahatthanatrakul 2012
17170	Rifampicin, 600 mg, PO, QD	Risperidone, PO	0.55	0.51	1.08	0.52	0.76	0.68	Kim 2018
17172	Rifampicin, 600 mg, PO, QD	9-Hydroxyrisperidone, PO	0.79	0.60	1.32	0.81	0.67	1.21	Kim 2018
17174	Rifampicin, 600 mg, PO, QD	Risperidone, PO	0.49	0.29	1.68	0.50	0.50	1.00	Mahatthanatrakul 2007
17176	Verapamil, 80/80/80 mg, PO, TID	Risperidone, PO	1.32	1.58	0.84	1.58	1.59	0.99	Nakagami 2005
17178	Verapamil, 80/80/80 mg, PO, TID	9-Hydroxyrisperidone, PO	1.23	1.58	0.78	1.27	1.59	0.80	Nakagami 2005

2.1 DDI Network Evaluations

Figure 1 shows the developed CYP2D6 DDGI network of interacting perpetrator and victim drugs. (a) Schematic illustration of the modeled interactions of CYP2D6 perpetrator and victim drugs. Black arrows indicate metabolism or transport, green arrows indicate induction, red solid lines indicate competitive inhibition, red dashed lines down-regulation (bupropion), noncompetitive inhibition (verapamil P-gp inhibition) or mechanism-based inactivation (others). (b-c) Drug-drug-(gene) interaction matrix for modeled interactions mediated by (a) CYP2D6 and (b) CYP3A4 and P-gp. Colors indicate categories according to the FDA Examples of Drugs that Interact with CYP Enzymes and Transporter Systems.⁴² Height of the grey ribbons indicates the number of clinical studies for the respective interaction covered by the network, numbers in brackets indicate the number of clinical interaction studies for the corresponding compound. CYP: cytochrome P450, P-gp: P-glycoprotein.

Figure 1: CYP2D6 DDGI modeling network

The following victim and/or perpetrator PBPK models were applied:

- **Alprazolam** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Alprazolam-Model/blob/master/Alprazolam-Model.json>
- **Atomoxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/blob/main/atomoxetine-model.json>
- **Bupropion** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Bupropion-Model/blob/main/Bupropion-Model.json>
- **Carbamazepine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Carbamazepine-Model/blob/main/Carbamazepine-Model.json>
- **Cimetidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Cimetidine-Model/blob/master/Cimetidine-Model.json>
- **Clarithromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/blob/master/Clarithromycin-Model.json>
- **(E)-Clomiphene** Model snapshot: [https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/\(E\)-clomiphene-DGI-Model.json](https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/(E)-clomiphene-DGI-Model.json)
- **Desipramine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Desipramine-Model/blob/main/Desipramine-Model.json>
- **Dextromethorphan** Model snapshot: https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/blob/main/dextromethorphan_aggregated_simulations.json
- **Digoxin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/blob/master/Digoxin.json>
- **Erythromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/blob/master/Erythromycin-Model.json>
- **Fluvoxamine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Fluvoxamine-Model/blob/master/Fluvoxamine-Model.json>
- **Itraconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/blob/master/Itraconazole-Model.json>
- **Ketoconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/blob/main/Ketoconazole-Model.json>
- **Metoprolol** Model file: <https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/blob/main/Metoprolol-Model.pksim5>
- **Mexiletine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Mexiletine-Model/blob/main/Mexiletine-Model.json>
- **Omeprazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Omeprazole-Model/blob/main/Omeprazole-Model.json>
- **Paroxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/blob/main/paroxetine-model.json>
- **Quinidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Quinidine-Model/blob/main/Quinidine-Model.json>
- **Rifampicin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/blob/master/Rifampicin-Model.json>
- **Risperidone** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Risperidone-Model/blob/main/risperidone-model.json>
- **Verapamil** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/blob/master/Verapamil-Model.json>

The following interaction scenarios were predicted and used to qualify the final DDGI network:

- Alprazolam as victim:
 - Paroxetine-alprazolam-DDI

- Atomoxetine as victim:
 - Bupropion-atomoxetine-DDGI
 - Fluvoxamine-atomoxetine-DDI
 - Paroxetine-atomoxetine-DDGI
- (E)-Clomiphene as victim:
 - Clarithromycin-clomiphene-DDGI
 - Paroxetine-clomiphene-DDGI
- Desipramine as victim:
 - Atomoxetine-desipramine-DDI
 - Bupropion-desipramine-DDI
 - Paroxetine-desipramine-DDGI
 - Quinidine-desipramine-DDI
- Dextromethorphan as victim:
 - Paroxetine-dextromethorphan-DDGI
 - Quinidine-dextromethorphan-DDI
- Digoxin as victim:
 - Quinidine-digoxin-DDI
- Metoprolol as victim:
 - Cimetidine-metoprolol-DDI
 - Paroxetine-metoprolol-DDI
 - Quinidine-metoprolol-DDGI
 - Rifampicin-metoprolol-DDI
- Mexiletine as victim:
 - Quinidine-mexiletine-DDGI
- Midazolam as victim:
 - Atomoxetine-midazolam-DDI
- Paroxetine as victim:
 - Itraconazole-paroxetine-DDI
 - Quinidine-paroxetine-DDI
- Quinidine as victim:
 - Carbamazepine-quinidine-DDI
 - Cimetidine-quinidine-DDI
 - Erythromycin-quinidine-DDI
 - Fluvoxamine-quinidine-DDI
 - Itraconazole-quinidine-DDI
 - Omeprazole-quinidine-DDI
 - Rifampicin-quinidine-DDI
 - Verapamil-quinidine-DDI
- Risperidone as victim:
 - Ketoconazole-risperidone-DDI
 - Rifampicin-risperidone-DDI
 - Verapamil-risperidone-DDI

The published DD(G)I studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report.

2.1.1 Paroxetine - Alprazolam DDI

The paroxetine-alprazolam interaction was evaluated using clinical DDI studies listed in [Table 1](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Calvo 2004	po	1 mg q.d.	European	f	22	+/- PAR, 20 mg q.d. po

Table 1: f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.2 Fluvoxamine - Atomoxetine DDI

The fluvoxamine-atomoxetine interaction was evaluated using clinical DDI studies listed in [Table 3](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Todor 2017	po	25 mg s.d.	European	m	18	+/- FLV, 50/100 mg q.d. po

Table 3: FLV: fluvoxamine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.3 Atomoxetine - Desipramine DDI

The atomoxetine-desipramine interaction was evaluated using clinical DDI studies listed in [Table 7](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Sauer 2004	po	50 mg s.d.	American	m	22	+/- ATO, 60 mg b.i.d. po

Table 7: ATO: atomoxetine, b.i.d.: twice daily, m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.4 Quinidine - Desipramine DDI

The quinidine-desipramine interaction was evaluated using clinical DDI studies listed in [Table 10](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Brösen 1989	po	100 mg s.d.	European	m	6	+/- QUI, 200 mg q.d. po

Table 10: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.5 Quinidine - Dextromethorphan DDI

The quinidine-dextromethorphan interaction was evaluated using clinical DDI studies listed in [Table 12](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Capon 1996	po	30 mg s.d.	European	m	6 NM	+/- QUI, 50 mg s.d. po
Schadel 1995	po	30 mg s.d.	American	f	5 NM	+/- QUI, 100 mg s.d. po

Table 12: f: female, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.6 Quinidine - Digoxin DDI

The quinidine-digoxin interaction was evaluated using clinical DDI studies listed in [Table 13](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Steiness 1980	iv	10 µg/kg s.d.	European	m	6	+/- QUI, 200 mg b.i.d. po
Ochs 1981	iv	1 mg s.d.	European	m	7	+/- QUI, 200 mg q.i.d. po

Table 13: b.i.d.: twice daily, m: male, iv: intravenous, N: number of study participants, po: oral, pop.: population used in simulations, q.i.d.: four times daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.7 Cimetidine - Metoprolol DDI

The cimetidine-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 14](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Chellingsworth 1988	po	100 mg s.d.	European	m	12	+/- CIM, 800 mg q.d. po
Chellingsworth 1988	po	100 mg b.i.d.	European	m	12	+/- CIM, 800 mg q.d. po
Toon 1988	po	100 mg b.i.d.	European	m	12	+/- CIM, 800 mg q.d. po
Kirch 1982	po	100 mg b.i.d.	European	m	6	+/- CIM, 200/400 mg q.i.d. po

Table 14: b.i.d.: twice daily, CIM: cimetidine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.8 Paroxetine - Metoprolol DDI

The paroxetine-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 15](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Stout 2011	po	50 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Stout 2011	po	100 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Parker 2011	po	100 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	100 mg b.i.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	200 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Hemeryck 2000	po	100 mg s.d.	European	m	8	+/- PAR, 10 mg b.i.d. po

Table 15: b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.9 Rifampicin - Metoprolol DDI

The rifampicin-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 17](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Bennett 1982	po	100 mg s.d.	European	m	12	+/- RIF, 600 mg q.d. po

Table 17: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.10 Atomoxetine - Midazolam DDI

The atomoxetine-midazolam interaction was evaluated using clinical DDI studies listed in [Table 19](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Sauer 2004	po	5 mg s.d.	American	f	8	+/- ATO, 60 mg b.i.d. po

Table 19: ATO: atomoxetine, b.i.d.: twice daily, f: female, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.11 Itraconazole - Paroxetine DDI

The itraconazole-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 20](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Yasui-Furukori 2007	po	20 mg s.d.	Japanese	f	13	+/- ITR, 100 mg b.i.d. po

Table 20: b.i.d.: twice daily, f: female, ITR: itraconazole, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.12 Quinidine - Paroxetine DDI

The quinidine-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 21](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Schoedel 2012	po	20 mg q.d.	American	m	14 NM	+/- QUI, 30 mg b.i.d. po

Table 21: m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.13 Carbamazepine - Quinidine DDI

The carbamazepine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 22](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Andreassen 2007	po	200 mg s.d.	European	m	10	+/- CBZ, 200/400 mg b.i.d. po

Table 22: b.i.d.: twice daily, CBZ: carbamazepine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.14 Cimetidine - Quinidine DDI

The cimetidine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 23](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kolb 1984	po	400 mg s.d.	American	m	9	+/- CIM, 300 mg q.d. po
Hardy 1988	po	400 mg s.d.	American	m	9	+/- CIM, 300 mg q.i.d. po

Table 23: CIM: cimetidine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.15 Erythromycin - Quinidine DDI

The erythromycin-quinidine interaction was evaluated using clinical DDI studies listed in [Table 24](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999b	po	200 mg s.d.	European	m	6	+/- ERY, 250 mg q.i.d. po

Table 24: ERY: erythromycin, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.16 Fluvoxamine - Quinidine DDI

The fluvoxamine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 25](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999a	po	200 mg s.d.	American	m	6	+/- FLV, 100 mg q.d. po

Table 25: FLV: fluvoxamine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.17 Itraconazole - Quinidine DDI

The itraconazole-quinidine interaction was evaluated using clinical DDI studies listed in [Table 26](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kaukonen 1997	po	100 mg s.d.	European	f	9	+/- ITR, 200 mg q.d. po
Damkier 1999b	po	200 mg s.d.	European	m	6	+/- ITR, 100 mg q.d. po

Table 26: f: female, ITR: itraconazole, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.18 Omeprazole - Quinidine DDI

The omeprazole-quinidine interaction was evaluated using clinical DDI studies listed in [Table 27](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Ching 1991	po	400 mg s.d.	European	m	8	+/- OME, 40 mg q.d. po

Table 27: m: male, N: number of study participants, OME: omeprazole, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.19 Rifampicin - Quinidine DDI

The rifampicin-quinidine interaction was evaluated using clinical DDI studies listed in [Table 28](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999	po	200 mg s.d.	European	m	6	+/- RIF, 600 mg q.d. po

Table 28: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.20 Verapamil - Quinidine DDI

The verapamil-quinidine interaction was evaluated using clinical DDI studies listed in [Table 29](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Edwards 1987	po	400 mg s.d.	European	m	6	+/- VER, 80 mg t.i.d. po
Edwards 1987	po	400 mg s.d.	European	m	6	+/- VER, 120 mg t.i.d. po

Table 29: m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose, t.i.d.: three times daily, VER: verapamil. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.21 Ketoconazole - Risperidone DDI

The ketoconazole-risperidone interaction was evaluated using clinical DDI studies listed in [Table 30](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mahatthanatrakul 2012	po	2 mg s.d.	Asian	m	10	+/- KET, 200 mg q.d. po

Table 30: KET: ketoconazole, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.22 Rifampicin - Risperidone DDI

The rifampicin-risperidone interaction was evaluated using clinical DDI studies listed in [Table 31](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kim 2008	po	1 mg s.d.	Asian	m	10	+/- RIF, 600 mg q.d. po
Mahatthanatrakul 2007	po	4 mg s.d.	Asian	m	10	+/- RIF, 600 mg q.d. po

Table 31: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.23 Verapamil - Risperidone DDI

The verapamil-risperidone interaction was evaluated using clinical DDI studies listed in [Table 32](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Nakagami 2005	po	1 mg s.d.	Japanese	m	12	+/- VER, 80 mg t.i.d. po

Table 32: m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose, t.i.d.: three times daily, VER: verapamil. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.2 DDI Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

2.2.1 Paroxetine - Alprazolam DDI

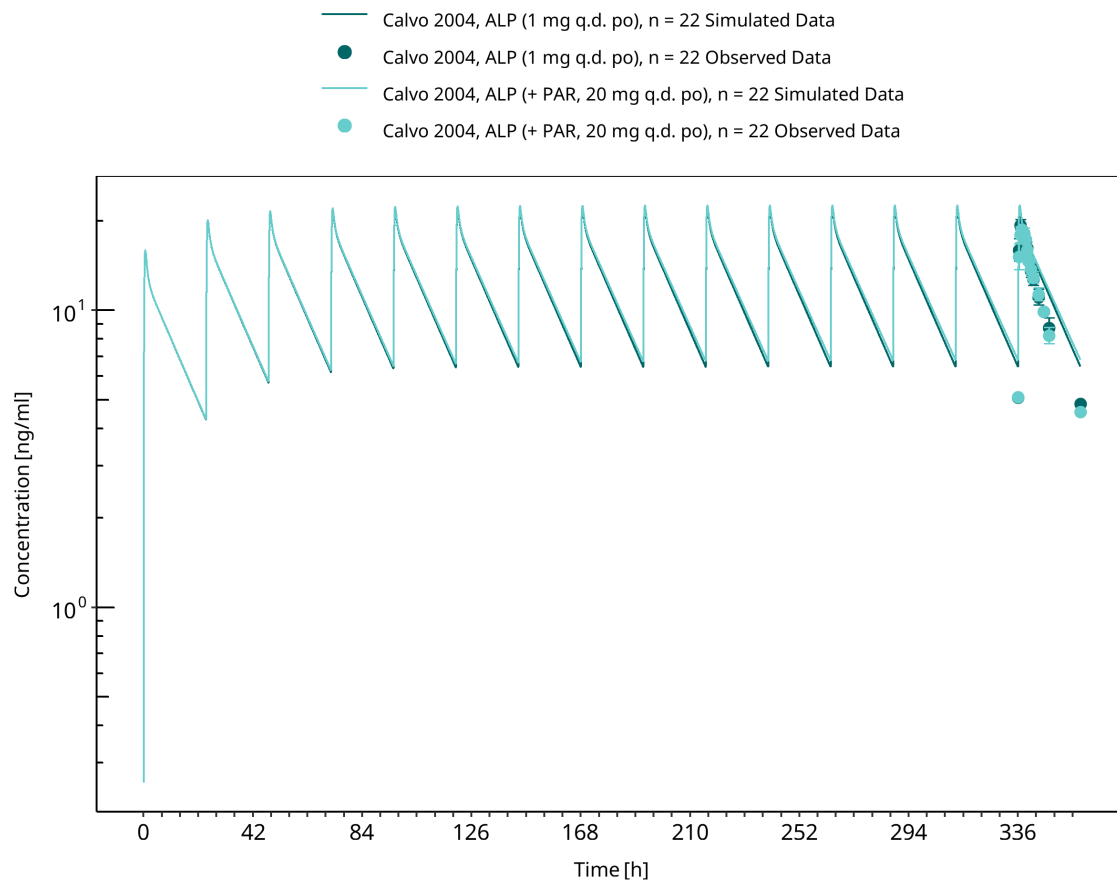


Figure 2-6: Paroxetine - Alprazolam DDI

2.2.2 Fluvoxamine - Atomoxetine DDI

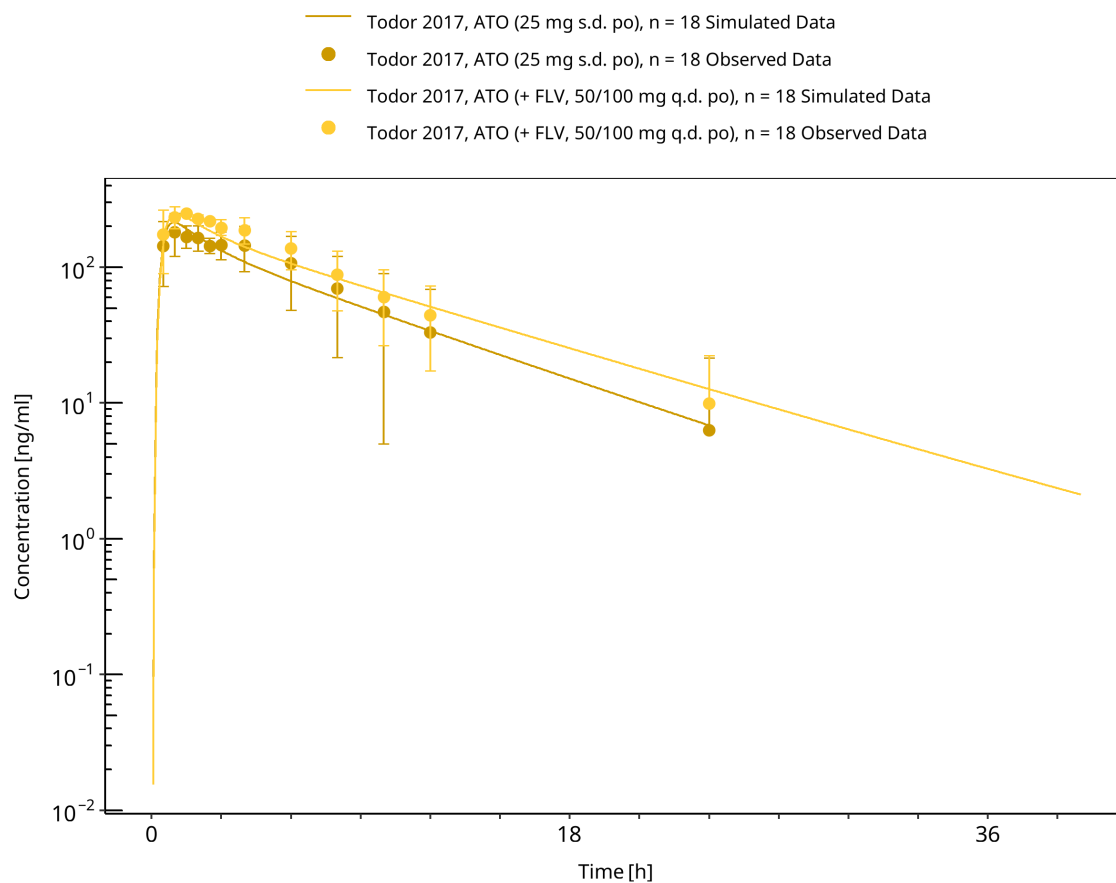


Figure 2-7: Fluvoxamine - Atomoxetine DDI

2.2.3 Atomoxetine - Desipramine DDI

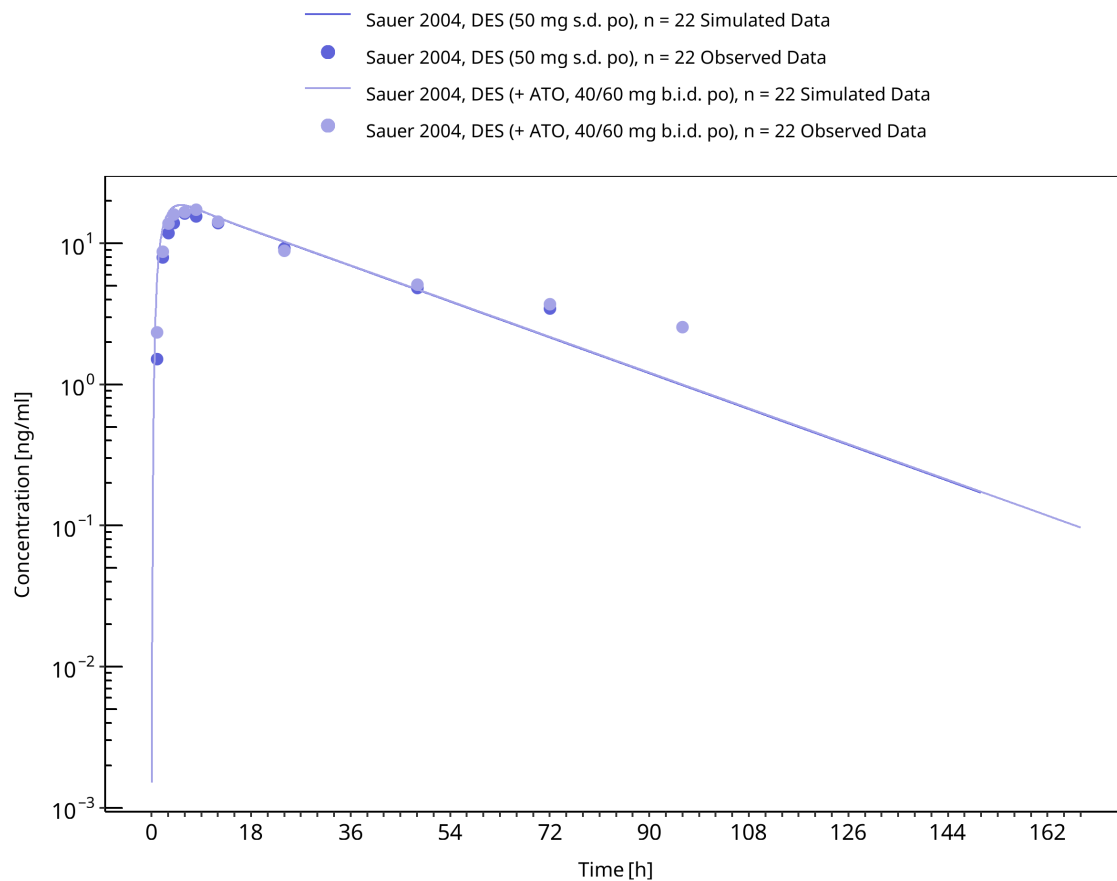


Figure 2-8: Atomoxetine - Desipramine DDI

2.2.4 Quinidine - Desipramine DDI

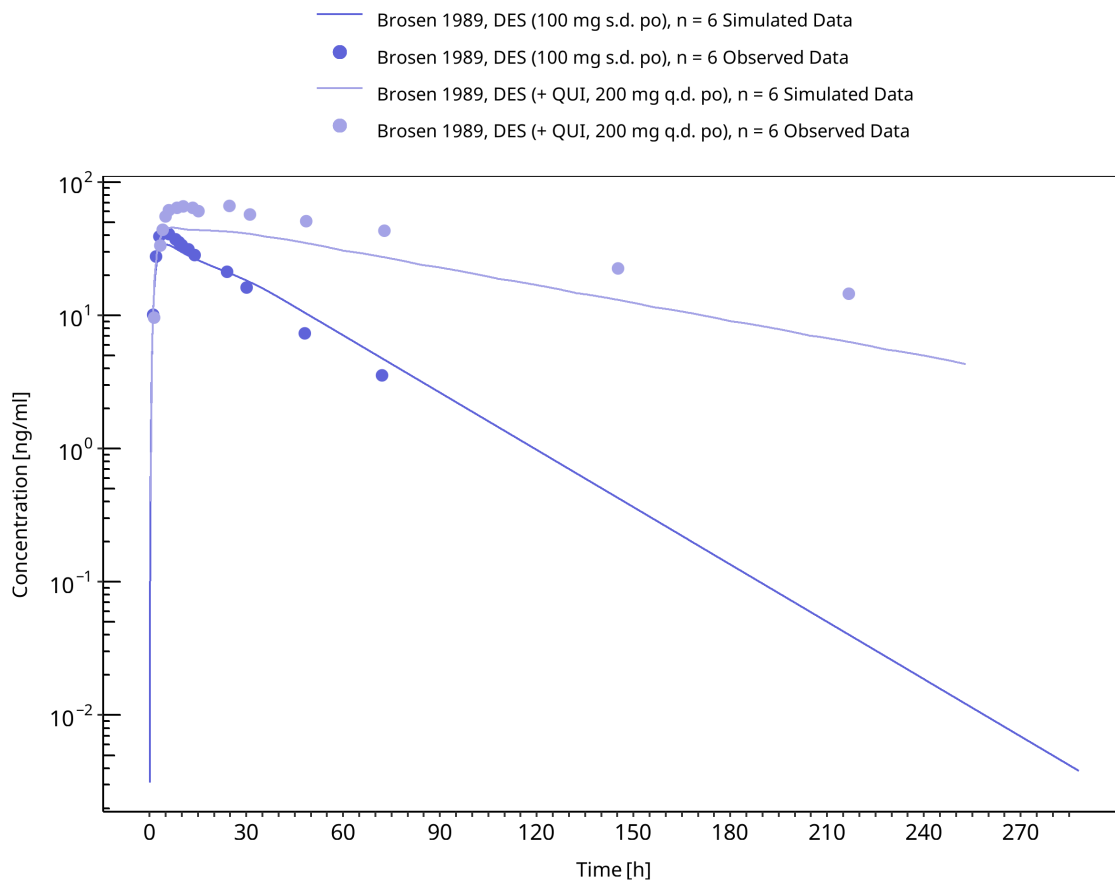


Figure 2-9: Quinidine - Desipramine DDI

2.2.5 Quinidine - Dextromethorphan DDI

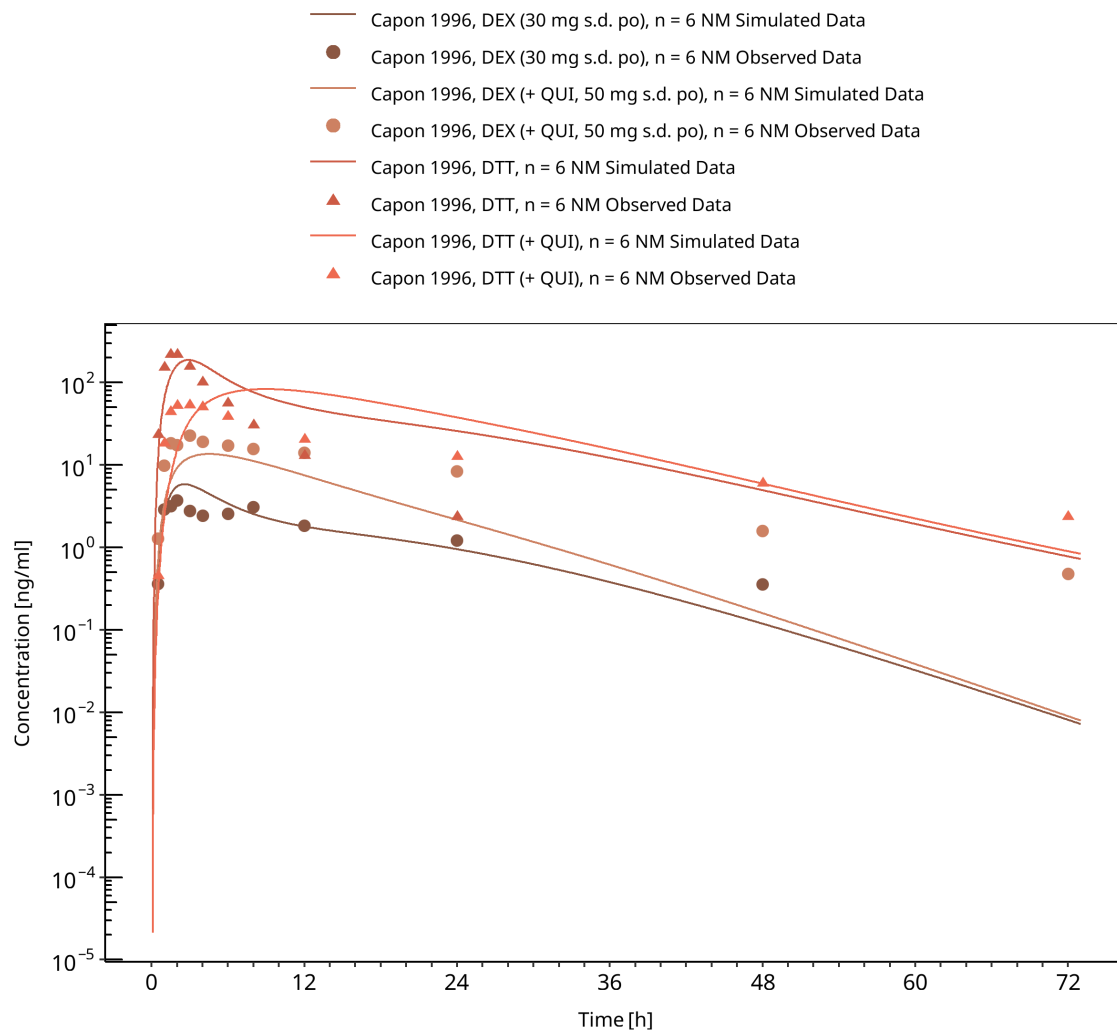


Figure 2-10: Quinidine - Dextromethorphan DDI

2.2.6 Quinidine - Digoxin DDI

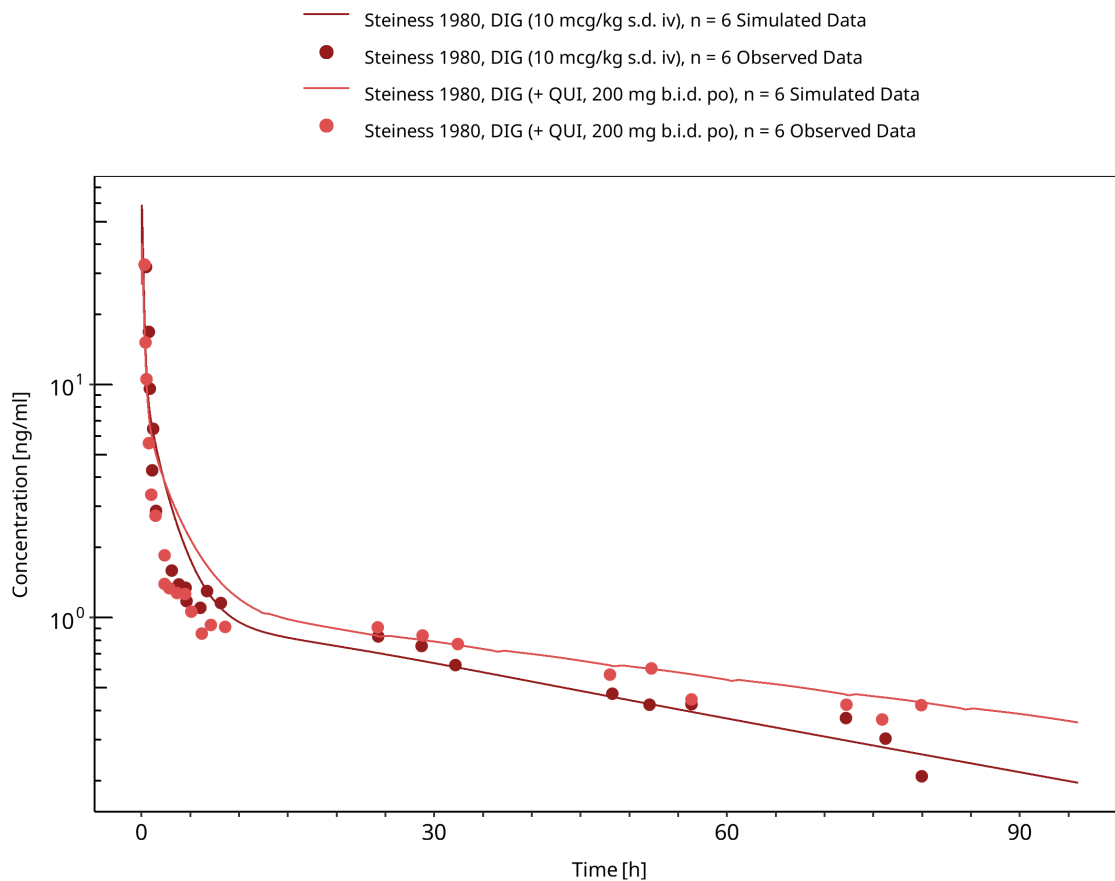


Figure 2-11: Quinidine - Digoxin DDI

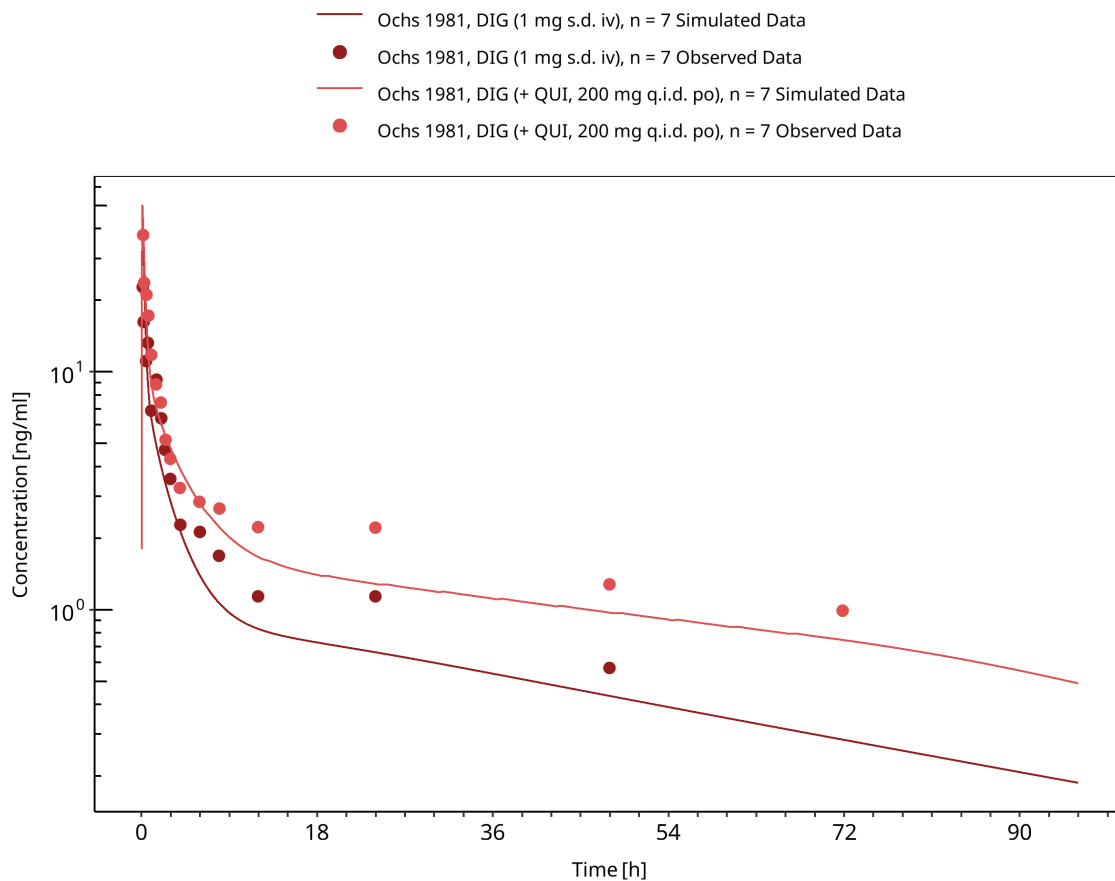


Figure 2-12: Quinidine - Digoxin DDI

2.2.7 Cimetidine - Metoprolol DDI

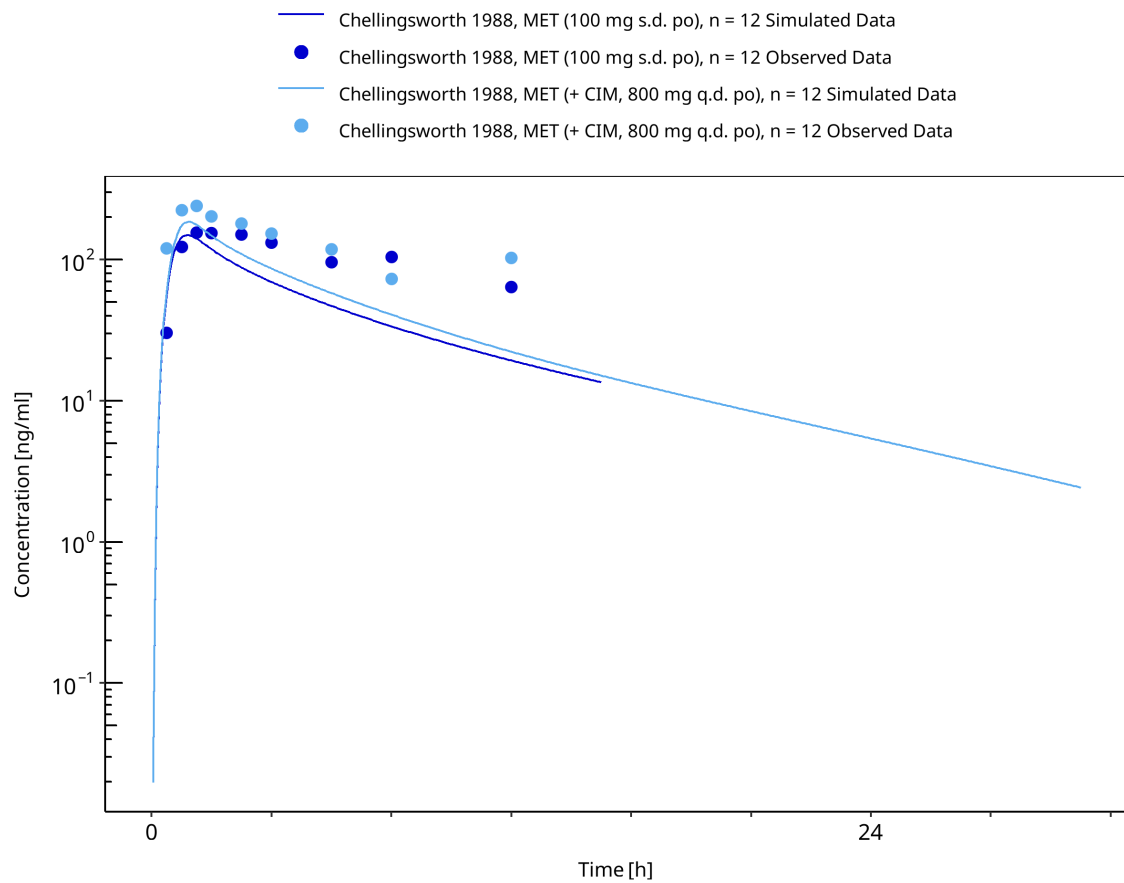


Figure 2-13: Cimetidine - Metoprolol DDI

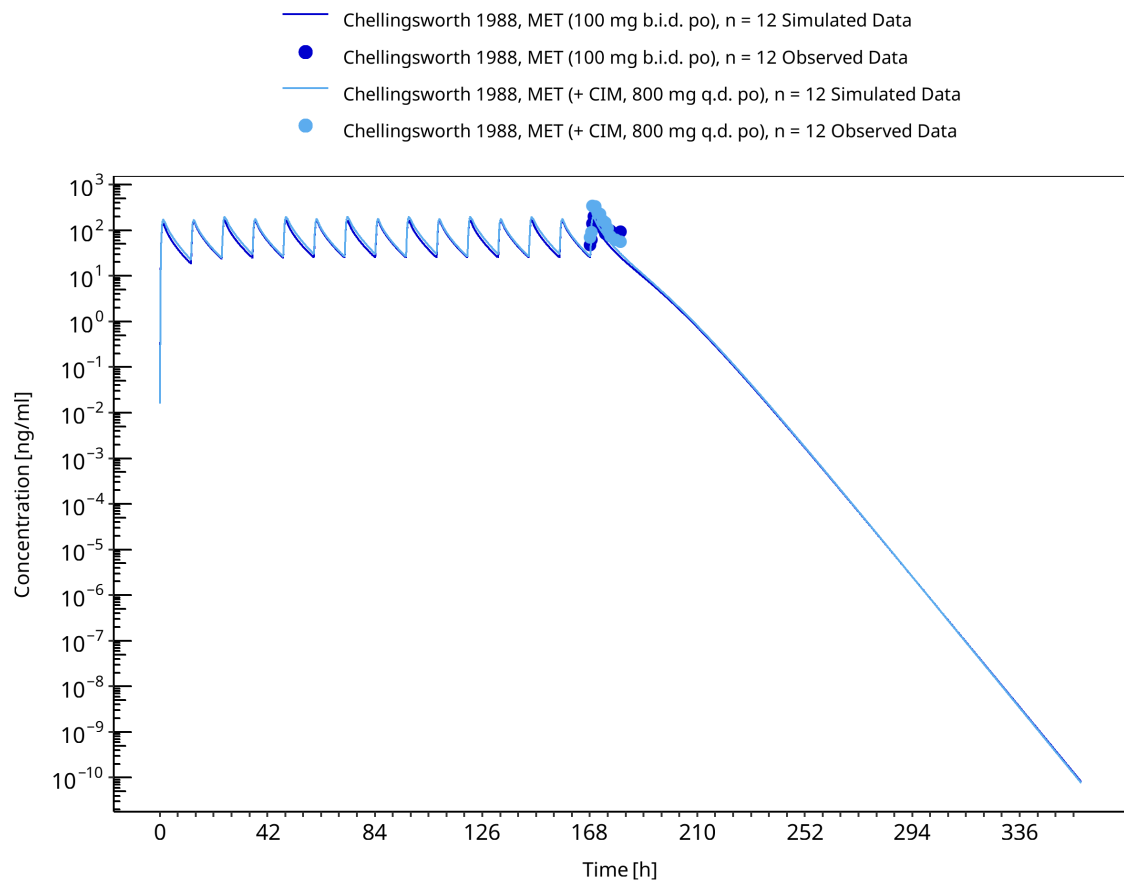


Figure 2-14: Cimetidine - Metoprolol DDI

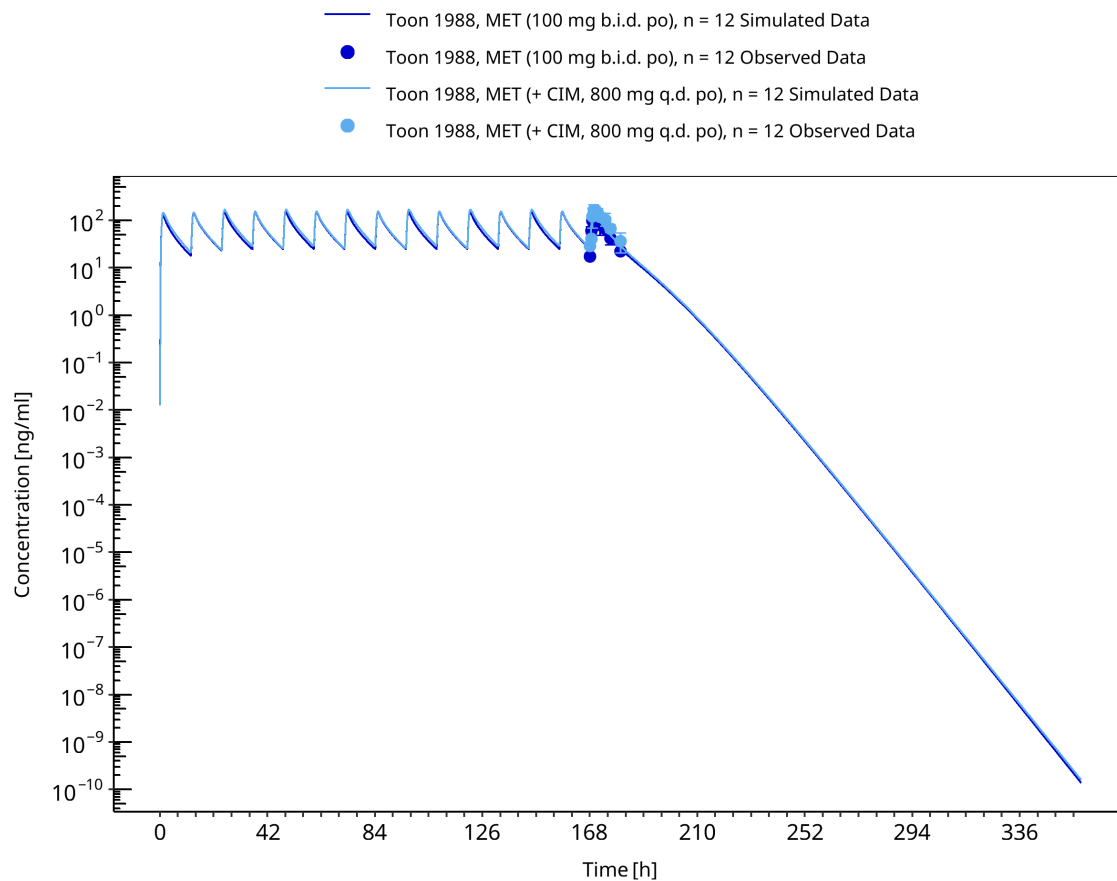


Figure 2-15: Cimetidine - Metoprolol DDI

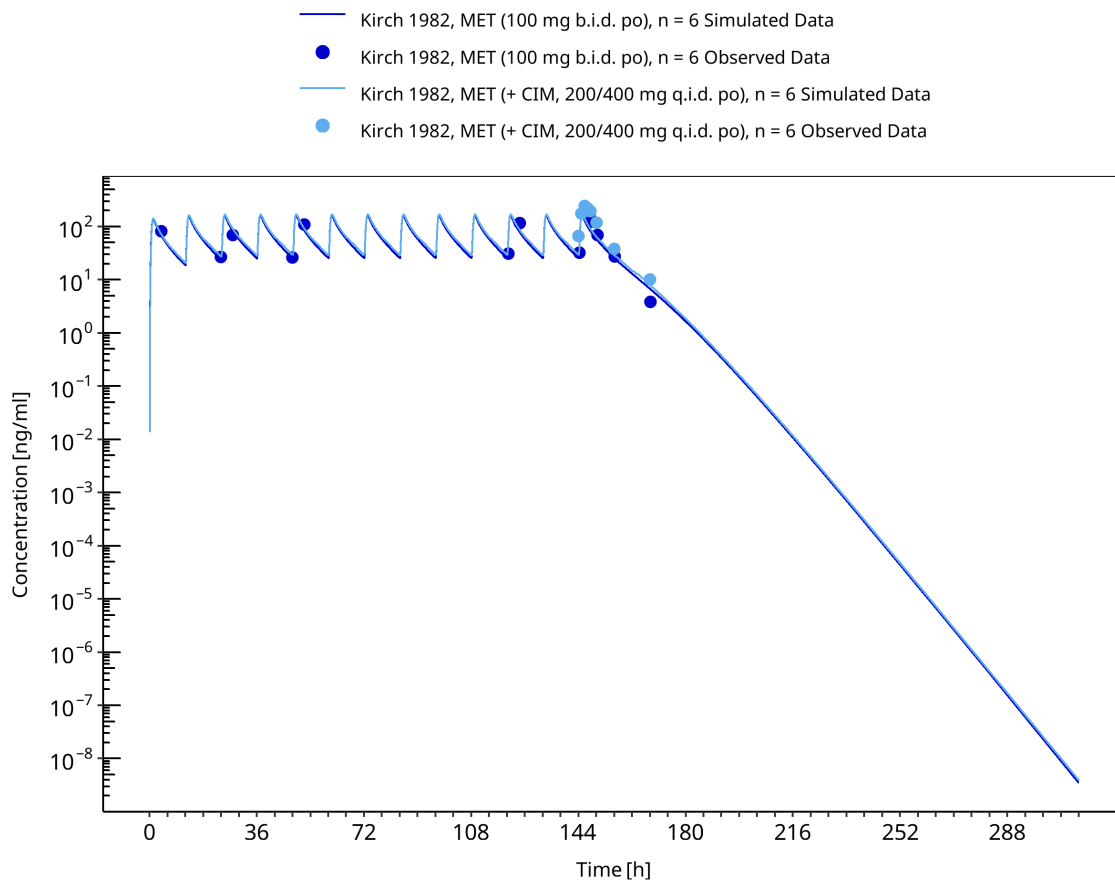


Figure 2-16: Cimetidine - Metoprolol DDI

2.2.8 Paroxetine - Metoprolol DDI

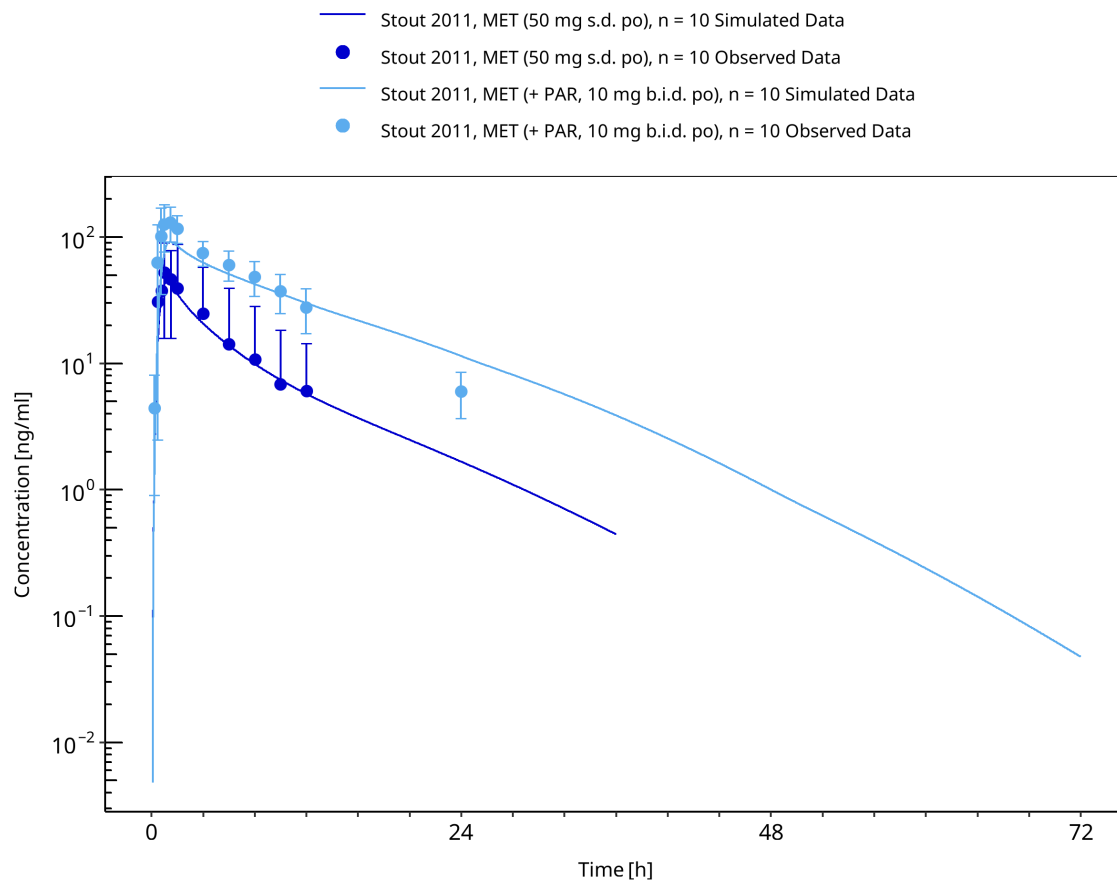


Figure 2-17: Paroxetine - Metoprolol DDI

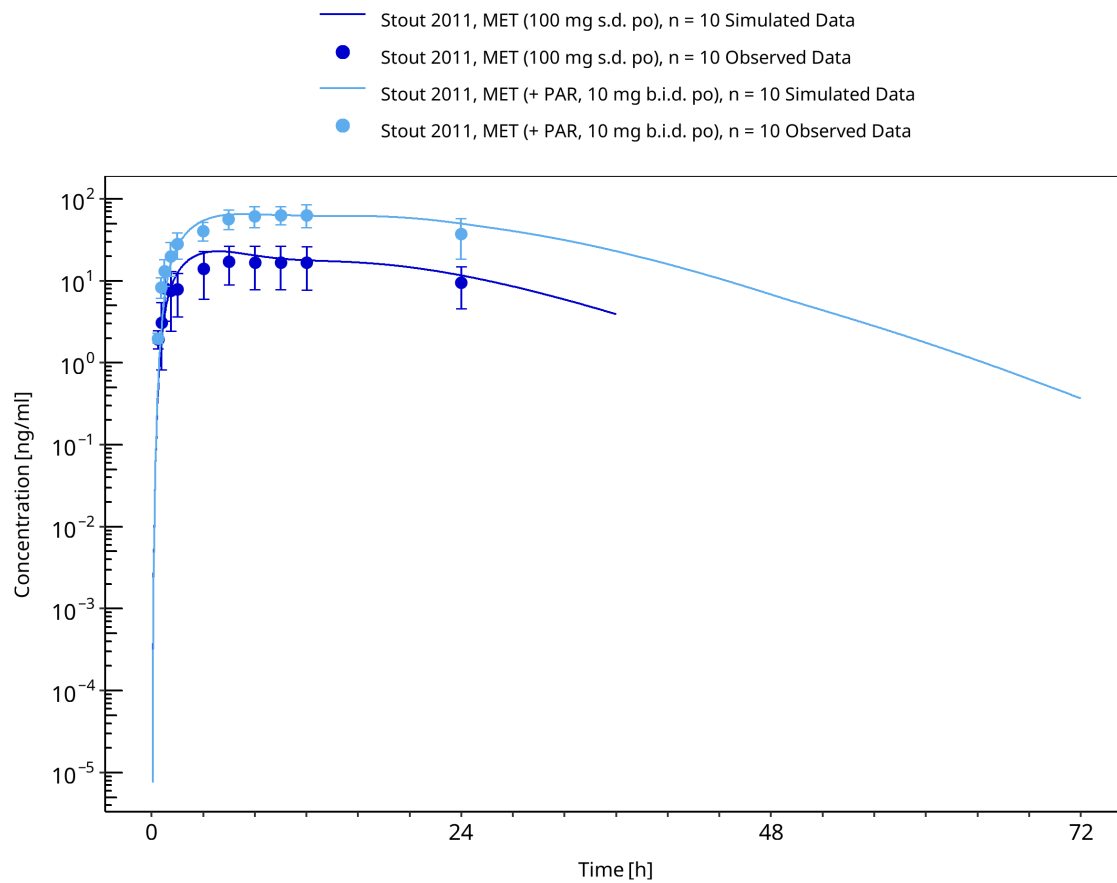


Figure 2-18: Paroxetine - Metoprolol DDI

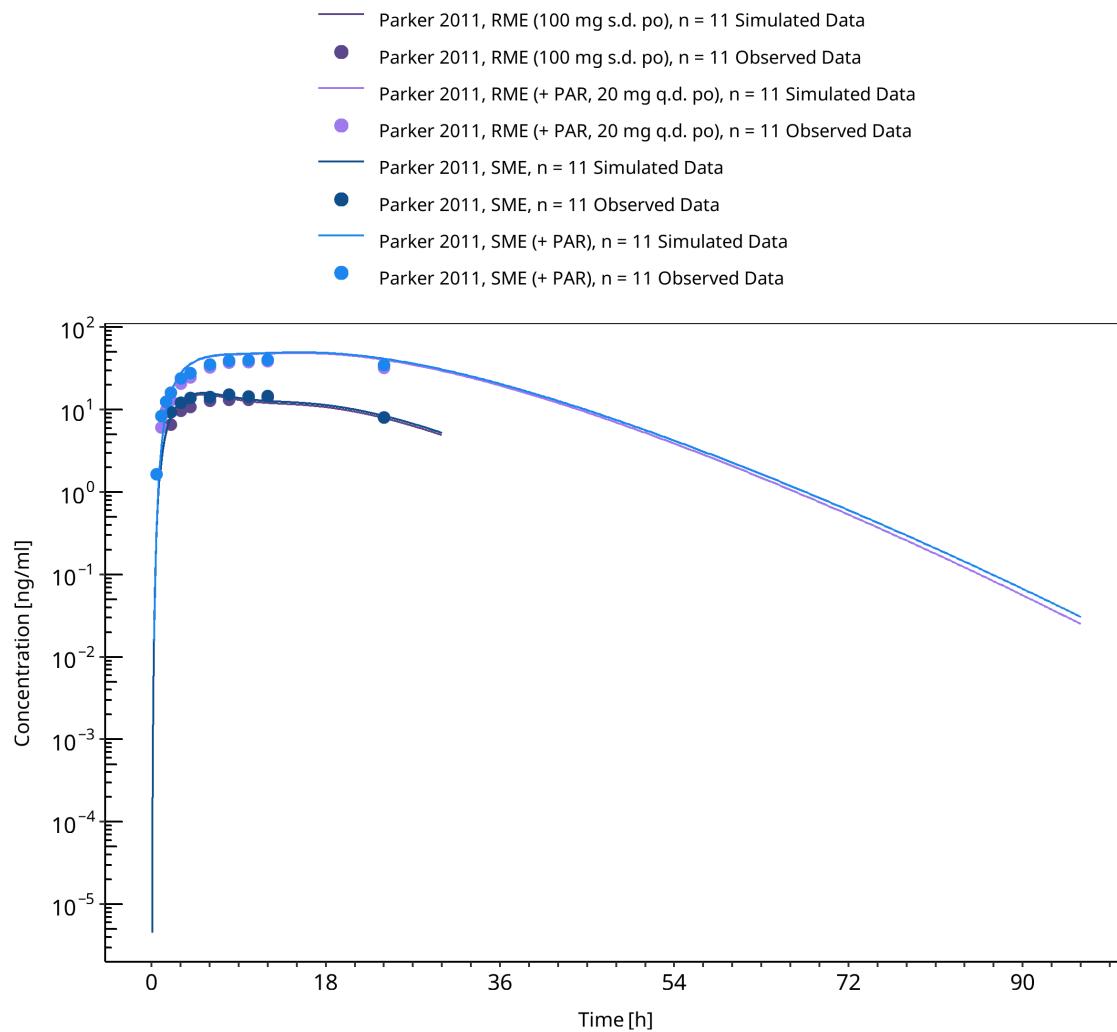


Figure 2-19: Paroxetine - Metoprolol DDI

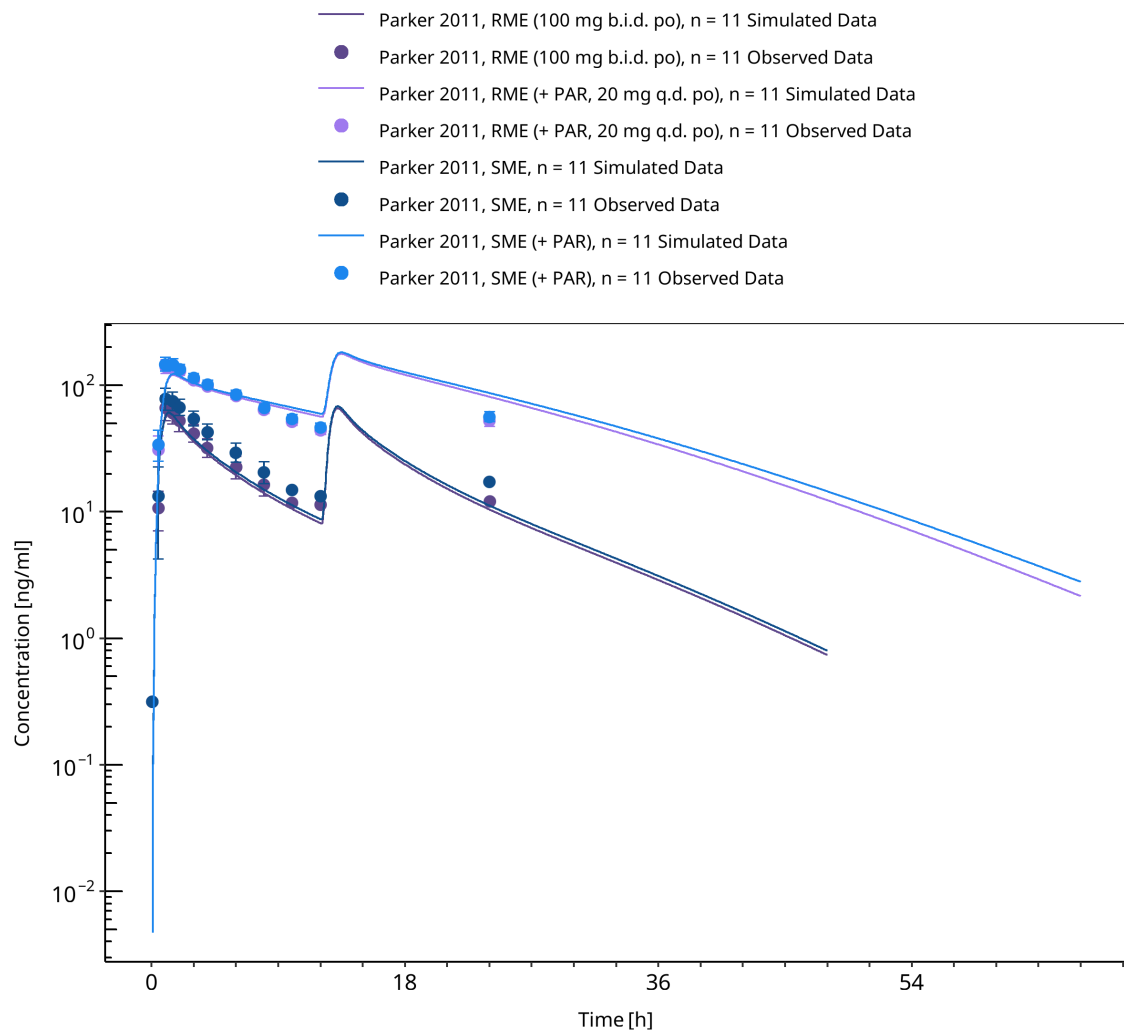


Figure 2-20: Paroxetine - Metoprolol DDI

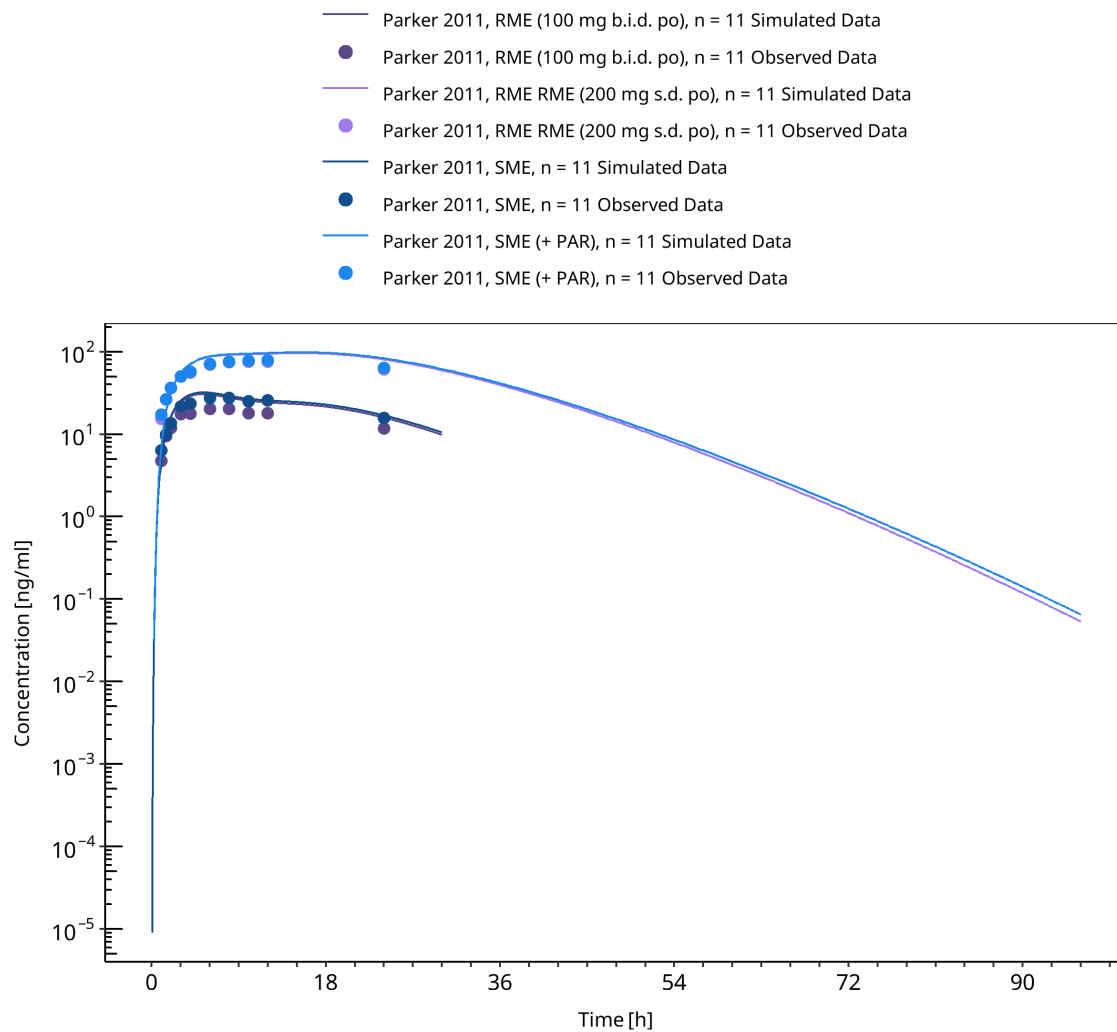


Figure 2-21: Paroxetine - Metoprolol DDI

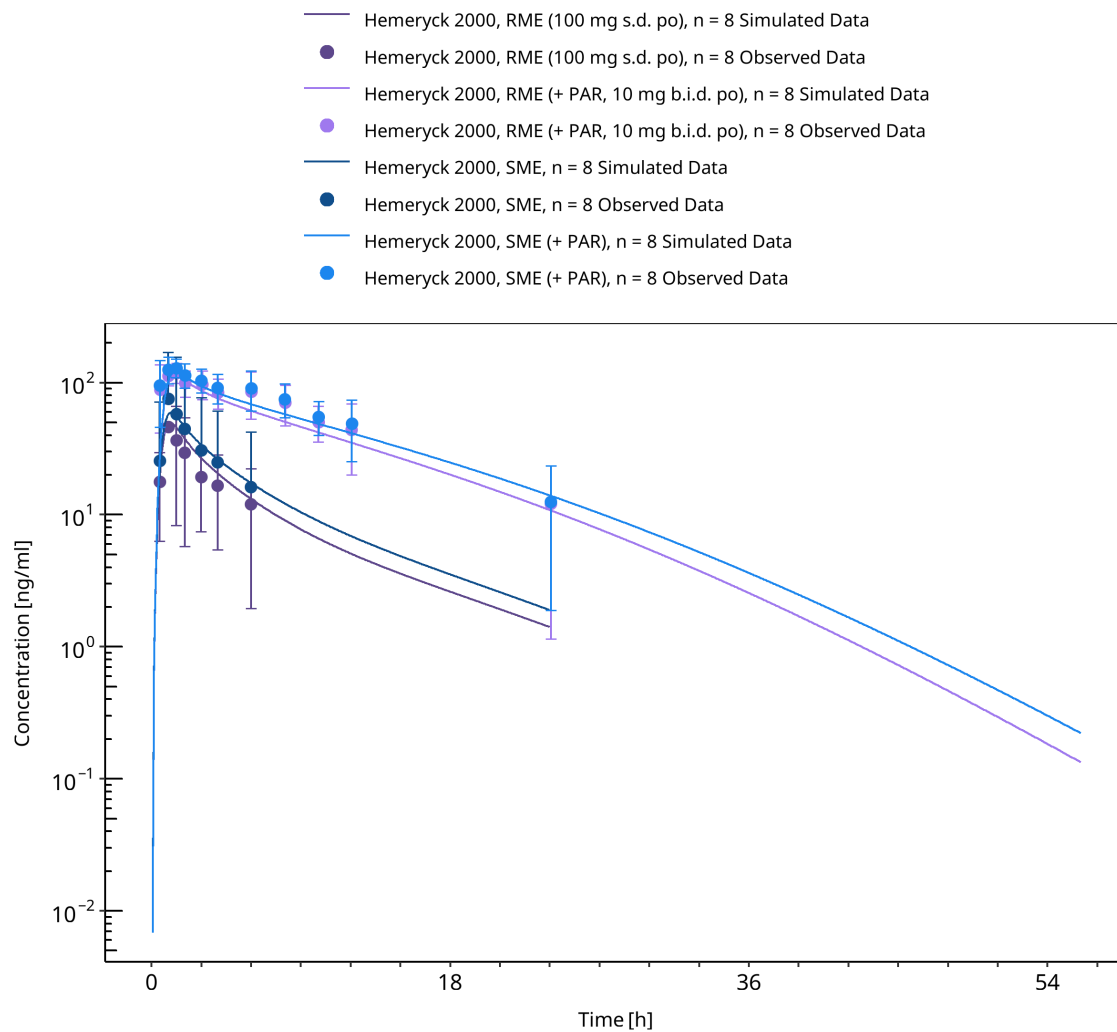


Figure 2-22: Paroxetine - Metoprolol DDI

2.2.9 Rifampicin - Metoprolol DDI

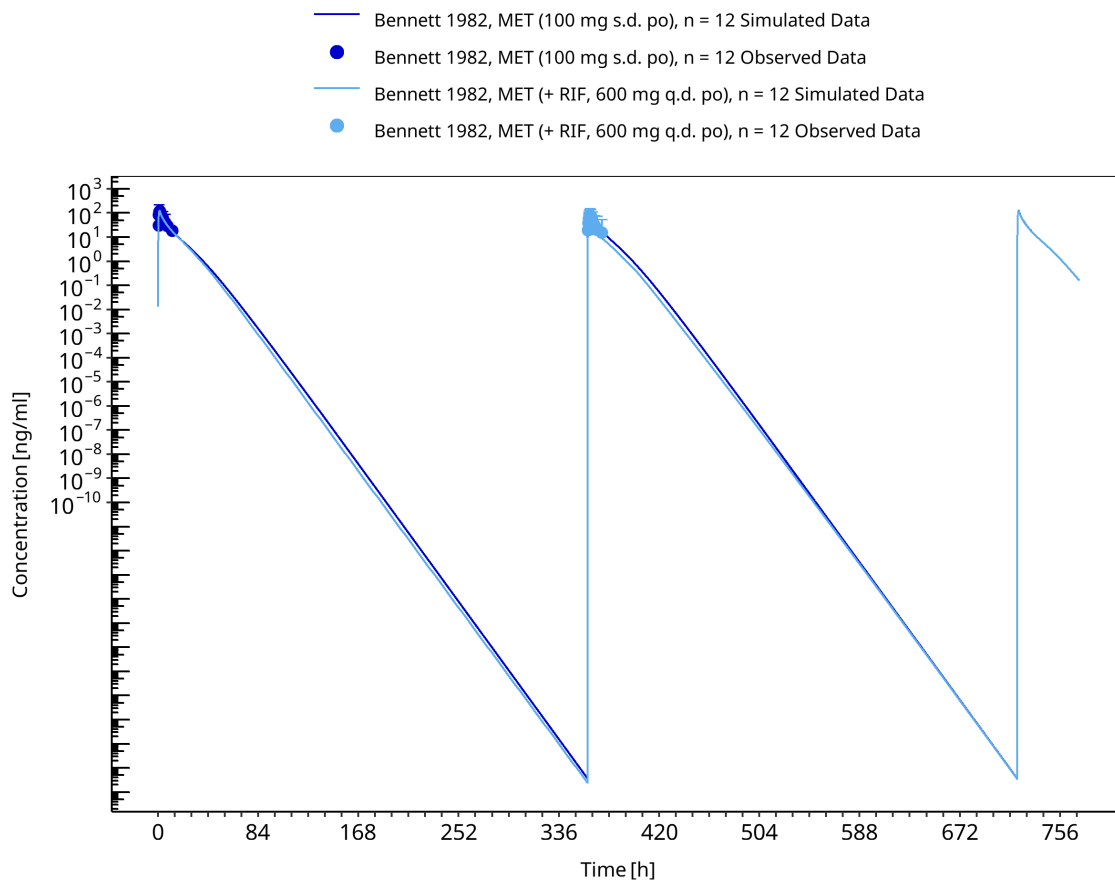


Figure 2-23: Rifampicin - Metoprolol DDI

2.2.10 Atomoxetine - Midazolam DDI

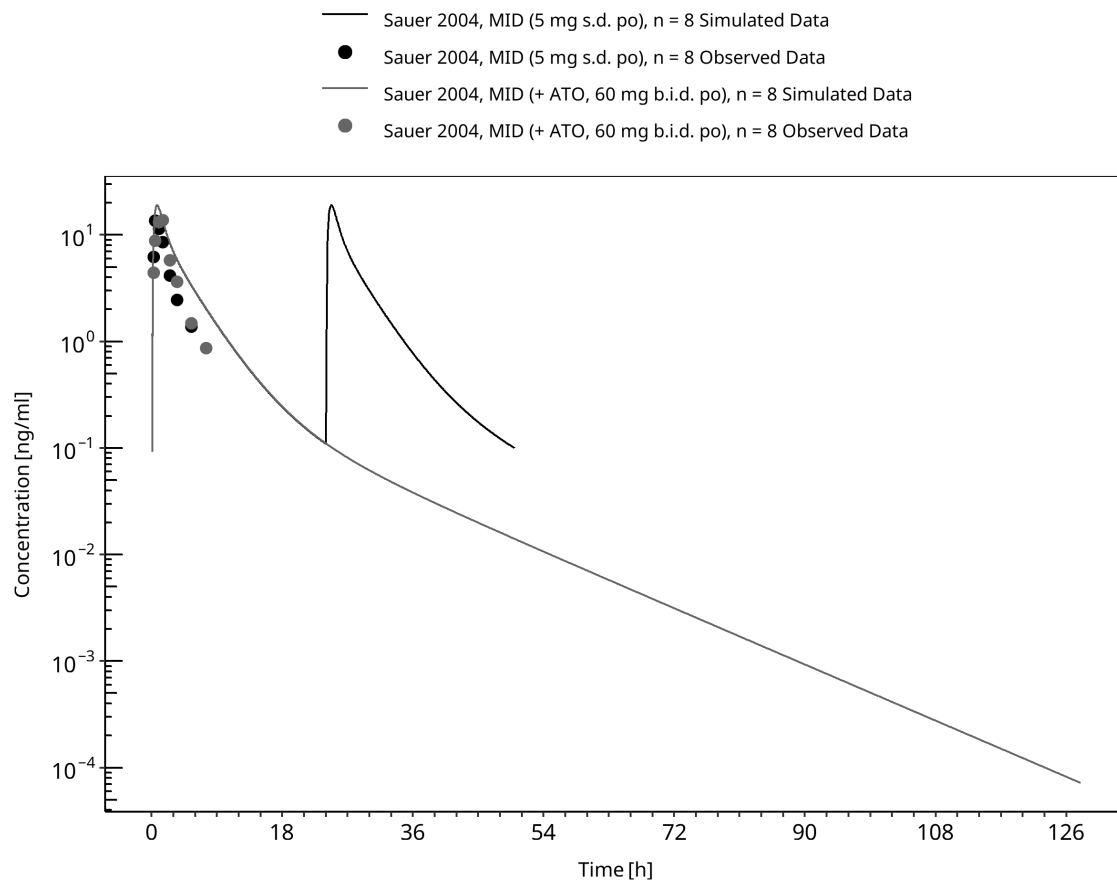


Figure 2-24: Atomoxetine - Midazolam DDI

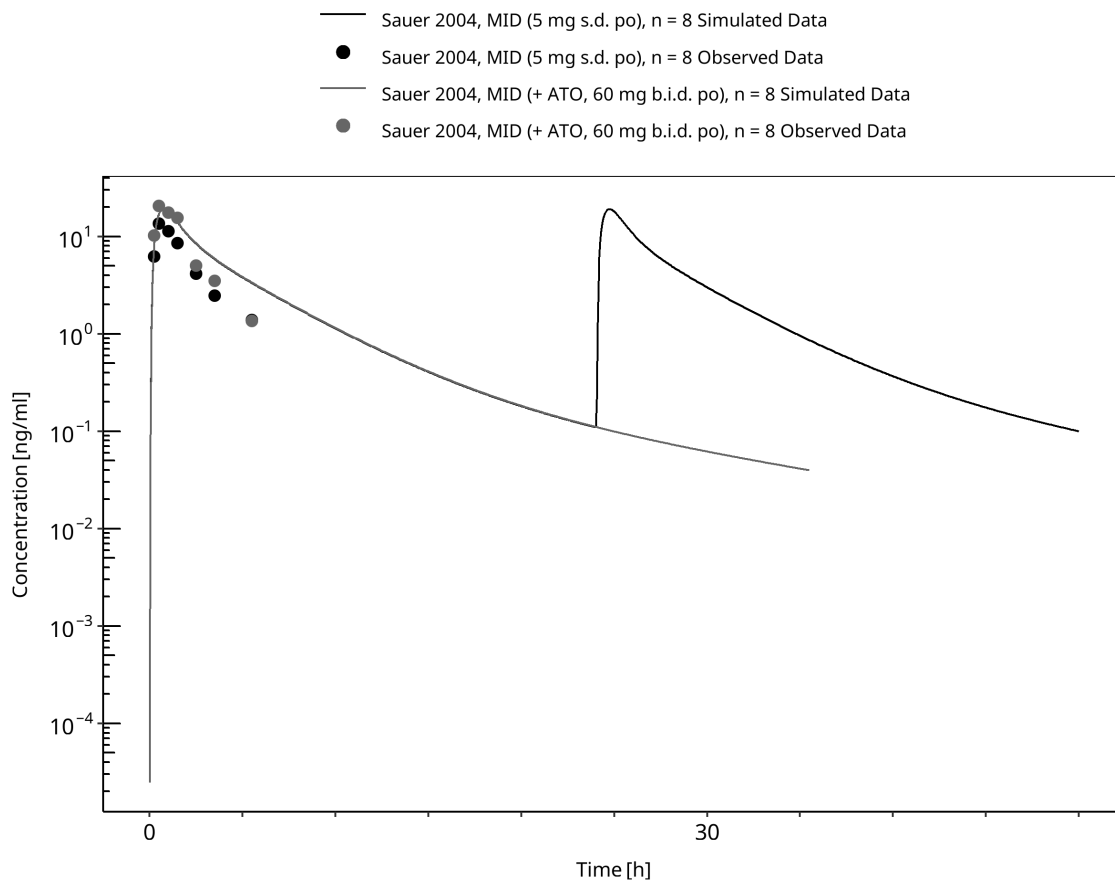


Figure 2-25: Atomoxetine - Midazolam DDI

2.2.11 Itraconazole - Paroxetine DDI

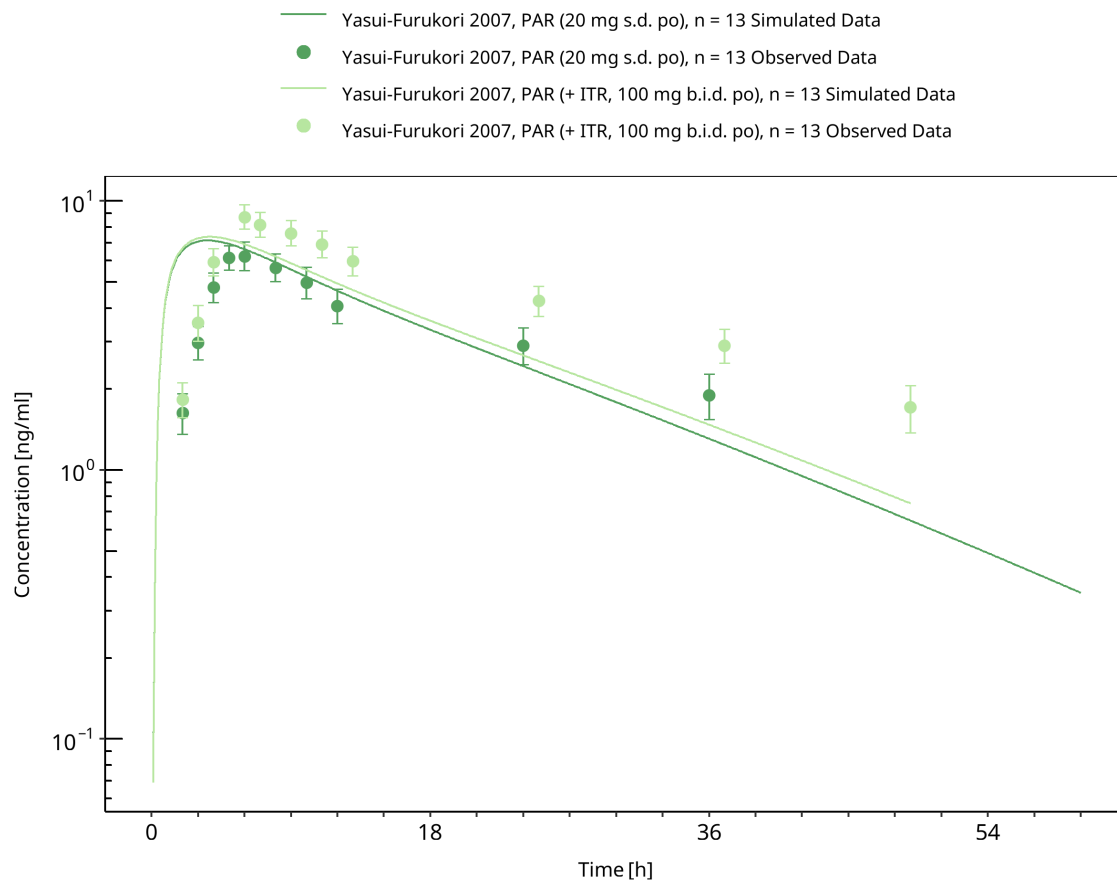


Figure 2-26: Itraconazole - Paroxetine DDI

2.2.12 Quinidine - Paroxetine DDI

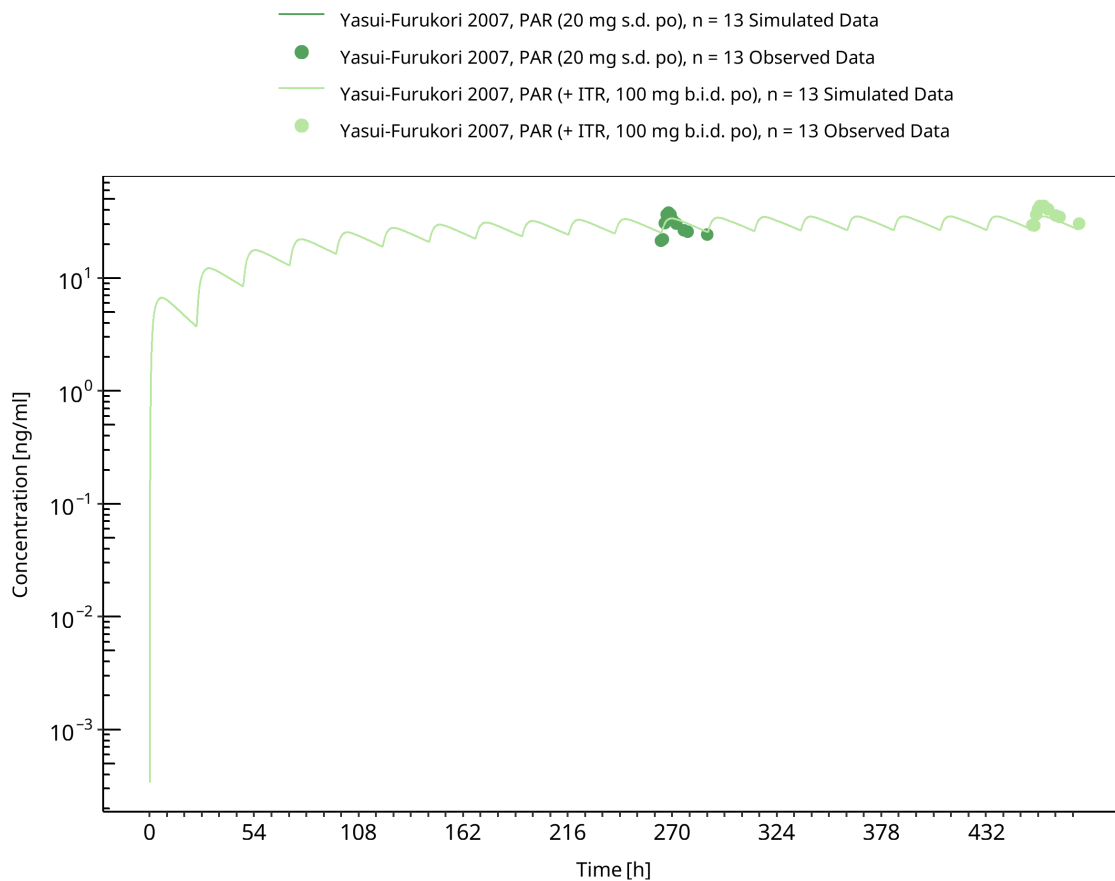


Figure 2-27: Quinidine - Paroxetine DDI

2.2.13 Carbamazepine - Quinidine DDI

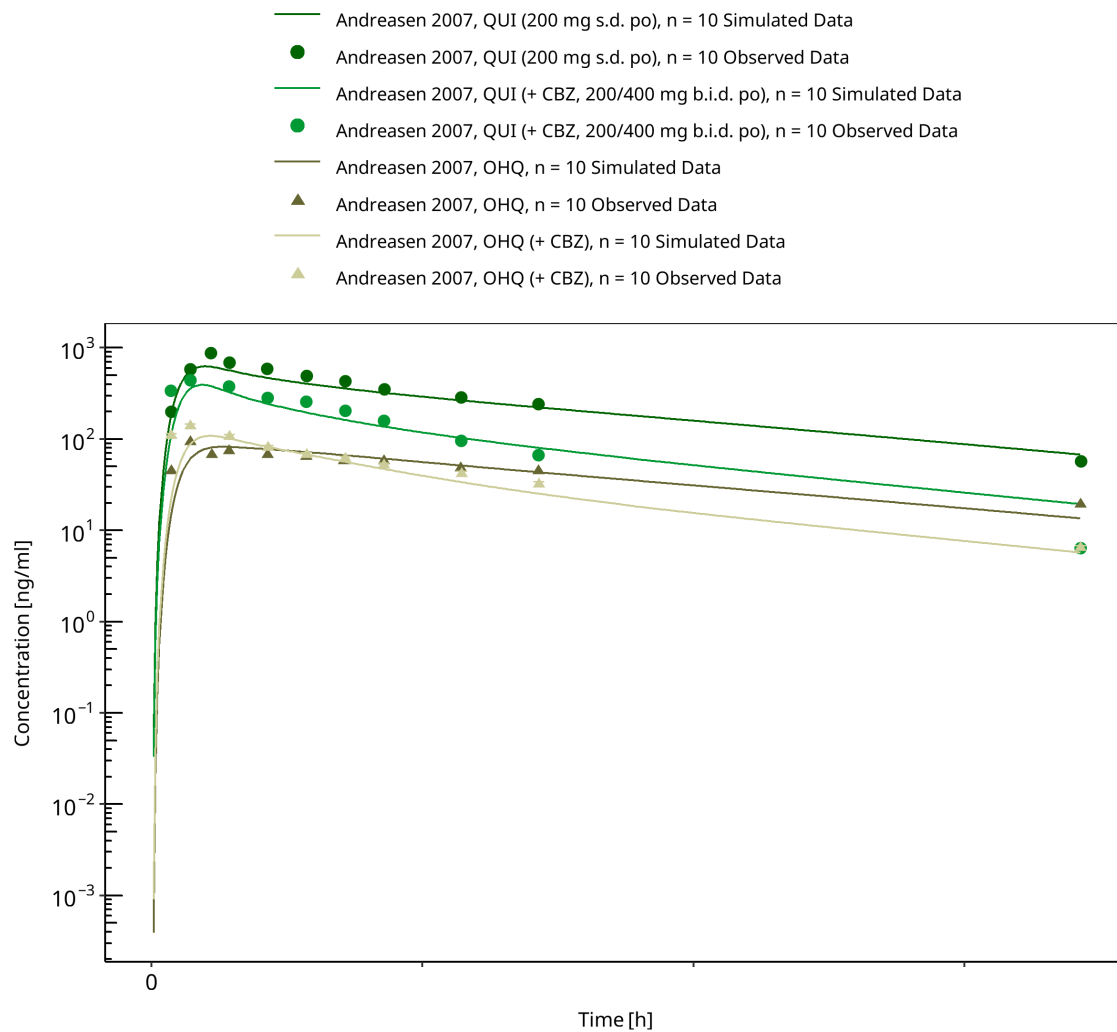


Figure 2-28: Carbamazepine - Quinidine DDI

2.2.14 Cimetidine - Quinidine DDI

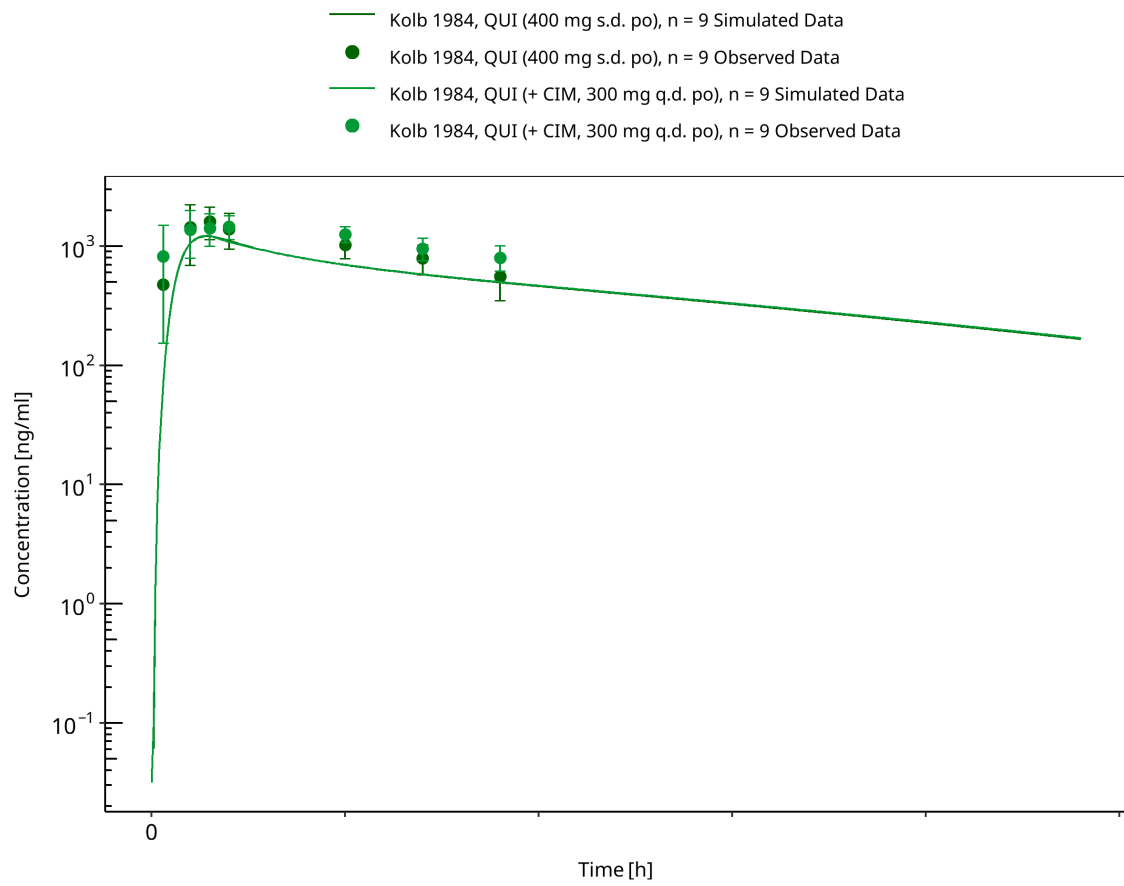


Figure 2-29: Cimetidine - Quinidine DDI

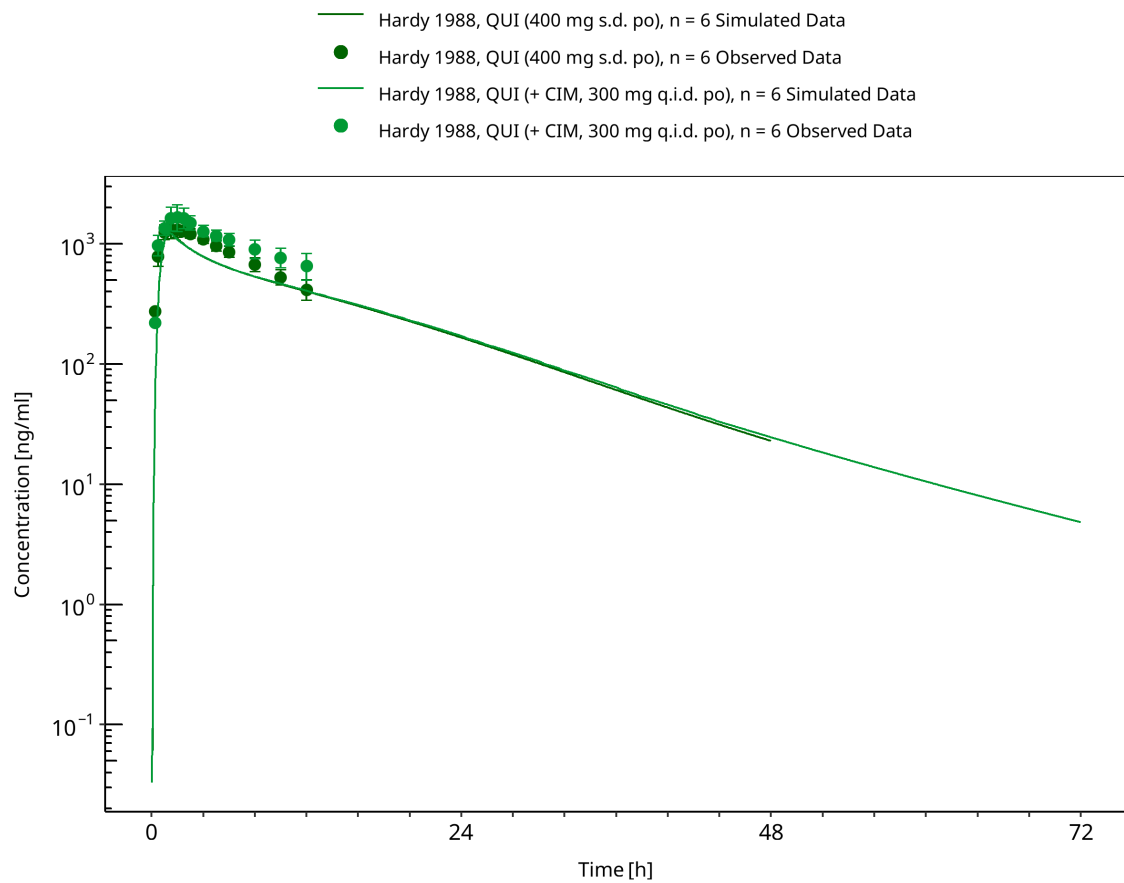


Figure 2-30: Cimetidine - Quinidine DDI

2.2.15 Erythromycin - Quinidine DDI

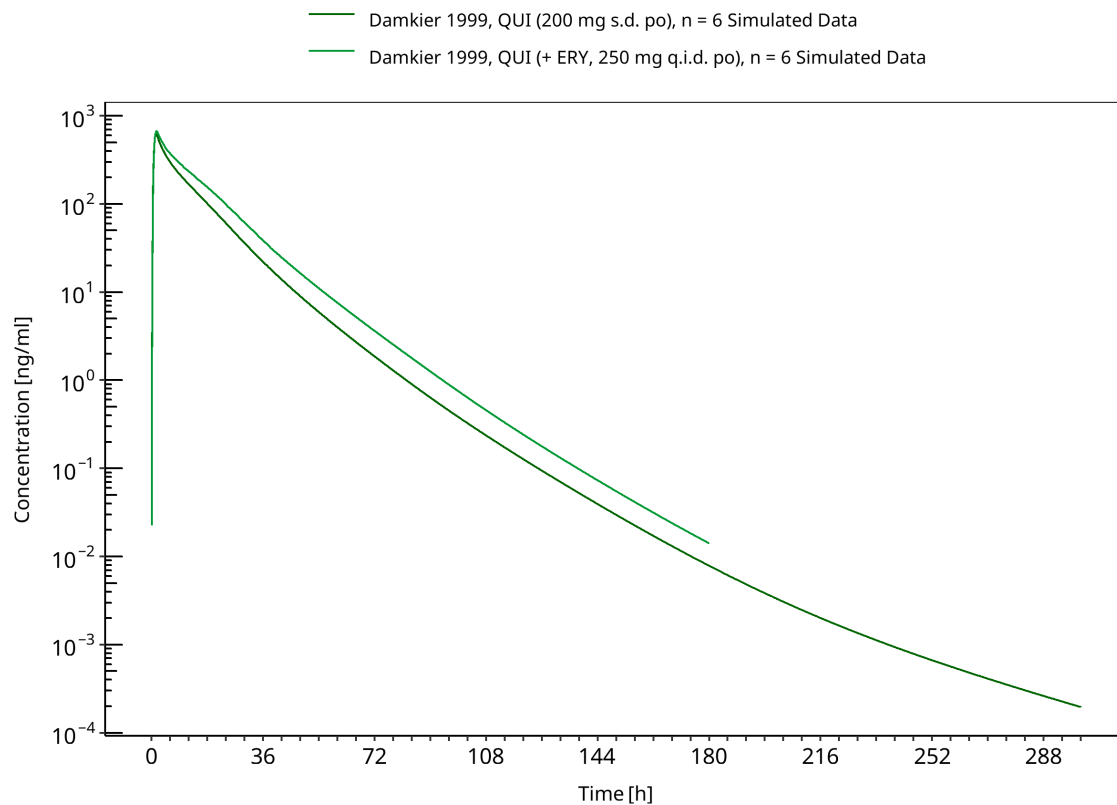


Figure 2-31: Erythromycin - Quinidine DDI

2.2.16 Fluvoxamine - Quinidine DDI

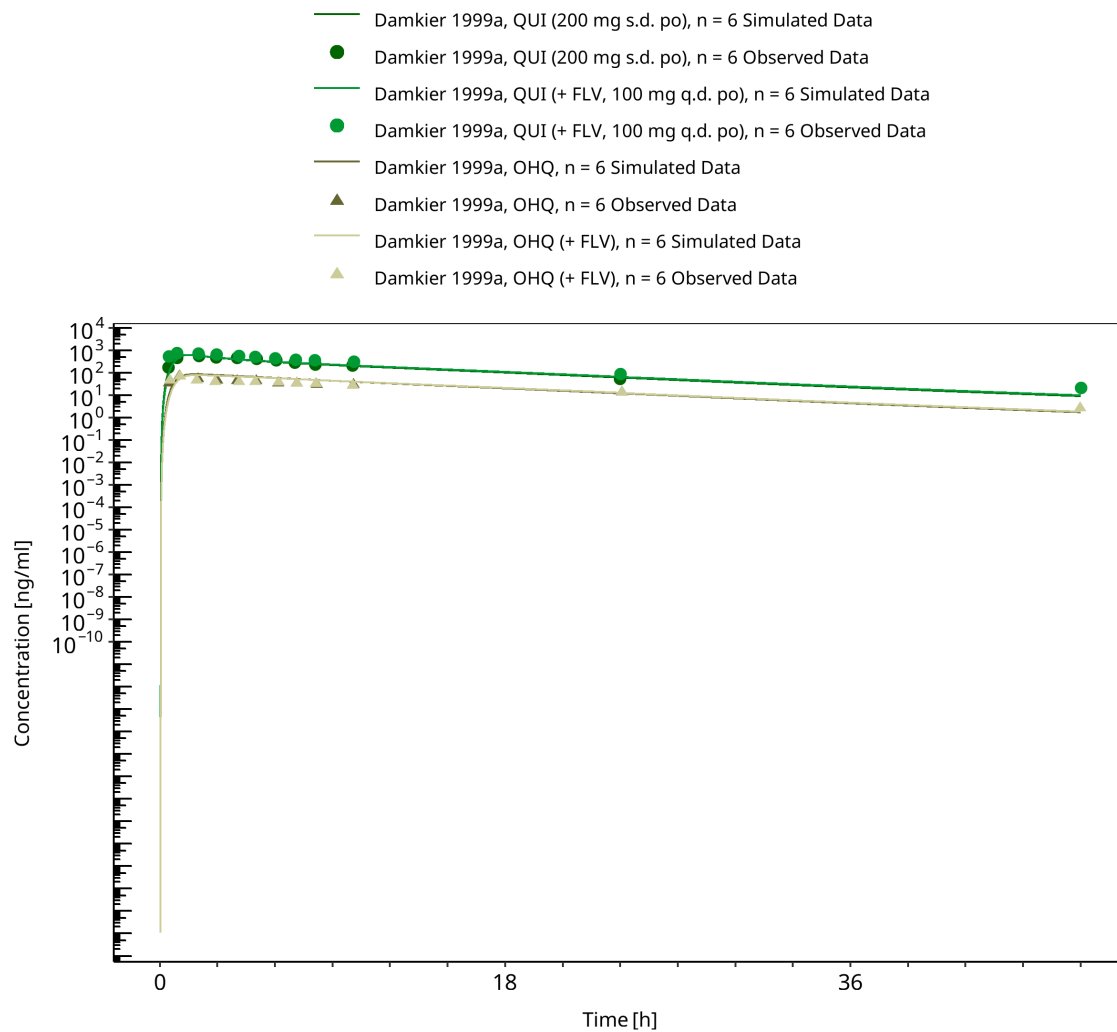


Figure 2-32: Fluvoxamine - Quinidine DDI

2.2.17 Itraconazole - Quinidine DDI

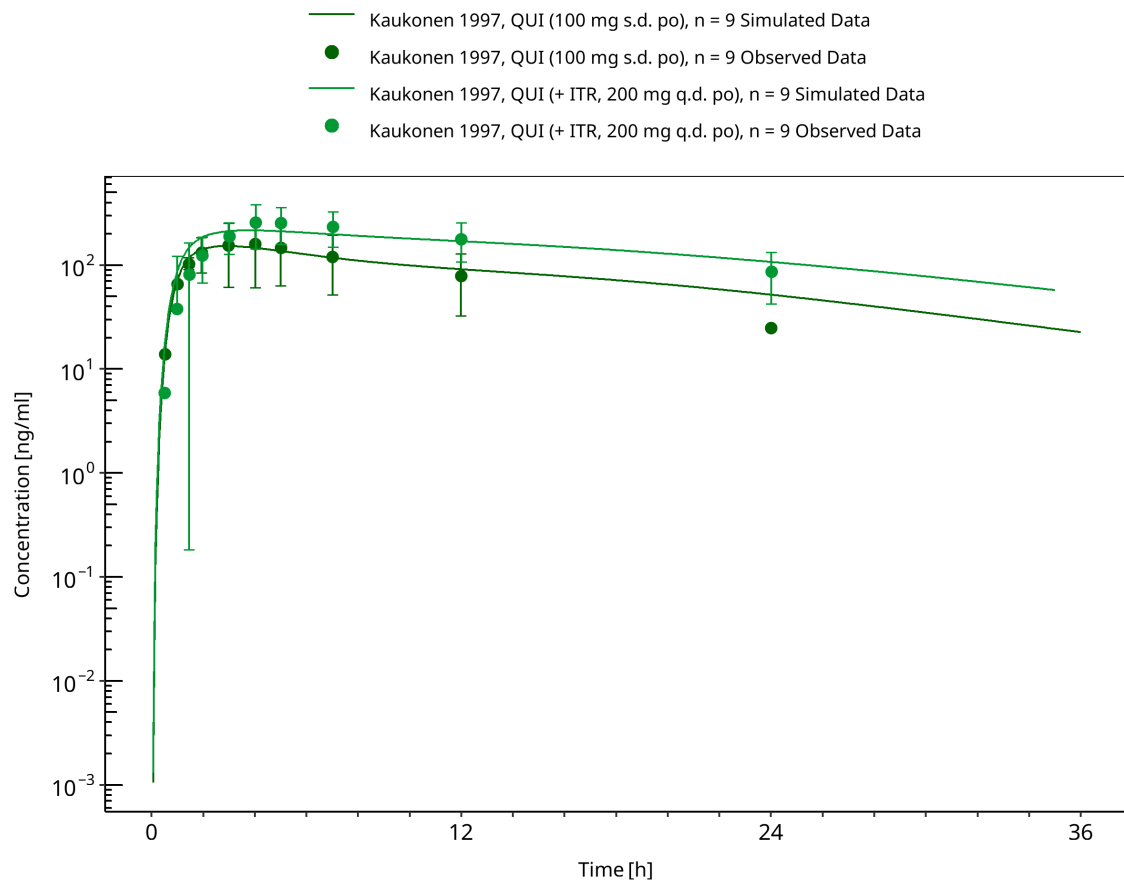


Figure 2-33: Itraconazole - Quinidine DDI

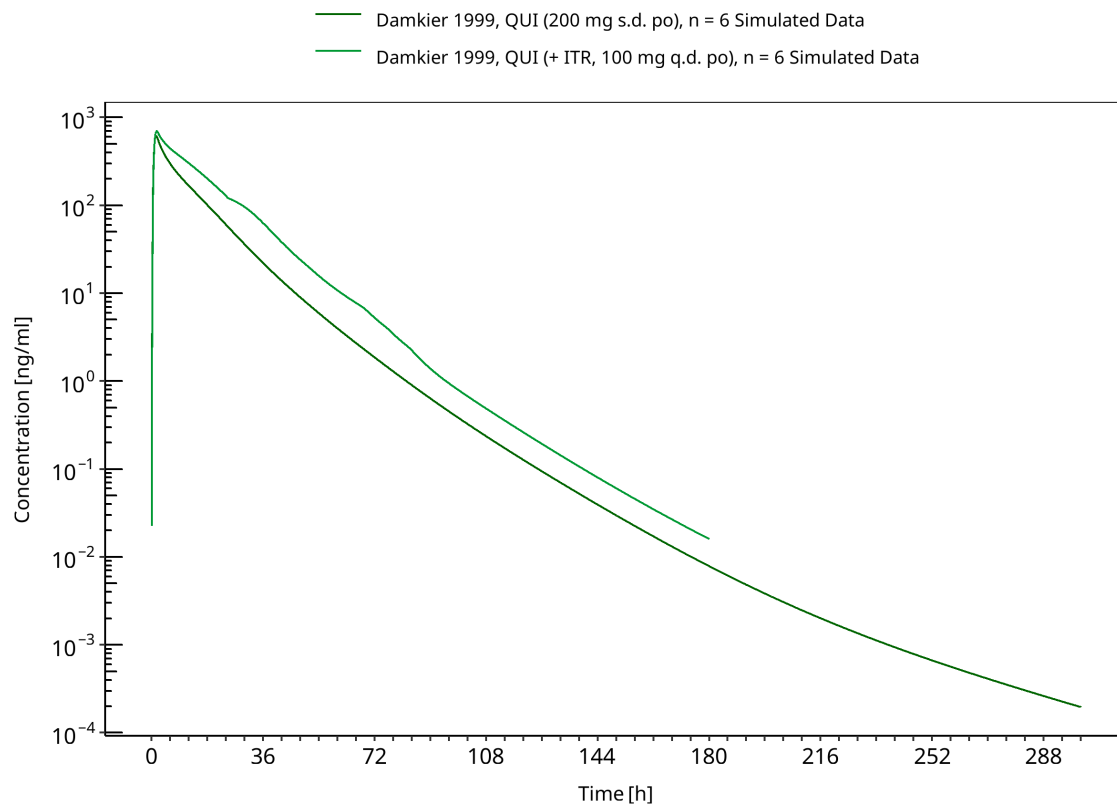


Figure 2-34: Itraconazole - Quinidine DDI

2.2.18 Omeprazole - Quinidine DDI

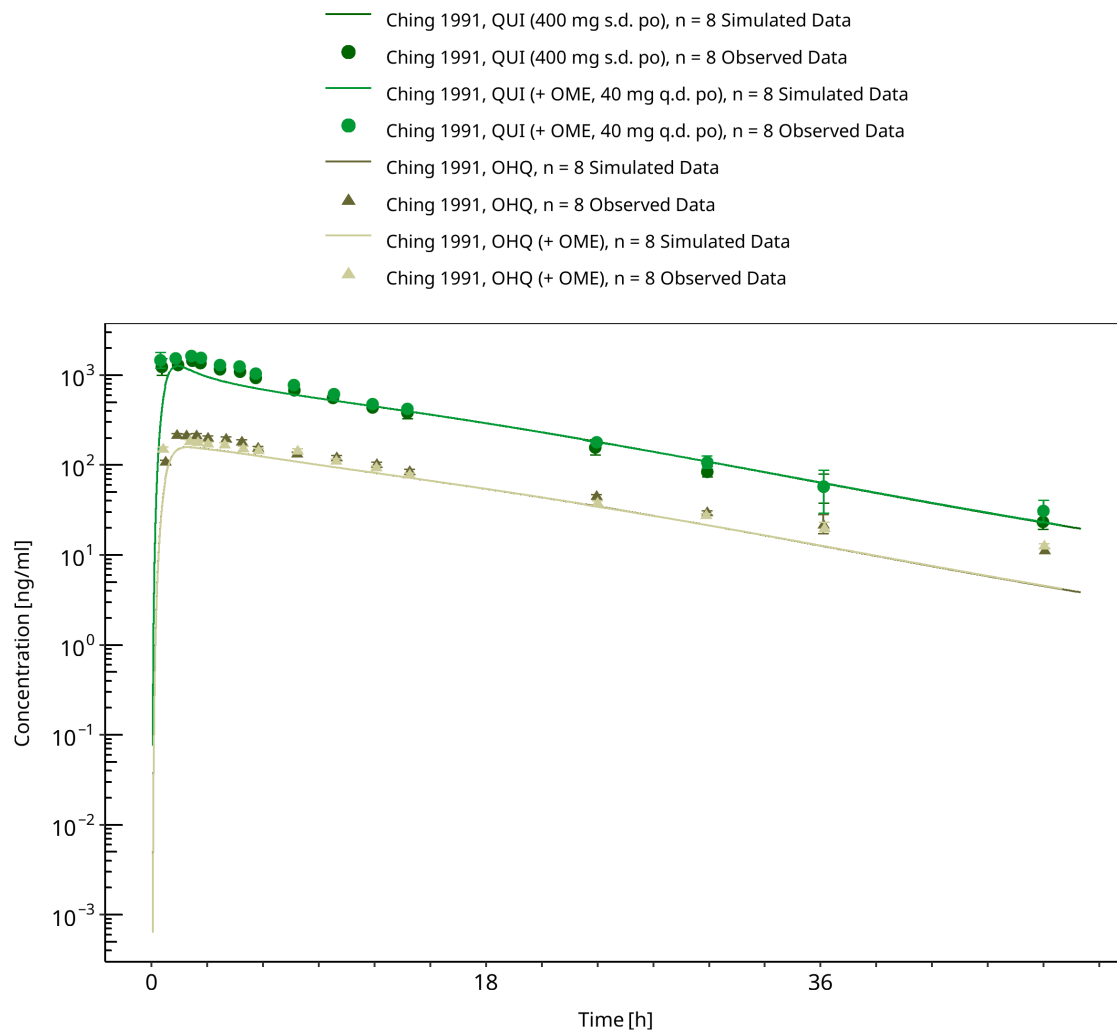


Figure 2-35: Omeprazole - Quinidine DDI

2.2.19 Rifampicin - Quinidine DDI

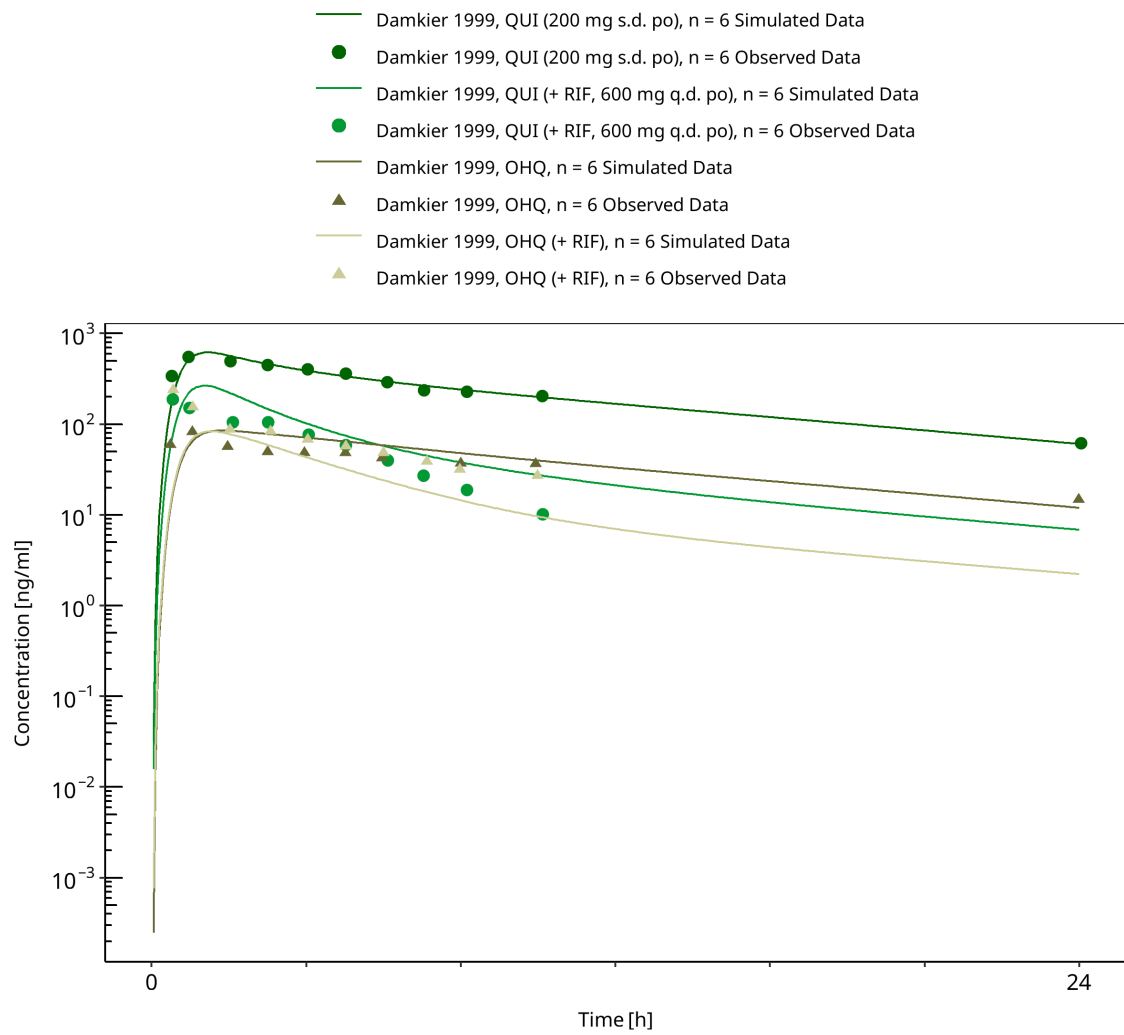


Figure 2-36: Rifampicin - Quinidine DDI

2.2.20 Verapamil - Quinidine DDI

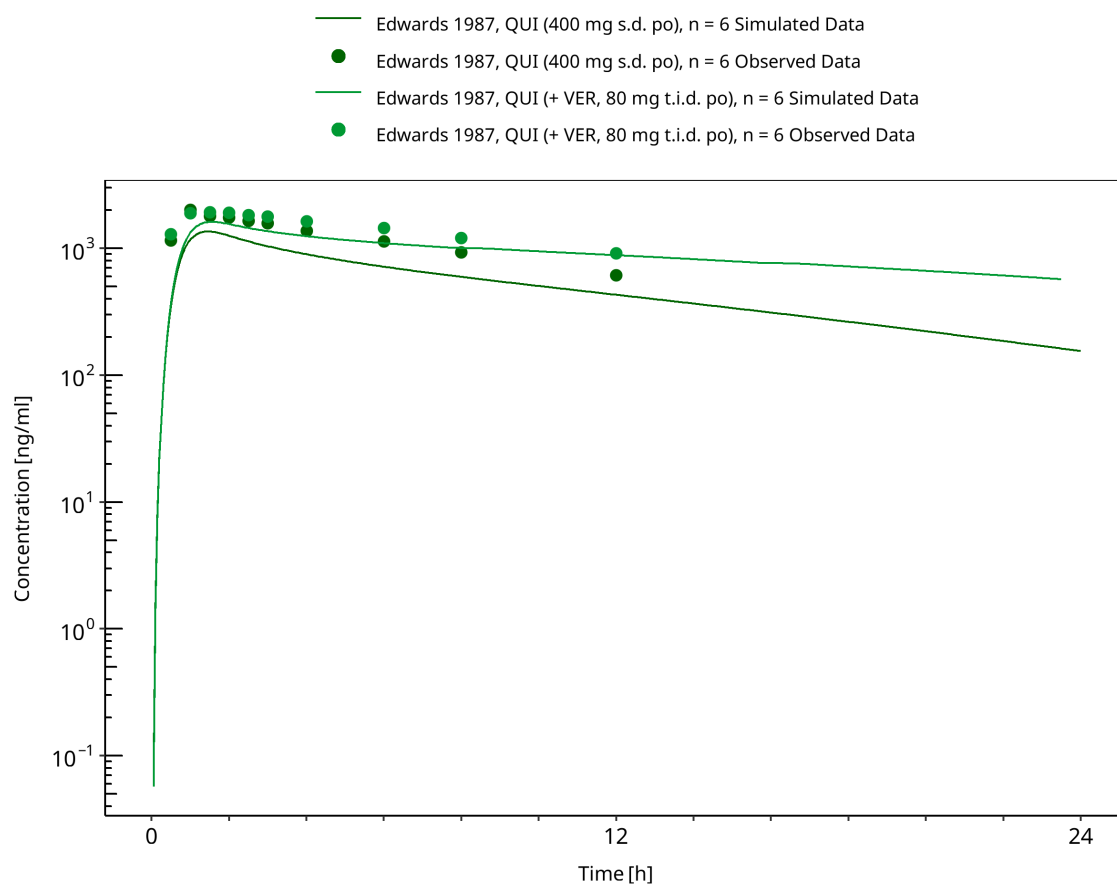


Figure 2-37: Verapamil - Quinidine DDI

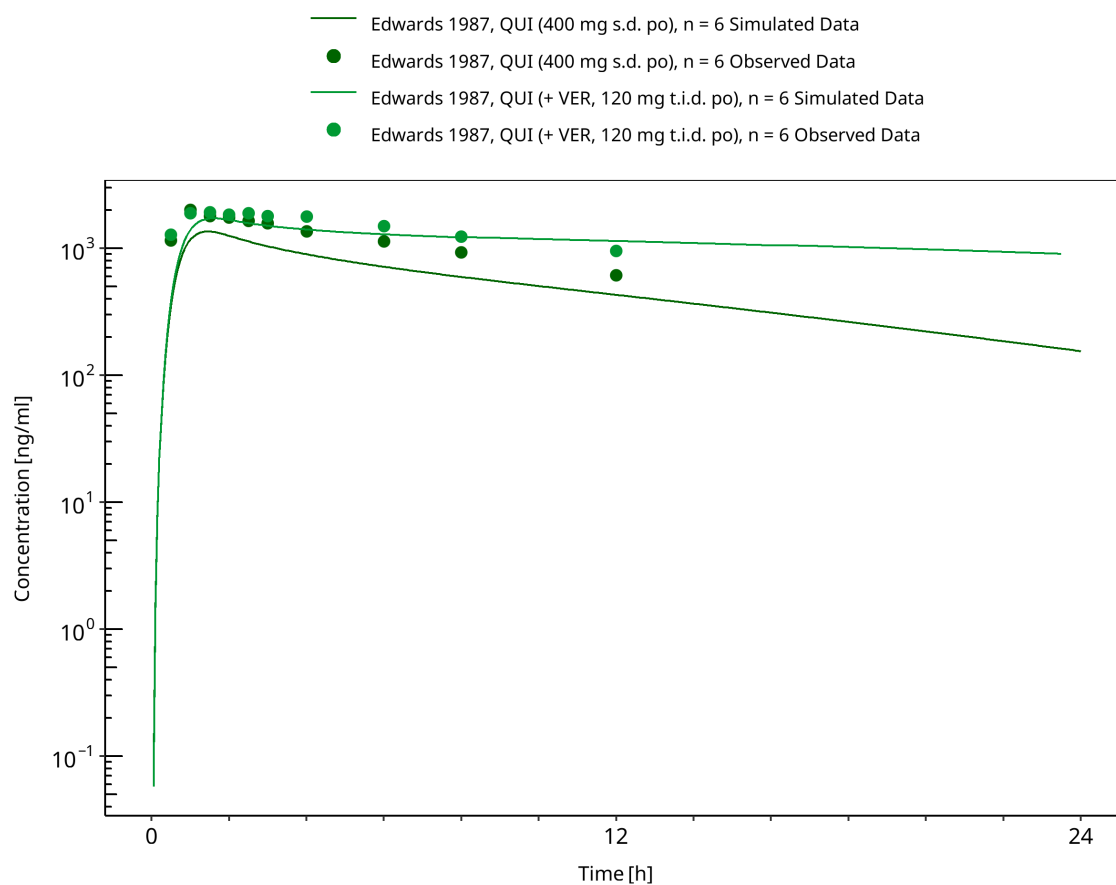


Figure 2-38: Verapamil - Quinidine DDI

2.2.21 Ketoconazole - Risperidone DDI

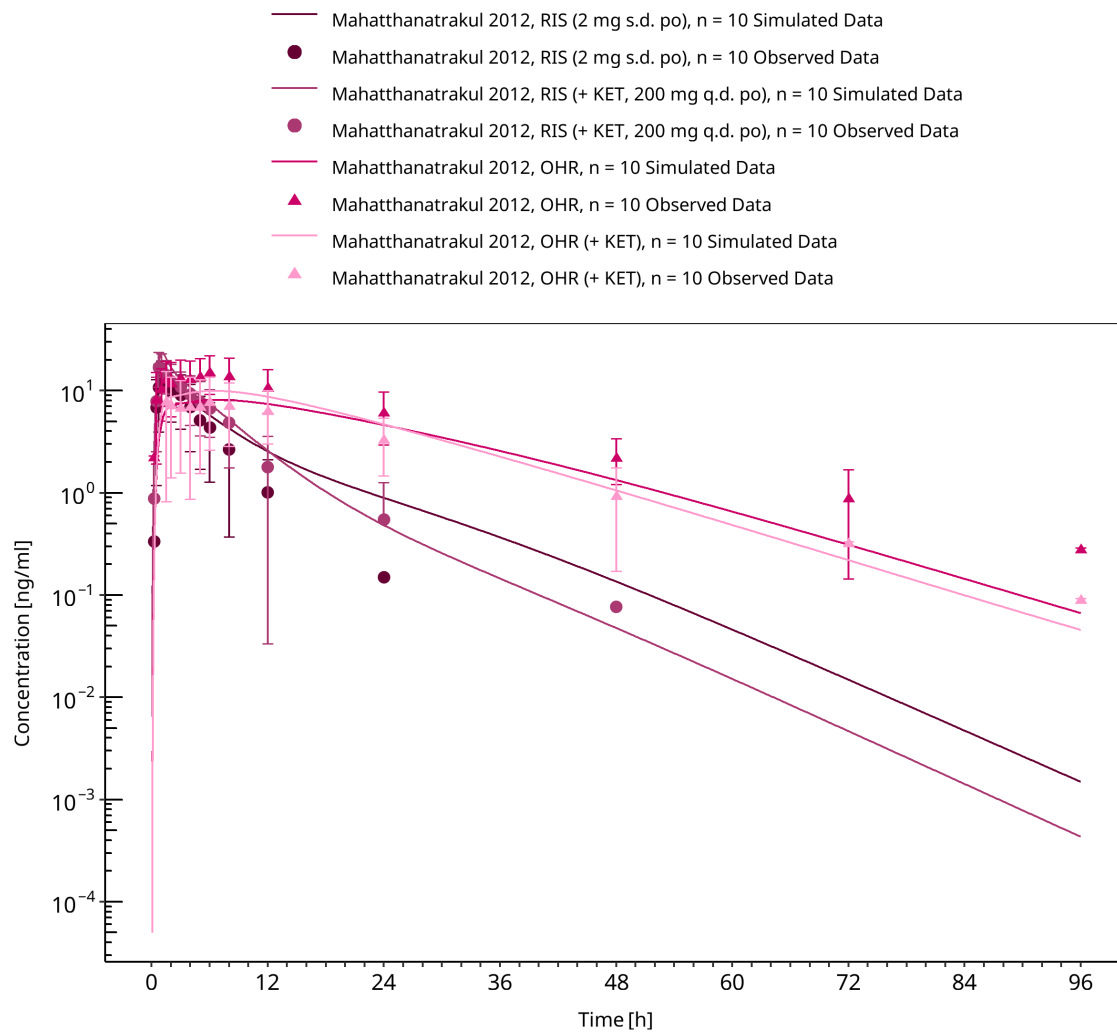


Figure 2-39: Ketoconazole - Risperidone DDI

2.2.22 Rifampicin - Risperidone DDI

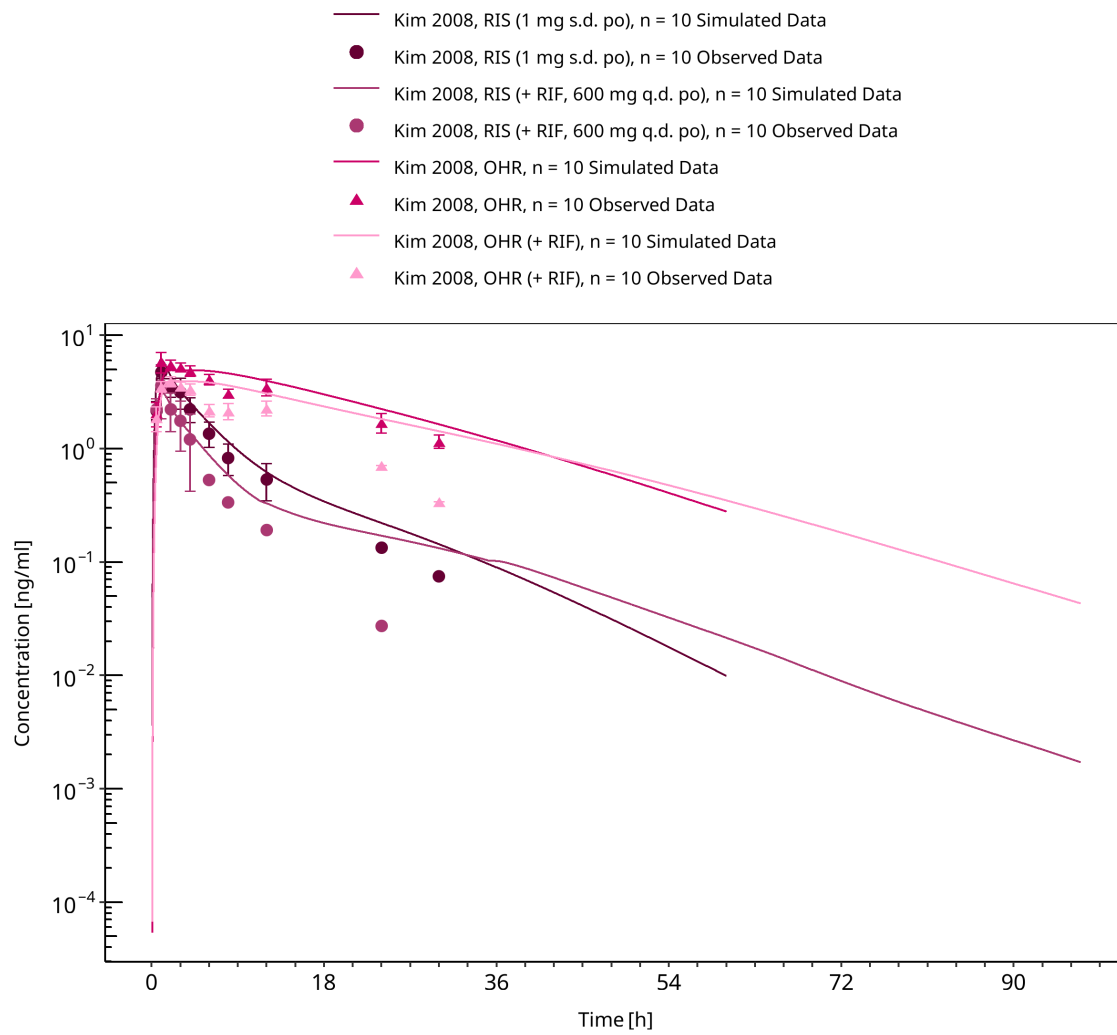


Figure 2-40: Rifampicin - Risperidone DDI

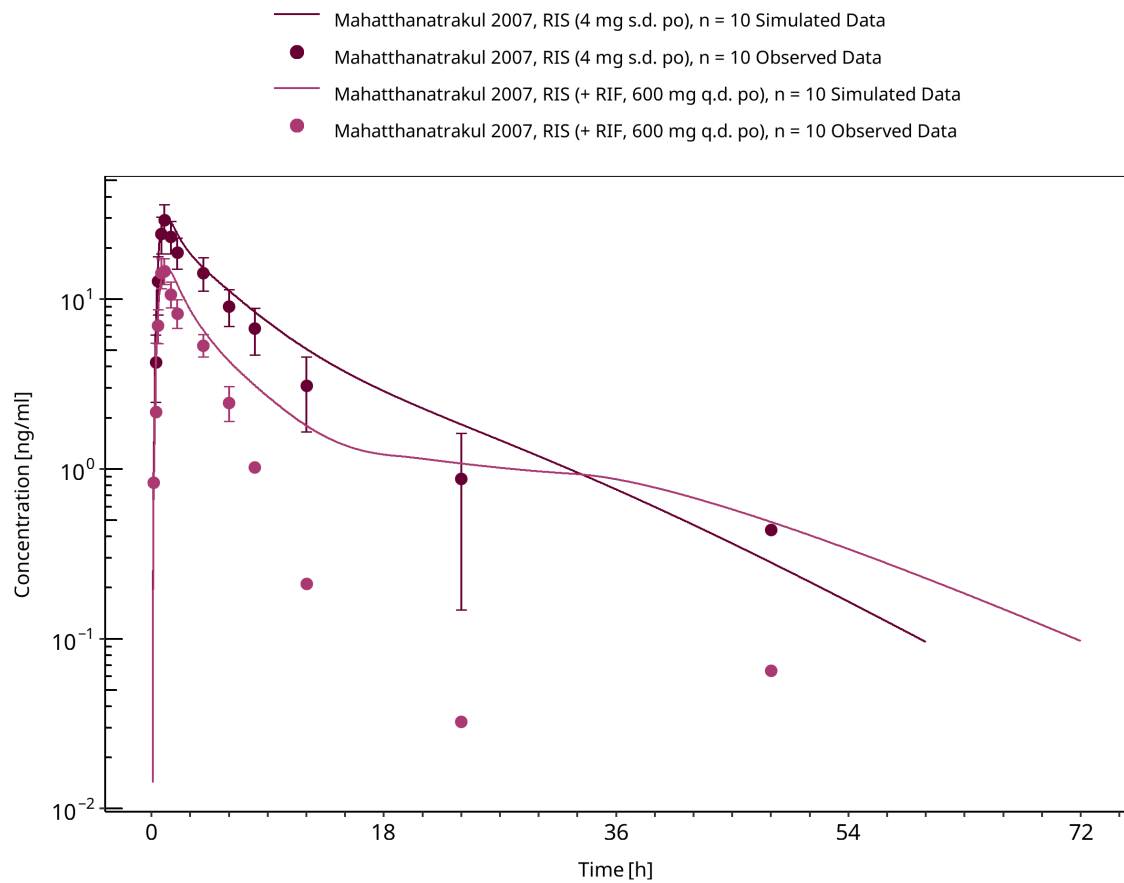


Figure 2-41: Rifampicin - Risperidone DDI

2.2.23 Verapamil - Risperidone DDI

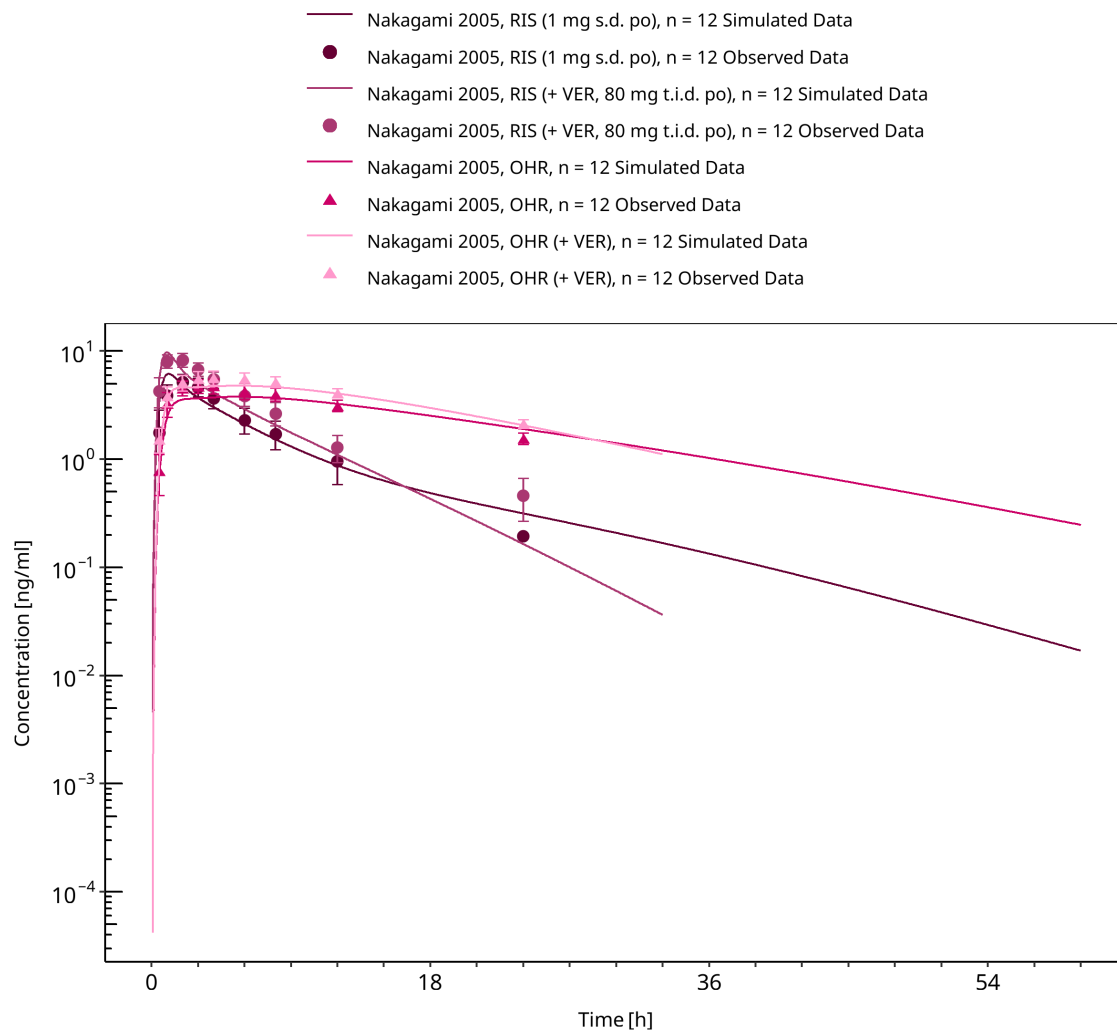


Figure 2-42: Verapamil - Risperidone DDI

3 DGI Evaluations

The following section evaluates drug-gene interaction (DGI) simulations without a perpetrator drug. These simulations isolate CYP2D6 activity-score effects on victim-drug and metabolite exposure and are interpreted separately from DDGI scenarios.

For the CYP2D6 DGI qualification, activity-score-stratified clinical concentration-time profiles were extracted from the local parent model repositories and their underlying publications or supplements where available. The DGI plots are therefore evaluated as parent-model evidence, not as interaction ratios.

3.1 DGI Parent-Model Evaluations

3.1.1 Clomiphenes DGI

The Clomiphenes DGI evaluation uses the local `Clomiphenes-Model` parent repository as source. The parent repository documents a CYP2D6 activity-score panel after oral single-dose clomiphenes citrate administration in healthy adult premenopausal women and links the dataset to Kovar et al. 2022 and the supporting CYP2D6 clomiphenes metabolism publication by Murdter et al. 2012.

The DGI simulations compare plasma concentration-time profiles for (E)-clomiphenes, (E)-4-hydroxyclophenes, (E)-N-desethylclomiphenes and (E)-4-hydroxy-N-desethylclomiphenes across CYP2D6 activity-score groups. No perpetrator drug is present in these simulations.

CYP2D6 activity score	Study size	Dose
0	6	42 mg clomiphenes citrate, po, single dose
0.5	4	42 mg clomiphenes citrate, po, single dose
0.75	1	42 mg clomiphenes citrate, po, single dose
1	2	42 mg clomiphenes citrate, po, single dose
2	3	42 mg clomiphenes citrate, po, single dose
3	3	42 mg clomiphenes citrate, po, single dose

The scientific check for this section is activity-score ordering and metabolite pattern consistency, not DDI ratio agreement. Parent and metabolite profiles are kept separate because CYP2D6 activity changes parent depletion and metabolite formation in different directions.

3.1.2 Atomoxetine DGI

The Atomoxetine DGI evaluation uses the local `Atomoxetine-Model` parent repository as source. The parent repository documents CYP2D6 activity-score-dependent atomoxetine clearance and links the activity-score framework to Ruedesheim et al. 2022.

Two activity-score datasets are included as separate DGI plot groups to avoid cross-study pooling:

Study	CYP2D6 activity-score groups	Endpoint
Byeon 2015	AS 0.5, 1.25 and 2	Atomoxetine plasma concentration
Kim 2018	AS 0.5 and 2	Atomoxetine plasma concentration

The scientific check for these plots is whether exposure decreases with increasing CYP2D6 activity within each study. Dose-normalized Byeon profiles and Kim profiles are not interpreted as a single pooled AS trend.

3.1.3 Risperidone DGI

The Risperidone DGI evaluation uses the local `Risperidone-Model` parent repository as source. The parent repository documents CYP2D6 activity-score-dependent metabolism of risperidone to 9-hydroxyrisperidone and links the model extension to Ruedesheim et al. 2022.

The Novalbos 2010 CYP2D6 activity-score dataset is used for the DGI concentration-time profile evaluation:

CYP2D6 activity score	Study size	Dose	Endpoints
0	6	1 mg risperidone, po	risperidone and 9-hydroxyrisperidone
1	26	1 mg risperidone, po	risperidone and 9-hydroxyrisperidone
2	33	1 mg risperidone, po	risperidone and 9-hydroxyrisperidone
3	6	1 mg risperidone, po	risperidone and 9-hydroxyrisperidone

The scientific check for these plots is parent-metabolite consistency across activity-score groups. Higher CYP2D6 activity should reduce parent exposure and increase formation-dependent metabolite behavior, while active-moiety interpretation must remain separate from parent-only interpretation.

3.2 DGI Ratio Plots

The following plots compare predicted and observed CYP2D6 activity-score exposure ratios for DGI scenarios without a perpetrator drug.

Ratios are only included where a matching observed ratio record and matching parent-model DGI simulations are available. The current DGI ratio section includes clomiphen, atomoxetine, and risperidone activity-score comparisons with AS 2 as the within-study reference group.

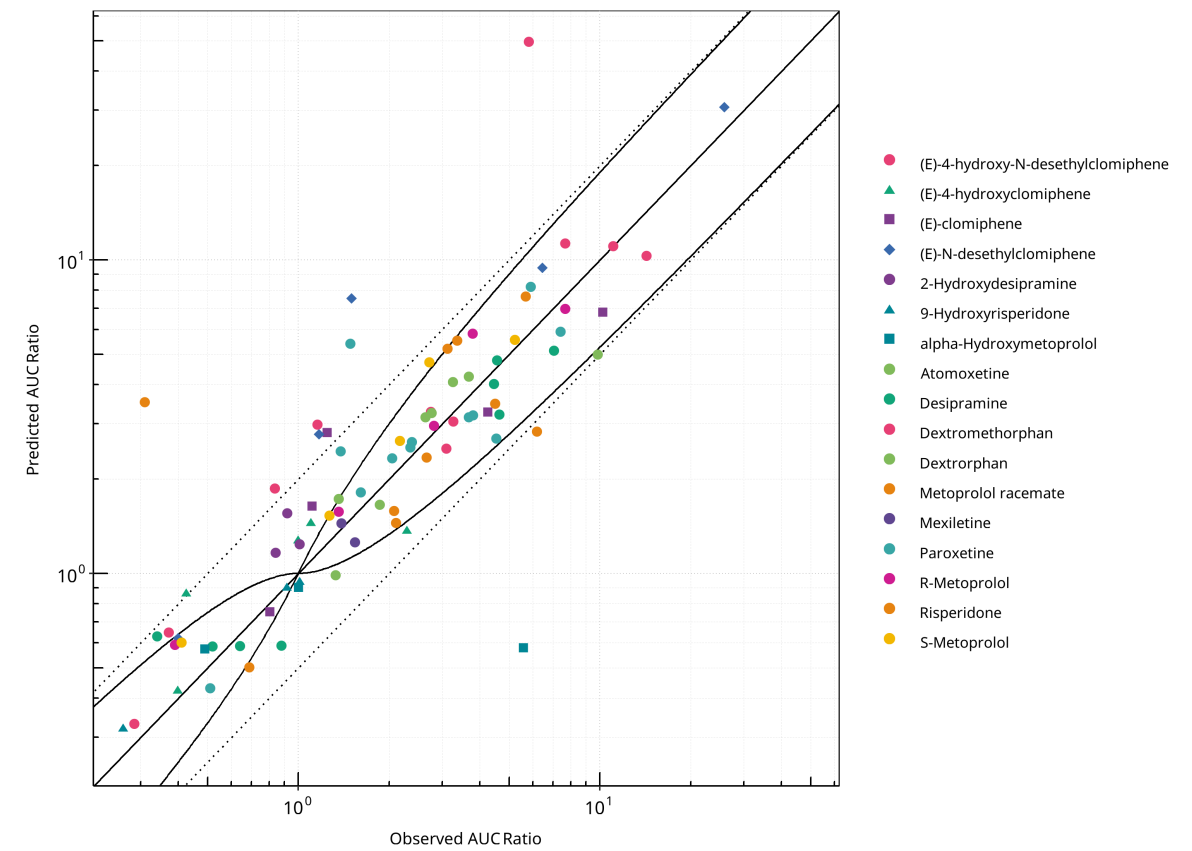


Figure 3-1: CYP2D6 DGIs. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest et al. formula)

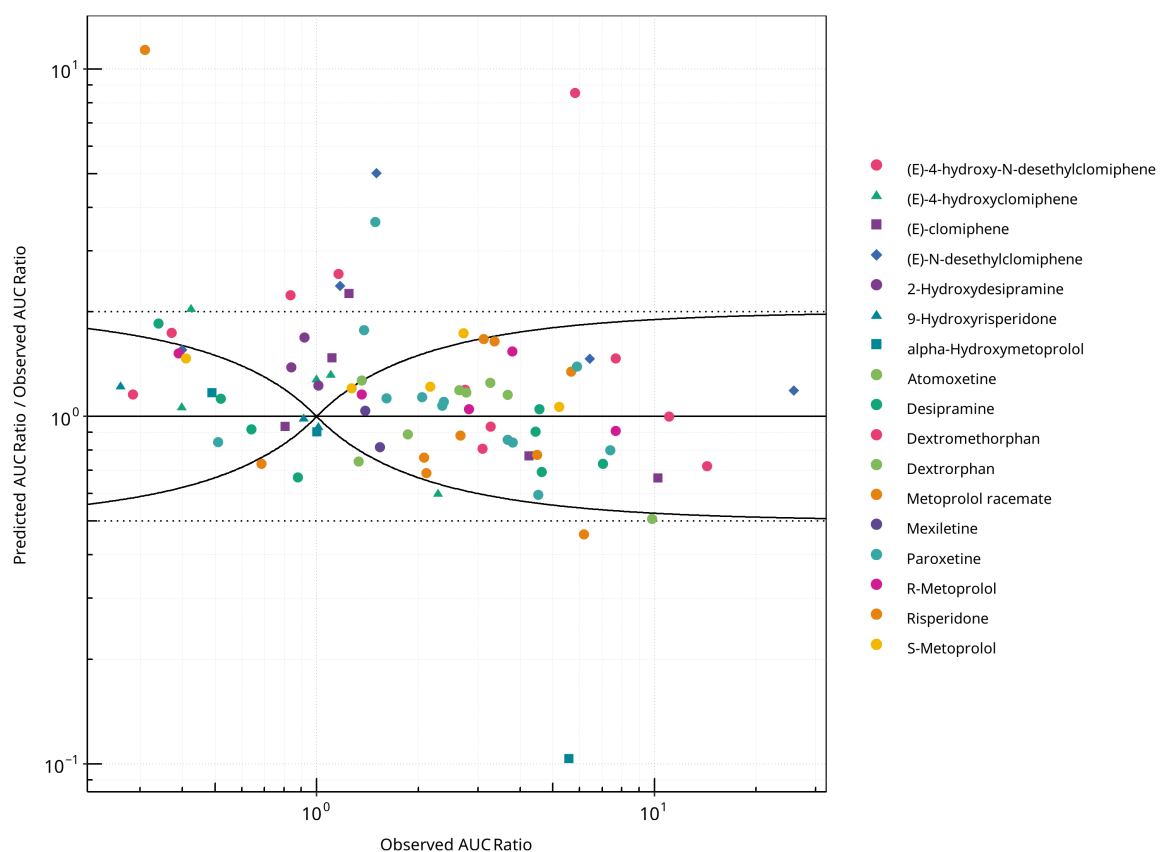


Figure 3-2: CYP2D6 DGIs. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

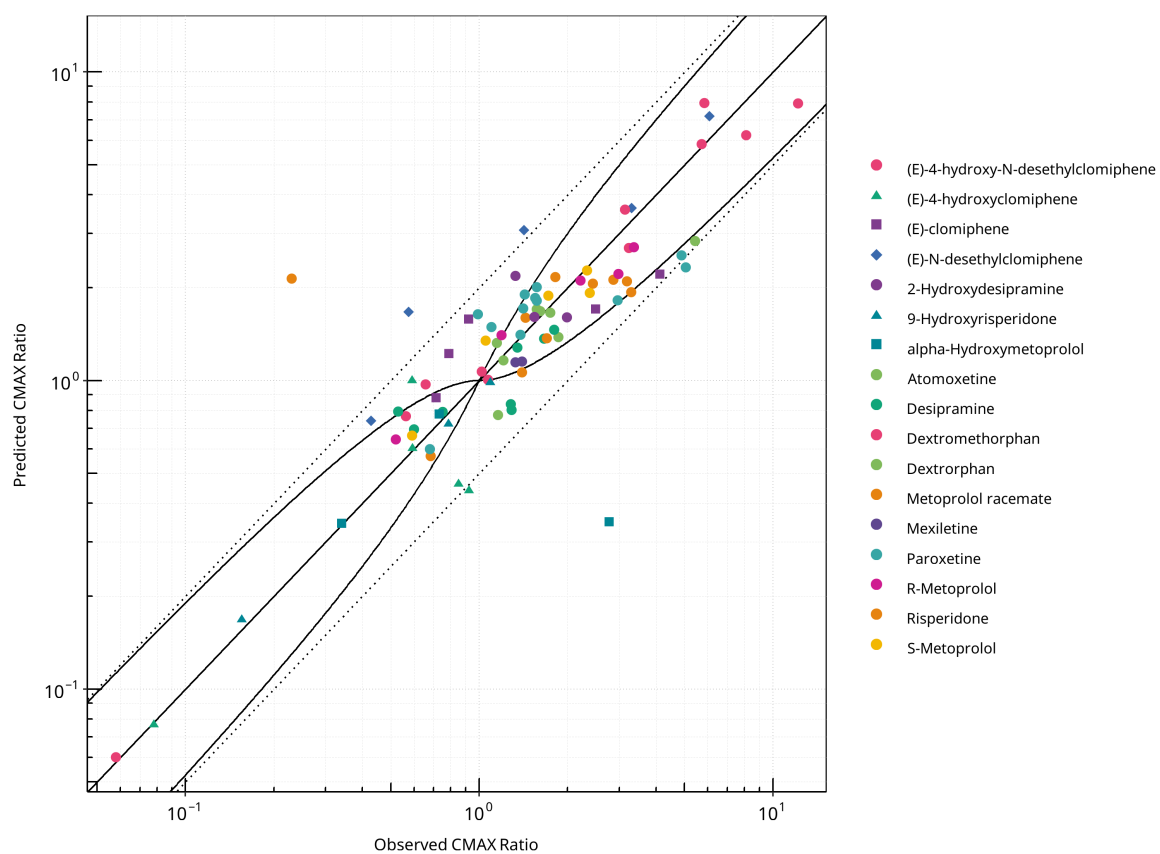


Figure 3-3: CYP2D6 DGIs. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

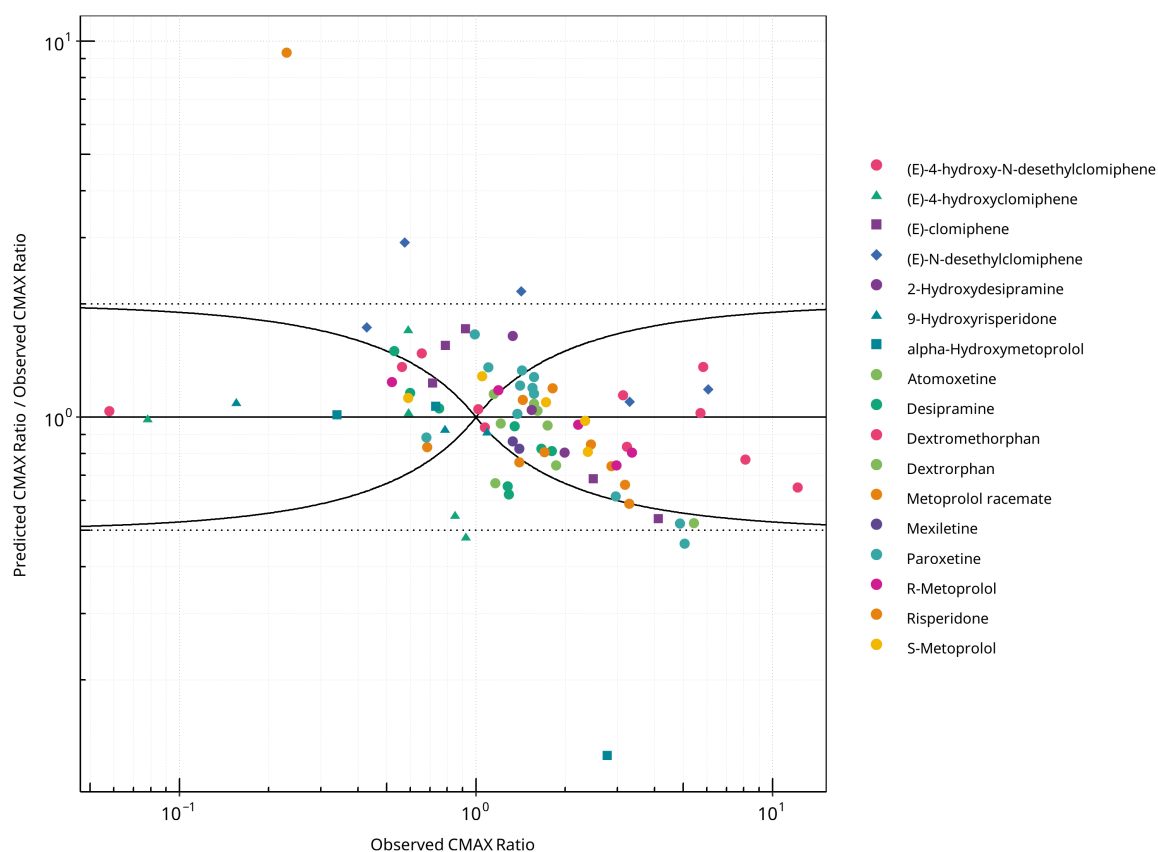


Figure 3-4: CYP2D6 DGIs. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 3-1: GMFE for CYP2D6 DGIs Ratio

PK parameter	GMFE
AUC	1.48
CMAX	1.36

Table 3-2: Summary table for CYP2D6 DGIs - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	85	-
Points within Guest <i>et al.</i>	56	65.88
Points within 2 fold	74	87.06

Table 3-3: Summary table for CYP2D6 DGIs - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

C _{MAX}	Number	Ratio [%]
Points total	85	-
Points within Guest <i>et al.</i>	56	65.88
Points within 2 fold	79	92.94

Table 3-4: Summary table for CYP2D6 DGIs

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
900001	- , 42 mg, - , -	(E)-clomiphene, PO	6.81	10.23	0.67	2.21	4.12	0.54	Kovar 2022
900002	- , 42 mg, - , -	(E)-clomiphene, PO	3.27	4.25	0.77	1.71	2.49	0.69	Kovar 2022
900003	- , 42 mg, - , -	(E)-clomiphene, PO	2.81	1.25	2.26	1.58	0.92	1.72	Kovar 2022
900004	- , 42 mg, - , -	(E)-clomiphene, PO	1.64	1.11	1.48	1.22	0.79	1.55	Kovar 2022
900005	- , 42 mg, - , -	(E)-clomiphene, PO	0.75	0.80	0.94	0.88	0.71	1.23	Kovar 2022
900006	- , 42 mg, - , -	(E)-4- hydroxyclophene, PO	0.43	0.40	1.07	0.08	0.08	1.00	Kovar 2022
900007	- , 42 mg, - , -	(E)-4- hydroxyclophene, PO	1.38	2.29	0.60	0.45	0.92	0.48	Kovar 2022
900008	- , 42 mg, - , -	(E)-4- hydroxyclophene, PO	1.46	1.10	1.33	0.47	0.85	0.55	Kovar 2022
900009	- , 42 mg, - , -	(E)-4- hydroxyclophene, PO	1.29	1.00	1.29	0.61	0.59	1.03	Kovar 2022
900010	- , 42 mg, - , -	(E)-4- hydroxyclophene, PO	0.87	0.42	2.06	1.01	0.59	1.72	Kovar 2022
900011	- , 42 mg, - , -	(E)-N- desethylclomiphene, PO	30.84	25.84	1.19	7.23	6.07	1.19	Kovar 2022
900012	- , 42 mg, - , -	(E)-N- desethylclomiphene, PO	9.47	6.44	1.47	3.64	3.30	1.10	Kovar 2022
900013	- , 42 mg, - , -	(E)-N- desethylclomiphene, PO	7.56	1.50	5.03	3.08	1.42	2.17	Kovar 2022
900014	- , 42 mg, - , -	(E)-N- desethylclomiphene, PO	2.79	1.17	2.39	1.68	0.57	2.92	Kovar 2022
900015	- , 42 mg, - , -	(E)-N- desethylclomiphene, PO	0.62	0.40	1.56	0.74	0.43	1.74	Kovar 2022
900016	- , 42 mg, - , -	(E)-4-hydroxy-N- desethylclomiphene, PO	0.33	0.29	1.16	0.06	0.06	1.04	Kovar 2022
900017	- , 42 mg, - , -	(E)-4-hydroxy-N- desethylclomiphene, PO	3.29	2.75	1.20	1.07	1.02	1.05	Kovar 2022

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
900018	- , 42 mg, - , -	(E)-4-hydroxy-N-desethylclomiphen, PO	3.00	1.16	2.58	1.01	1.07	0.94	Kovar 2022
900019	- , 42 mg, - , -	(E)-4-hydroxy-N-desethylclomiphen, PO	1.88	0.84	2.24	0.98	0.66	1.48	Kovar 2022
900020	- , 42 mg, - , -	(E)-4-hydroxy-N-desethylclomiphen, PO	0.65	0.37	1.75	0.77	0.56	1.37	Kovar 2022
900021	- , 40 mg, - , -	Atomoxetine, PO	3.16	2.64	1.20	1.67	1.74	0.95	Byeon 2015
900022	- , 40 mg, - , -	Atomoxetine, PO	1.73	1.36	1.27	1.33	1.15	1.16	Byeon 2015
900023	- , 20 mg, - , -	Atomoxetine, PO	3.26	2.77	1.18	1.69	1.61	1.05	Kim 2018
900024	- , 1 mg, - , -	Risperidone, PO	7.67	5.67	1.35	2.18	1.81	1.20	Novalbos 2010
900025	- , 1 mg, - , -	Risperidone, PO	2.35	2.66	0.88	1.60	1.44	1.12	Novalbos 2010
900026	- , 1 mg, - , -	Risperidone, PO	0.50	0.69	0.73	0.57	0.68	0.84	Novalbos 2010
900027	- , 1 mg, - , -	9-Hydroxyrisperidone, PO	0.32	0.26	1.23	0.17	0.15	1.10	Novalbos 2010
900028	- , 1 mg, - , -	9-Hydroxyrisperidone, PO	0.91	0.91	1.00	0.73	0.78	0.94	Novalbos 2010
900029	- , 1 mg, - , -	9-Hydroxyrisperidone, PO	0.95	1.01	0.94	1.00	1.09	0.92	Novalbos 2010
900030	- , 20 mg, - , -	Atomoxetine, PO	4.09	3.26	1.26	1.71	1.57	1.09	Jung 2020
900031	- , 20 mg, - , -	Atomoxetine, PO	1.66	1.86	0.89	1.17	1.21	0.97	Jung 2020
900032	- , 20 mg, - , -	Atomoxetine, PO	5.01	9.84	0.51	2.85	5.43	0.52	Sauer 2003
900033	- , 25 mg, - , -	Atomoxetine, PO	4.27	3.68	1.16	1.39	1.86	0.75	Todor 2016
900034	- , 50 mg, - , -	Desipramine, IV	4.04	4.45	0.91	0.81	1.29	0.62	Brösen 1988
900035	- , 50 mg, - , -	Desipramine, IV	0.59	0.88	0.67	0.84	1.28	0.66	Brösen 1988

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted CMAX Ratio	Observed CMAX Ratio	Pred/Obs CMAX Ratio	Reference
900036	- , 25 mg, - , -	Desipramine, PO	3.23	4.64	0.70	1.47	1.80	0.82	Spina 1987
900037	- , 100 mg, - , -	Desipramine, PO	4.81	4.56	1.05	1.37	1.66	0.83	Brösen 1993
900038	- , 100 mg, - , -	Desipramine, PO	0.63	0.34	1.86	0.80	0.53	1.51	Brösen 1993
900039	- , 100 mg, - , -	2-Hydroxydesipramine, PO	1.17	0.84	1.39	1.61	1.54	1.05	Brösen 1993
900040	- , 100 mg, - , -	Desipramine, PO	0.59	0.52	1.13	0.70	0.60	1.16	Bergmann 2001
900041	- , 100 mg, - , -	2-Hydroxydesipramine, PO	1.56	0.92	1.70	2.20	1.33	1.65	Bergmann 2001
900042	- , 100 mg, - , -	Desipramine, PO	5.16	7.04	0.73	1.28	1.35	0.95	Brösen 1986
900043	- , 100 mg, - , -	Desipramine, PO	0.59	0.64	0.92	0.79	0.75	1.06	Brösen 1986
900044	- , 100 mg, - , -	2-Hydroxydesipramine, PO	1.25	1.01	1.23	1.61	1.99	0.81	Brösen 1986
900045	- , 5 mg, - , -	Dextromethorphan, PO	3.07	3.27	0.94	3.60	3.13	1.15	Storelli 2018
900046	- , 5 mg, - , -	Dextrophan, PO	0.99	1.33	0.75	0.78	1.16	0.67	Storelli 2018
900047	- , 15 mg, - , -	Dextromethorphan, PO	11.11	11.06	1.00	7.95	12.17	0.65	Qiu 2016
900048	- , 15 mg, - , -	Dextromethorphan, PO	2.51	3.10	0.81	2.70	3.23	0.84	Qiu 2016
900049	- , 30 mg, - , -	Dextromethorphan, PO	11.32	7.67	1.48	7.97	5.84	1.37	Yamazaki 2017
900050	- , 30 mg, - , -	Dextromethorphan, PO	49.85	5.82	8.57	5.87	5.71	1.03	Capon 1996
900051	- , 30 mg, - , -	Dextromethorphan, PO	10.33	14.29	0.72	6.28	8.11	0.77	Gorski 2004
900052	- , 20 mg, - , -	Metoprolol racemate, IV	1.46	2.11	0.69	1.07	1.40	0.76	Leemann 1993
900053	- , 100 mg, - , -	Metoprolol racemate, PO	3.53	0.31	11.39	2.15	0.23	9.36	Bae 2014
900054	- , 100 mg, - , -	alpha-Hydroxymetoprolol, PO	0.58	5.58	0.10	0.35	2.77	0.13	Bae 2014

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted CMAX Ratio	Observed CMAX Ratio	Pred/Obs CMAX Ratio	Reference
900055	-, 100 mg, -, -	Metoprolol racemate, PO	3.49	4.49	0.78	2.11	3.18	0.66	Jin 2008
900056	-, 100 mg, -, -	alpha-Hydroxymetoprolol, PO	0.57	0.49	1.17	0.35	0.34	1.01	Jin 2008
900057	-, 100 mg, -, -	Metoprolol racemate, PO	1.59	2.08	0.76	1.38	1.70	0.81	Jin 2008
900058	-, 100 mg, -, -	alpha-Hydroxymetoprolol, PO	0.90	1.00	0.90	0.78	0.73	1.07	Jin 2008
900059	-, 100 mg, -, -	Metoprolol racemate, PO	5.54	3.36	1.65	2.13	2.86	0.74	Hamelin 2000
900060	-, 100 mg, -, -	Metoprolol racemate, PO	5.23	3.12	1.68	2.07	2.44	0.85	Sharma 2005
900061	-, 100 mg, -, -	R-Metoprolol, PO	5.85	3.79	1.54	2.23	2.98	0.75	Sharma 2005
900062	-, 100 mg, -, -	S-Metoprolol, PO	4.74	2.72	1.74	1.93	2.38	0.81	Sharma 2005
900063	-, 100 mg, -, -	R-Metoprolol, PO	2.97	2.82	1.05	2.12	2.21	0.96	Huang 1999
900064	-, 100 mg, -, -	S-Metoprolol, PO	2.66	2.17	1.22	1.89	1.72	1.10	Huang 1999
900065	-, 100 mg, -, -	R-Metoprolol, PO	1.58	1.36	1.16	1.41	1.19	1.18	Huang 1999
900066	-, 100 mg, -, -	S-Metoprolol, PO	1.54	1.27	1.21	1.36	1.05	1.29	Huang 1999
900067	-, 100 mg, -, -	R-Metoprolol, PO	7.01	7.67	0.91	2.72	3.36	0.81	Seeringer 2008
900068	-, 100 mg, -, -	S-Metoprolol, PO	5.58	5.22	1.07	2.29	2.33	0.98	Seeringer 2008
900069	-, 100 mg, -, -	R-Metoprolol, PO	0.59	0.39	1.52	0.65	0.52	1.24	Seeringer 2008
900070	-, 100 mg, -, -	S-Metoprolol, PO	0.60	0.41	1.47	0.67	0.59	1.13	Seeringer 2008
900071	-, 100 mg, -, -	Mexiletine, PO	1.26	1.54	0.82	1.16	1.40	0.83	Labbe 2000
900072	-, 200 mg, -, -	Mexiletine, PO	1.45	1.39	1.04	1.15	1.33	0.86	Abolfathi 1993
900073	-, 25 mg, -, -	Paroxetine, PO	5.93	7.40	0.80	2.34	5.05	0.46	Chen 2015
900074	-, 25 mg, -, -	Paroxetine, PO	3.17	3.68	0.86	1.83	2.96	0.62	Chen 2015

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
900075	-, 25 mg, -, -	Paroxetine, PO	1.82	1.61	1.13	1.50	1.10	1.36	Chen 2015
900076	-, 30 mg, -, -	Paroxetine, PO	2.53	2.35	1.08	1.86	1.55	1.20	Sindrup 1992
900077	-, 40 mg, -, -	Paroxetine, PO	3.21	3.80	0.84	2.02	1.57	1.28	Mürdter 2016
900078	-, 40 mg, -, -	Paroxetine, PO	2.64	2.38	1.11	1.72	1.41	1.22	Mürdter 2016
900079	-, 40 mg, -, -	Paroxetine, PO	2.71	4.53	0.60	1.82	1.57	1.16	Mürdter 2016
900080	-, 40 mg, -, -	Paroxetine, PO	2.34	2.05	1.14	1.65	0.99	1.67	Mürdter 2016
900081	-, 40 mg, -, -	Paroxetine, PO	0.43	0.51	0.85	0.60	0.68	0.89	Mürdter 2016
900082	-, 40 mg, -, -	Paroxetine, PO	8.23	5.89	1.40	2.56	4.88	0.52	Yoon 2000
900083	-, 40 mg, -, -	Paroxetine, PO	5.43	1.49	3.64	1.91	1.43	1.34	Yoon 2000
900084	-, 40 mg, -, -	Paroxetine, PO	2.46	1.38	1.78	1.41	1.38	1.02	Yoon 2000
900085	-, 2 mg, -, -	Risperidone, PO	2.84	6.18	0.46	1.94	3.29	0.59	Bondolfi 2002

3.3 DGI Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

3.3.1 Clomiphenes DGI

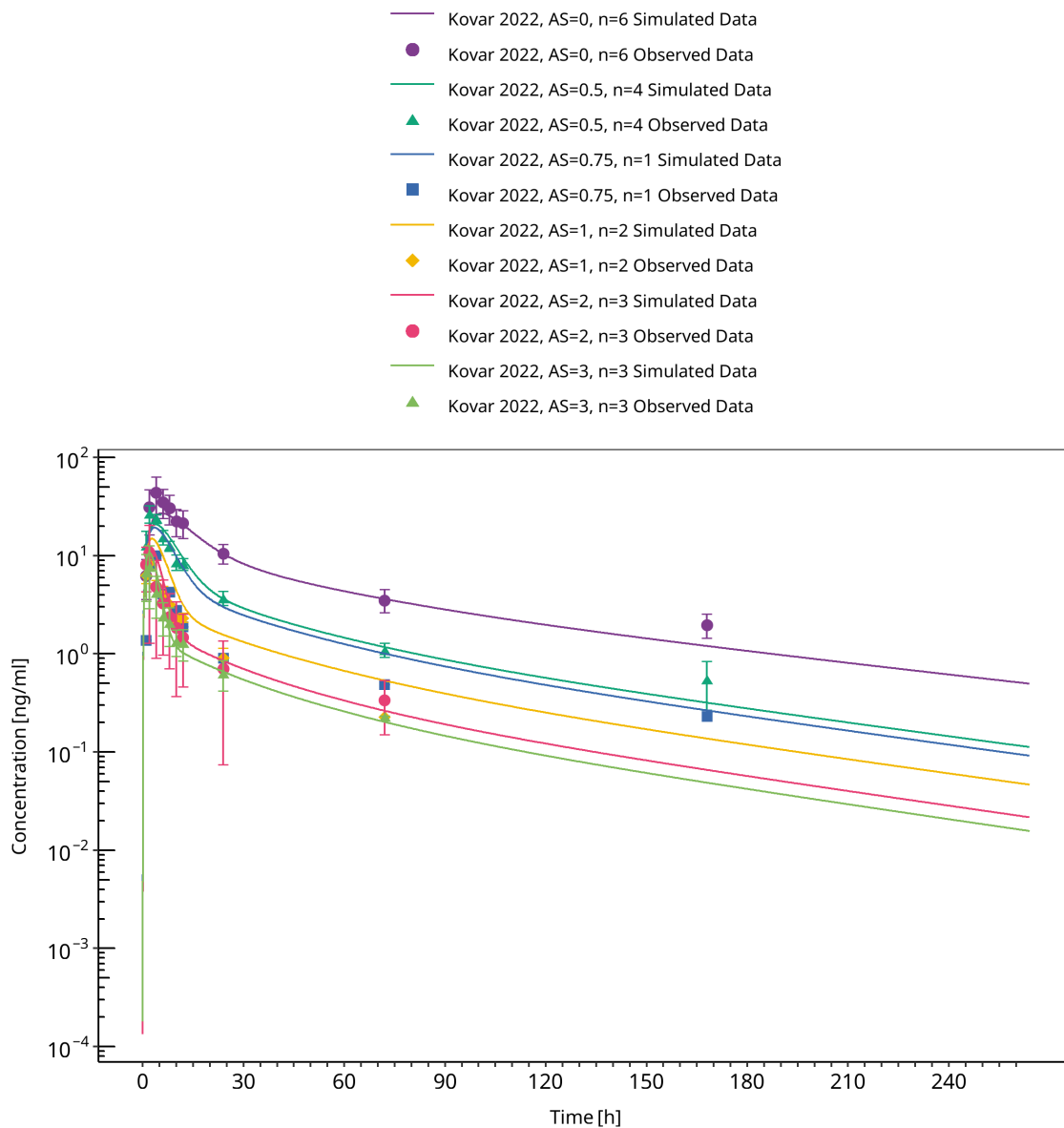


Figure 3-5: Clomiphene DGI - (E)-clomiphene

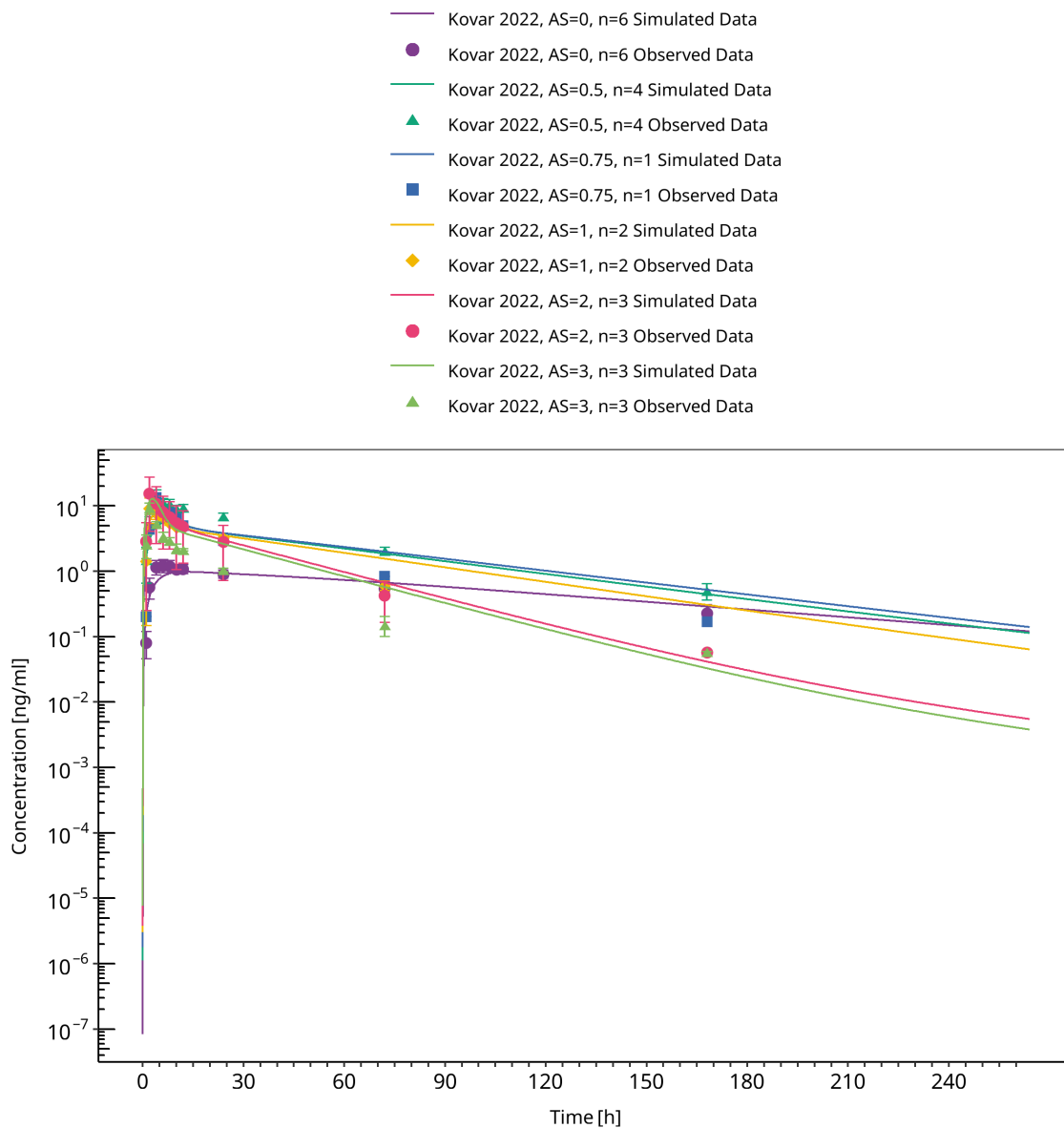


Figure 3-6: Clomiphene DGI - (E)-4-hydroxyclophene

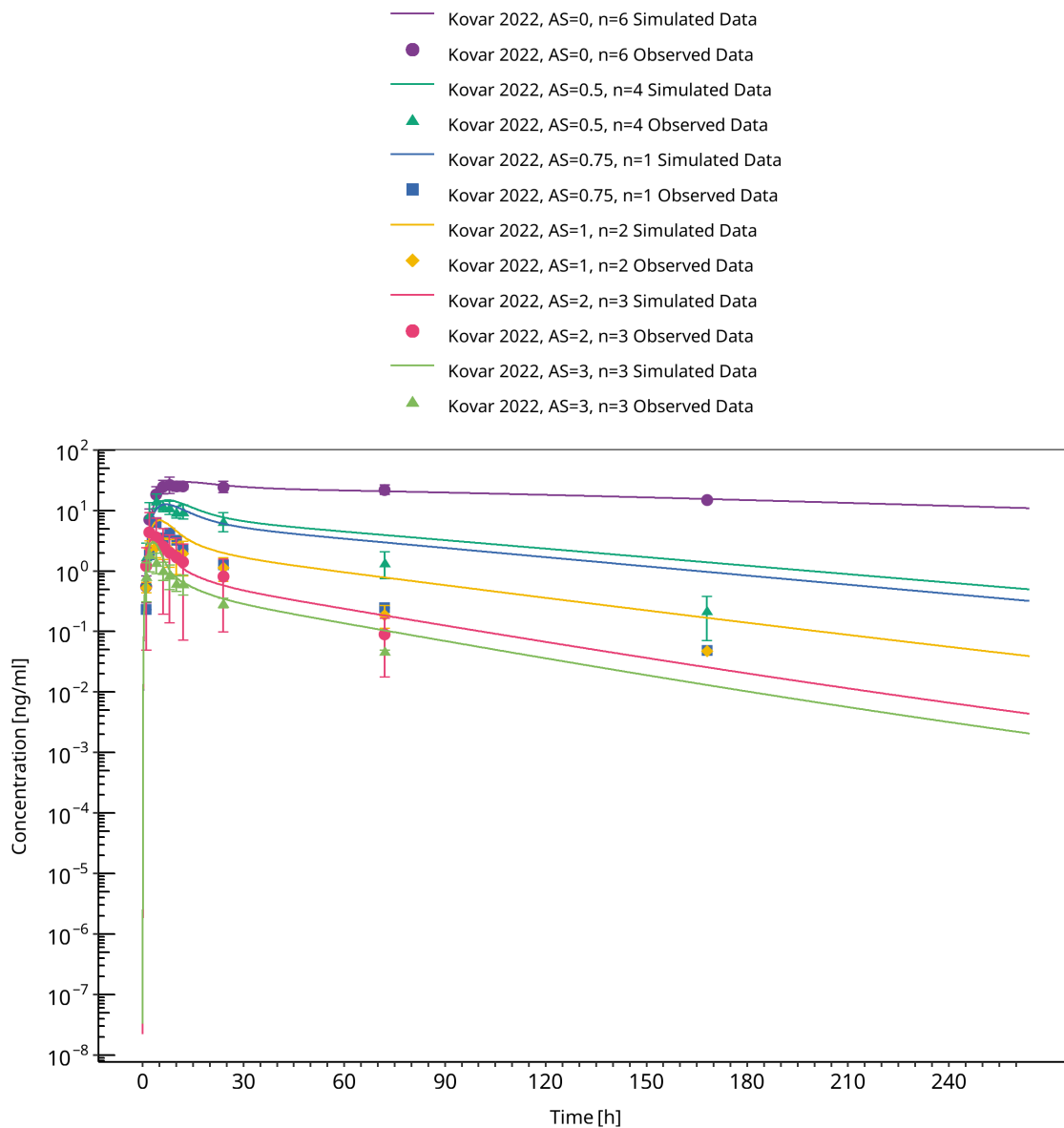


Figure 3-7: Clomiphene DGI - (E)-N-desethylclomiphene

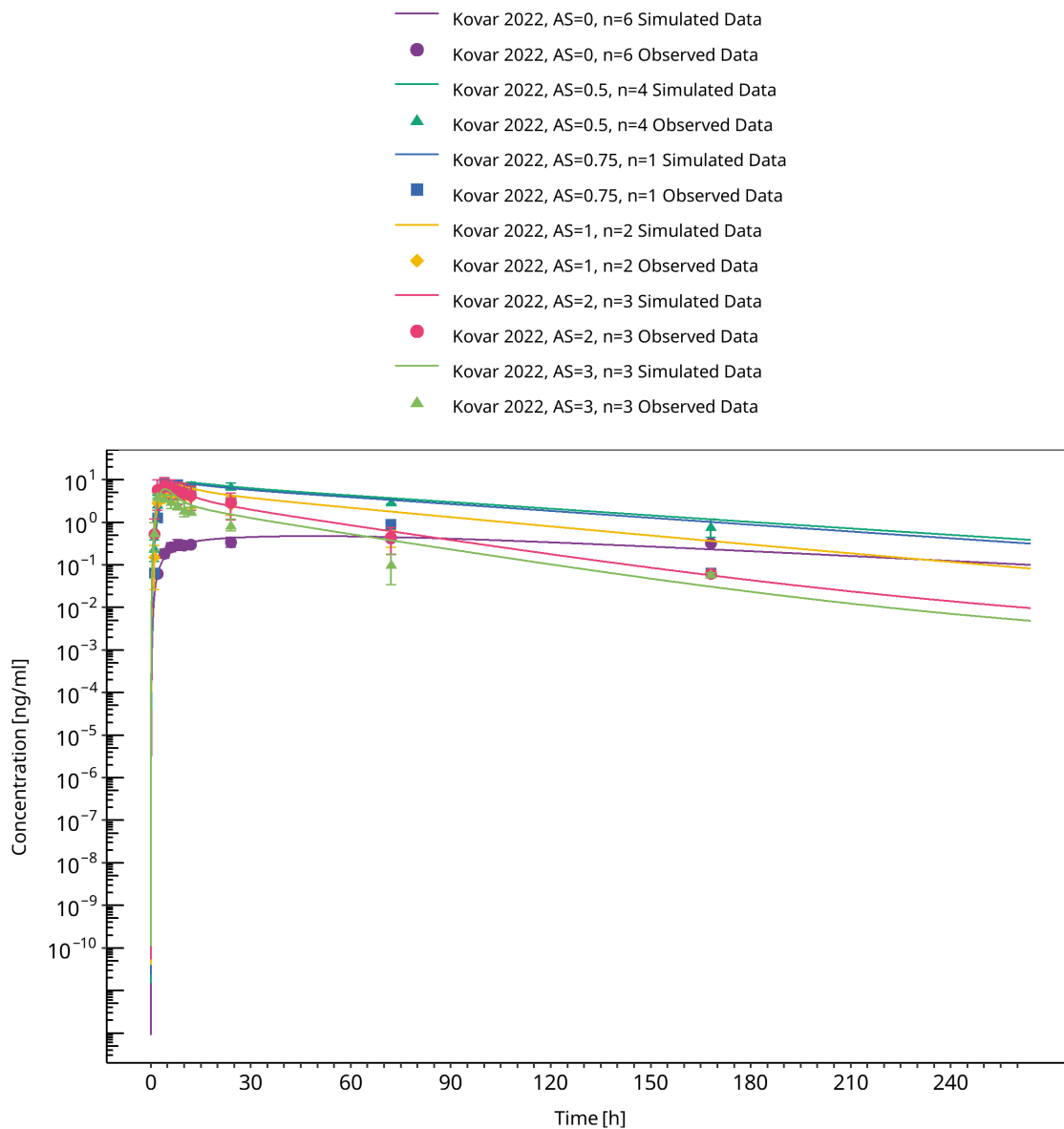


Figure 3-8: Clomiphenes DGI - (E)-4-hydroxy-N-desethylclomiphenes

3.3.2 Atomoxetine DGI

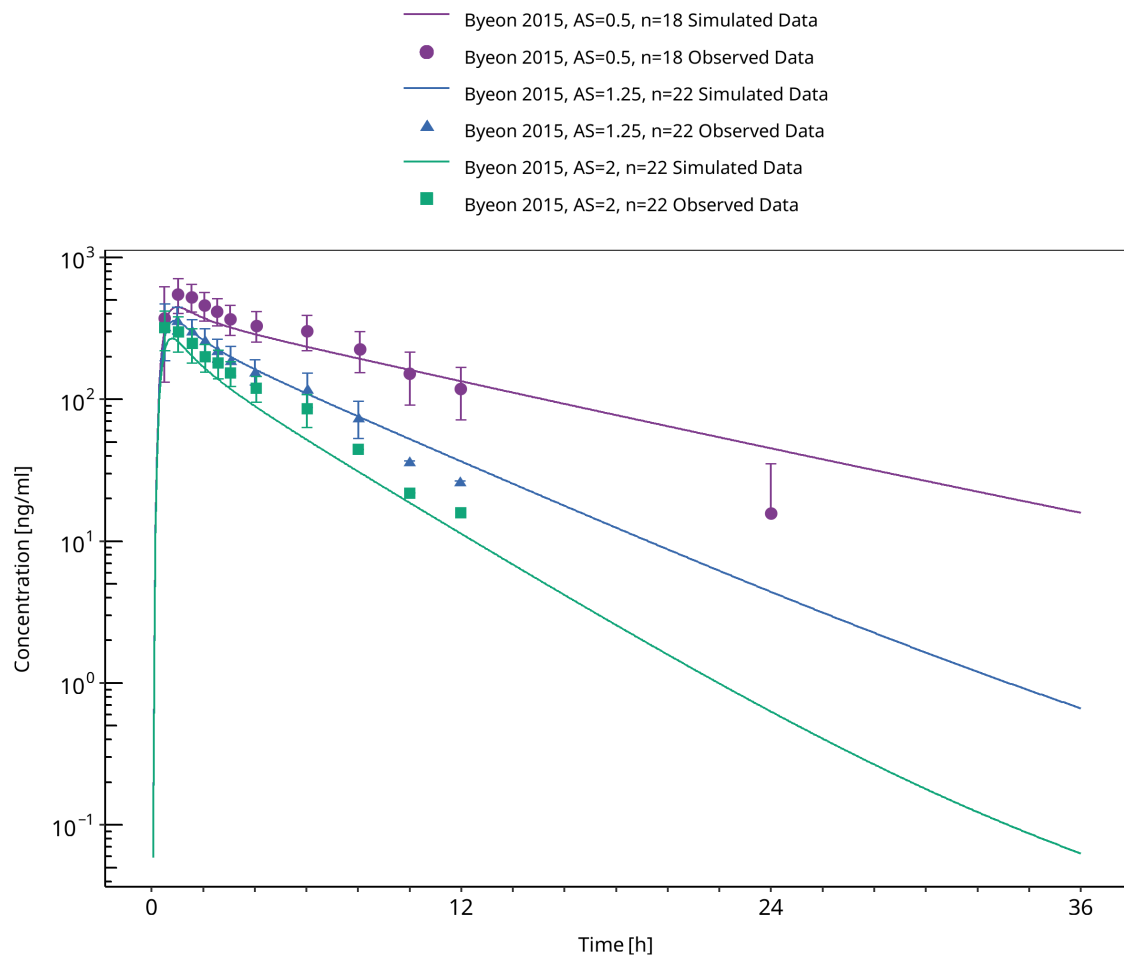


Figure 3-9: Atomoxetine DGI - Byeon 2015

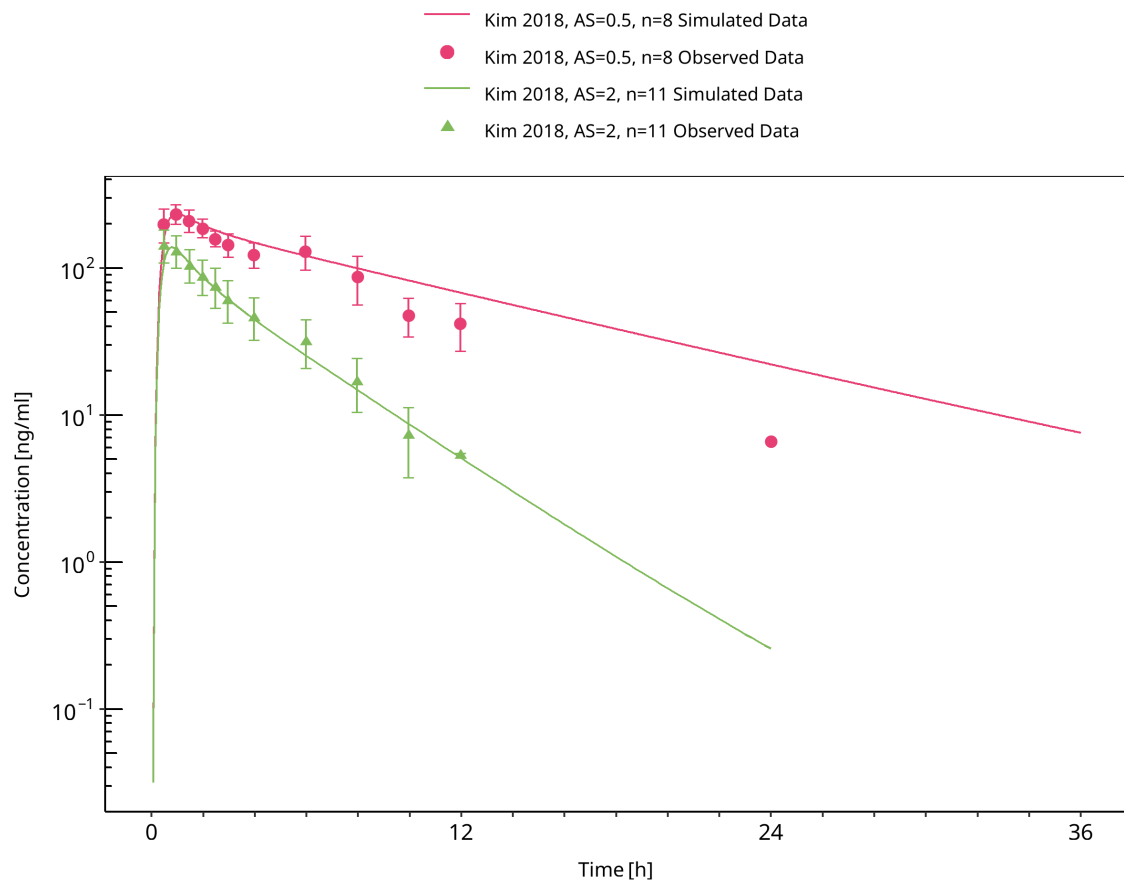


Figure 3-10: Atomoxetine DGI - Kim 2018

3.3.3 Risperidone DGI

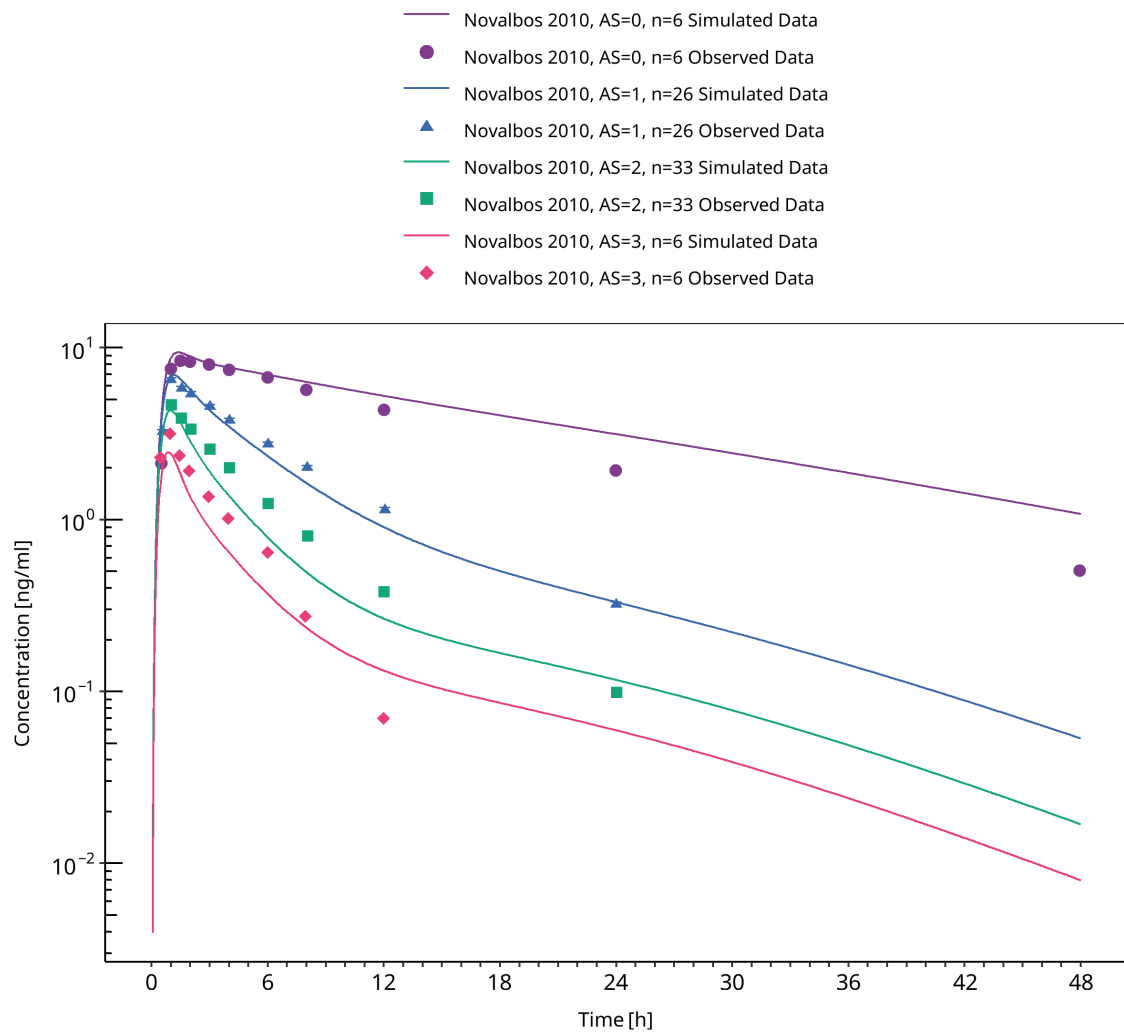


Figure 3-11: Risperidone DGI - Risperidone

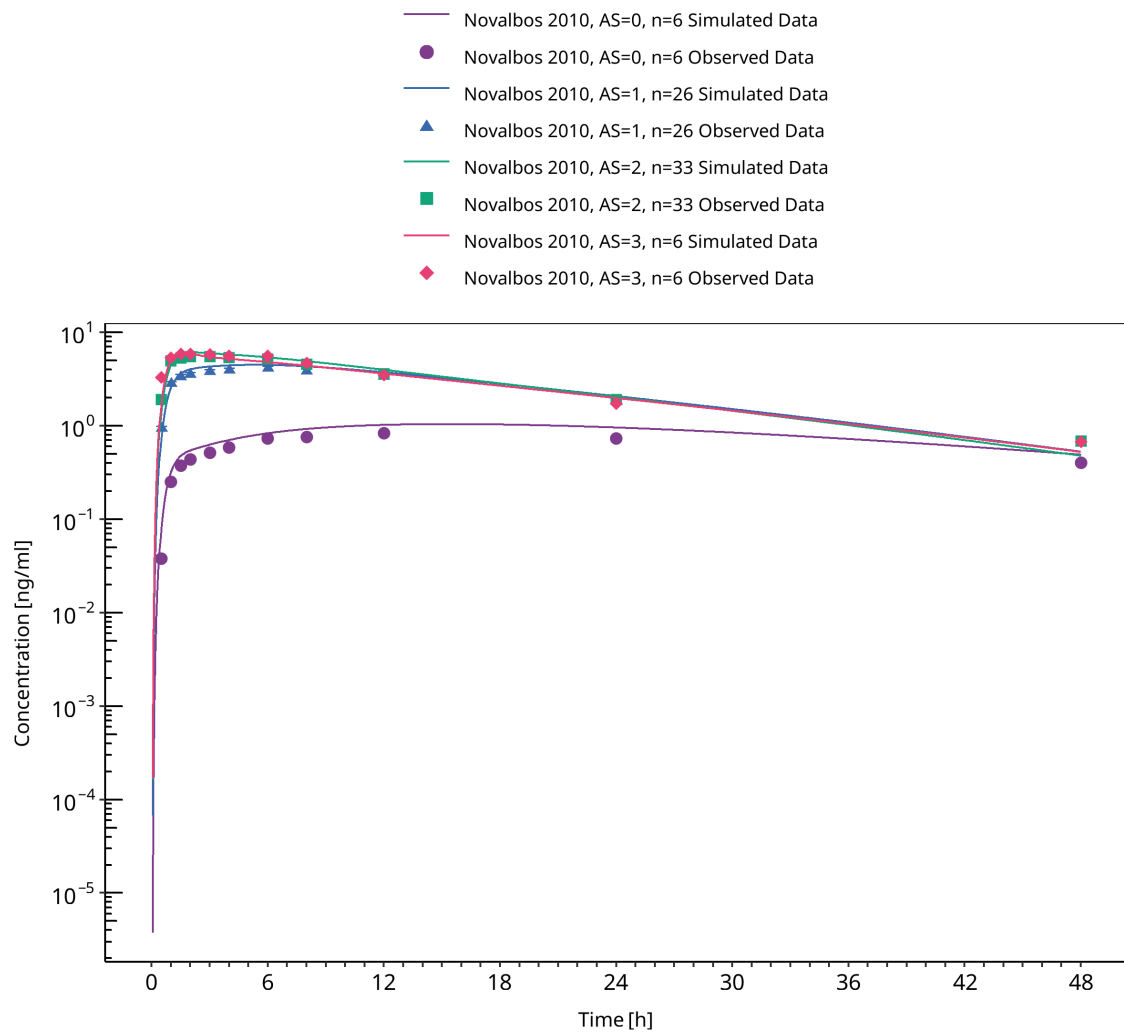


Figure 3-12: Risperidone DGI - 9-Hydroxyrisperidone

4 DDGI Evaluations

The following section shows the correlations between observed and model-predicted AUC_{last} and C_{max} ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and **C_{max} ratio (CmaxR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
- the solid lines denote the limits suggested by [Guest et al. 2011](#),
- the bold solid line denotes the unity line,
- each color represents one combination of perpetrator and victim (parent drug or metabolite)

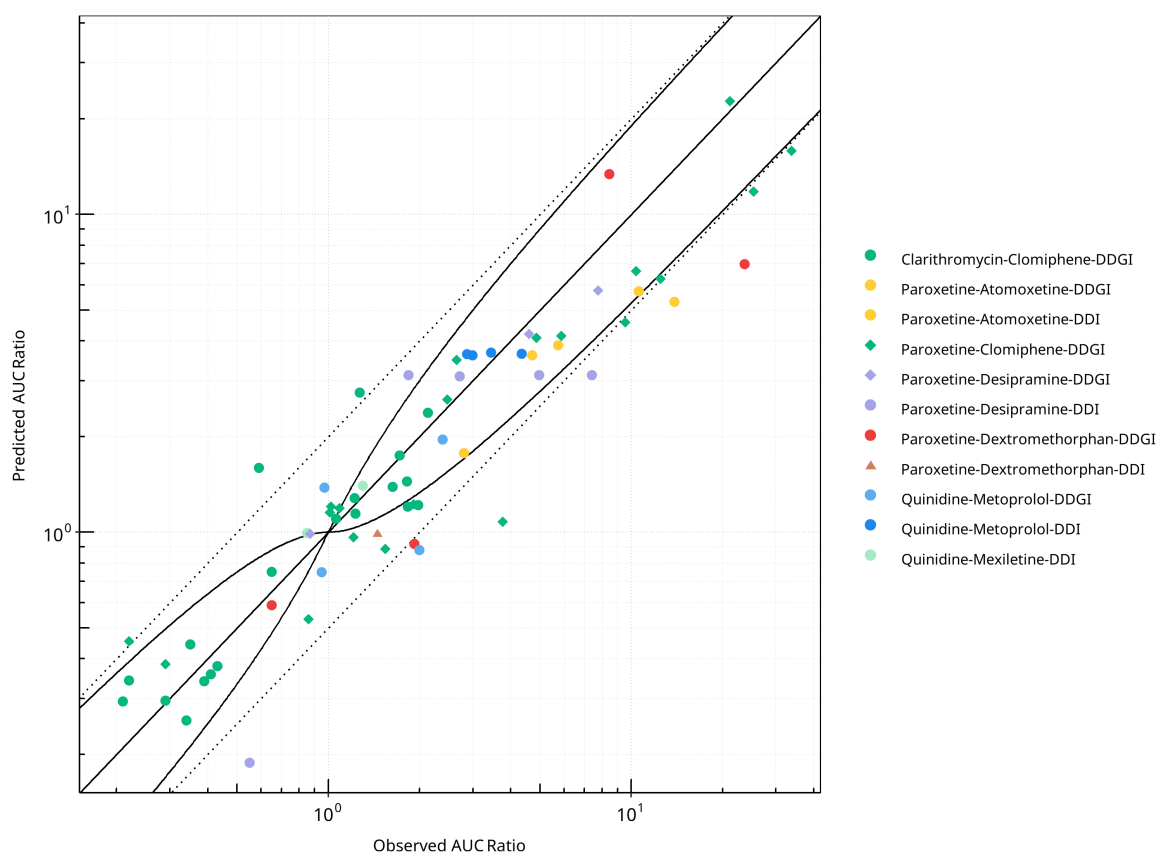


Figure 4-1: CYP2D6 DDGIs. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

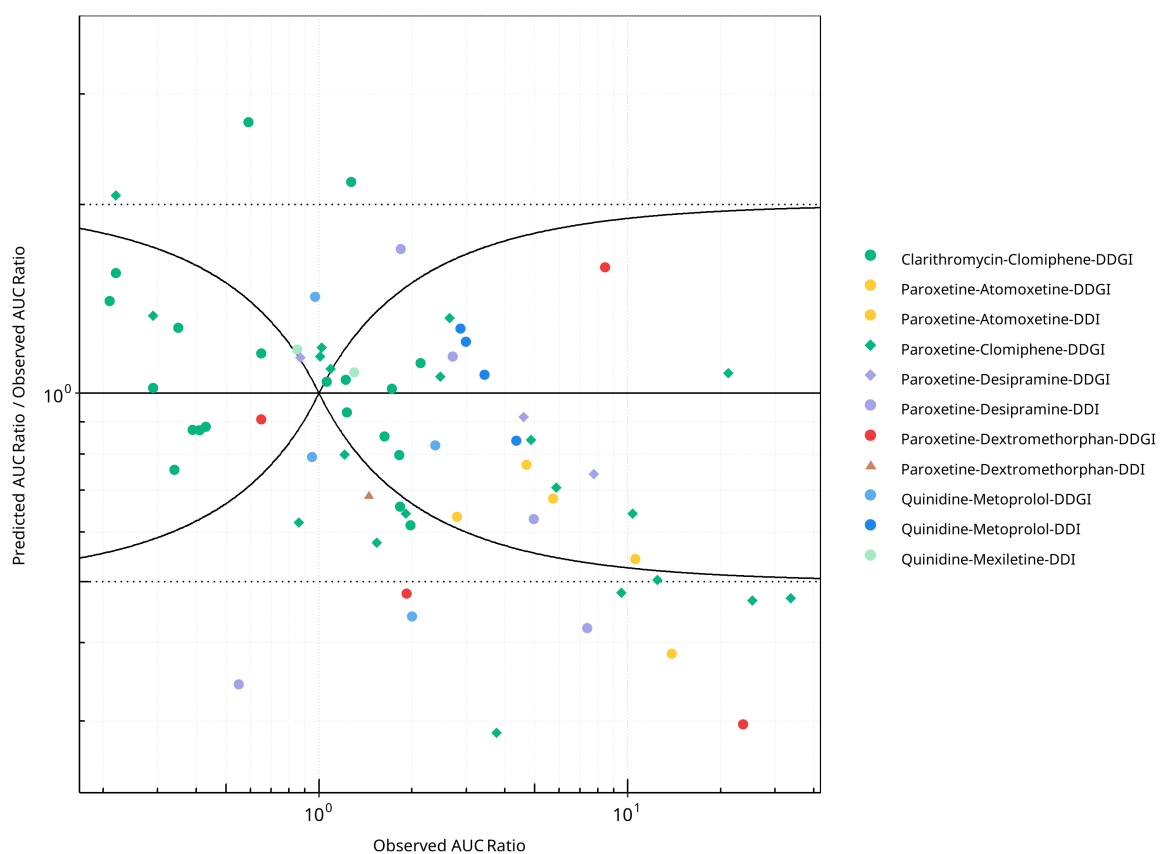


Figure 4-2: CYP2D6 DDGIs. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

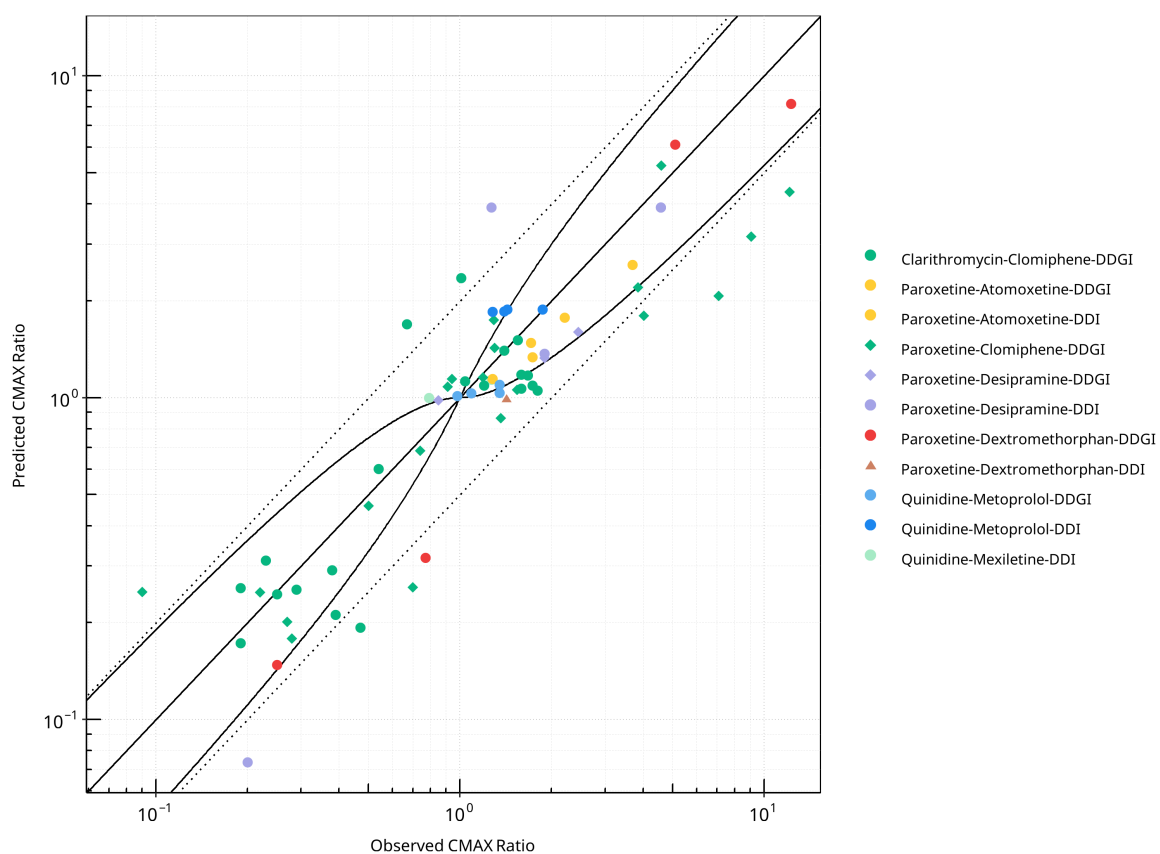


Figure 4-3: CYP2D6 DDGIs. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

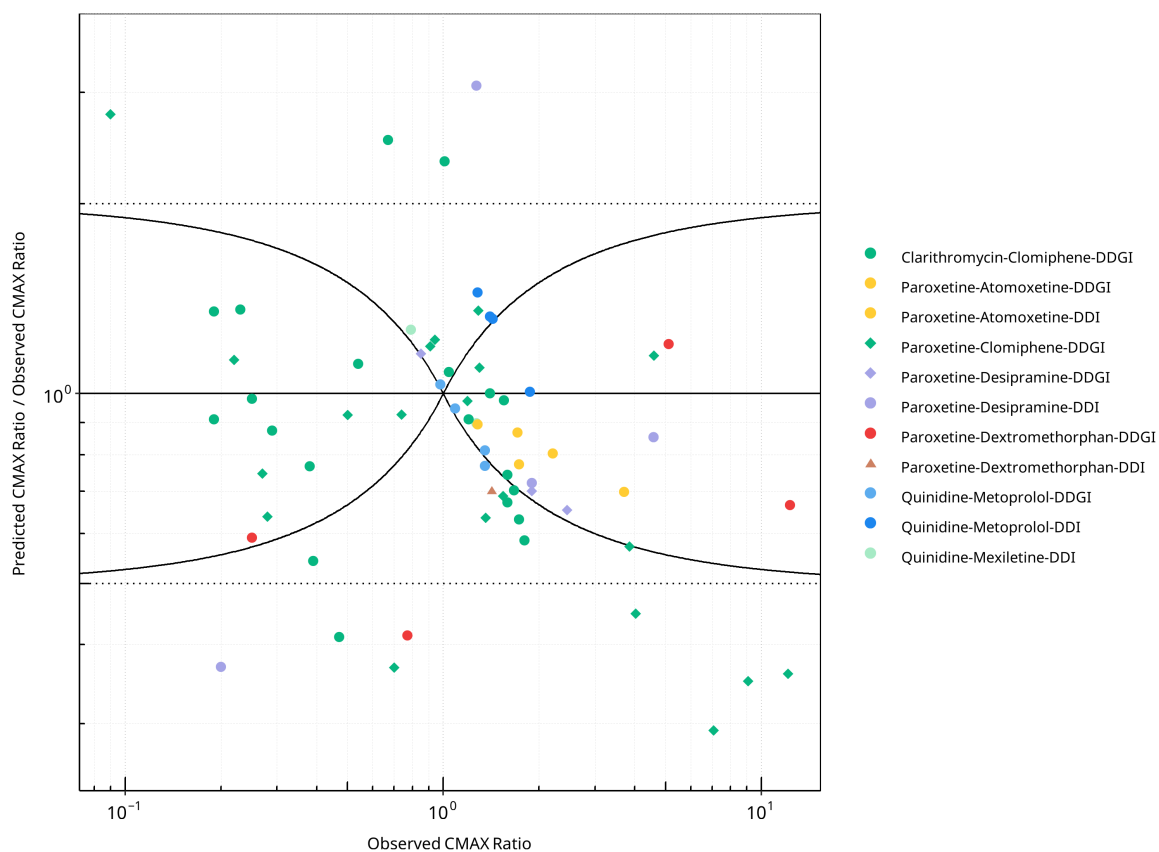


Figure 4-4: CYP2D6 DDGIs. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 4-1: GMFE for CYP2D6 DDGIs Ratio

PK parameter	GMFE
AUC	1.45
CMAX	1.45

Table 4-2: Summary table for CYP2D6 DDGIs - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	68	-
Points within Guest <i>et al.</i>	39	57.35
Points within 2 fold	55	80.88

Table 4-3: Summary table for CYP2D6 DDGIs - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

C _{MAX}	Number	Ratio [%]
Points total	68	-
Points within Guest <i>et al.</i>	36	52.94
Points within 2 fold	56	82.35

Table 4-4: Summary table for CYP2D6 DDGIs

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16950	Quinidine, 50 mg, PO, QID	Mexiletine, PO	1.00	0.85	1.18	1.00	0.79	1.27	Abolfathi 1993
16952	Quinidine, 50 mg, PO, QID	Mexiletine, PO	1.41	1.30	1.08	1.14	1.27	0.90	Abolfathi 1993
16988	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	1.78	2.80	0.64	1.15	1.28	0.90	Jung 2020
16990	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	3.62	4.70	0.77	1.49	1.71	0.87	Jung 2020
16992	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.76	10.57	0.54	1.78	2.21	0.81	Jung 2020
16994	Paroxetine, 20/20 mg, PO, BID/QD	Atomoxetine, PO	3.90	5.73	0.68	1.34	1.73	0.77	Todor 2015
16996	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.34	13.87	0.38	2.59	3.70	0.70	Belle 2002
16998	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	2.38	2.13	1.12	1.52	1.55	0.98	Mürdter 2016
17000	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxyclophene, PO	2.77	1.27	2.18	2.36	1.01	2.34	Mürdter 2016
17002	Clarithromycin, 500 mg, PO, BID	(E)-N-Desethylclomiphene, PO	0.75	0.65	1.16	0.60	0.54	1.12	Mürdter 2016
17004	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N-desethylclomiphene, PO	1.60	0.59	2.71	1.70	0.67	2.53	Mürdter 2016
17006	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.45	1.82	0.80	1.18	1.59	0.75	Mürdter 2016
17008	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxyclophene, PO	1.75	1.72	1.02	1.41	1.40	1.00	Mürdter 2016
17010	Clarithromycin, 500 mg, PO, BID	(E)-N-Desethylclomiphene, PO	0.30	0.29	1.02	0.17	0.19	0.91	Mürdter 2016
17012	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N-desethylclomiphene, PO	0.45	0.35	1.27	0.25	0.25	0.98	Mürdter 2016
17014	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.22	1.98	0.62	1.10	1.73	0.63	Mürdter 2016
17016	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxyclophene, PO	1.39	1.63	0.86	1.18	1.67	0.70	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17018	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.26	0.34	0.76	0.19	0.47	0.41	Mürdter 2016
17020	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.38	0.43	0.89	0.21	0.39	0.54	Mürdter 2016
17022	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.15	1.23	0.93	1.07	1.59	0.67	Mürdter 2016
17024	Clarithromycin, 500 mg, PO, BID	(E)-4- Hydroxyclophene, PO	1.28	1.22	1.05	1.13	1.04	1.09	Mürdter 2016
17026	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.30	0.21	1.41	0.26	0.19	1.35	Mürdter 2016
17028	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.34	0.22	1.56	0.25	0.29	0.88	Mürdter 2016
17030	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.11	1.06	1.05	1.06	1.80	0.59	Mürdter 2016
17032	Clarithromycin, 500 mg, PO, BID	(E)-4- Hydroxyclophene, PO	1.21	1.83	0.66	1.09	1.20	0.91	Mürdter 2016
17034	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.34	0.39	0.88	0.31	0.23	1.36	Mürdter 2016
17036	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.36	0.41	0.88	0.29	0.38	0.77	Mürdter 2016
17038	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	1.19	1.09	1.10	1.08	0.91	1.19	Mürdter 2016
17040	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	1.21	1.02	1.19	1.16	1.19	0.98	Mürdter 2016
17042	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	0.97	1.21	0.80	0.87	1.36	0.64	Mürdter 2016
17044	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	1.16	1.01	1.15	1.15	0.94	1.22	Mürdter 2016
17046	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	2.63	2.47	1.06	1.43	1.30	1.10	Mürdter 2016
17048	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	0.46	0.22	2.07	0.25	0.09	2.78	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17050	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	4.10	4.86	0.84	1.75	1.29	1.36	Mürdter 2016
17052	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.39	0.29	1.33	0.25	0.22	1.13	Mürdter 2016
17054	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	4.59	9.53	0.48	1.81	4.02	0.45	Mürdter 2016
17056	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	0.54	0.86	0.62	0.18	0.28	0.64	Mürdter 2016
17058	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	11.85	25.32	0.47	3.18	9.07	0.35	Mürdter 2016
17060	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	1.23	1.91	0.64	0.46	0.50	0.93	Mürdter 2016
17062	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	6.66	10.35	0.64	2.21	3.85	0.57	Mürdter 2016
17064	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	0.89	1.54	0.58	0.20	0.27	0.75	Mürdter 2016
17066	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	22.80	21.14	1.08	5.28	4.59	1.15	Mürdter 2016
17068	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	3.50	2.65	1.32	0.69	0.74	0.93	Mürdter 2016
17070	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	6.29	12.47	0.50	2.08	7.08	0.29	Mürdter 2016
17072	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	1.08	3.76	0.29	0.26	0.70	0.37	Mürdter 2016
17074	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	15.92	33.72	0.47	4.38	12.14	0.36	Mürdter 2016
17076	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	4.16	5.87	0.71	1.06	1.54	0.69	Mürdter 2016
17082	Paroxetine, 20 mg, PO, QD	Desipramine, PO	3.11	2.71	1.15	1.37	1.90	0.72	Nichols 2009
17084	Paroxetine, 20 mg, PO, QD	2- Hydroxydesipramine, PO	0.19	0.55	0.34	0.07	0.20	0.37	Nichols 2009
17086	Paroxetine, 20 mg, PO, QD	Desipramine, PO	0.99	0.87	1.14	0.99	0.85	1.16	Brösen 1993

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17088	Paroxetine, 20 mg, PO, QD	Desipramine, PO	4.22	4.60	0.92	1.33	1.90	0.70	Brösen 1993
17090	Paroxetine, 20 mg, PO, QD	Desipramine, PO	5.78	7.76	0.74	1.61	2.45	0.66	Brösen 1993
17092	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	1.84	1.70	3.91	1.27	3.08	Aldermann 1997
17093	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	4.96	0.63	3.91	4.58	0.85	Aldermann 1997
17094	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	7.40	0.42	3.91	4.58	0.85	Aldermann 1997
17098	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	13.46	8.45	1.59	6.14	5.11	1.20	Storelli 2018
17100	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.59	0.65	0.91	0.15	0.25	0.59	Storelli 2018
17102	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	7.01	23.61	0.30	8.21	12.29	0.67	Storelli 2018
17104	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.92	1.92	0.48	0.32	0.77	0.42	Storelli 2018
17106	Paroxetine, 20 mg, PO, QD	Dextromethorphan, PO	1.00	1.45	0.69	1.00	1.42	0.70	Schoedel 2012
17136	Quinidine, 50 mg, PO, SD	Metoprolol, IV	0.75	0.95	0.79	1.02	0.98	1.04	Leemann 1993
17138	Quinidine, 50 mg, PO, SD	Metoprolol, IV	0.88	2.00	0.44	1.04	1.35	0.77	Leemann 1993
17140	Quinidine, 250 mg, PO, BID	Metoprolol, IV	1.39	0.97	1.43	1.04	1.09	0.95	Leemann 1993
17142	Quinidine, 250 mg, PO, BID	Metoprolol, IV	1.97	2.38	0.83	1.10	1.35	0.81	Leemann 1993
17144	Quinidine, 100 mg, PO, QD	R-Metoprolol, PO	3.69	3.44	1.07	1.88	1.43	1.32	Johnson 1996
17146	Quinidine, 100 mg, PO, QD	S-Metoprolol, PO	3.65	2.87	1.27	1.86	1.28	1.45	Johnson 1996
17148	Quinidine, 100 mg, PO, QD	R-Metoprolol, PO	3.66	4.35	0.84	1.89	1.87	1.01	Johnson 1996
17150	Quinidine, 100 mg, PO, QD	S-Metoprolol, PO	3.62	2.99	1.21	1.86	1.40	1.33	Johnson 1996

4.1 DDGI Network Evaluations

Figure 1 shows the developed CYP2D6 DDGI network of interacting perpetrator and victim drugs. (a) Schematic illustration of the modeled interactions of CYP2D6 perpetrator and victim drugs. Black arrows indicate metabolism or transport, green arrows indicate induction, red solid lines indicate competitive inhibition, red dashed lines down-regulation (bupropion), noncompetitive inhibition (verapamil P-gp inhibition) or mechanism-based inactivation (others). (b-c) Drug-drug-(gene) interaction matrix for modeled interactions mediated by (a) CYP2D6 and (b) CYP3A4 and P-gp. Colors indicate categories according to the FDA Examples of Drugs that Interact with CYP Enzymes and Transporter Systems.⁴² Height of the grey ribbons indicates the number of clinical studies for the respective interaction covered by the network, numbers in brackets indicate the number of clinical interaction studies for the corresponding compound. CYP: cytochrome P450, P-gp: P-glycoprotein.

Figure 1: CYP2D6 DDGI modeling network

The following victim and/or perpetrator PBPK models were applied:

- **Alprazolam** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Alprazolam-Model/blob/master/Alprazolam-Model.json>
- **Atomoxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/blob/main/atomoxetine-model.json>
- **Bupropion** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Bupropion-Model/blob/main/Bupropion-Model.json>
- **Carbamazepine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Carbamazepine-Model/blob/main/Carbamazepine-Model.json>
- **Cimetidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Cimetidine-Model/blob/master/Cimetidine-Model.json>
- **Clarithromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/blob/master/Clarithromycin-Model.json>
- **(E)-Clomiphene** Model snapshot: [https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/\(E\)-clomiphene-DGI-Model.json](https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/(E)-clomiphene-DGI-Model.json)
- **Desipramine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Desipramine-Model/blob/main/Desipramine-Model.json>
- **Dextromethorphan** Model snapshot: https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/blob/main/dextromethorphan_aggregated_simulations.json
- **Digoxin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/blob/master/Digoxin.json>
- **Erythromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/blob/master/Erythromycin-Model.json>
- **Fluvoxamine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Fluvoxamine-Model/blob/master/Fluvoxamine-Model.json>
- **Itraconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/blob/master/Itraconazole-Model.json>
- **Ketoconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/blob/main/Ketoconazole-Model.json>
- **Metoprolol** Model file: <https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/blob/main/Metoprolol-Model.pksim5>
- **Mexiletine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Mexiletine-Model/blob/main/Mexiletine-Model.json>
- **Omeprazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Omeprazole-Model/blob/main/Omeprazole-Model.json>
- **Paroxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/blob/main/paroxetine-model.json>

- **Quinidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Quinidine-Model/blob/main/Quinidine-Model.json>
- **Rifampicin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/blob/master/Rifampicin-Model.json>
- **Risperidone** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Risperidone-Model/blob/main/risperidone-model.json>
- **Verapamil** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/blob/master/Verapamil-Model.json>

The following interaction scenarios were predicted and used to qualify the final DDGI network:

- Alprazolam as victim:
 - Paroxetine-alprazolam-DDI
- Atomoxetine as victim:
 - Bupropion-atomoxetine-DDGI
 - Fluvoxamine-atomoxetine-DDI
 - Paroxetine-atomoxetine-DDGI
- (E)-Clomiphene as victim:
 - Clarithromycin-clomiphene-DDGI
 - Paroxetine-clomiphene-DDGI
- Desipramine as victim:
 - Atomoxetine-desipramine-DDI
 - Bupropion-desipramine-DDI
 - Paroxetine-desipramine-DDGI
 - Quinidine-desipramine-DDI
- Dextromethorphan as victim:
 - Paroxetine-dextromethorphan-DDGI
 - Quinidine-dextromethorphan-DDI
- Digoxin as victim:
 - Quinidine-digoxin-DDI
- Metoprolol as victim:
 - Cimetidine-metoprolol-DDI
 - Paroxetine-metoprolol-DDI
 - Quinidine-metoprolol-DDGI
 - Rifampicin-metoprolol-DDI
- Mexiletine as victim:
 - Quinidine-mexiletine-DDGI
- Midazolam as victim:
 - Atomoxetine-midazolam-DDI
- Paroxetine as victim:
 - Itraconazole-paroxetine-DDI
 - Quinidine-paroxetine-DDI
- Quinidine as victim:
 - Carbamazepine-quinidine-DDI
 - Cimetidine-quinidine-DDI
 - Erythromycin-quinidine-DDI
 - Fluvoxamine-quinidine-DDI
 - Itraconazole-quinidine-DDI

- Omeprazole-quinidine-DDI
- Rifampicin-quinidine-DDI
- Verapamil-quinidine-DDI
- Risperidone as victim:
 - Ketoconazole-risperidone-DDI
 - Rifampicin-risperidone-DDI
 - Verapamil-risperidone-DDI

The published DD(G)I studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report.

4.1.1 Paroxetine - Atomoxetine DDGI

The paroxetine-atomoxetine interaction was evaluated using clinical DDGI studies listed in [Table 4](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Jung 2020	po	20 mg s.d.	Asian	m	7 AS0	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	9 AS1.25	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	10 AS2	+/- PAR, 20 mg q.d. po
Todor 2015	po	25 mg s.d.	European	m	22	+/- PAR, 20 mg b.i.d./q.d. po
Belle 2002	po	20 mg b.i.d.	American	m	22	+/- PAR, 20 mg q.d. po

Table 4: AS: CYP2D6 activity score, b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

The Jung 2020 cohorts are reported by CYP2D6 activity score. These simulations therefore retain activity-score-specific atomoxetine CYP2D6 k_{cat} settings. The remaining atomoxetine studies use the NM/EM setting unless a PM cohort is explicitly reported.

4.1.2 Clarithromycin - Clomiphene DDGI

The clarithromycin-clomiphene interaction was evaluated using clinical DDGI studies listed in [Table 5](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mürdter 2016	po	42 mg s.d.	European	f	5 AS0	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0.5	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS1	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS2	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS3	+/- CLA, 500 mg b.i.d. po

Table 5: AS: CYP2D6 activity score, b.i.d.: twice daily, CLA: clarithromycin, f: female, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.1.3 Paroxetine - Clomiphene DDGI

The paroxetine-clomiphene interaction was evaluated using clinical DDGI studies listed in [Table 6](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0.5	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS1	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS2	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS3	+/- PAR, 40 mg q.d. po

Table 6: AS: CYP2D6 activity score, b.i.d.: twice daily, f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.1.4 Paroxetine - Desipramine DDGI

The paroxetine-desipramine interaction was evaluated using clinical DDGI studies listed in [Table 9](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Nichols 2009	po	50 mg s.d.	American	m	20	+/- PAR, 20 mg q.d. po
Brösen 1993	po	100 mg s.d.	European	m	8 PM	+/- PAR, 20 mg q.d. po
Brösen 1993	po	100 mg s.d.	European	m	5 NM	+/- PAR, 20 mg q.d. po
Brösen 1993	po	100 mg s.d.	European	m	4 fast NM	+/- PAR, 20 mg q.d. po
Aldermann 1997	po	50 mg q.d.	American	m	6	+/- PAR, 20/30 mg q.d. po

Table 9: m: male, N: number of study participants, PAR: paroxetine, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.1.5 Paroxetine - Dextromethorphan DDGI

The paroxetine-dextromethorphan interaction was evaluated using clinical DDGI studies listed in [Table 11](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Storelli 2018	po	5 mg s.d.	European	f	16 IM	+/- PAR, 20 mg b.i.d. po
Storelli 2018	po	5 mg s.d.	European	f	17 NM	+/- PAR, 20 mg b.i.d. po
Schoedel 2012	po	30 mg b.i.d.	American	m	13	+/- PAR, 30 mg q.d. po

Table 11: b.i.d.: twice daily, f: female, IM: CYP2D6 intermediate metabolizer, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, QUI: quinidine, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.1.6 Quinidine - Metoprolol DDGI

The quinidine-metoprolol interaction was evaluated using clinical DDGI studies listed in [Table 16](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Leemann 1993	iv	20 mg s.d.	European	m	3 PM	+/- QUI, 50 mg s.d. po
Leemann 1993	iv	20 mg s.d.	European	m	4 NM	+/- QUI, 50 mg s.d. po
Leemann 1993	iv	20 mg s.d.	European	m	3 PM	+/- QUI, 250 mg b.i.d. po
Leemann 1993	iv	20 mg s.d.	European	m	4 NM	+/- QUI, 250 mg b.i.d. po
Johnson 1996	po	200 mg s.d.	American	m	10	+/- QUI, 100 mg q.d. po
Johnson 1996	po	200 mg s.d.	American	m	10	+/- QUI, 100 mg q.d. po

Table 16: b.i.d.: twice daily, iv: intravenous, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.1.7 Quinidine - Mexiletine DDGI

The quinidine-mexiletine interaction was evaluated using clinical DDGI studies listed in [Table 18](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Abolfathi 1993	po	200 mg s.d.	American	m	4 PM	+/- QUI, 50 mg q.i.d. po
Abolfathi 1993	po	200 mg s.d.	American	m	10 NM	+/- QUI, 50 mg q.i.d. po

Table 18: m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.i.d.: four times daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

4.2 DDGI Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

4.2.1 Paroxetine - Atomoxetine DDGI

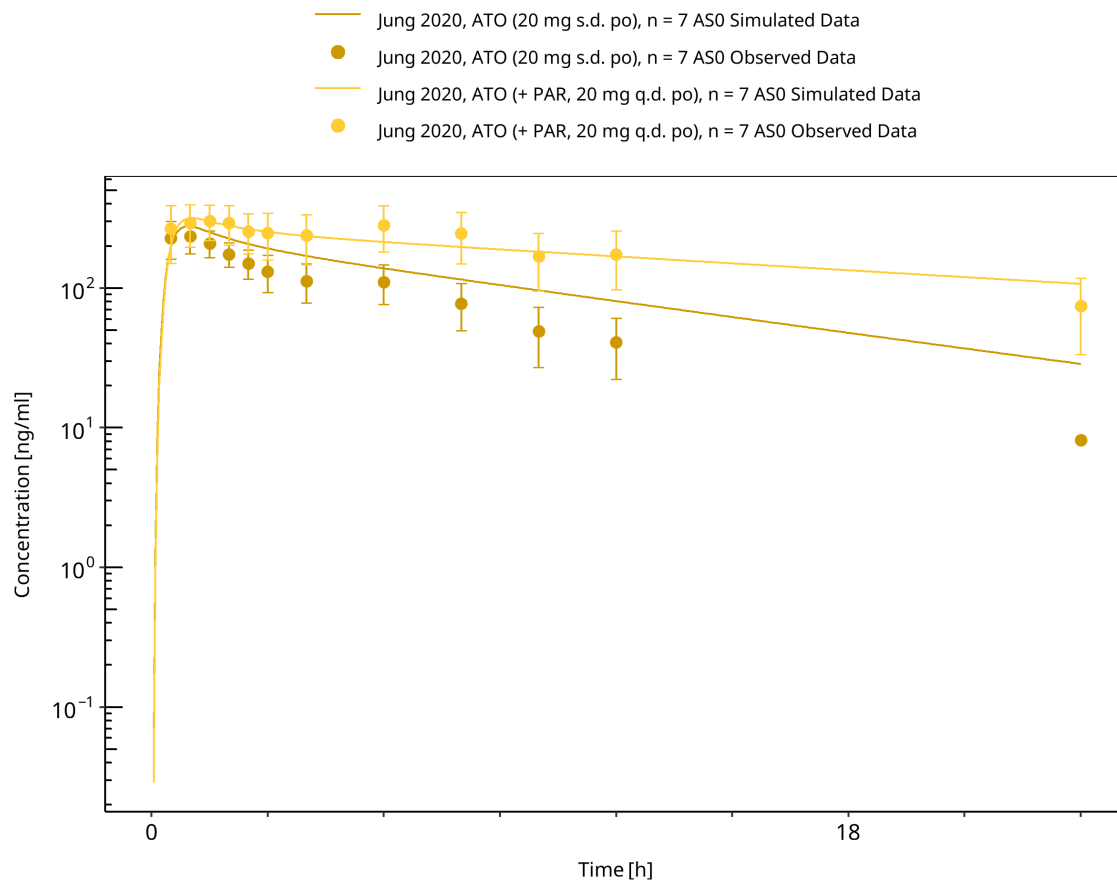


Figure 4-6: Paroxetine - Atomoxetine DDGI

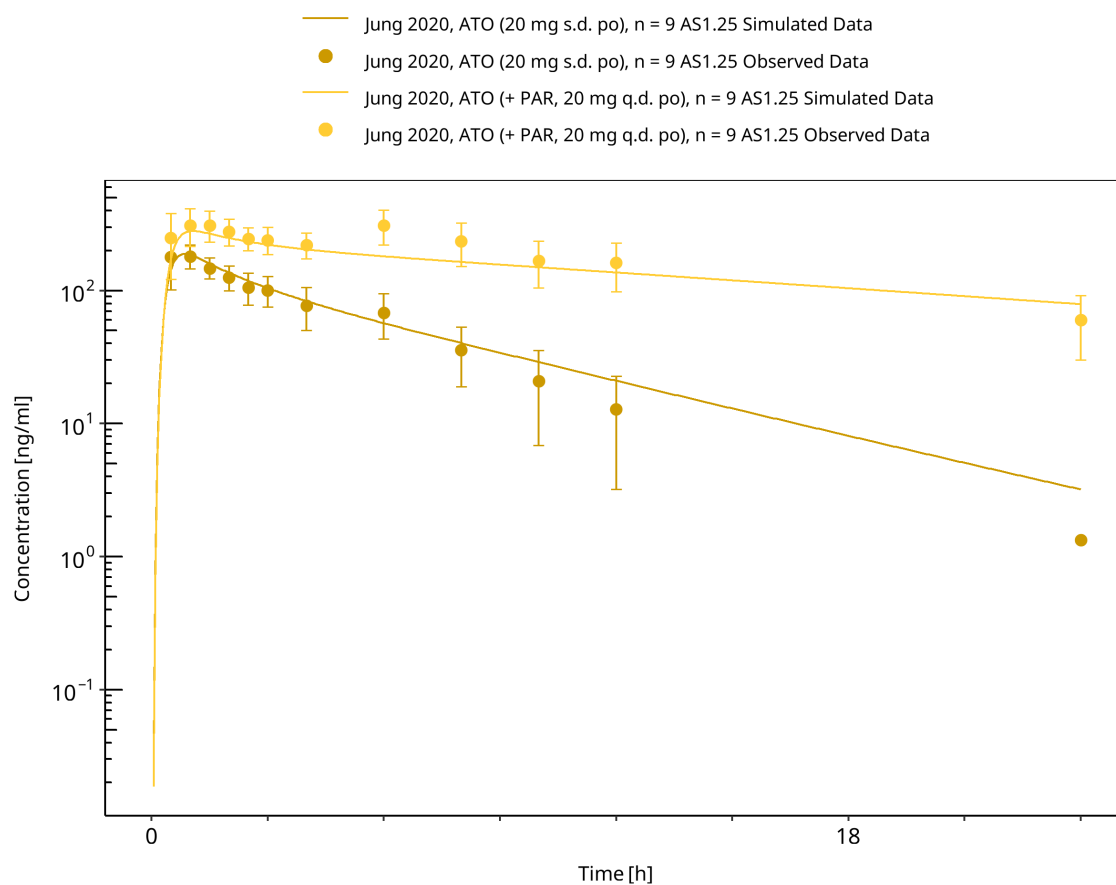


Figure 4-7: Paroxetine - Atomoxetine DDGI

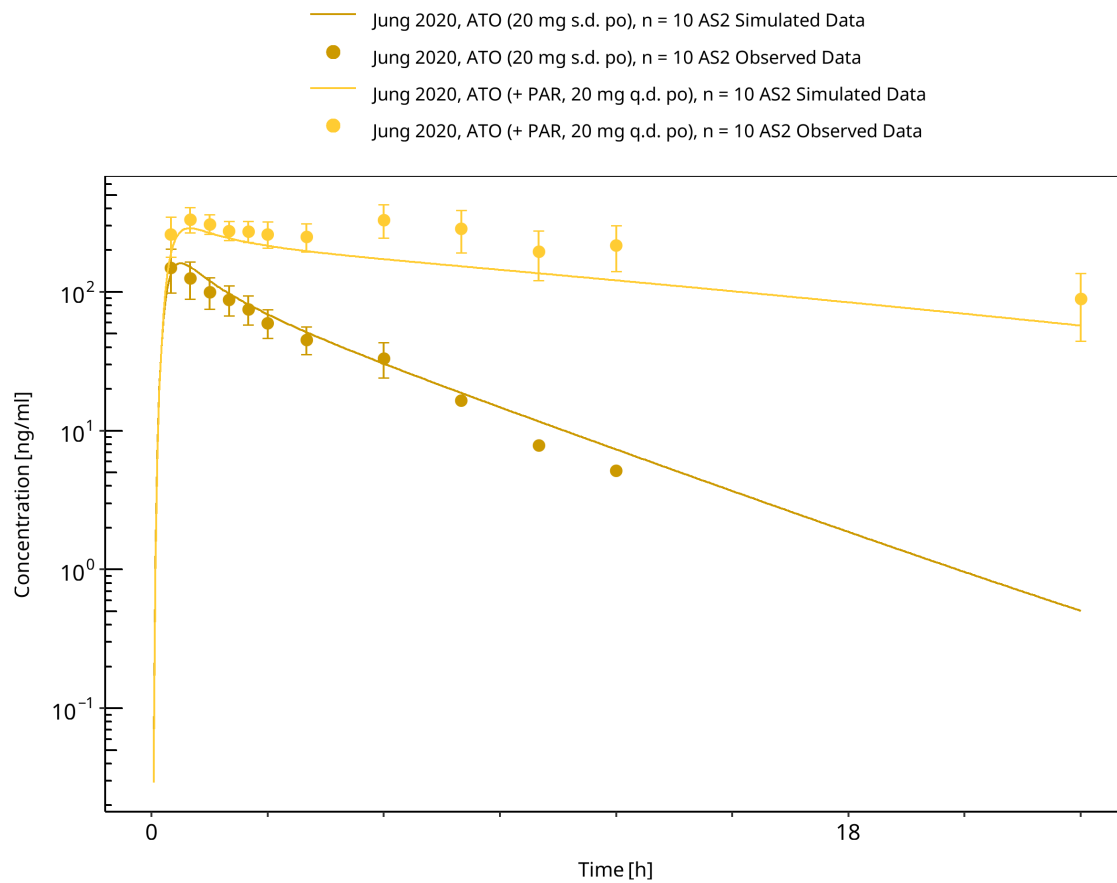


Figure 4-8: Paroxetine - Atomoxetine DDGI

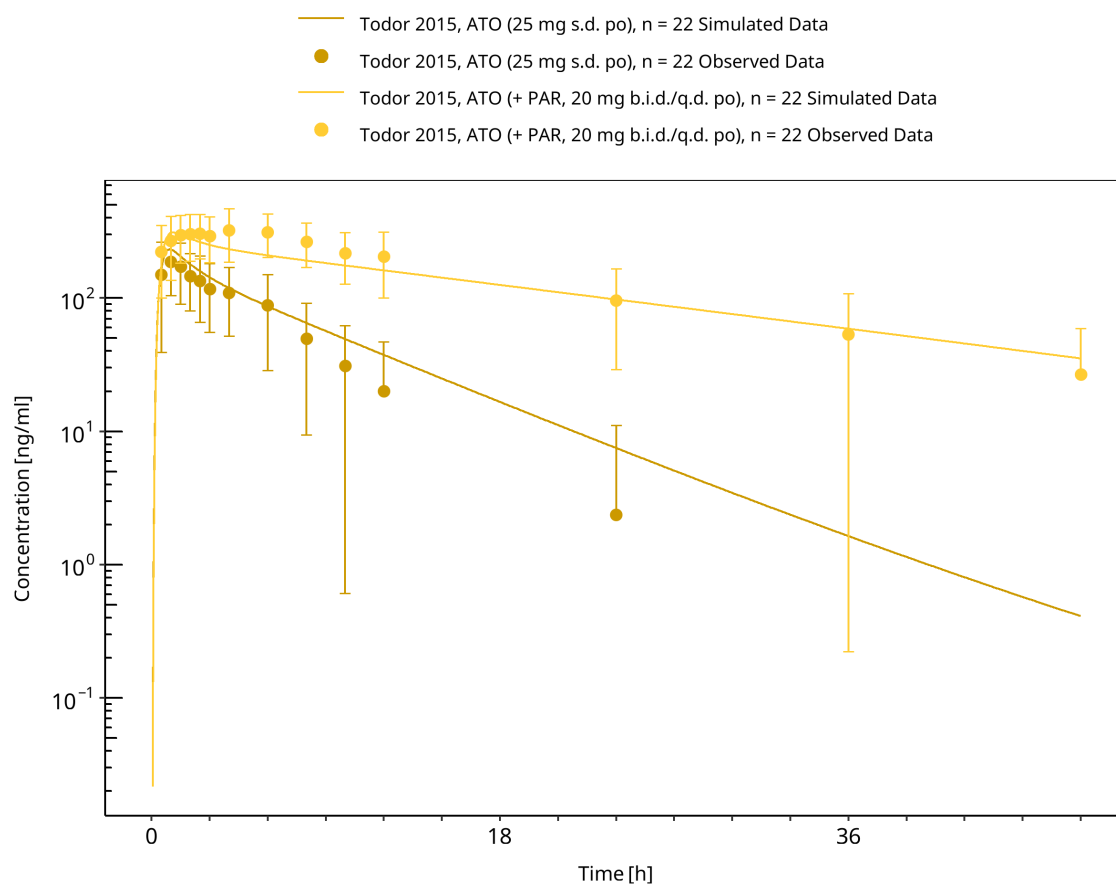


Figure 4-9: Paroxetine - Atomoxetine DDI

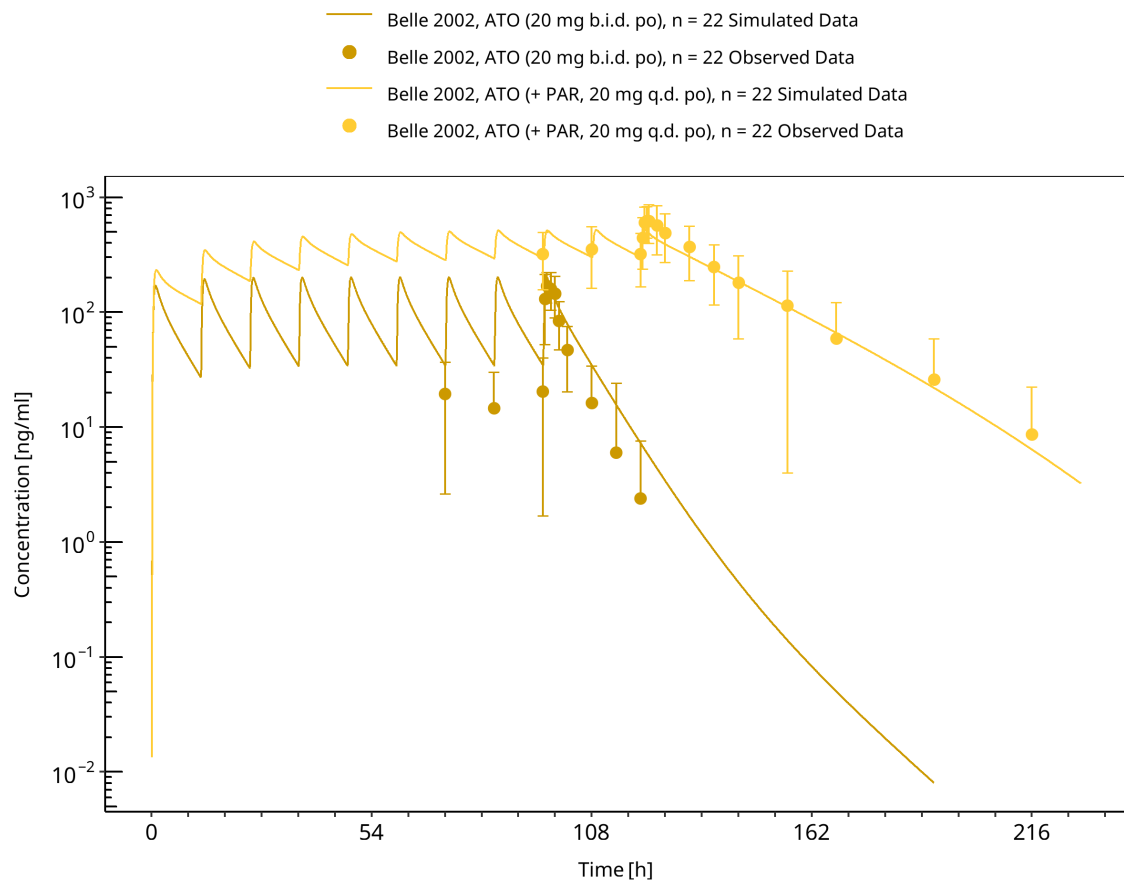


Figure 4-10: Paroxetine - Atomoxetine DDI

4.2.2 Clarithromycin - Clomiphene DDGI

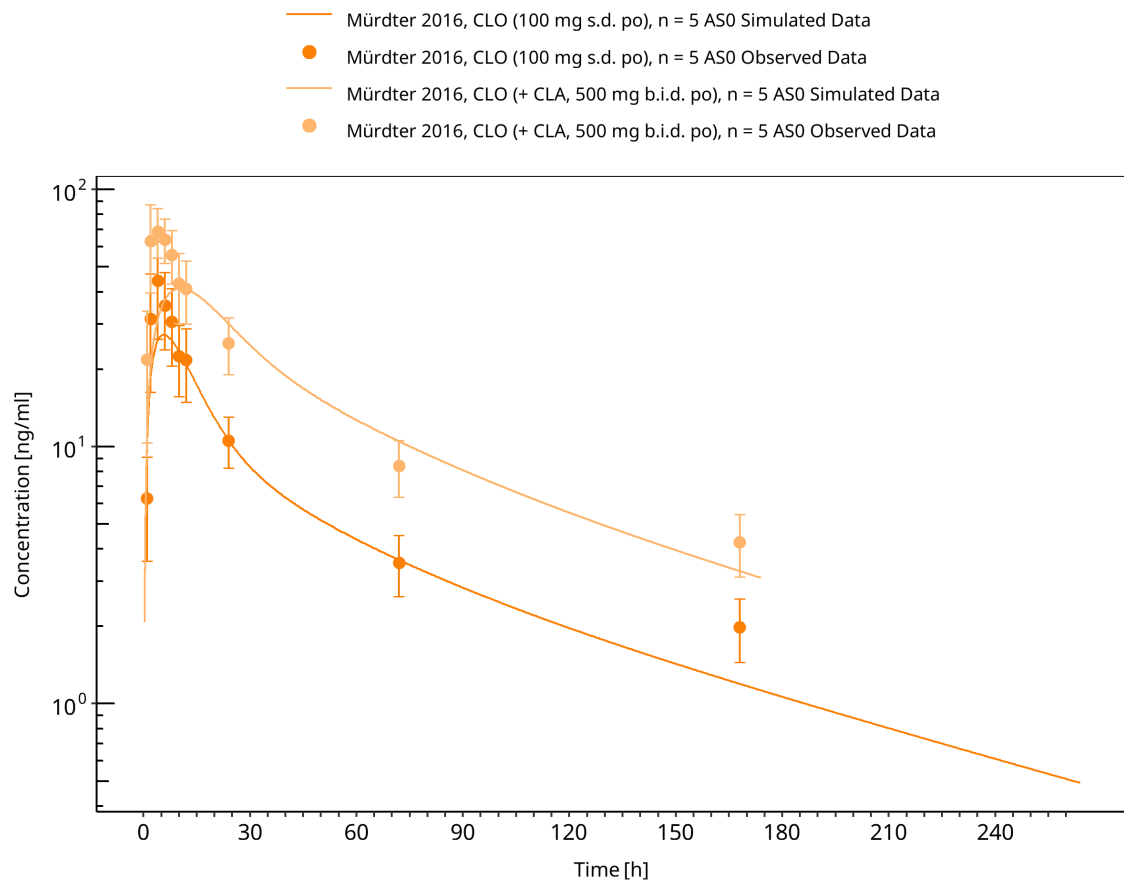


Figure 4-11: Clarithromycin - Clomiphene DDGI

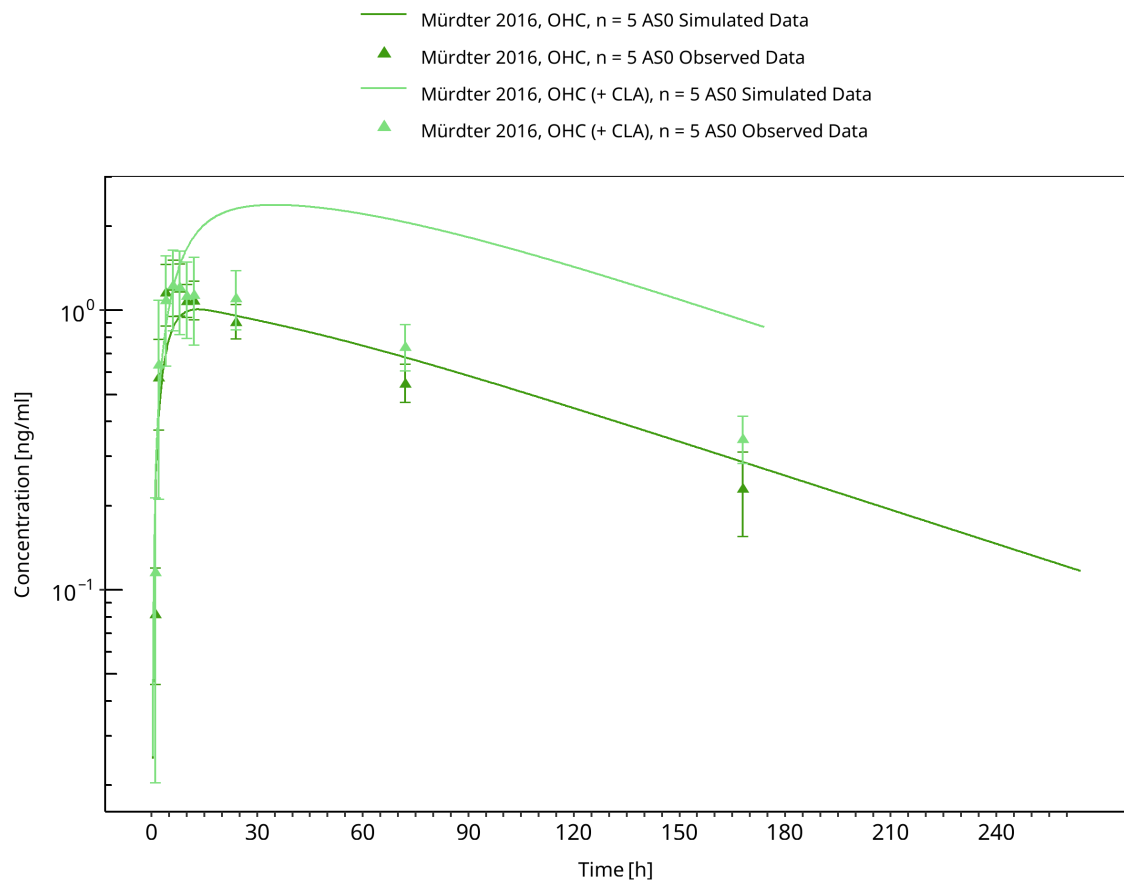


Figure 4-12: Clarithromycin - Clomiphene DDGI

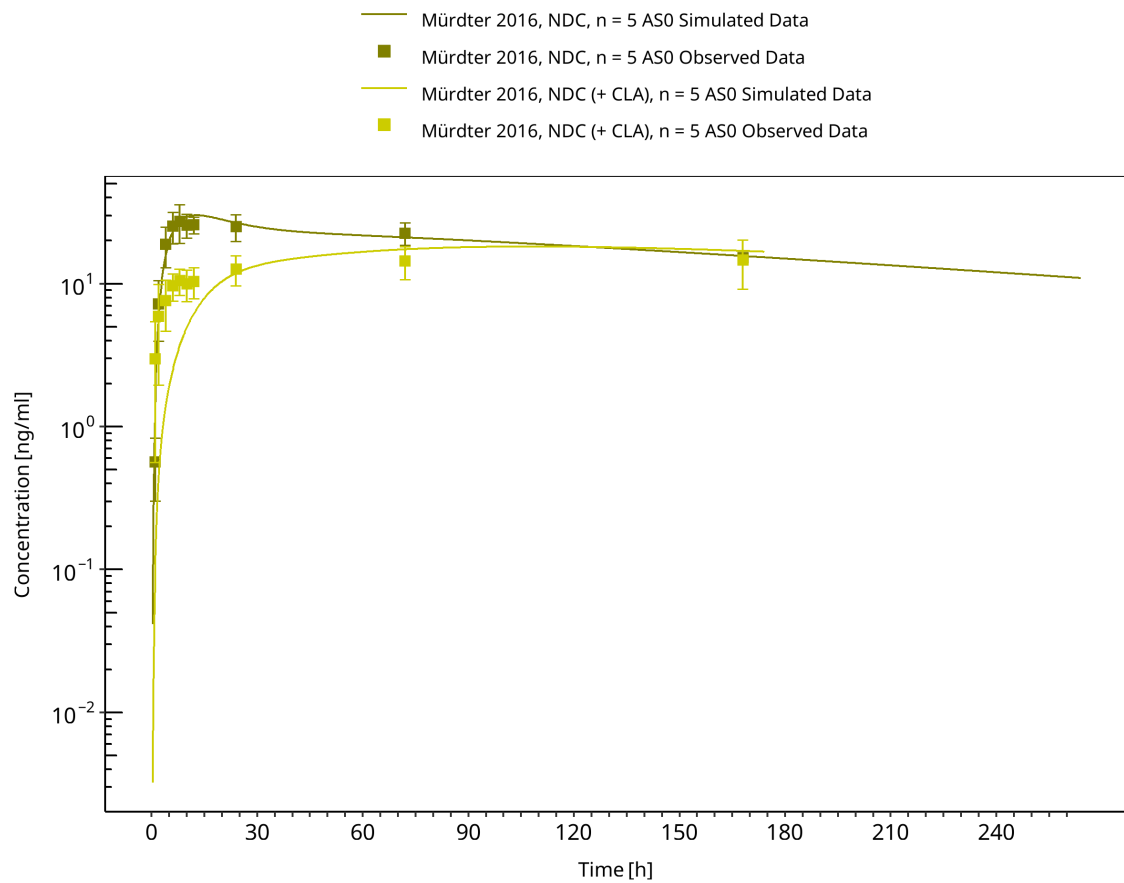


Figure 4-13: Clarithromycin - Clomiphene DDGI

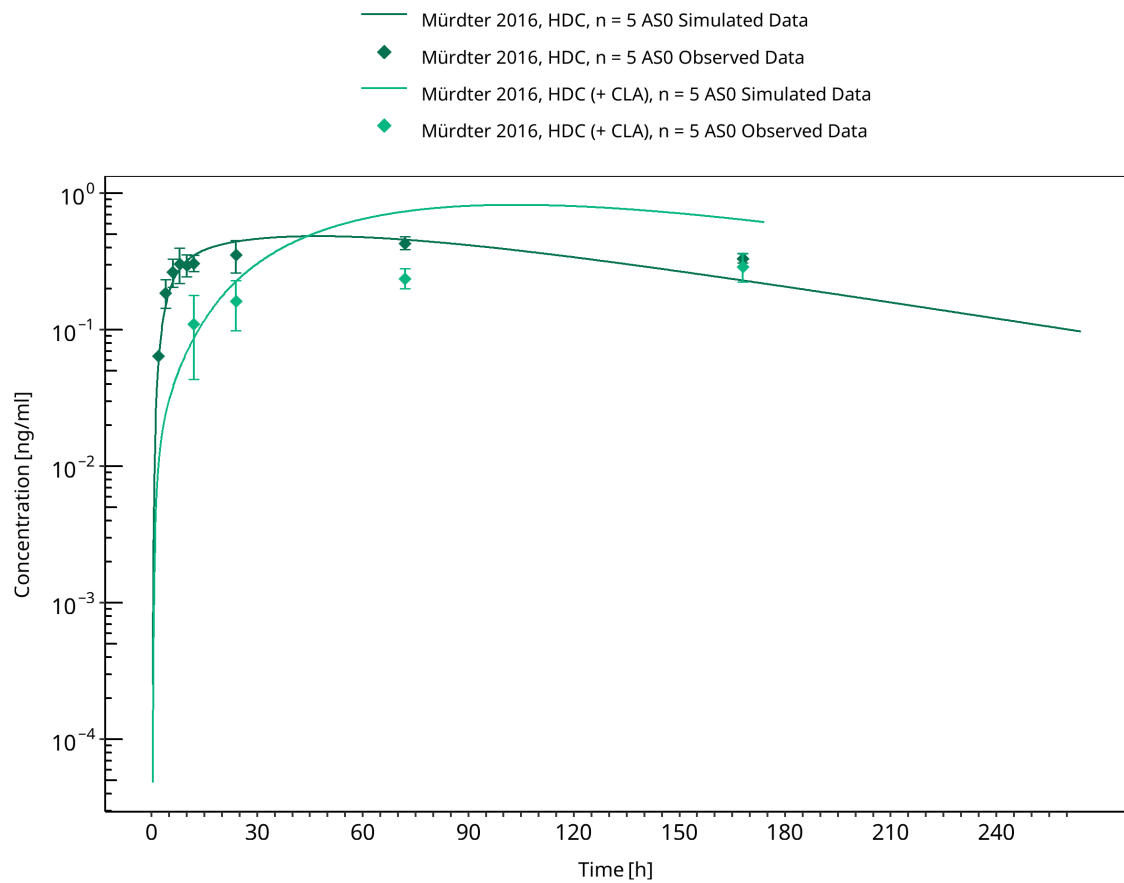


Figure 4-14: Clarithromycin - Clomiphene DDGI

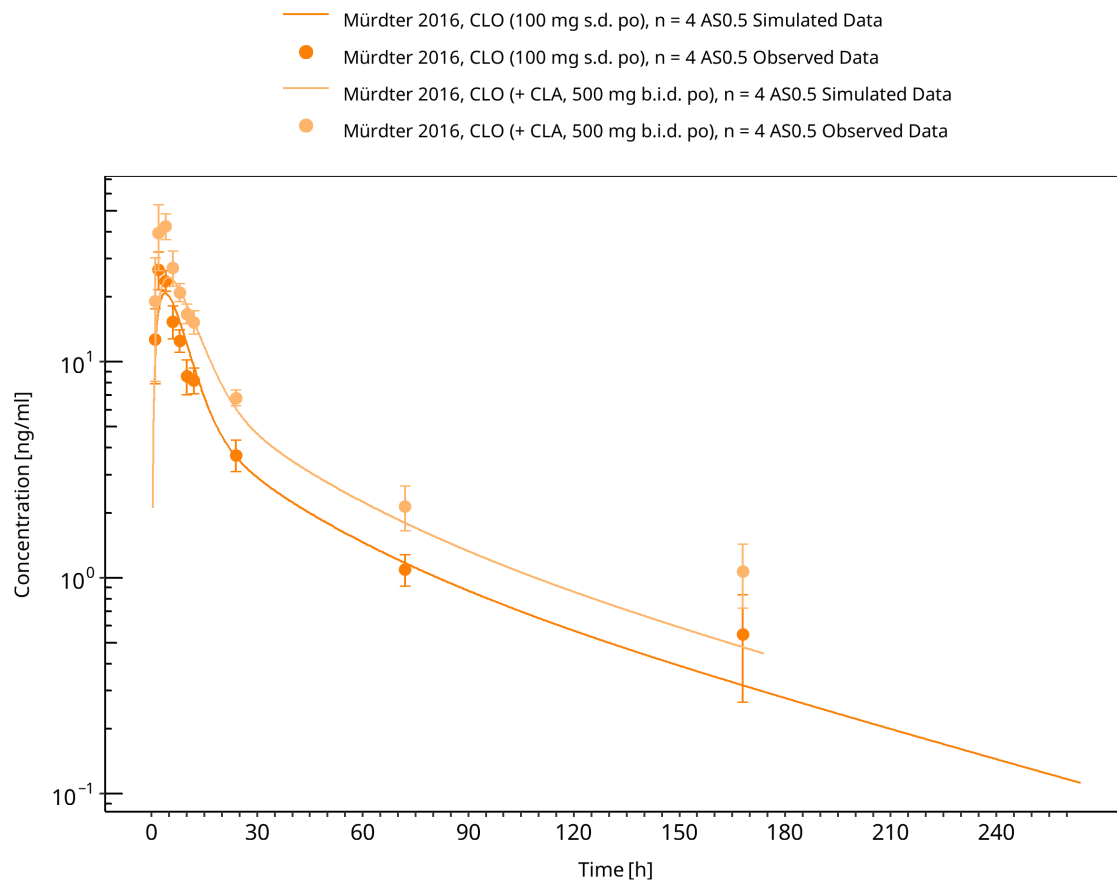


Figure 4-15: Clarithromycin - Clomiphene DDGI

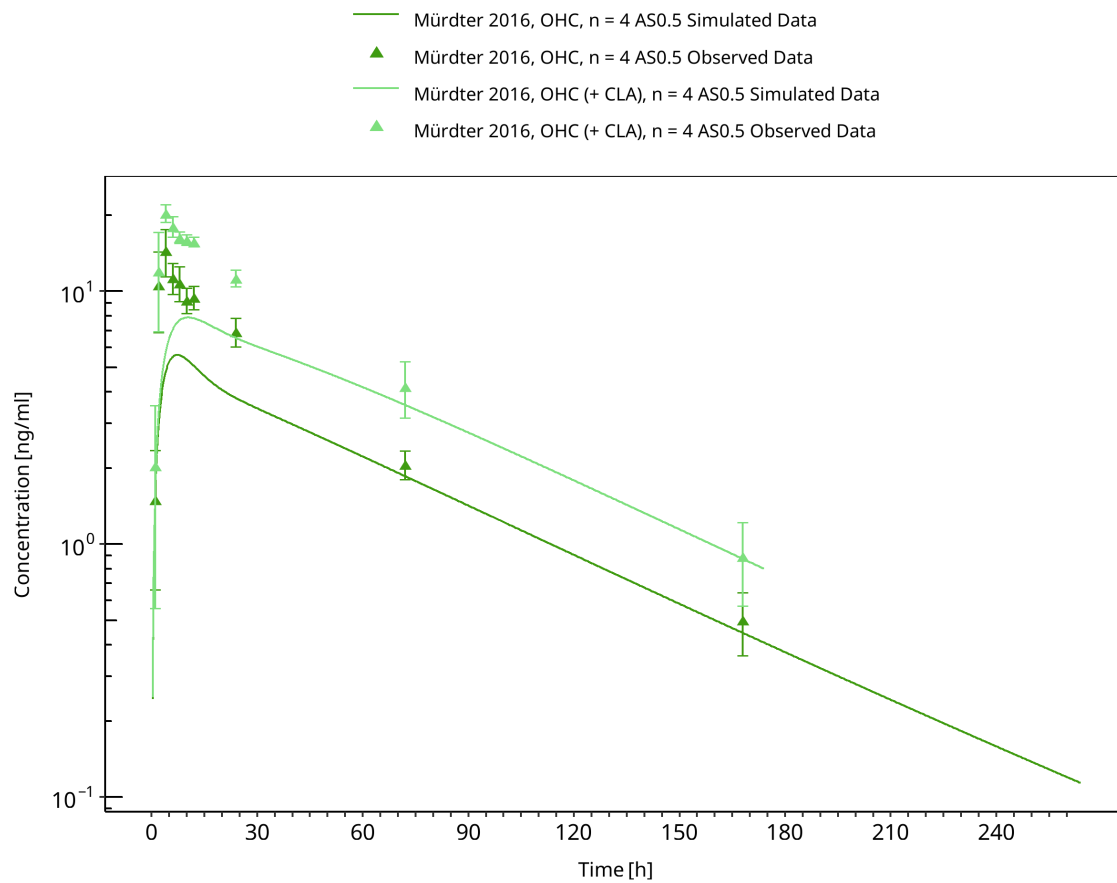


Figure 4-16: Clarithromycin - Clomiphene DDGI

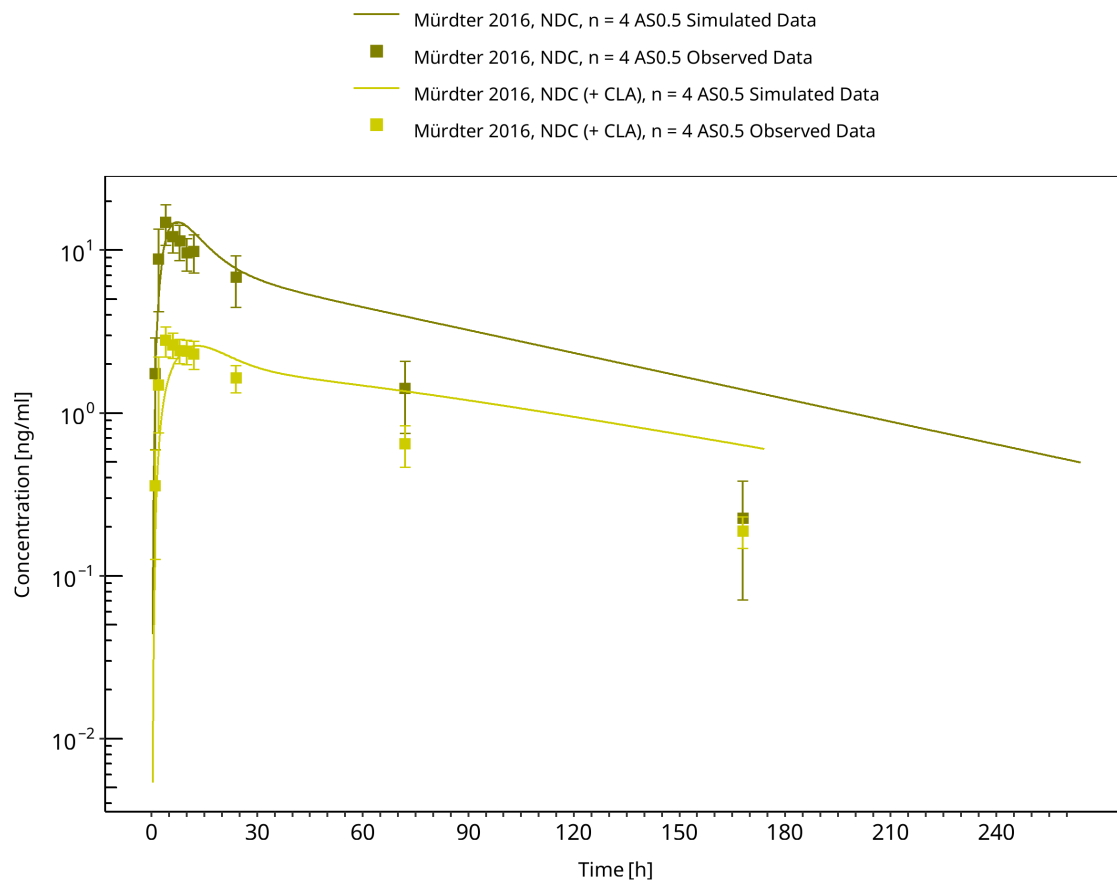


Figure 4-17: Clarithromycin - Clomiphene DDGI

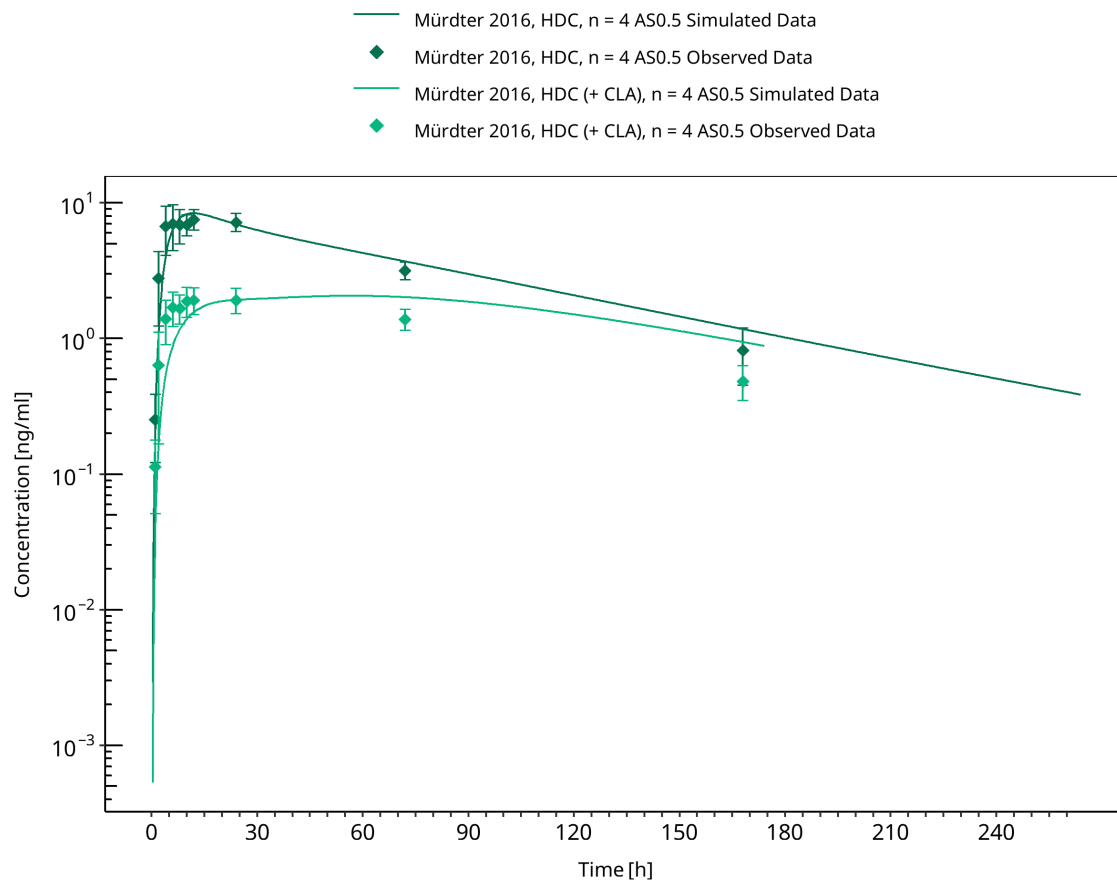


Figure 4-18: Clarithromycin - Clomiphene DDGI

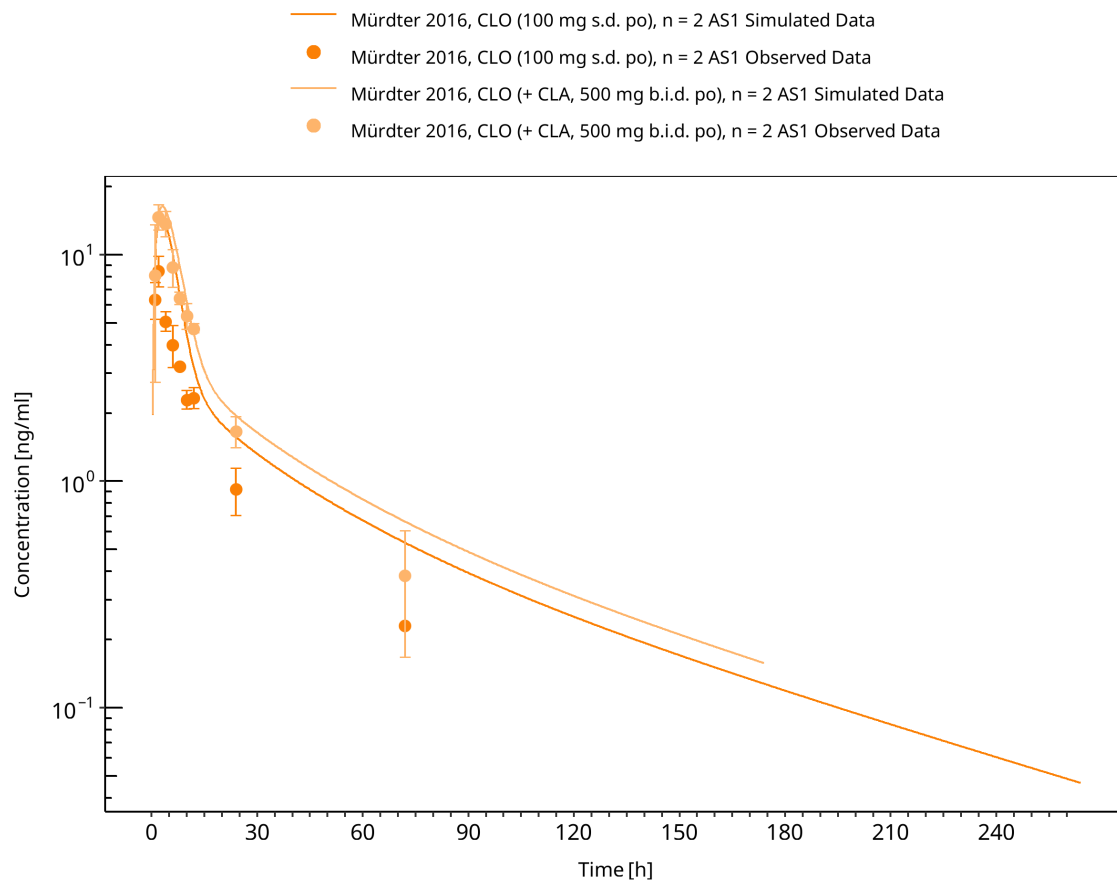


Figure 4-19: Clarithromycin - Clomiphene DDGI

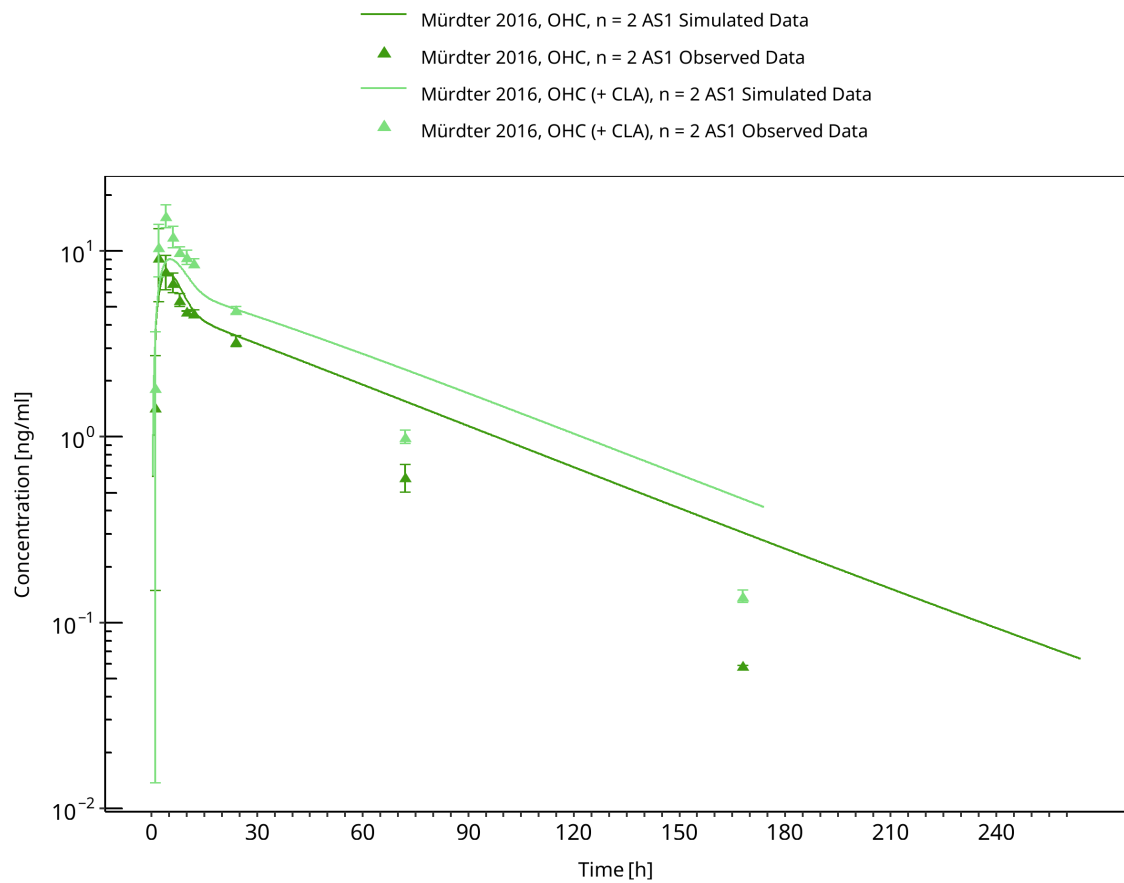


Figure 4-20: Clarithromycin - Clomiphene DDGI

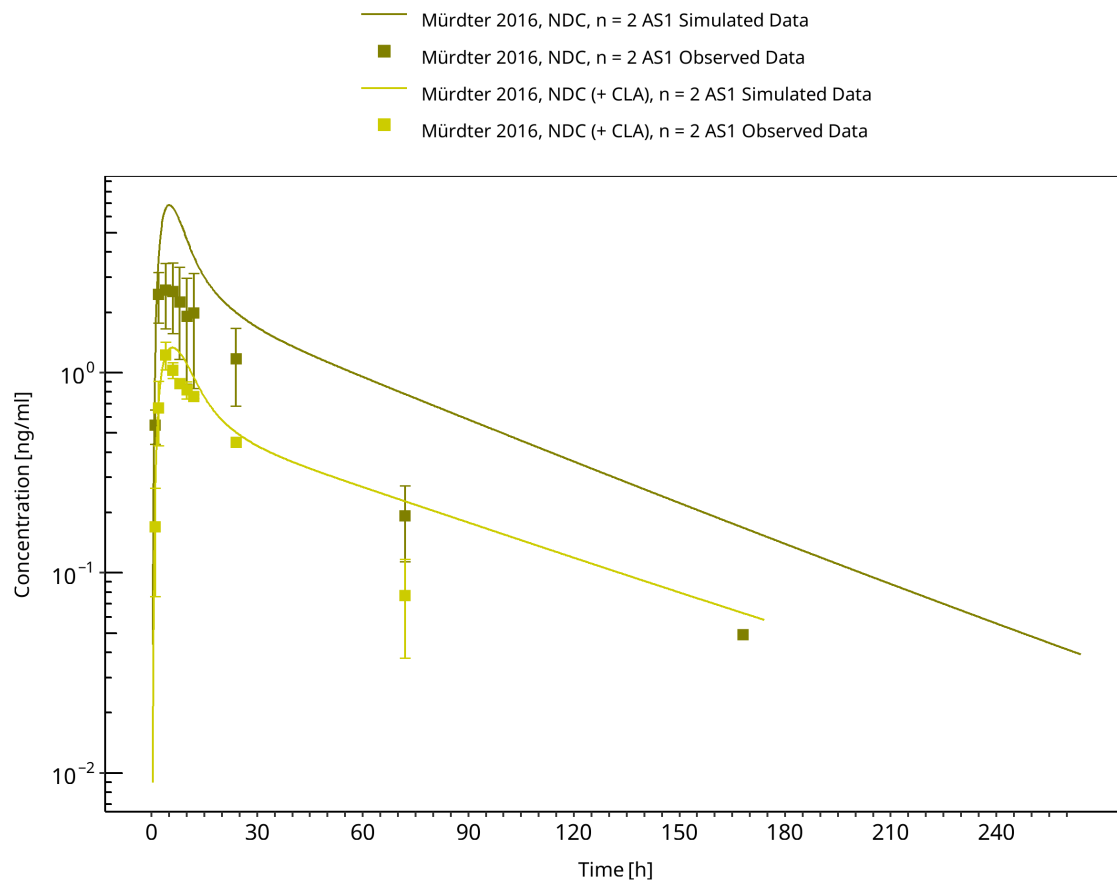


Figure 4-21: Clarithromycin - Clomiphene DDGI

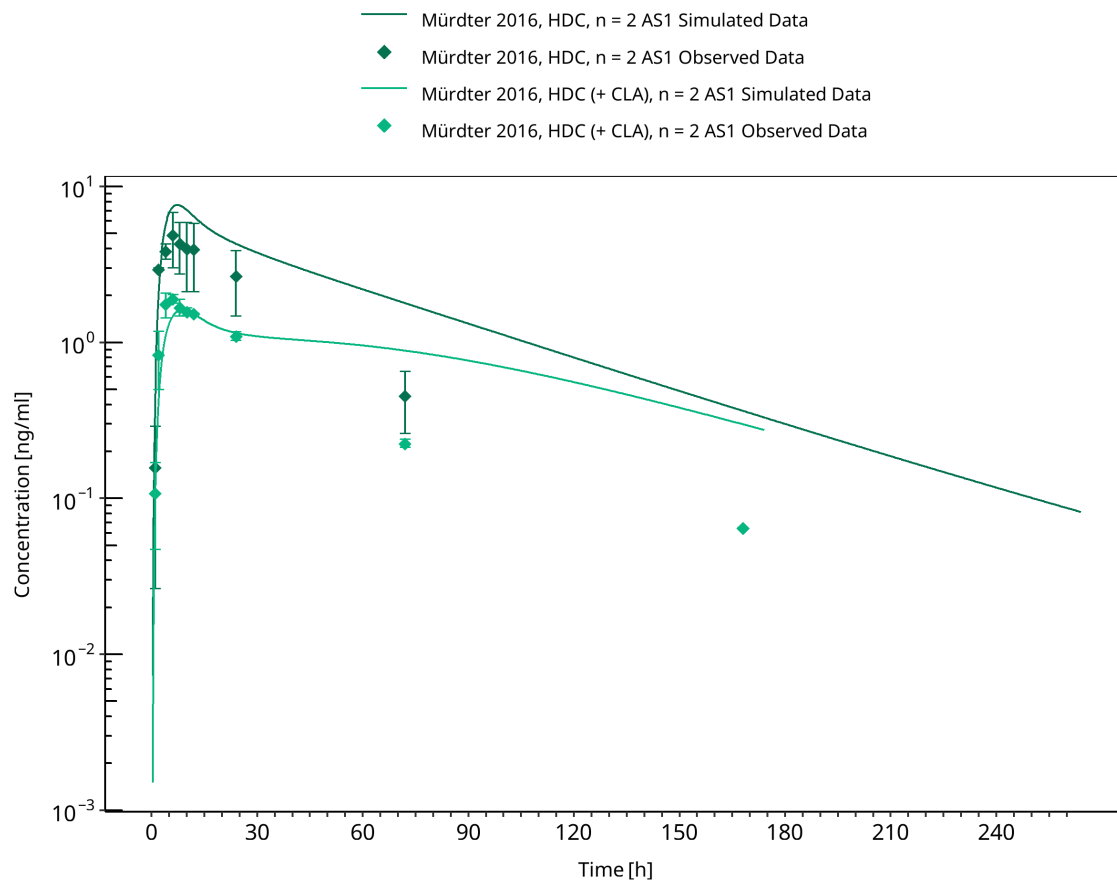


Figure 4-22: Clarithromycin - Clomiphene DDGI

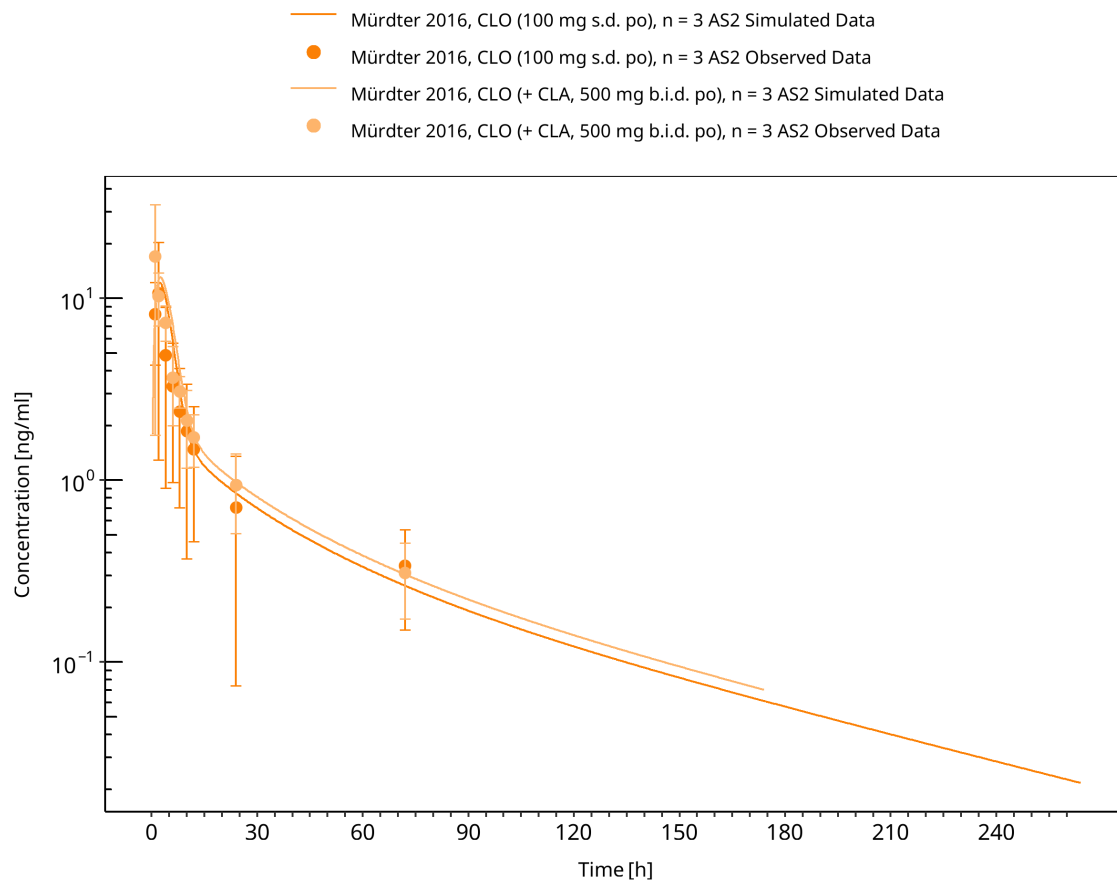


Figure 4-23: Clarithromycin - Clomiphene DDGI

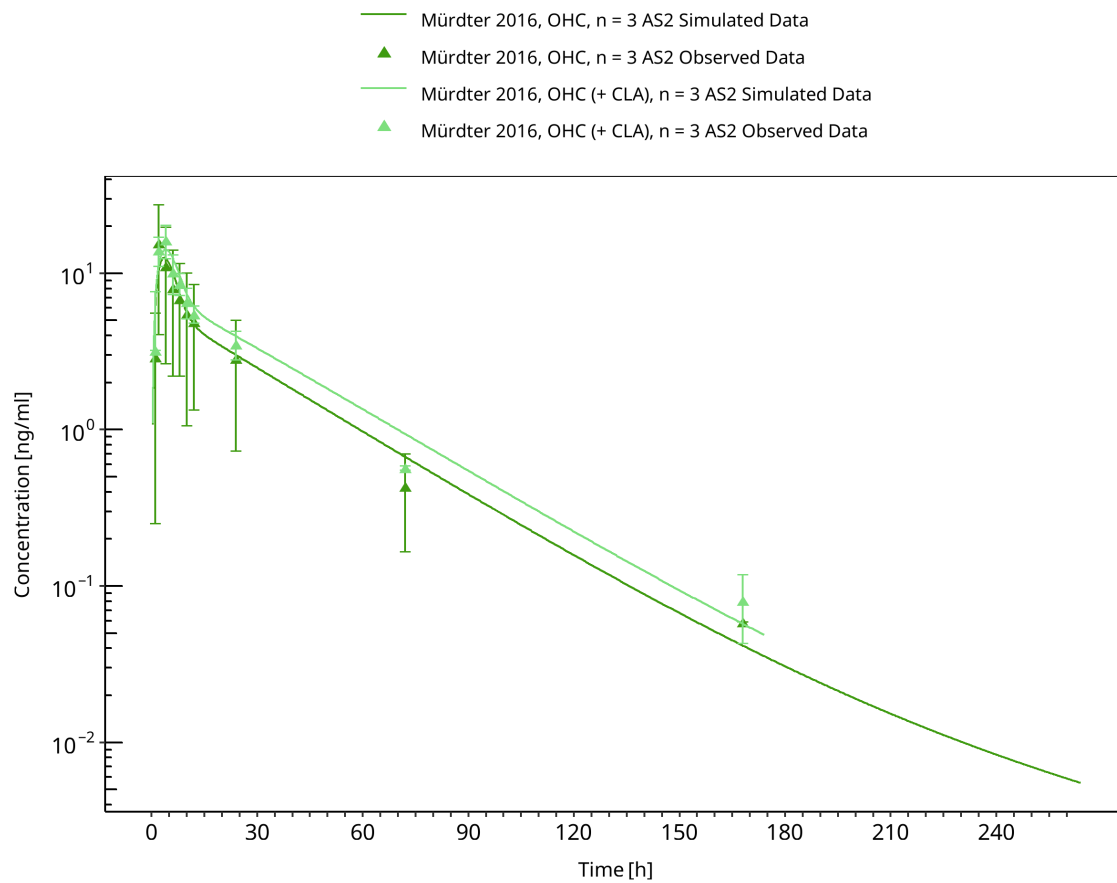


Figure 4-24: Clarithromycin - Clomiphene DDGI

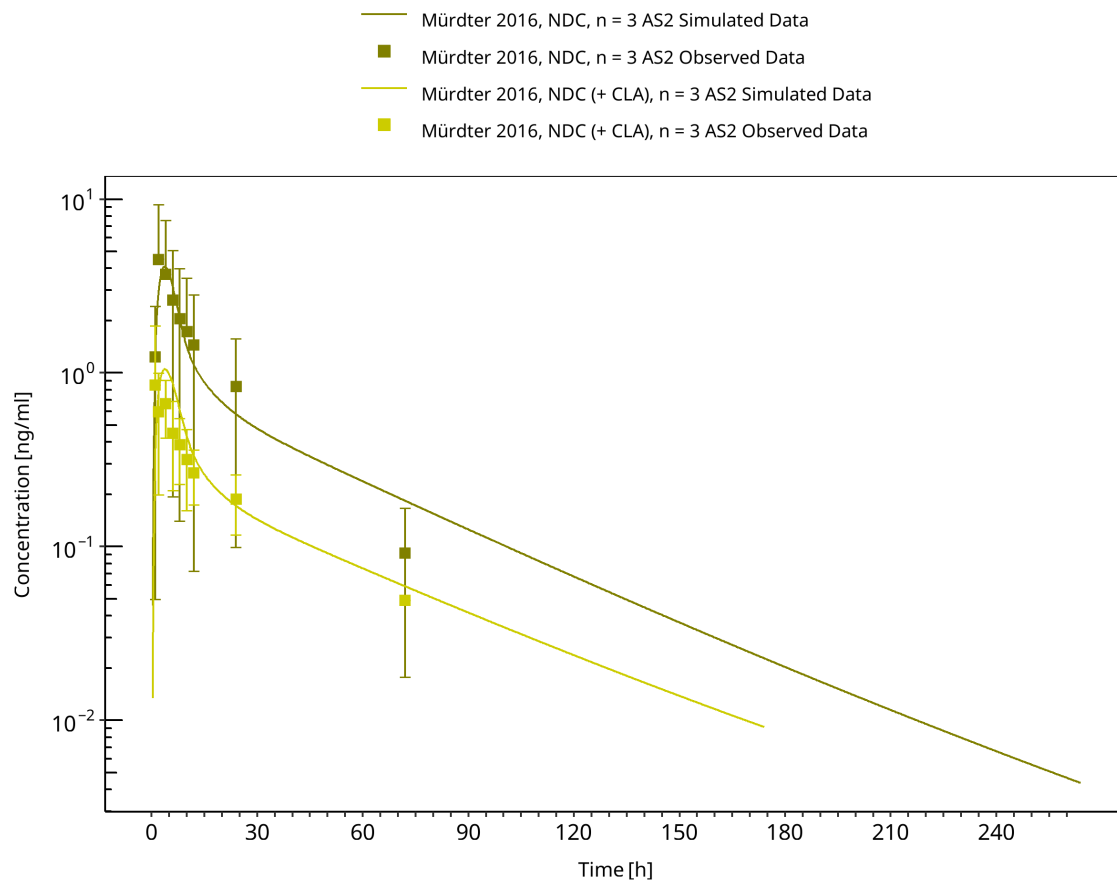


Figure 4-25: Clarithromycin - Clomiphene DDGI

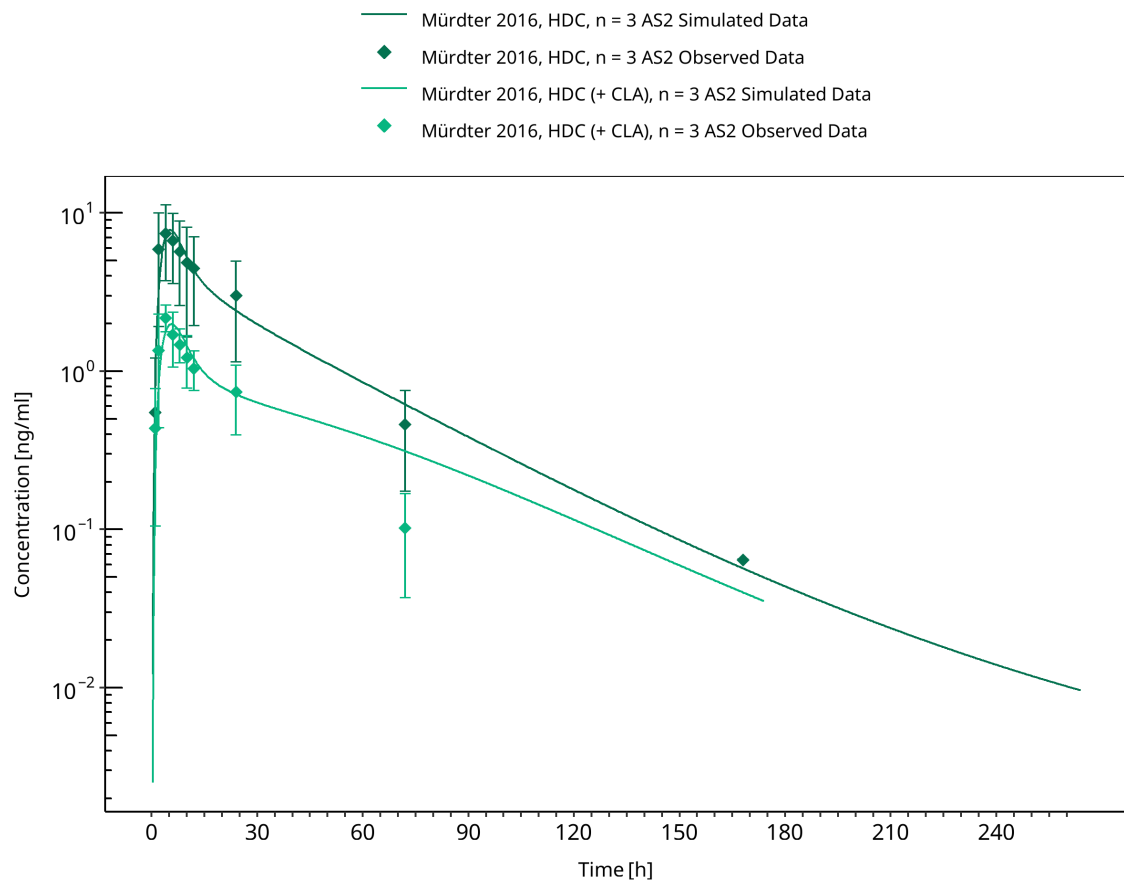


Figure 4-26: Clarithromycin - Clomiphene DDGI

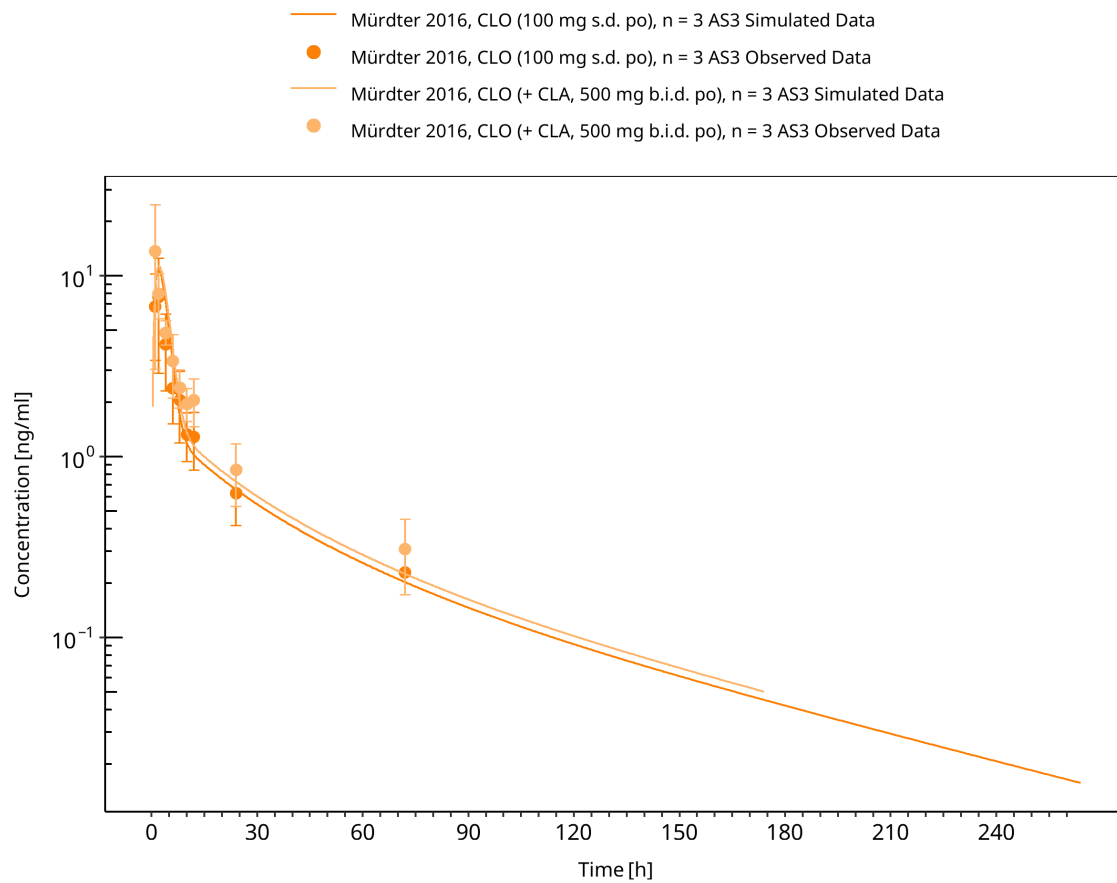


Figure 4-27: Clarithromycin - Clomiphene DDGI

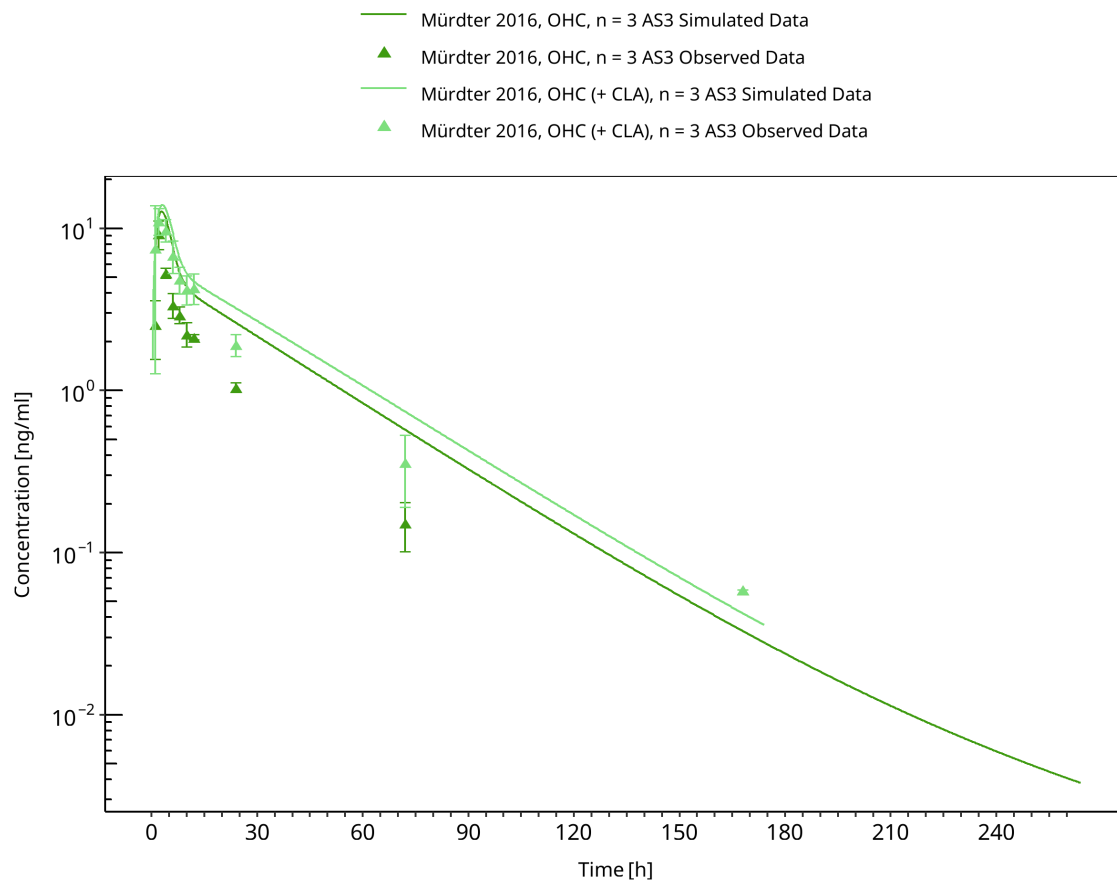


Figure 4-28: Clarithromycin - Clomiphene DDGI

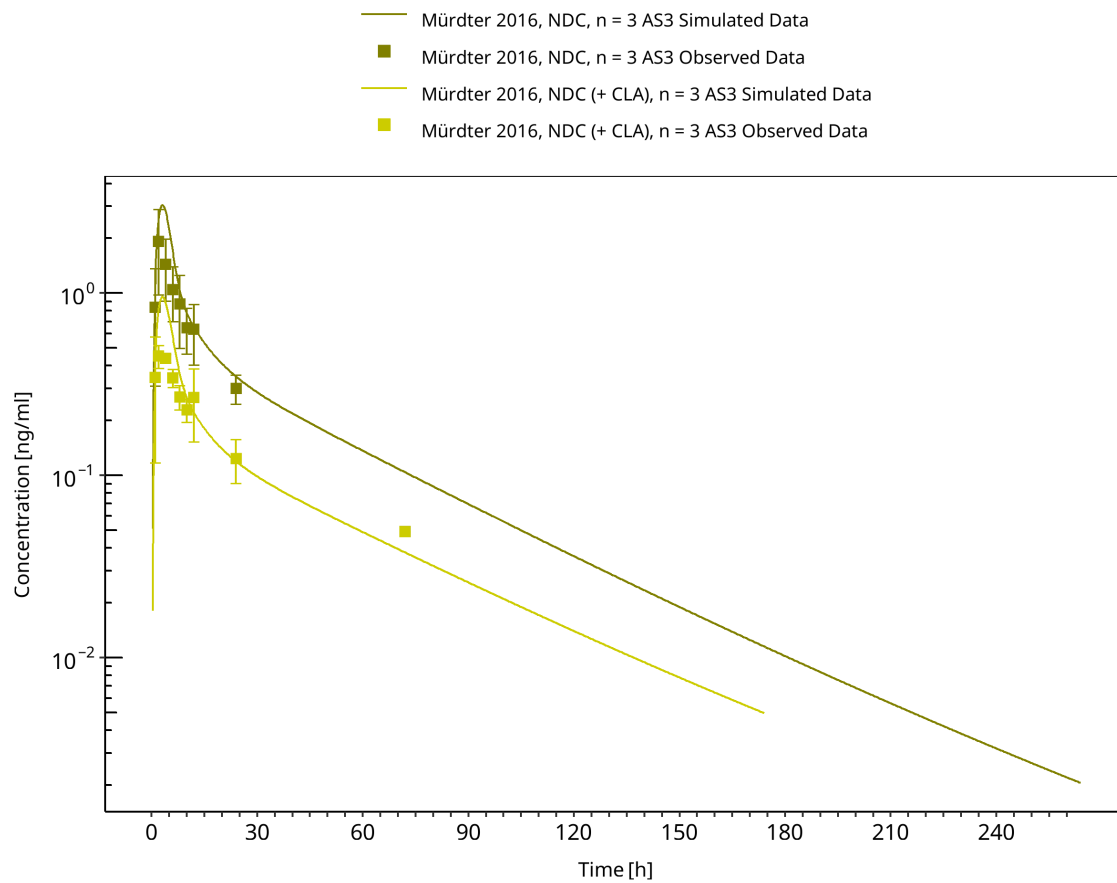


Figure 4-29: Clarithromycin - Clomiphene DDGI

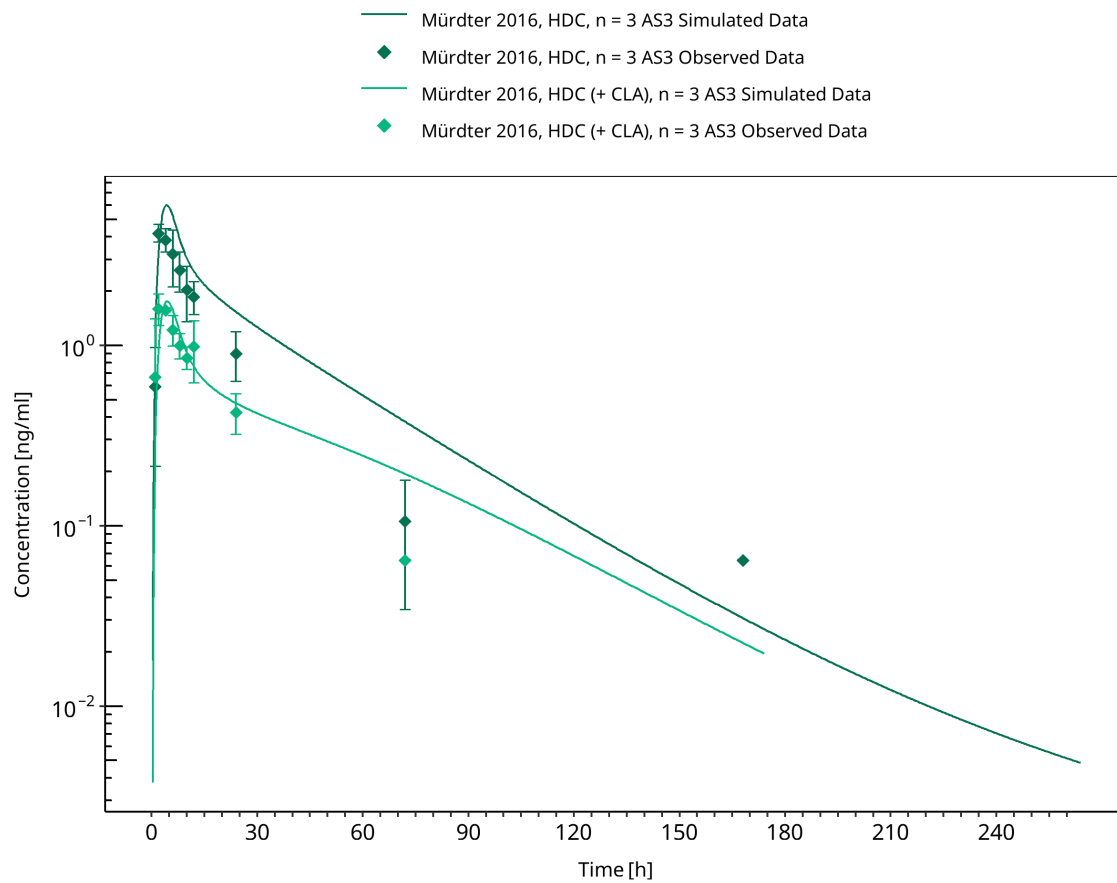


Figure 4-30: Clarithromycin - Clomiphene DDGI

4.2.3 Paroxetine - Clomiphene DDGI

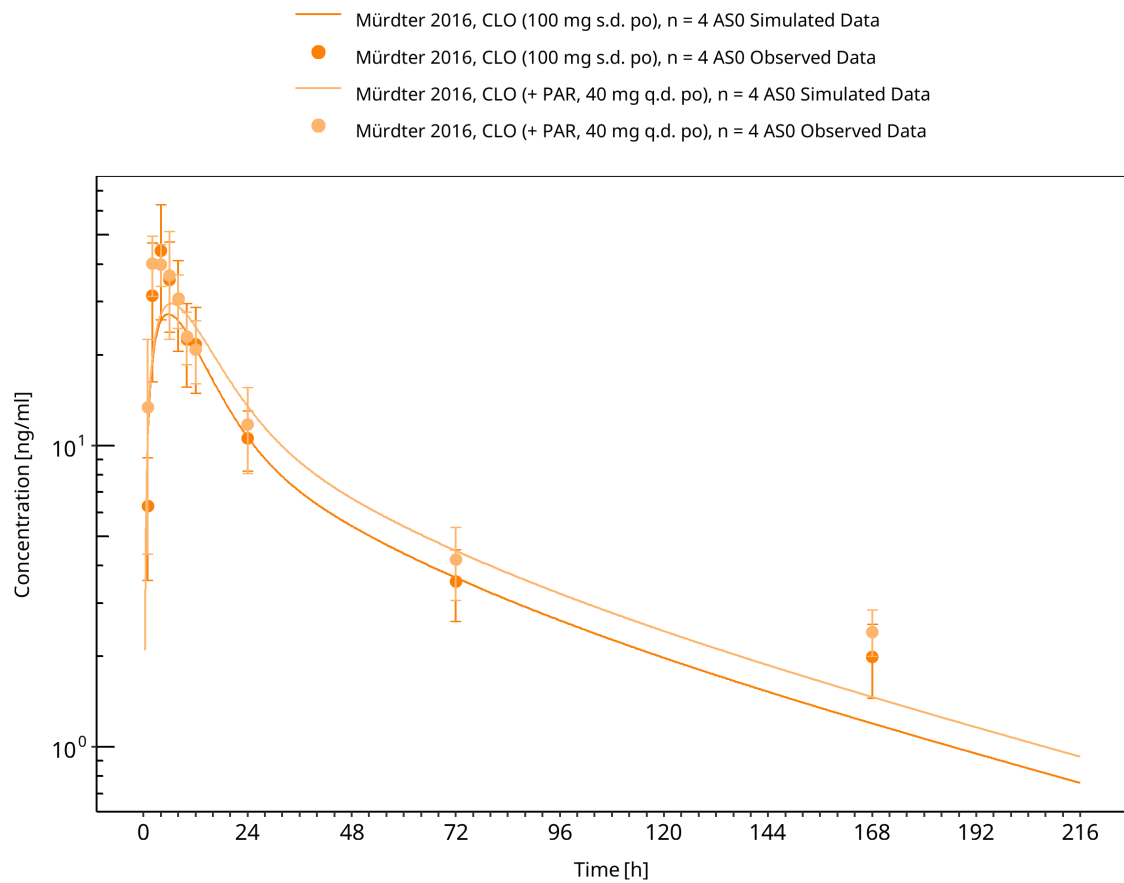


Figure 4-31: Paroxetine - Clomiphene DDGI

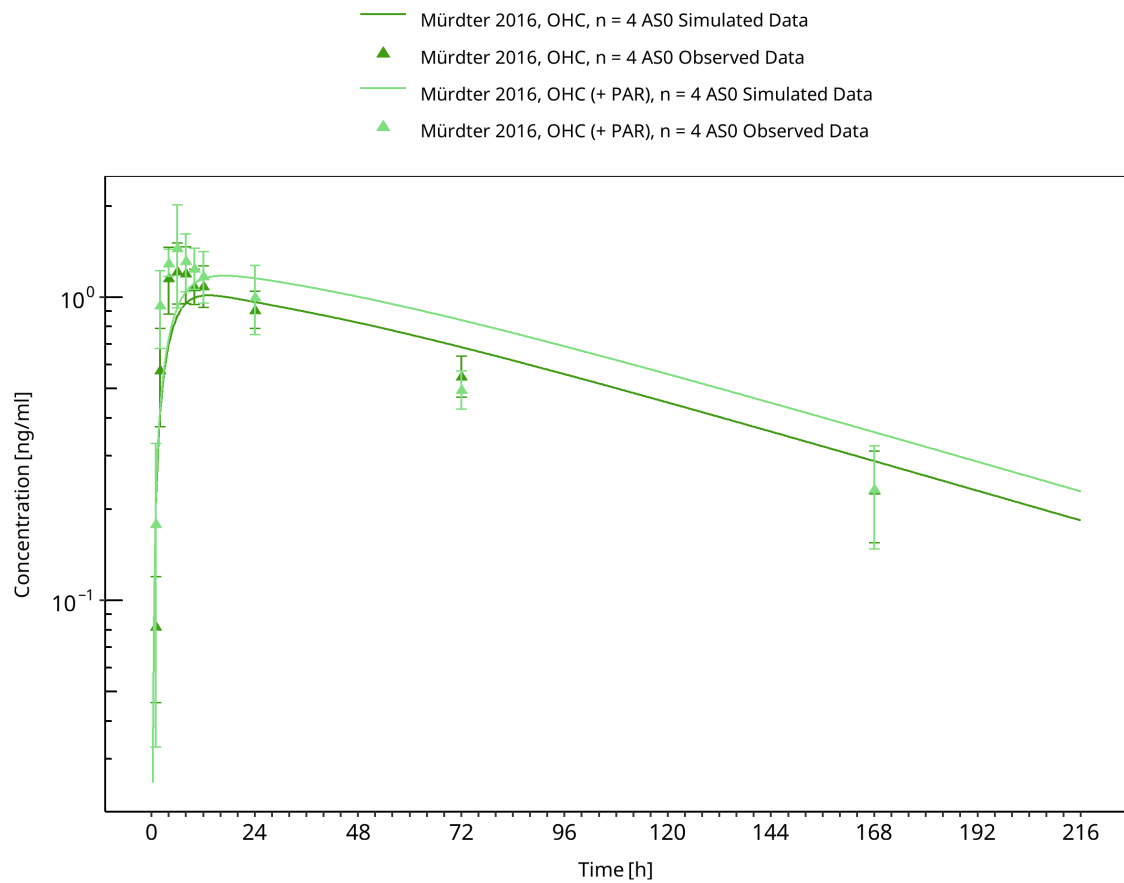


Figure 4-32: Paroxetine - Clomiphene DDGI

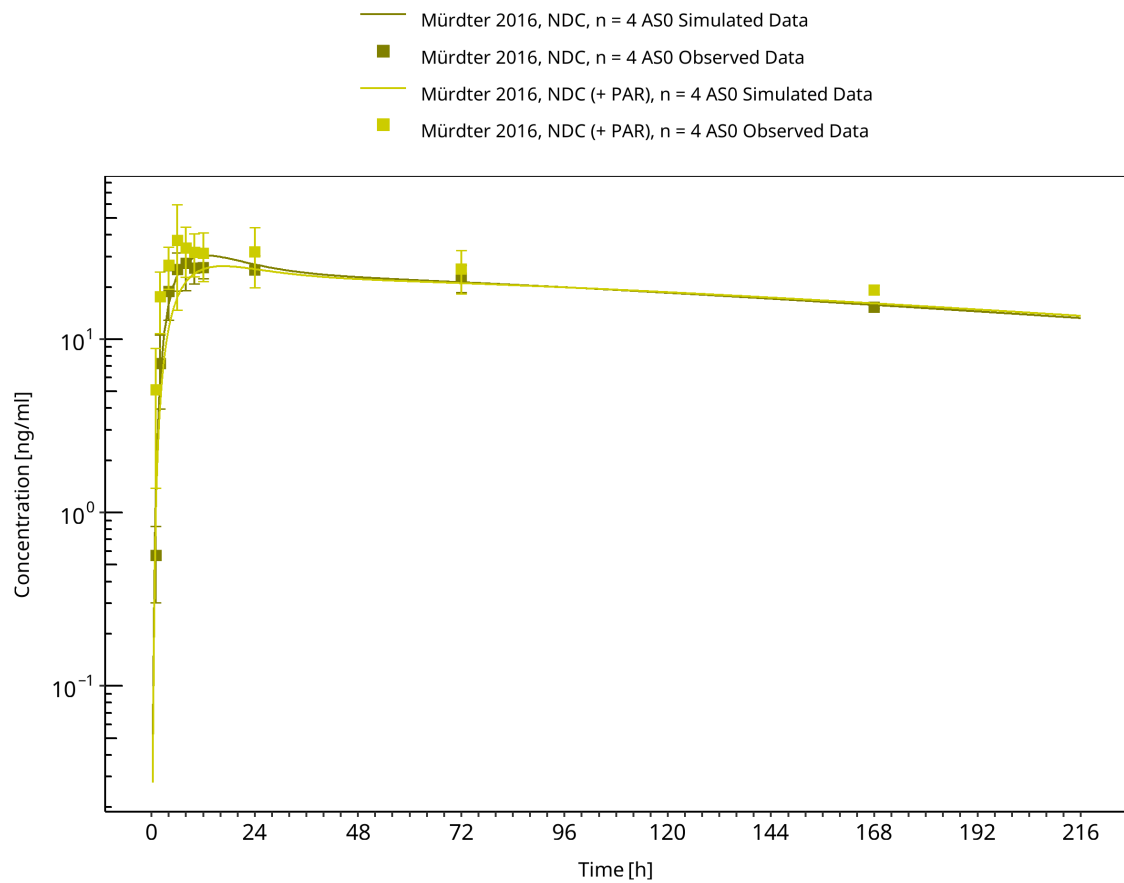


Figure 4-33: Paroxetine - Clomiphene DDGI

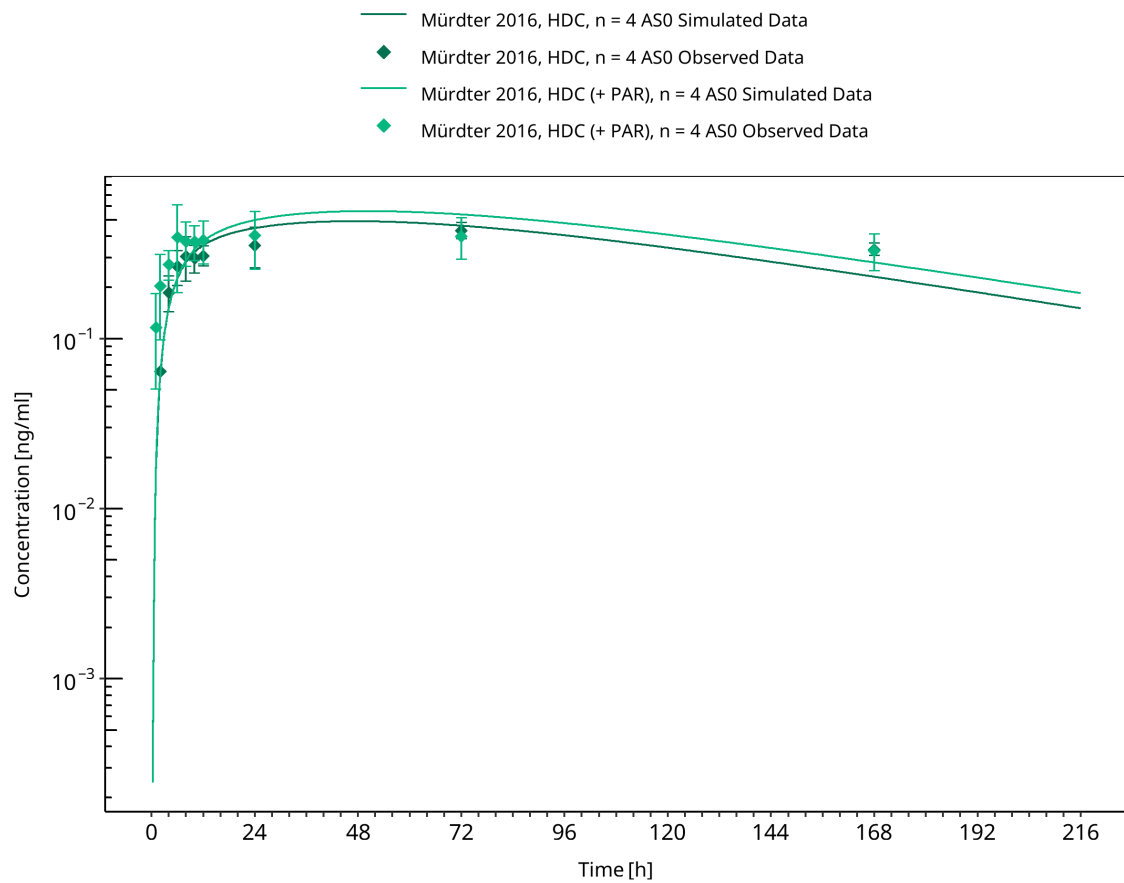


Figure 4-34: Paroxetine - Clomiphen DDGI

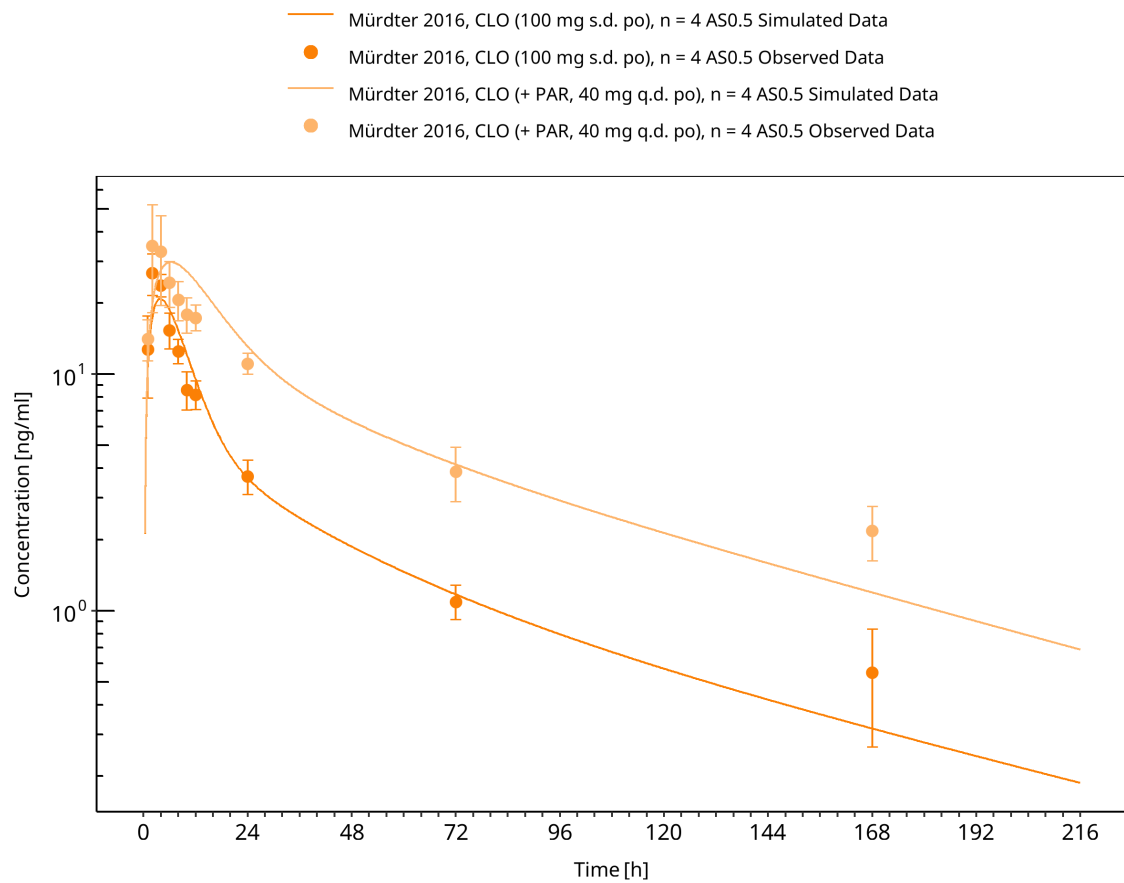


Figure 4-35: Paroxetine - Clomiphenes DDGI

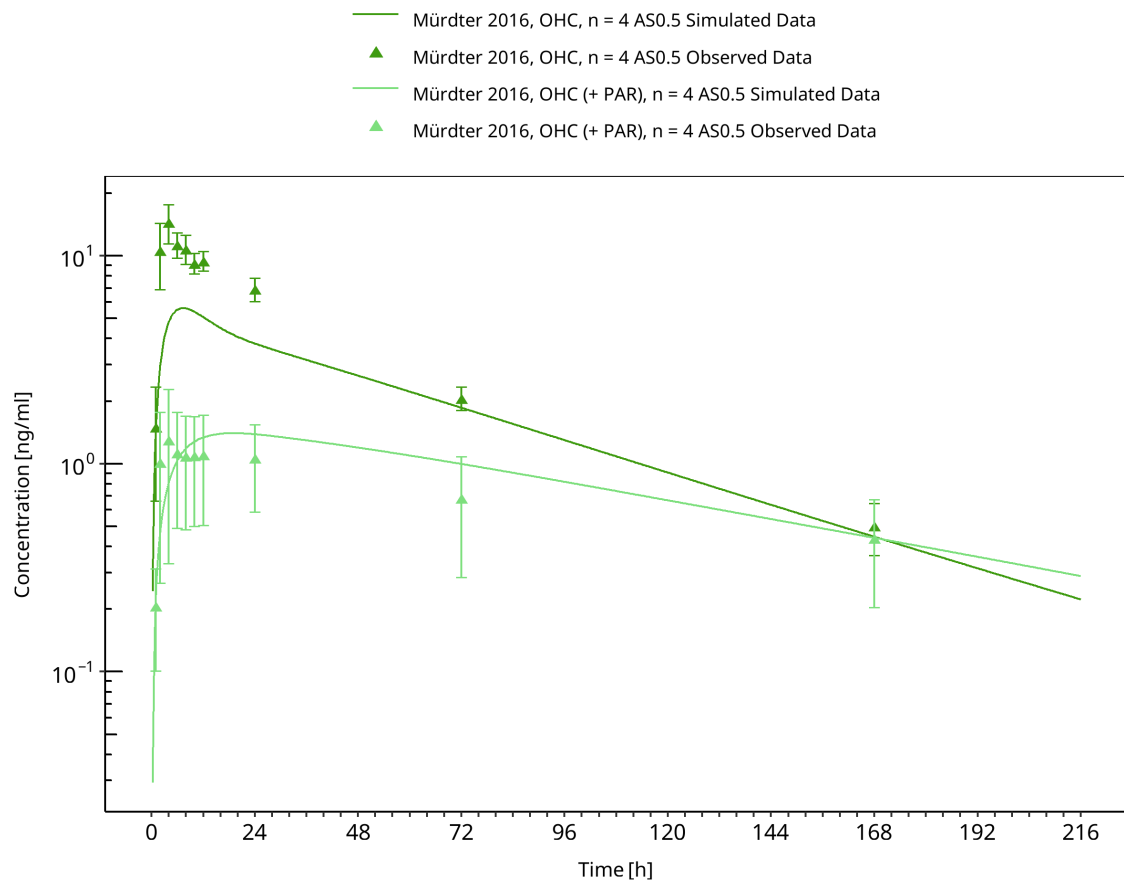


Figure 4-36: Paroxetine - Clomiphene DDGI

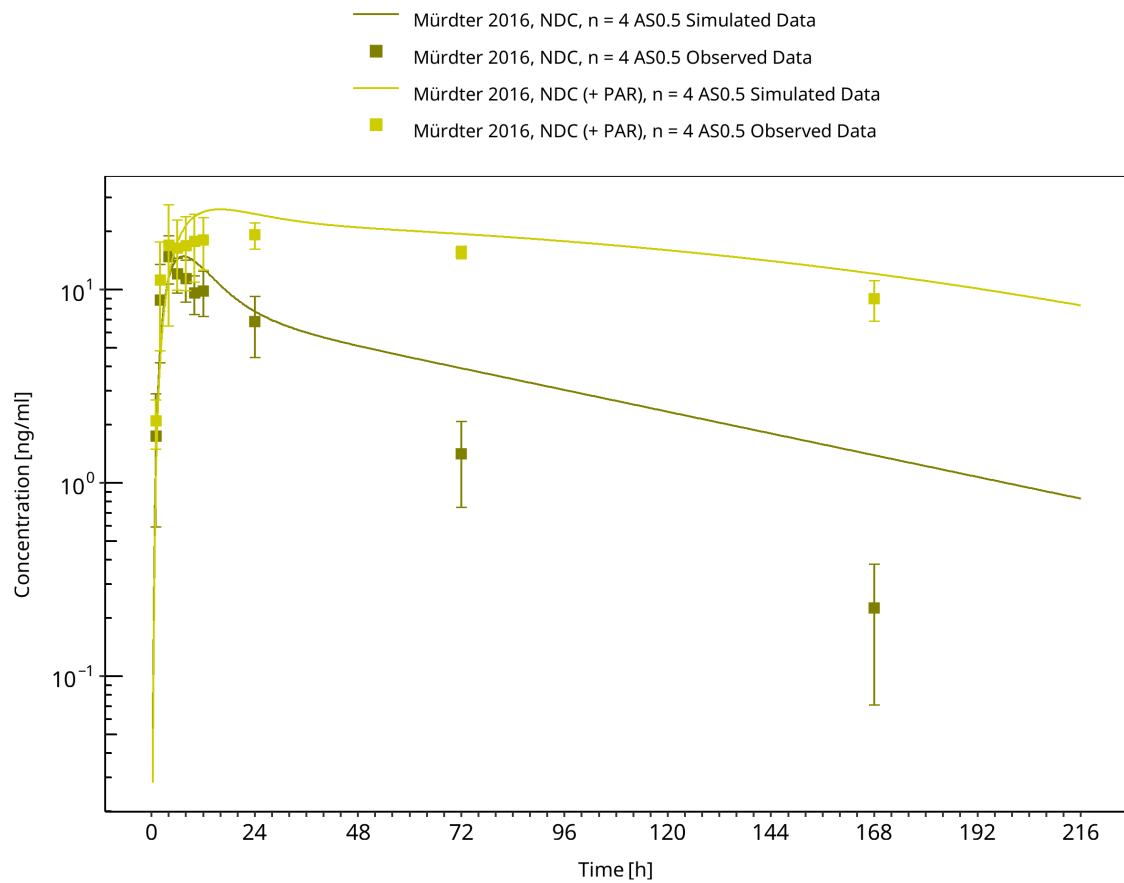


Figure 4-37: Paroxetine - Clomiphene DDGI

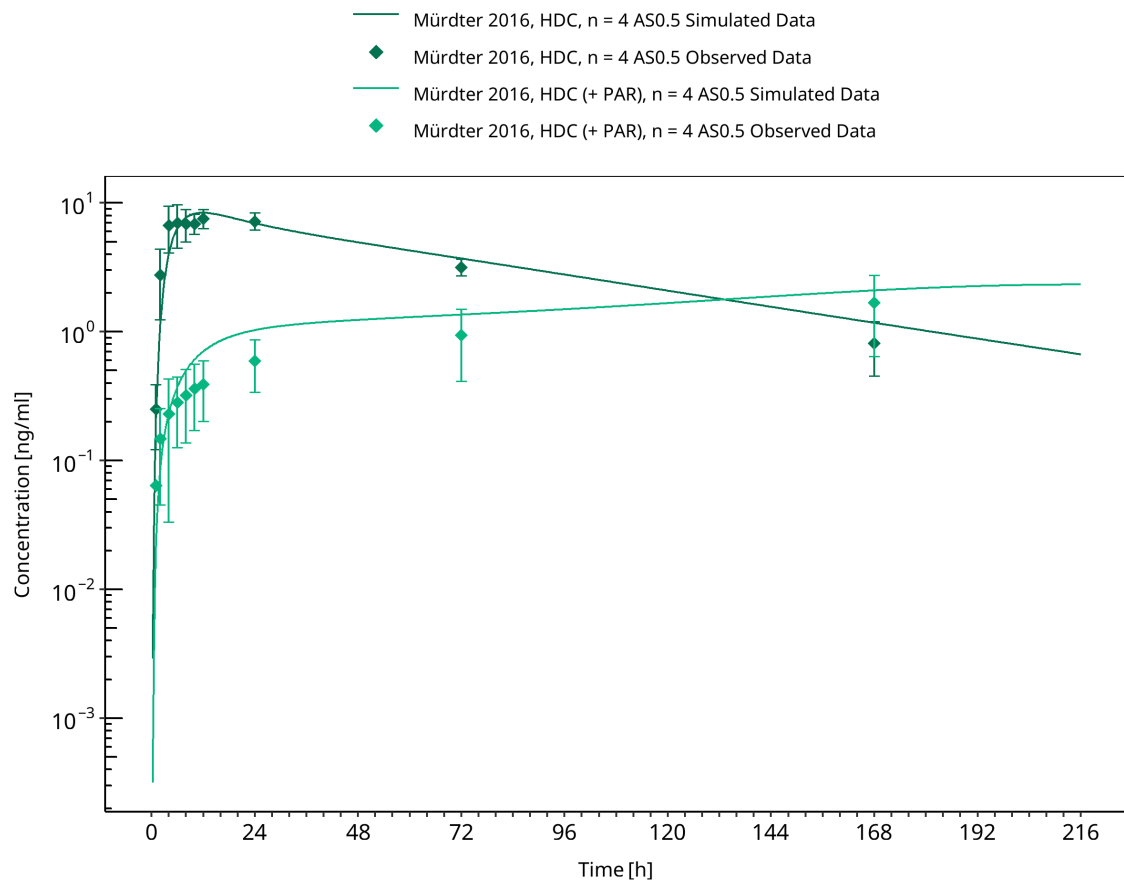


Figure 4-38: Paroxetine - Clomiphene DDGI

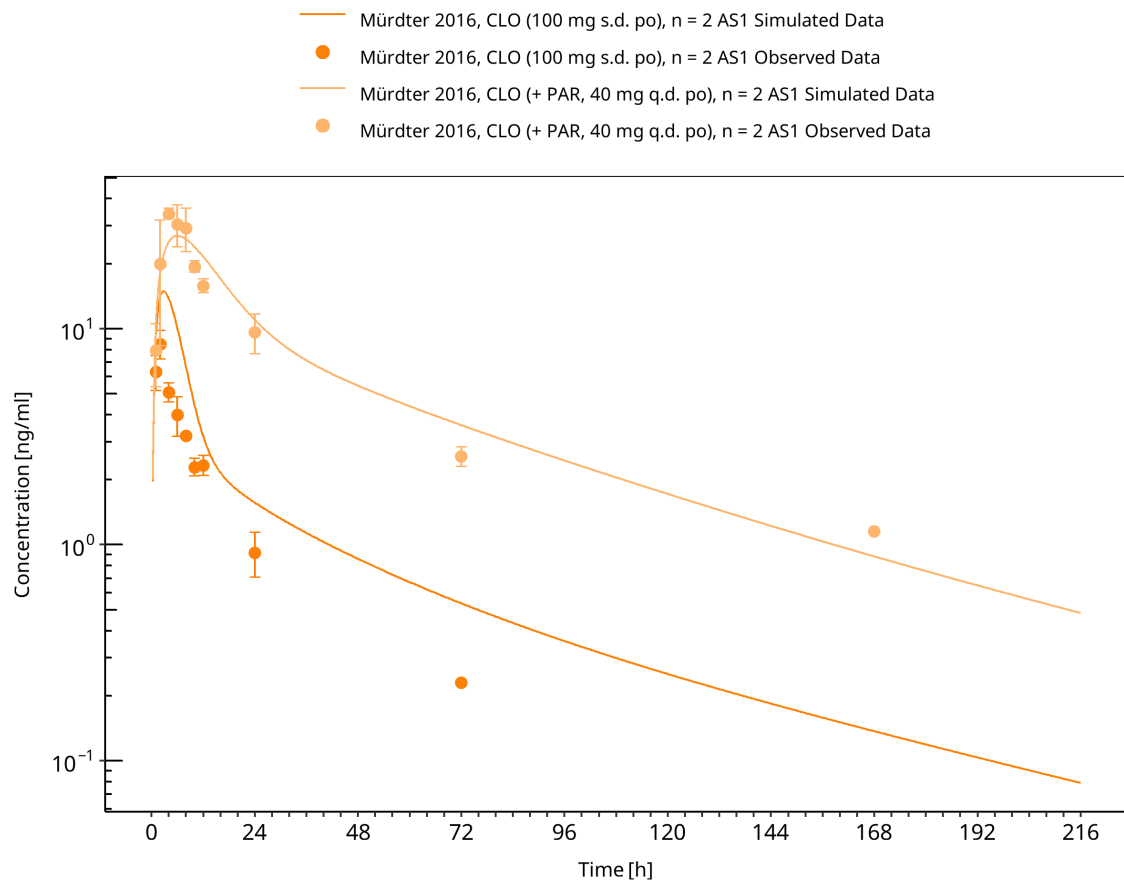


Figure 4-39: Paroxetine - Clomiphene DDGI

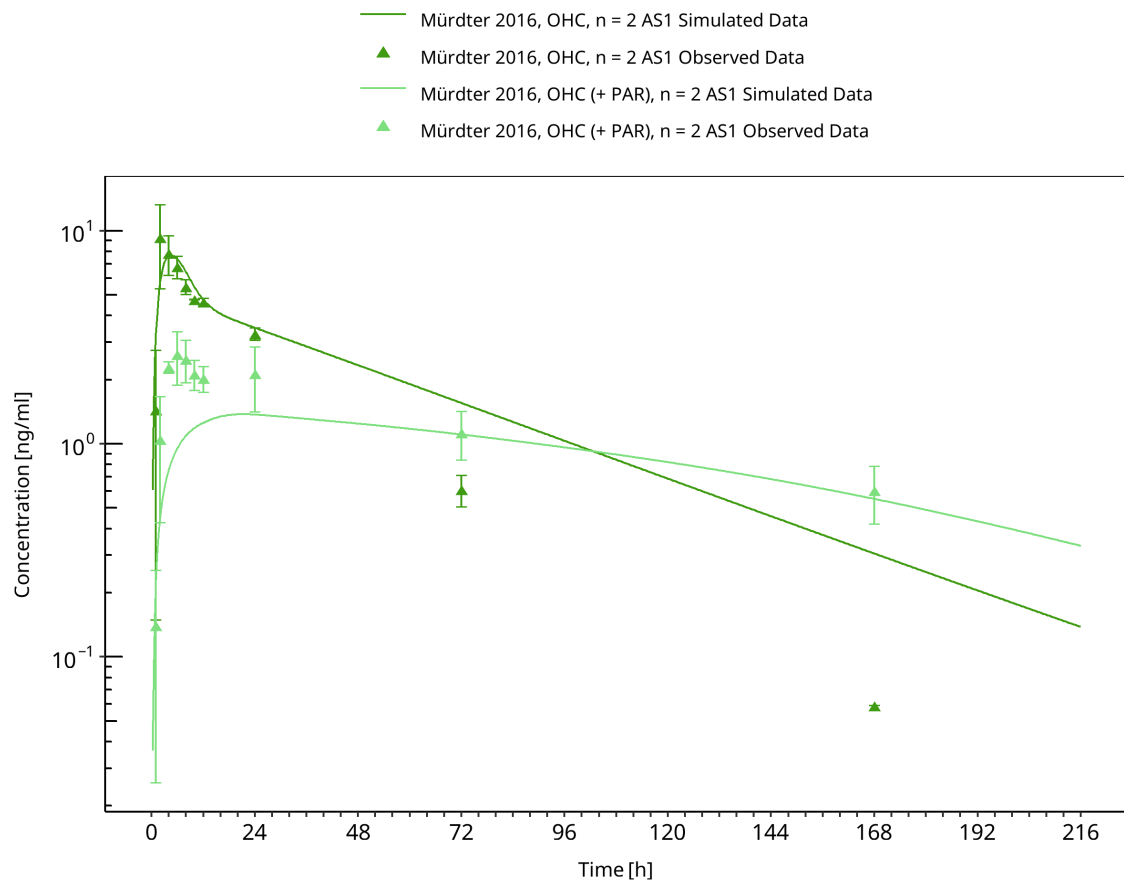


Figure 4-40: Paroxetine - Clomiphen DDGI

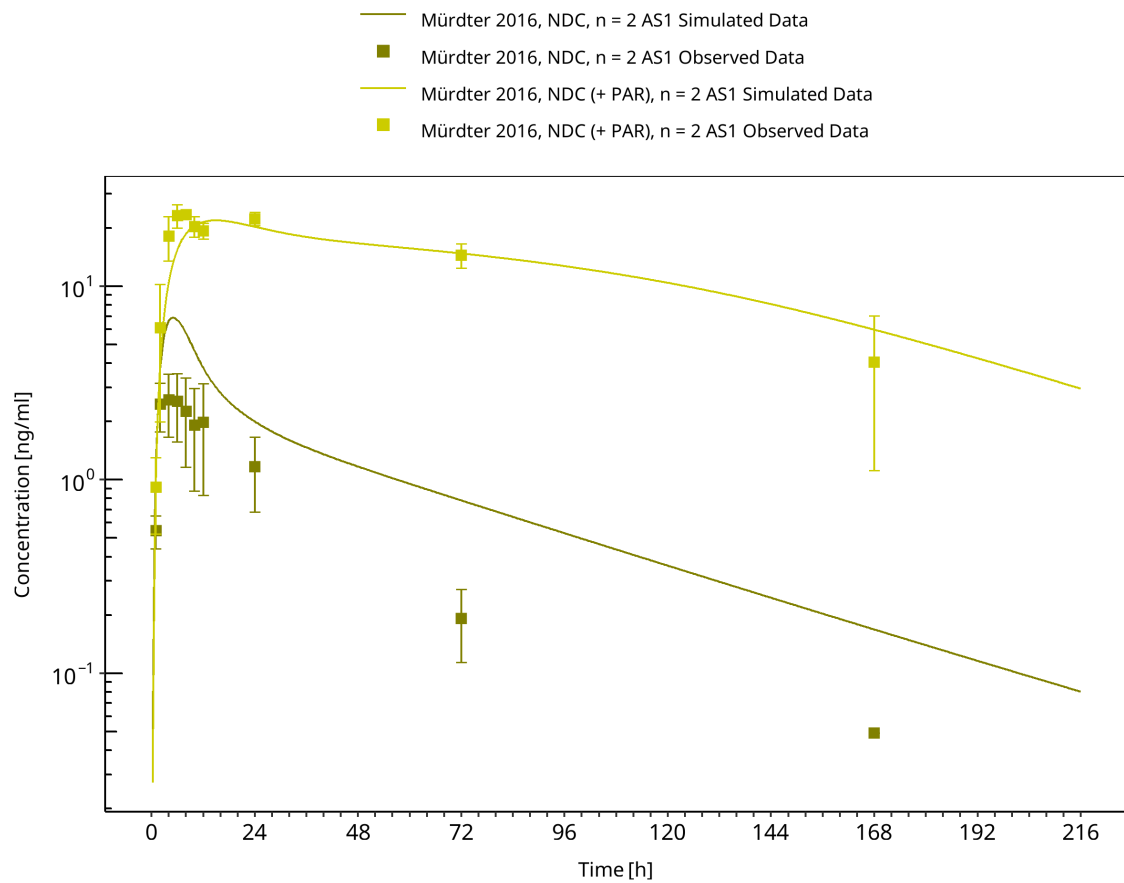


Figure 4-41: Paroxetine - Clomiphene DDGI

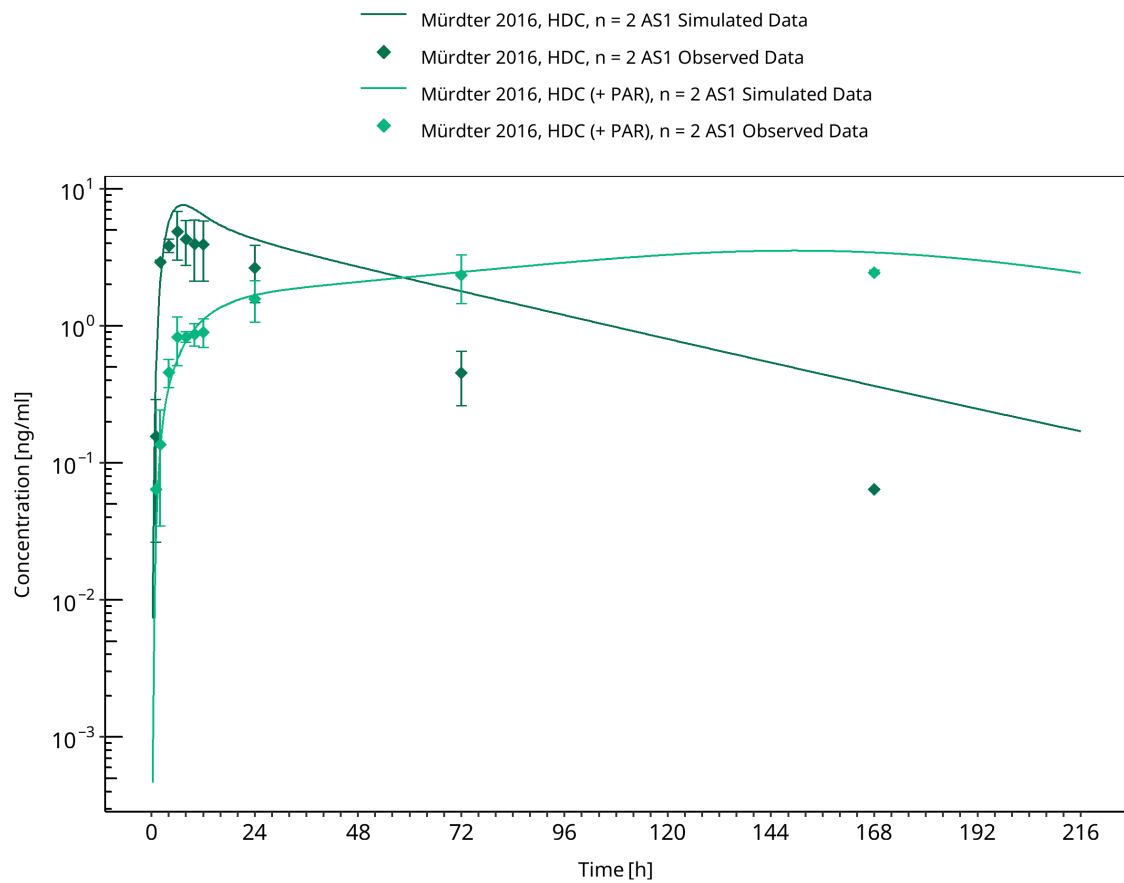


Figure 4-42: Paroxetine - Clomiphene DDGI

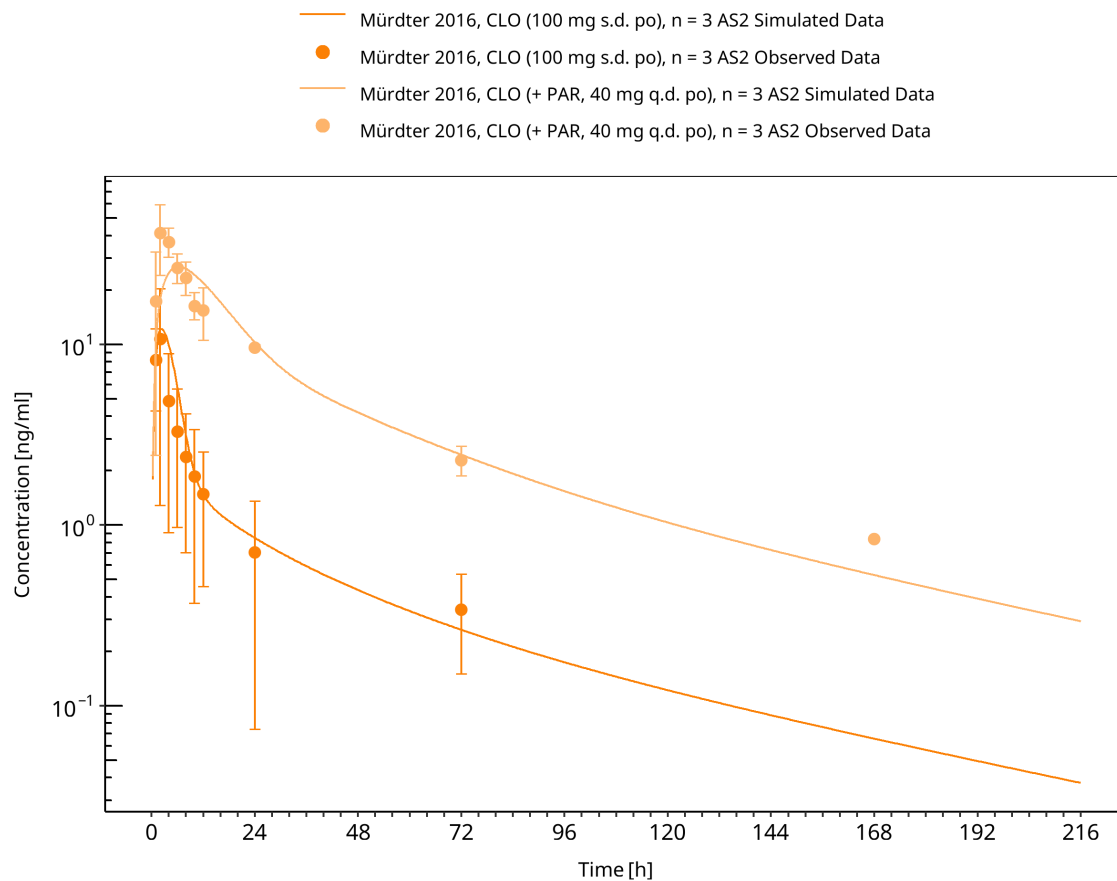


Figure 4-43: Paroxetine - Clomiphenes DDGI

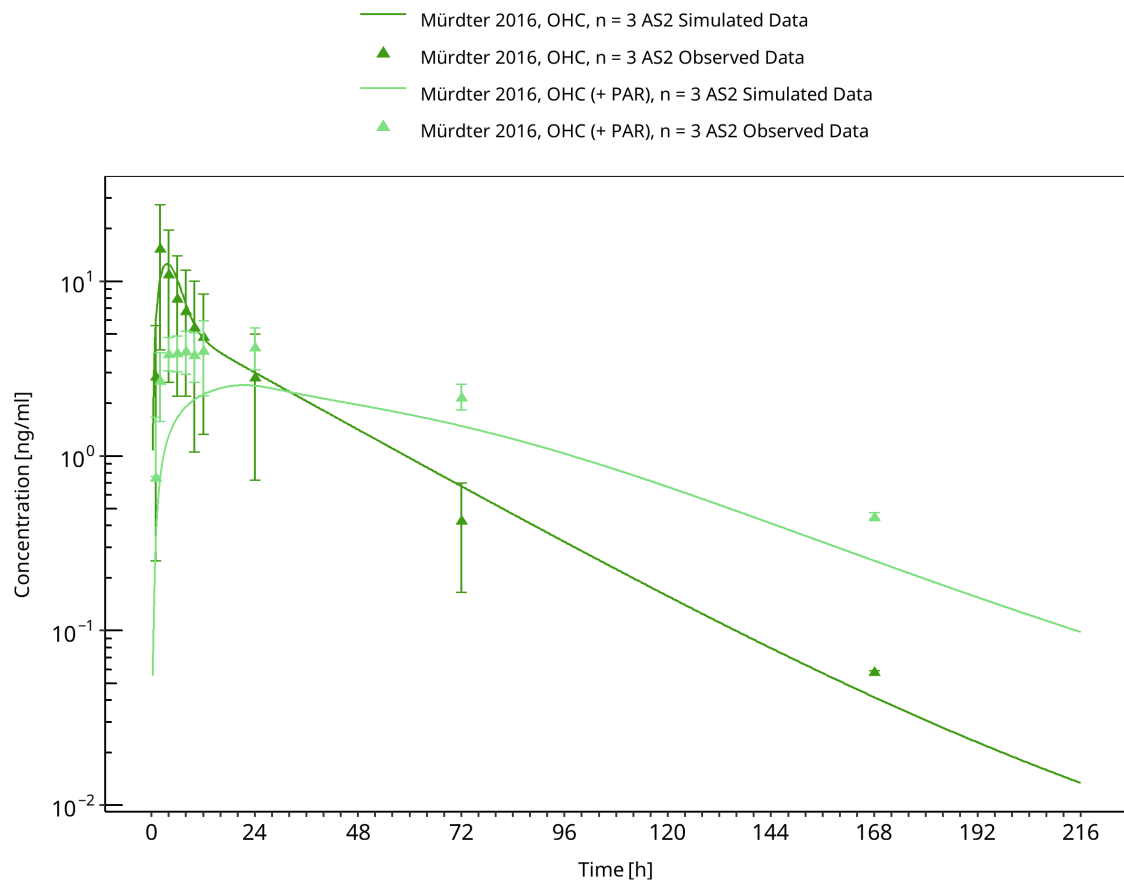


Figure 4-44: Paroxetine - Clomiphen DDGI

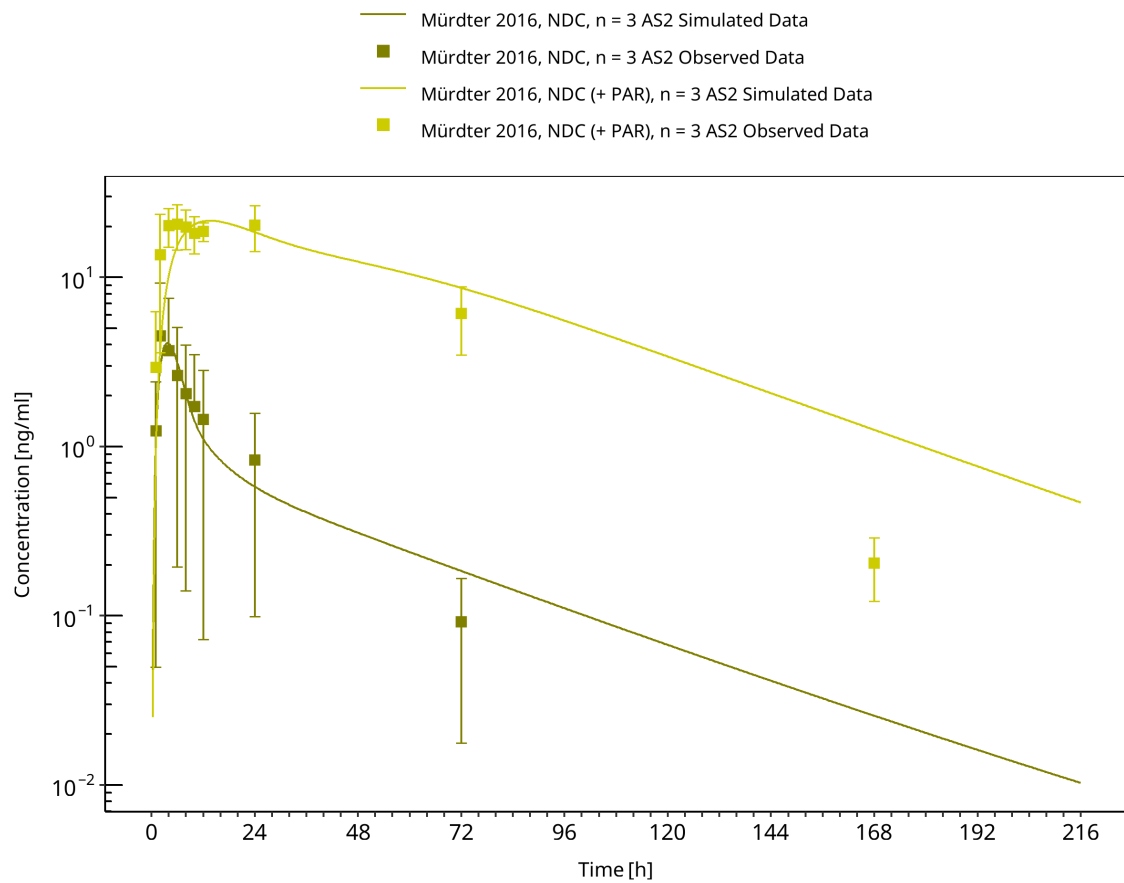


Figure 4-45: Paroxetine - Clomiphene DDGI

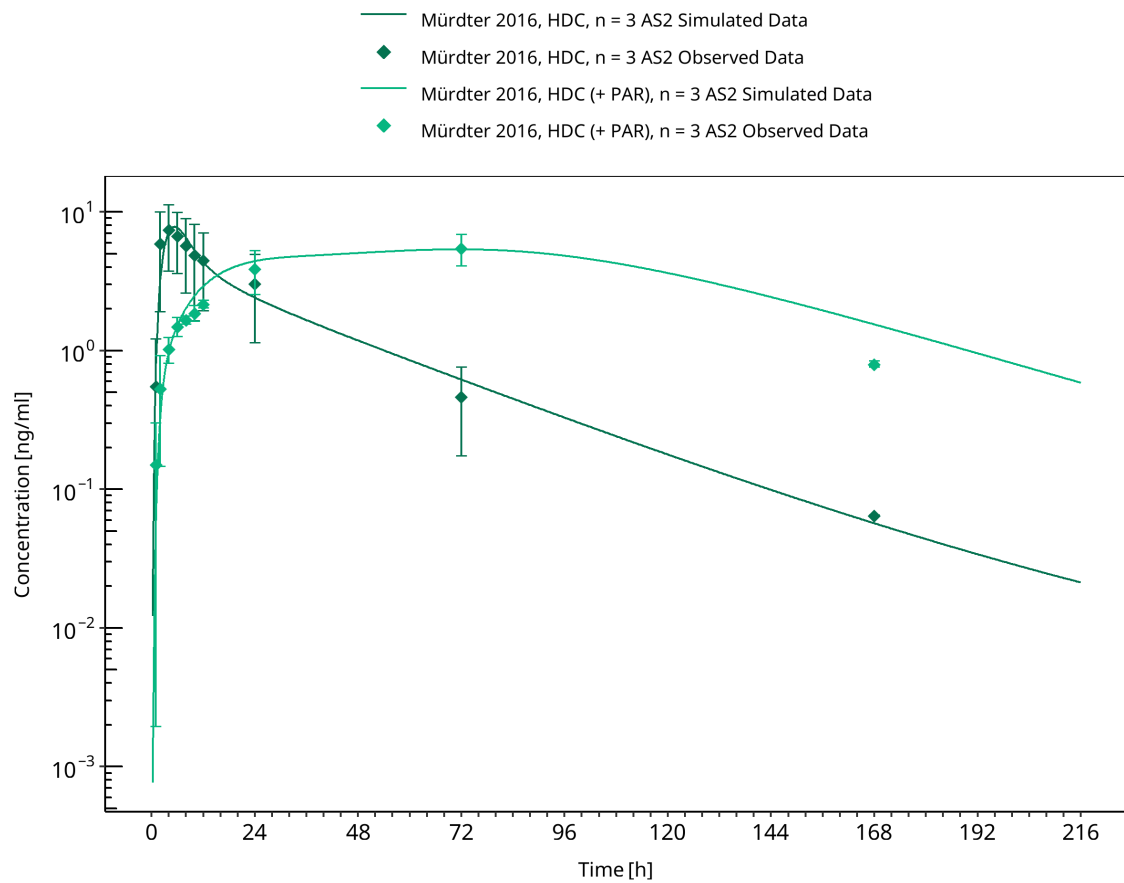


Figure 4-46: Paroxetine - Clomiphene DDGI

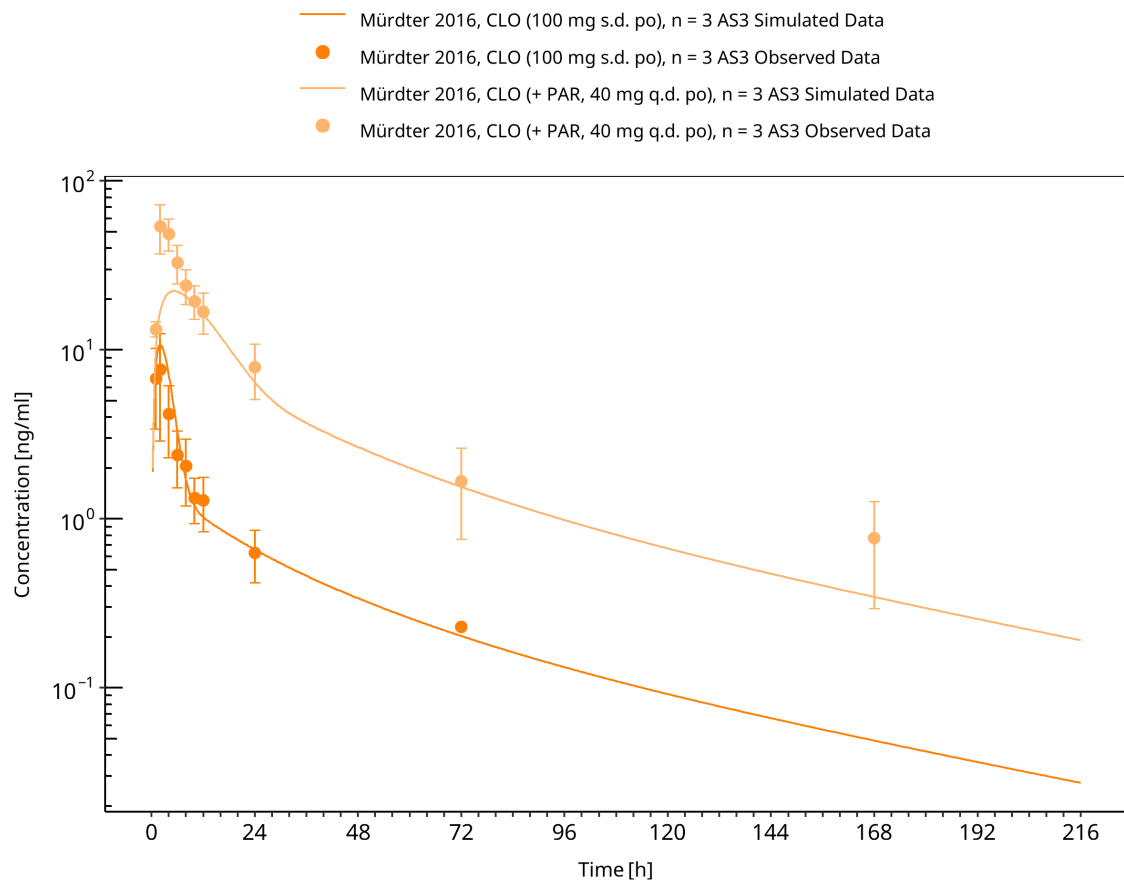


Figure 4-47: Paroxetine - Clomiphene DDGI

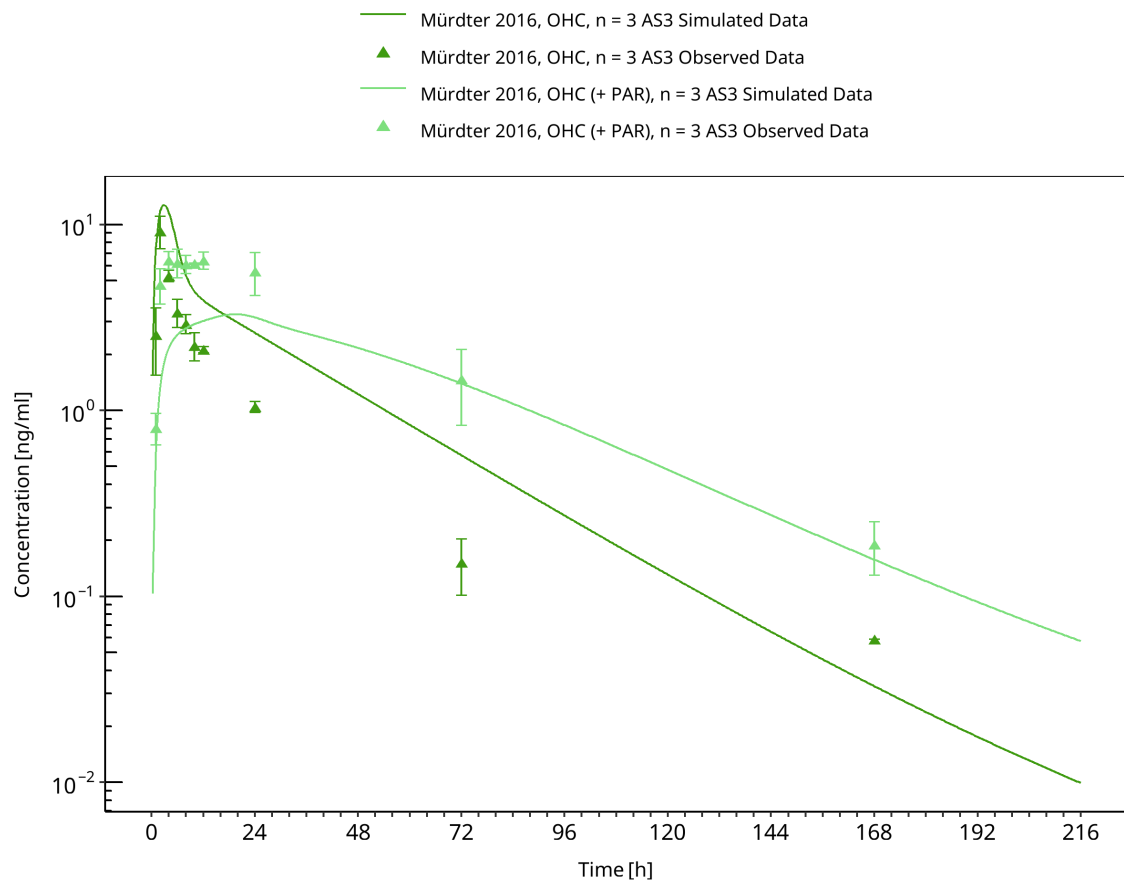


Figure 4-48: Paroxetine - Clomiphene DDGI

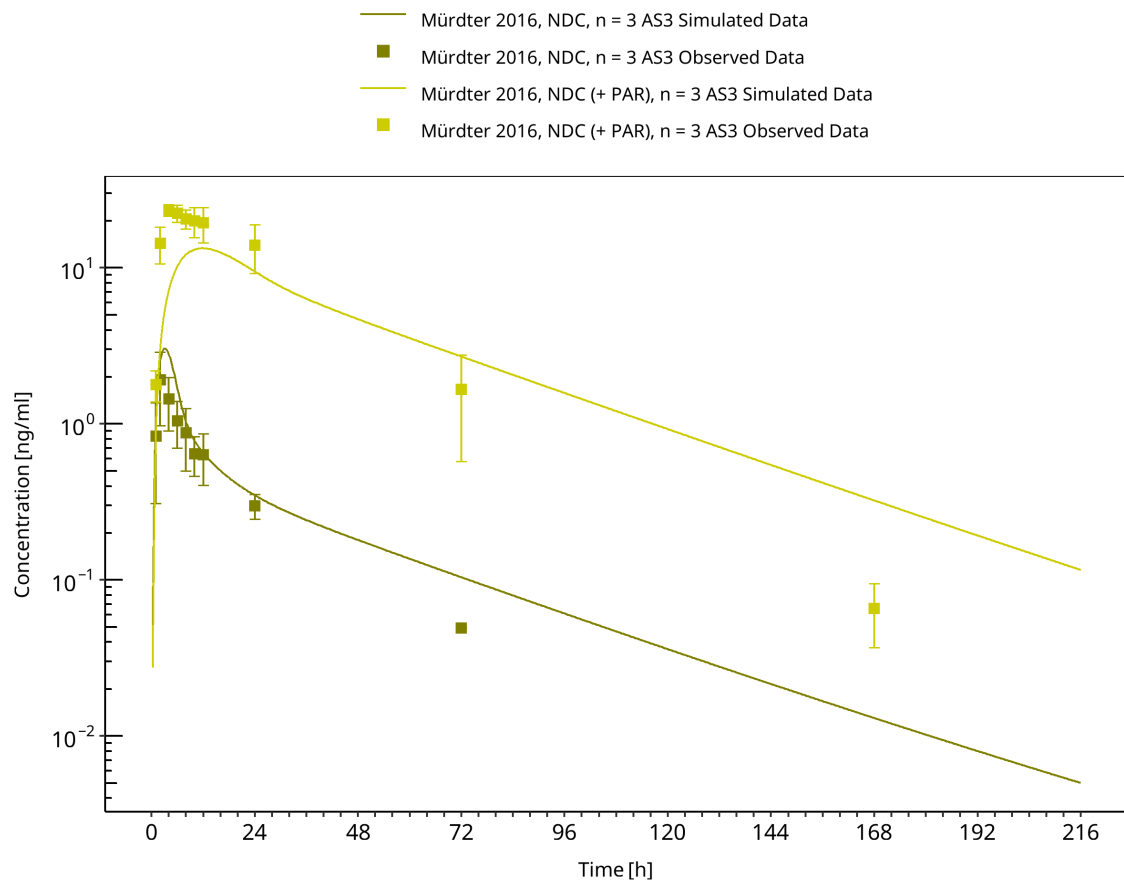


Figure 4-49: Paroxetine - Clomiphen DDGI

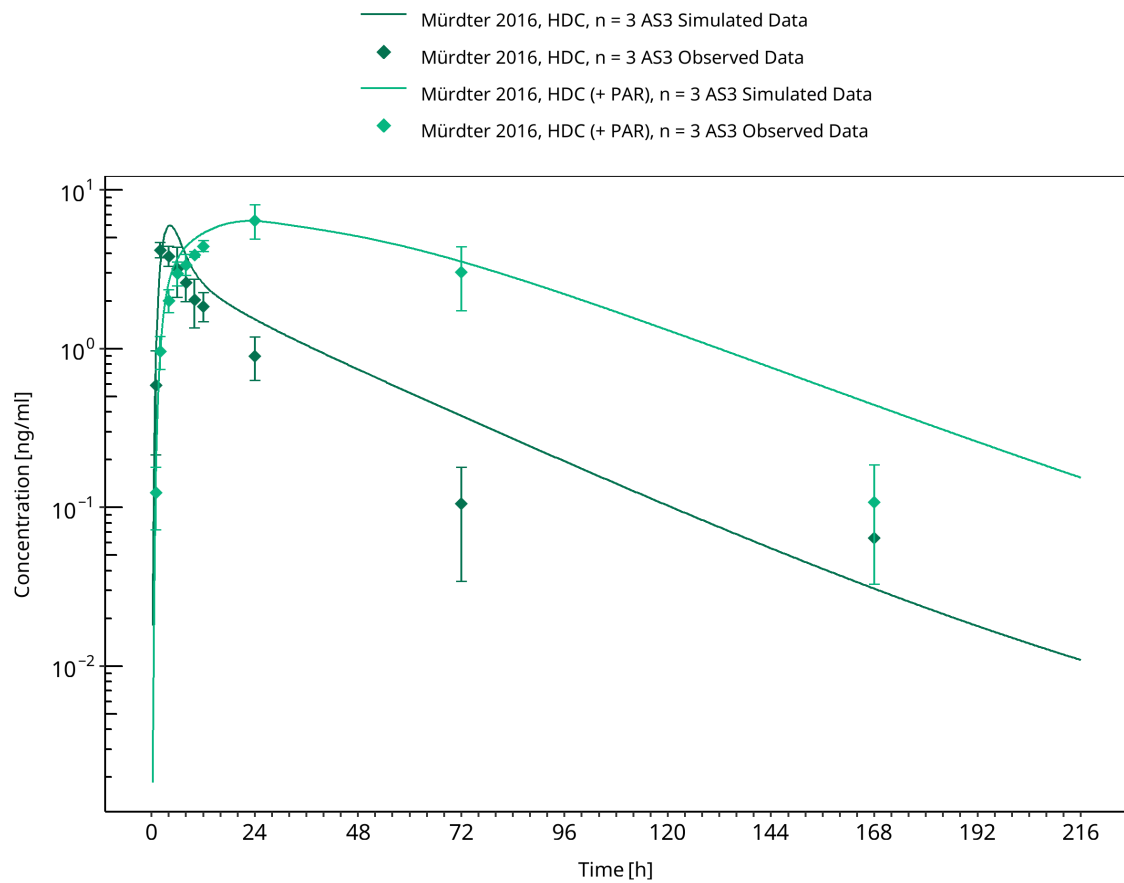


Figure 4-50: Paroxetine - Clomiphen DDGI

4.2.4 Paroxetine - Desipramine DDGI

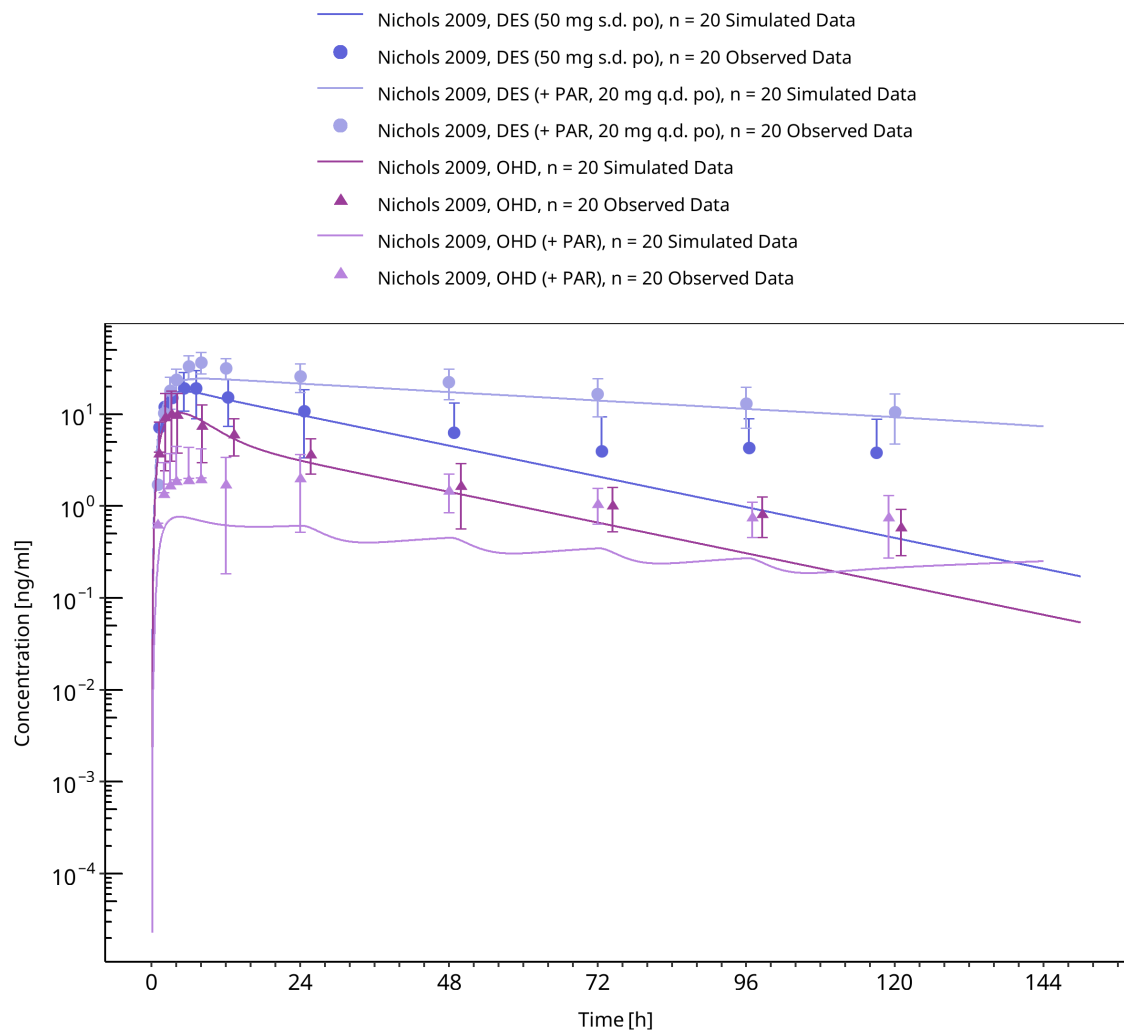


Figure 4-51: Paroxetine - Desipramine DDI

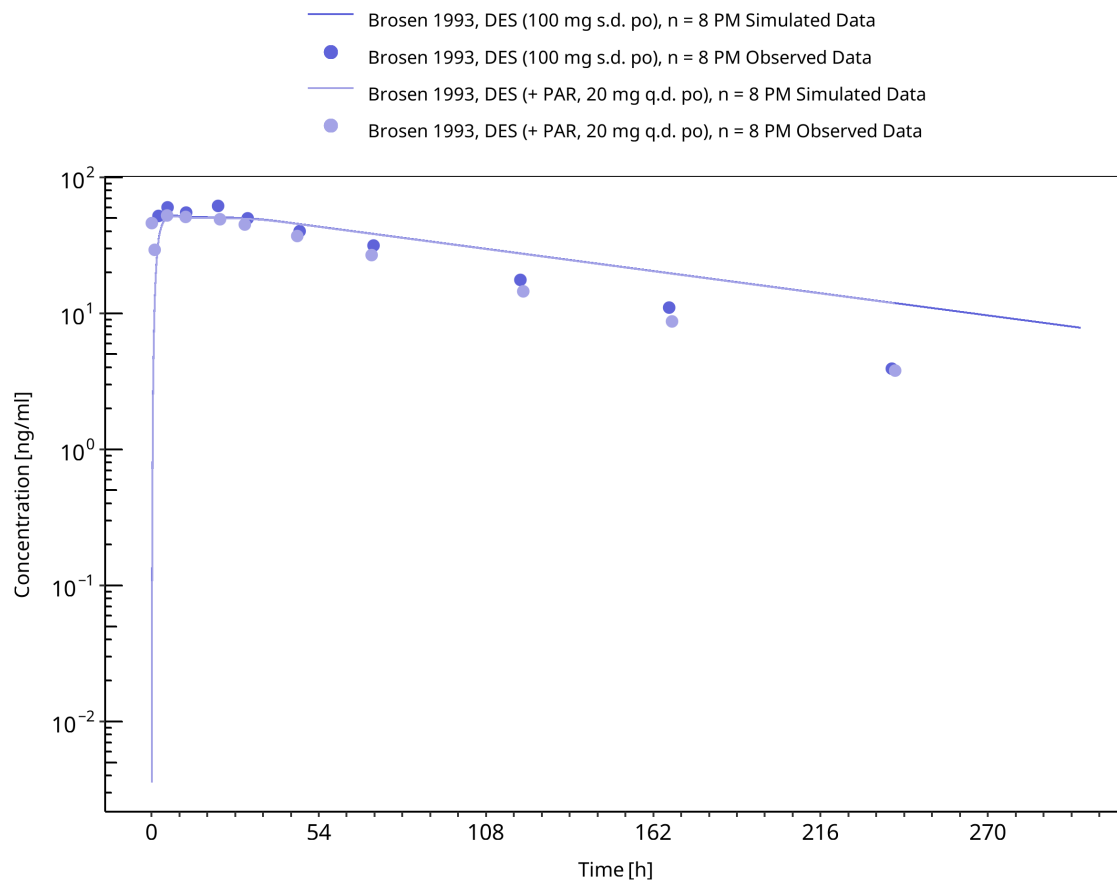


Figure 4-52: Paroxetine - Desipramine DDGI

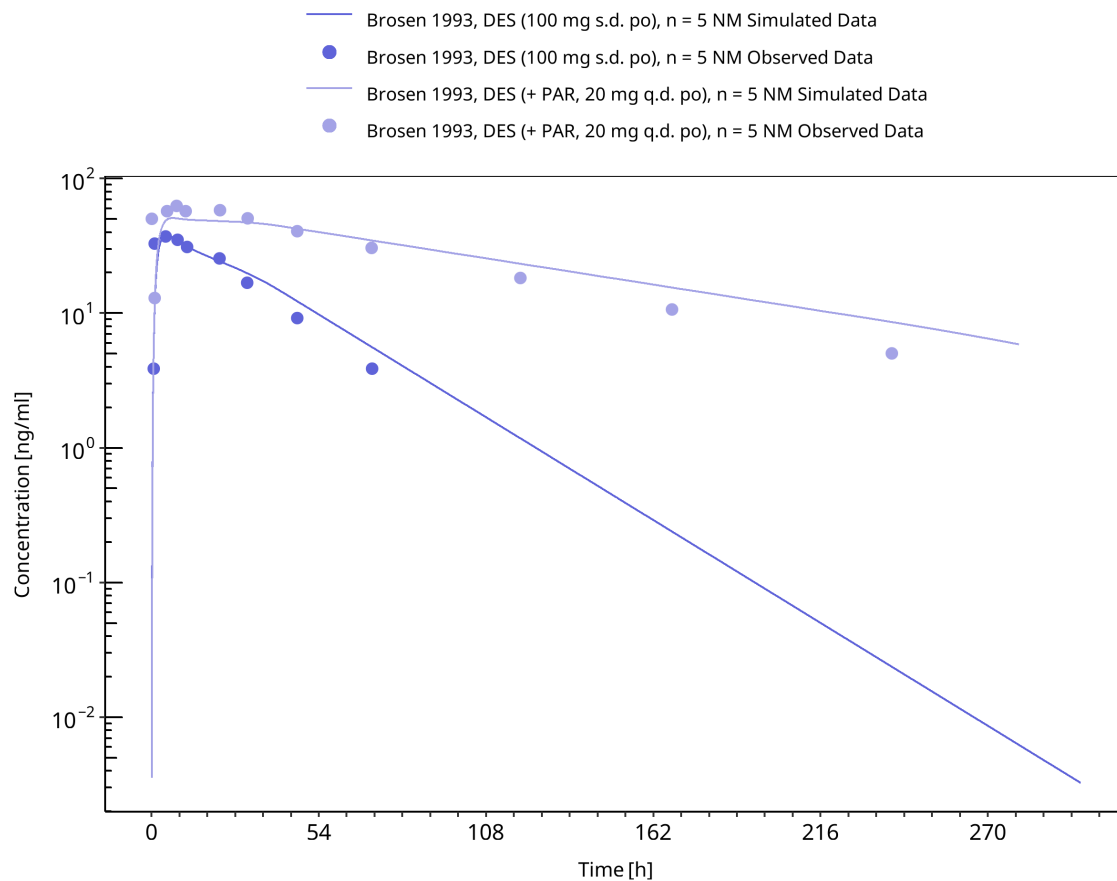


Figure 4-53: Paroxetine - Desipramine DDGI

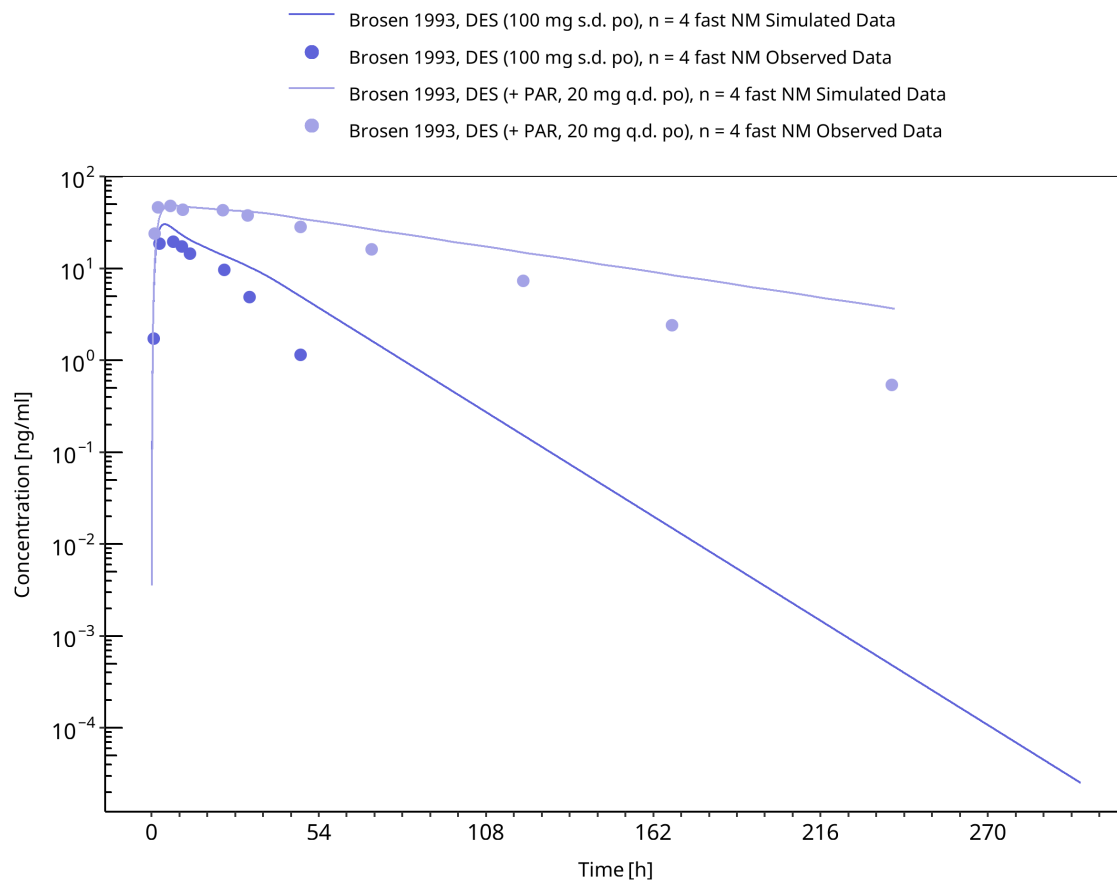


Figure 4-54: Paroxetine - Desipramine DDGI

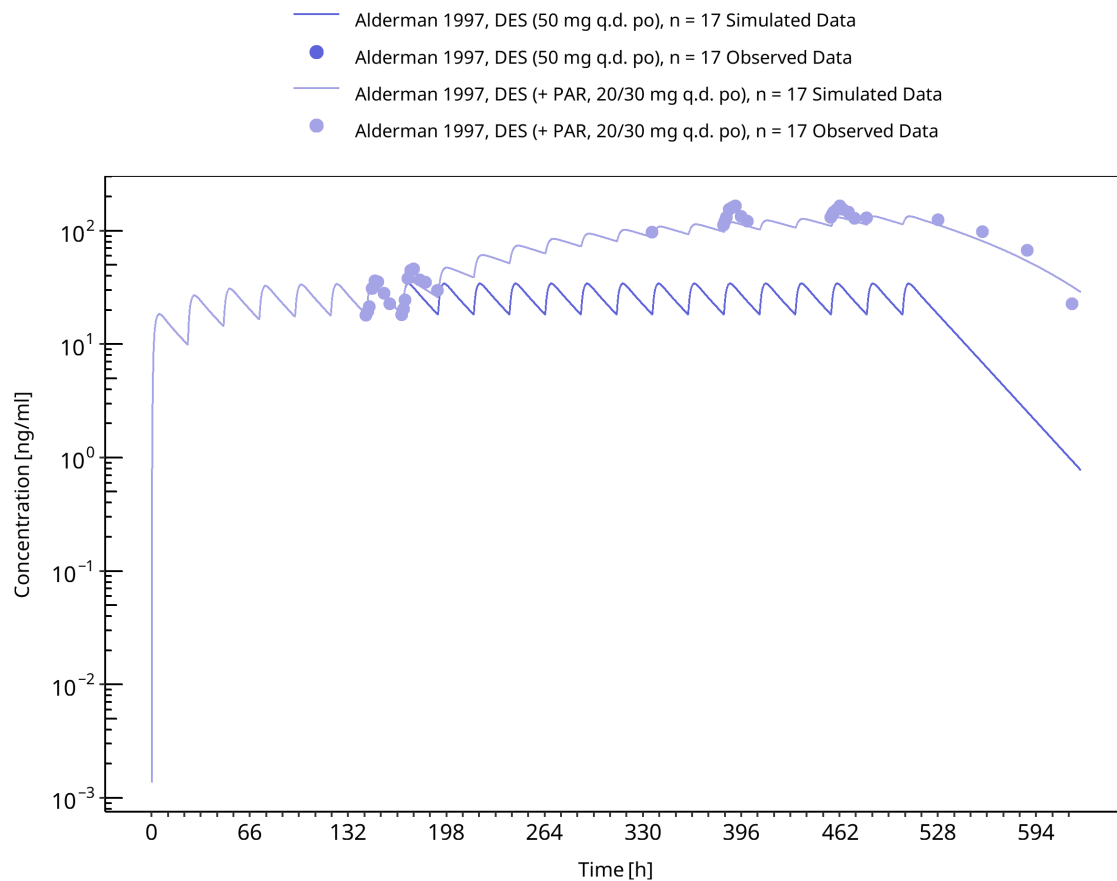


Figure 4-55: Paroxetine - Desipramine DDI

4.2.5 Paroxetine - Dextromethorphan DDGI

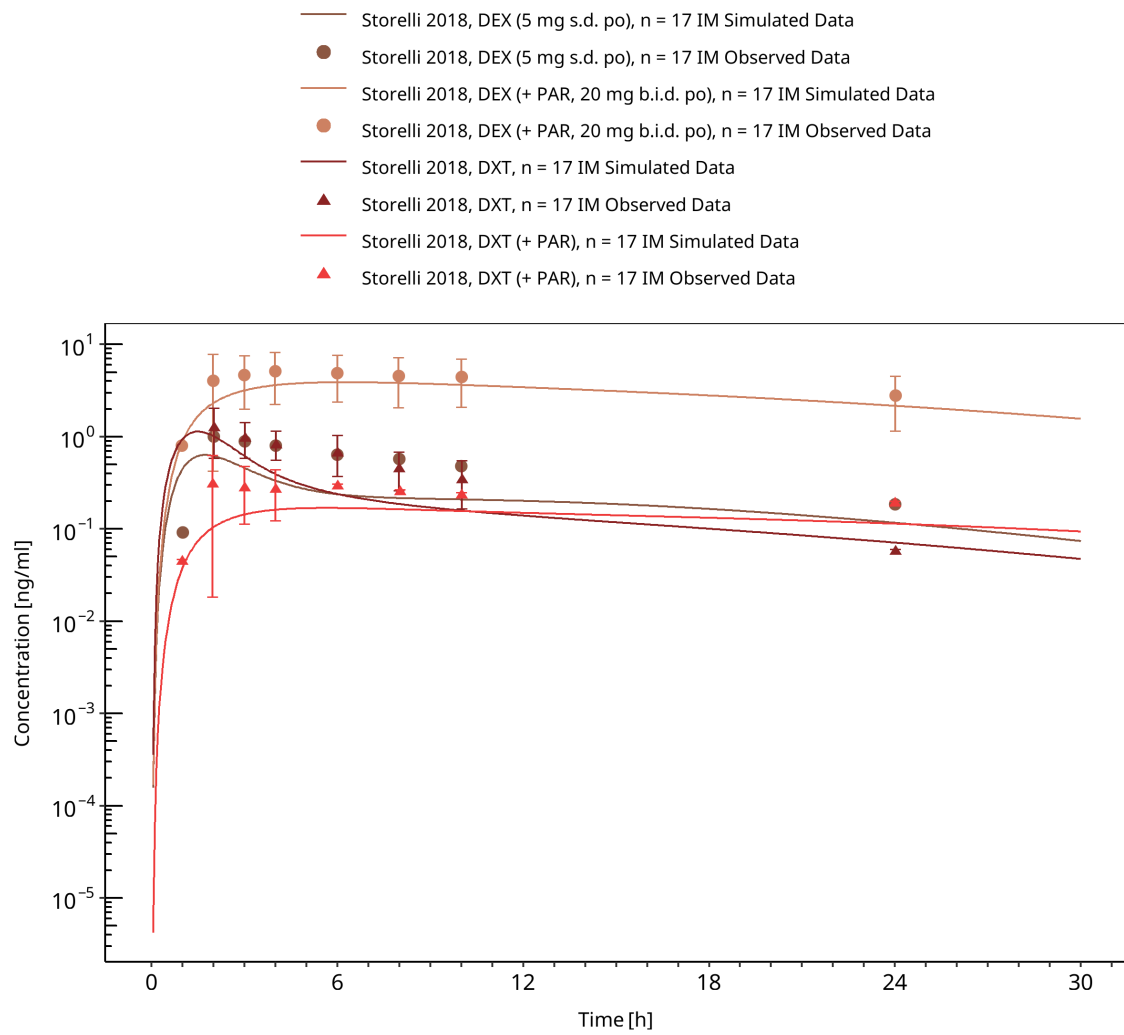


Figure 4-56: Paroxetine - Dextromethorphan DDGI

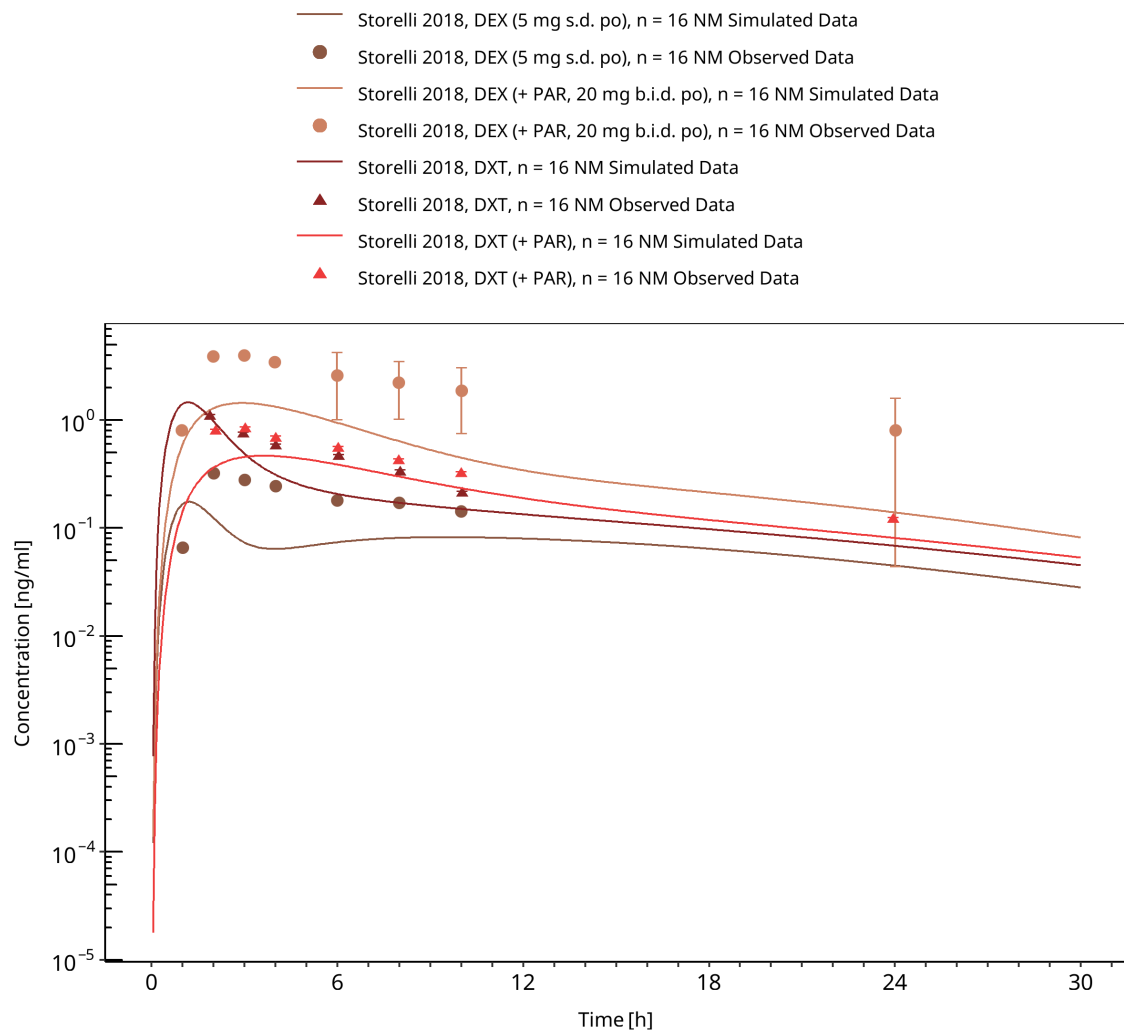


Figure 4-57: Paroxetine - Dextromethorphan DDGI

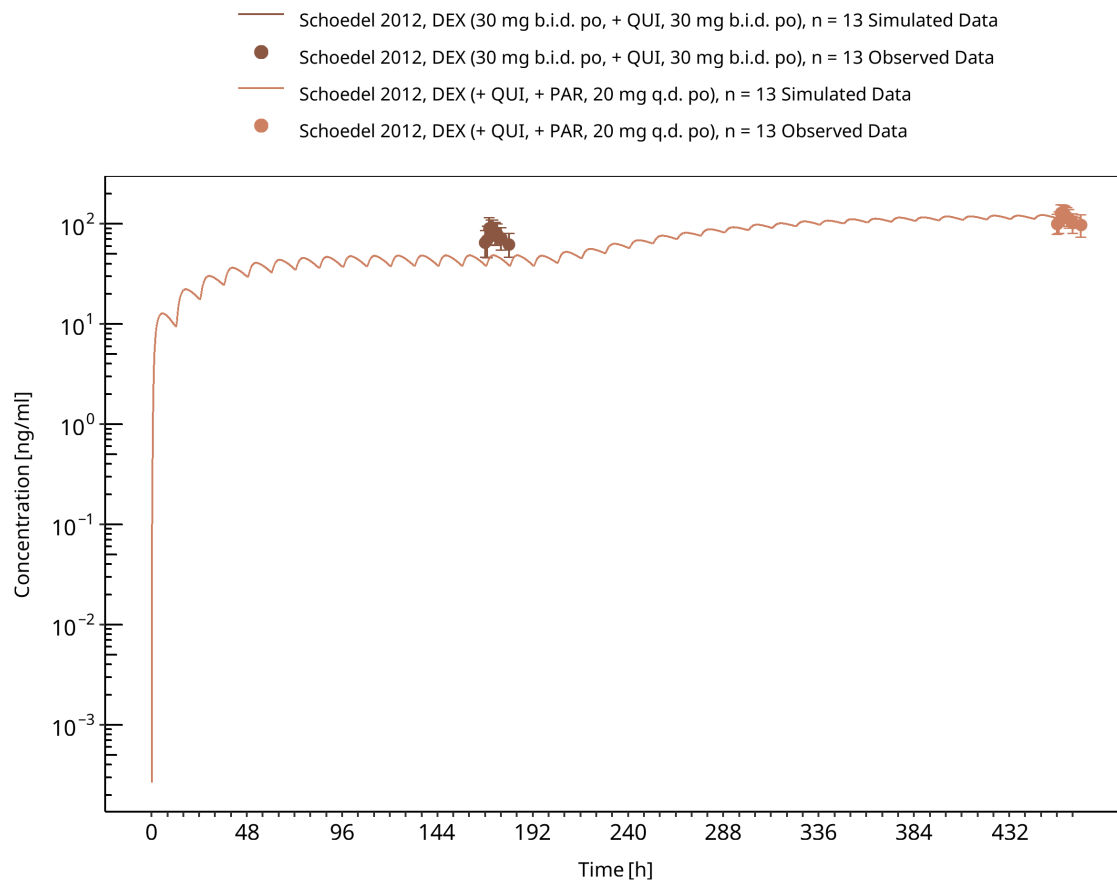


Figure 4-58: Paroxetine - Dextromethorphan DDI

4.2.6 Quinidine - Metoprolol DDGI

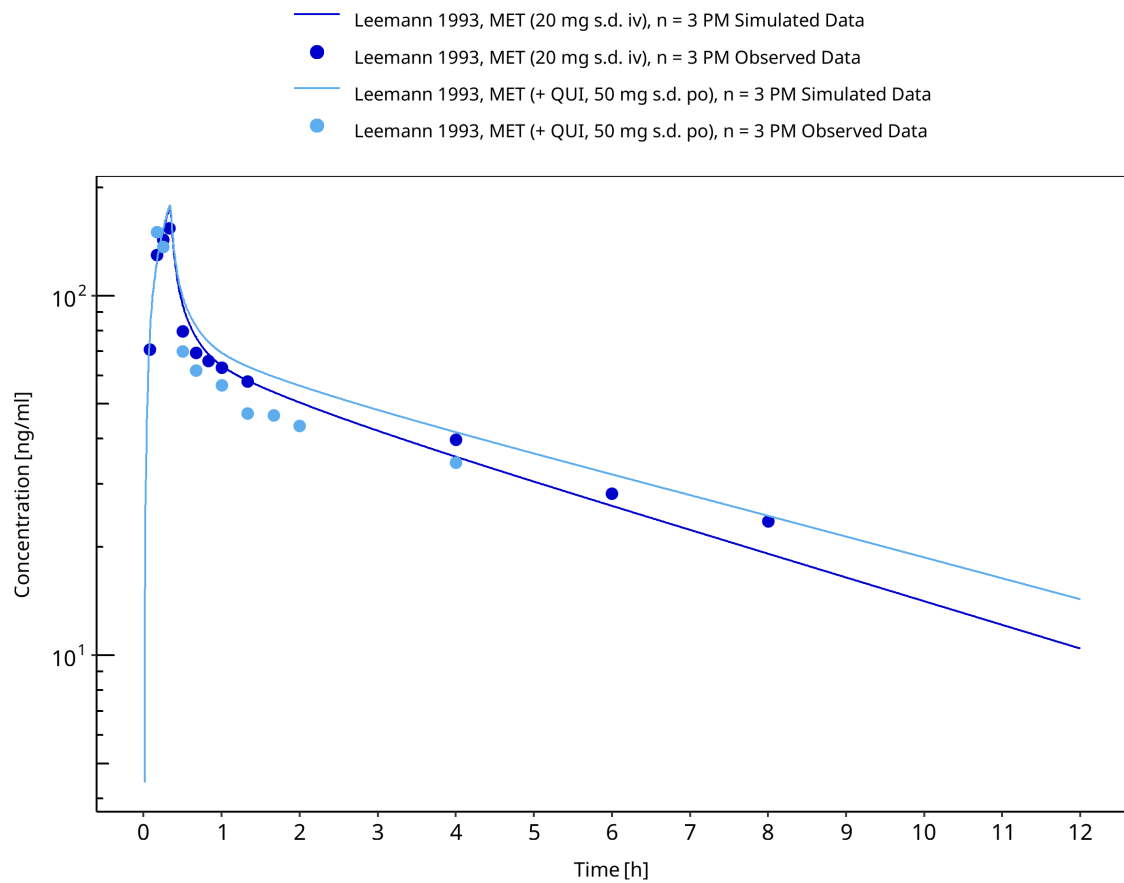


Figure 4-59: Quinidine - Metoprolol DDGI

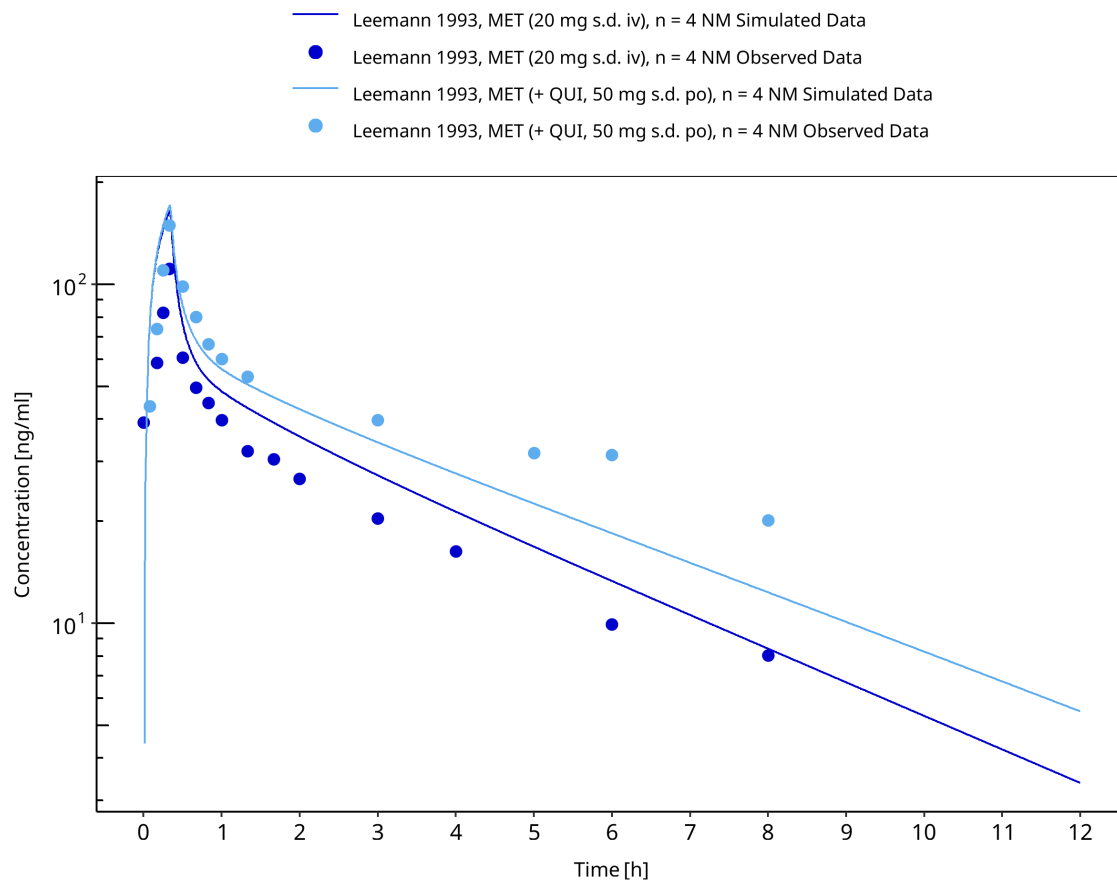


Figure 4-60: Quinidine - Metoprolol DDGI

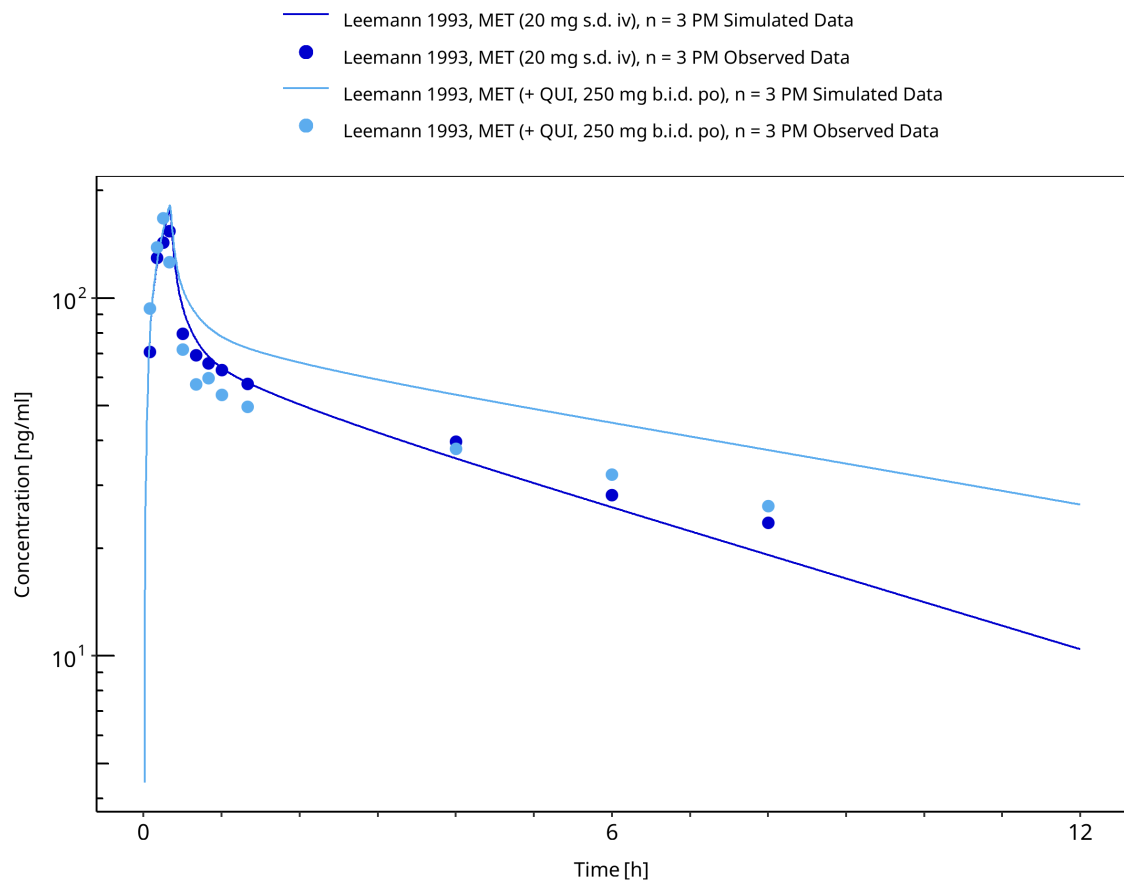


Figure 4-61: Quinidine - Metoprolol DDGI

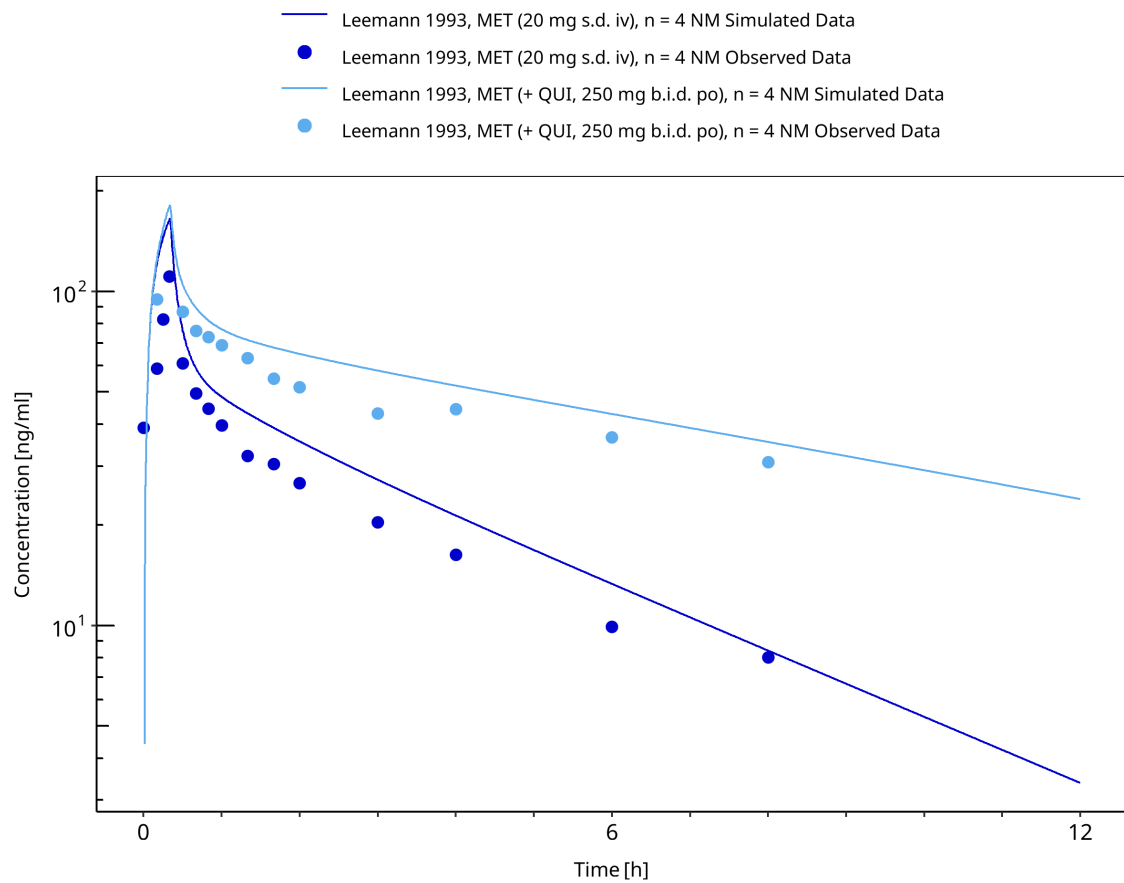


Figure 4-62: Quinidine - Metoprolol DDGI

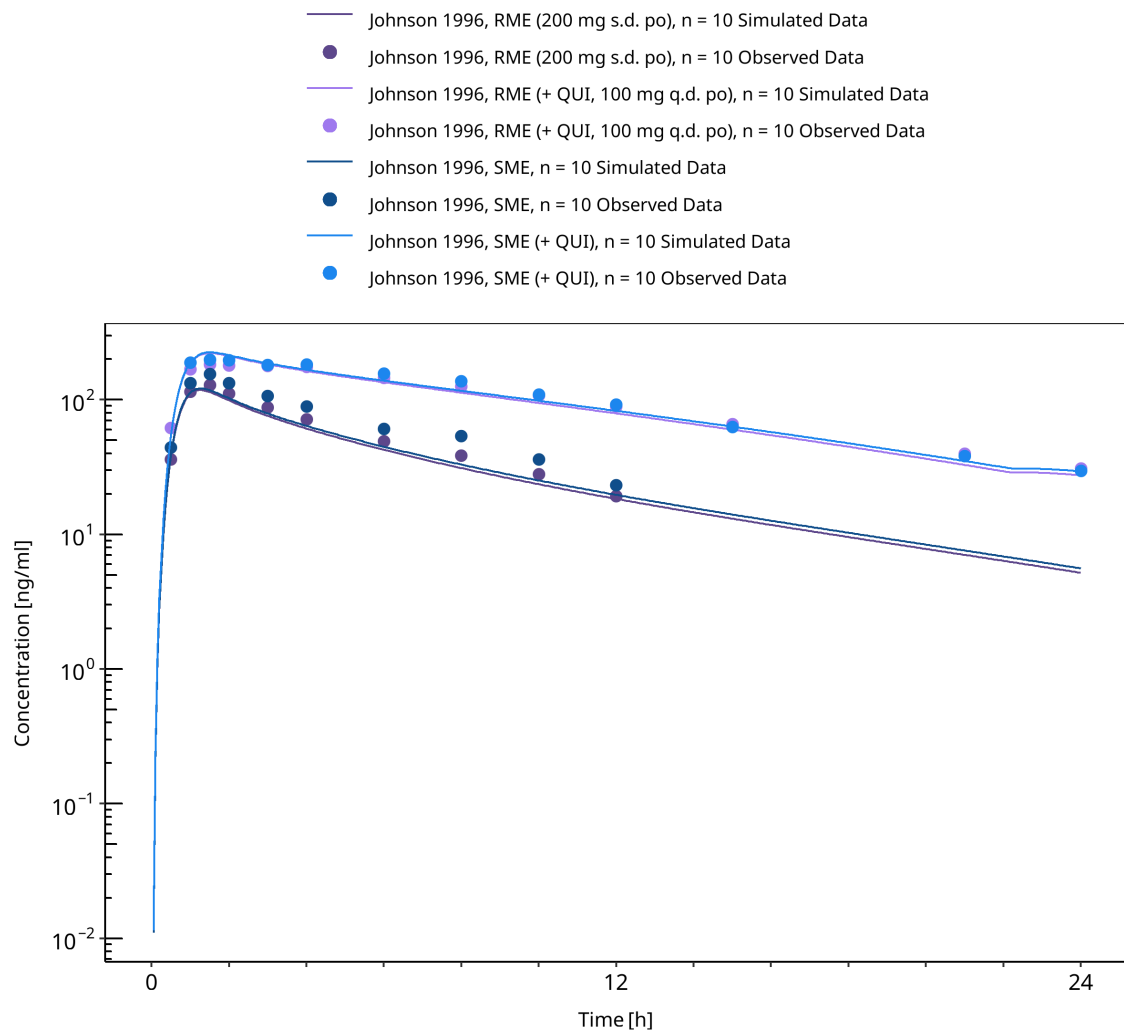


Figure 4-63: Quinidine - Metoprolol DDI

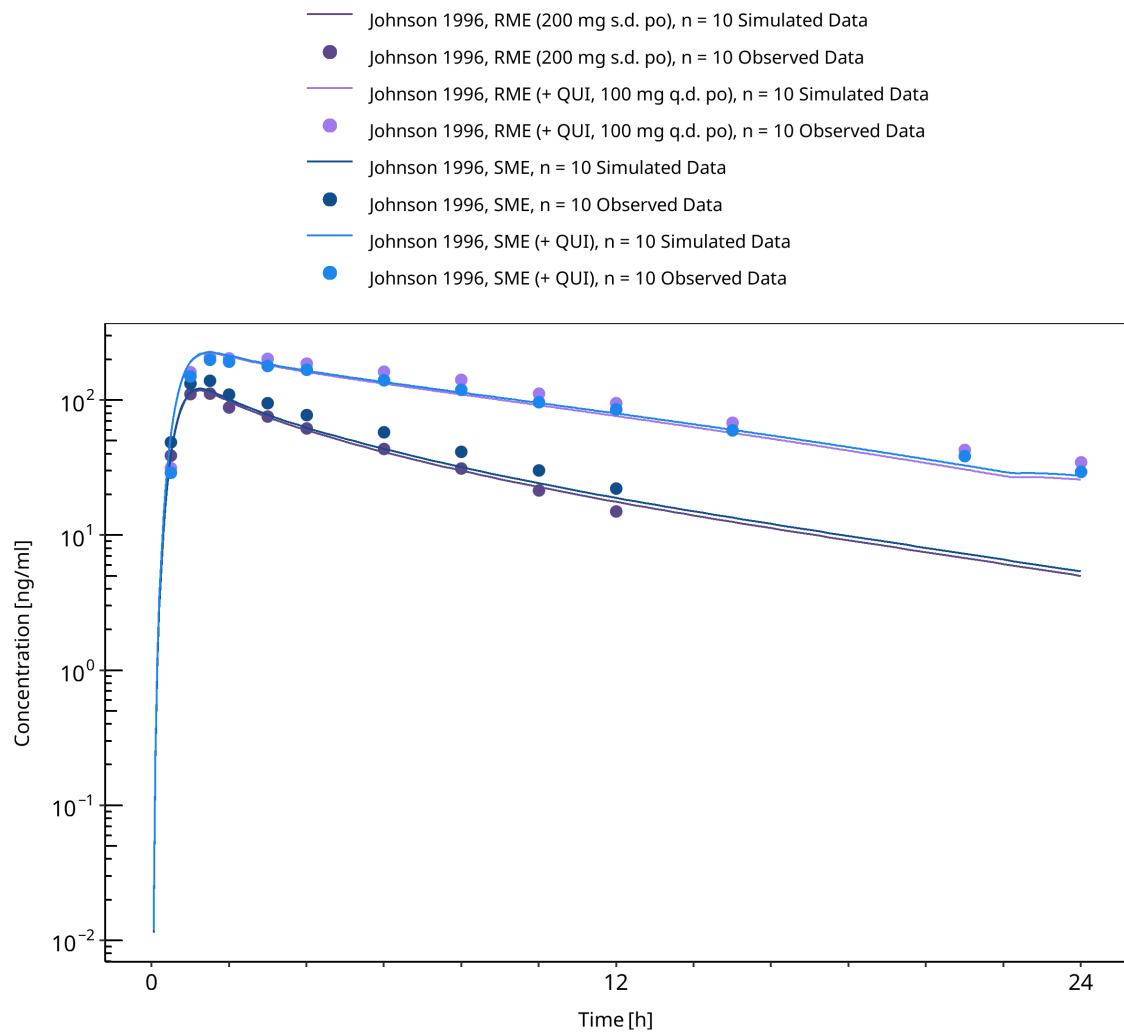


Figure 4-64: Quinidine - Metoprolol DDGI

4.2.7 Quinidine - Mexiletine DDGI

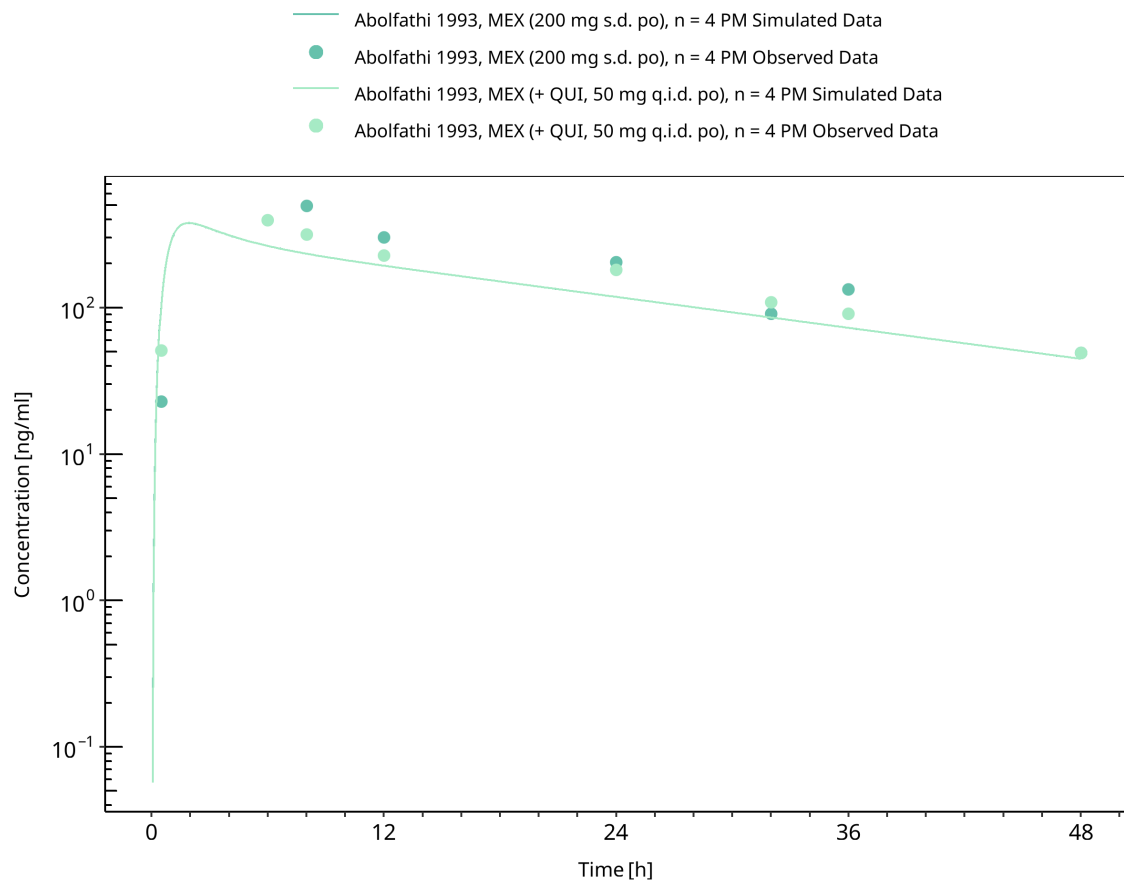


Figure 4-65: Quinidine - Mexiletine DDGI

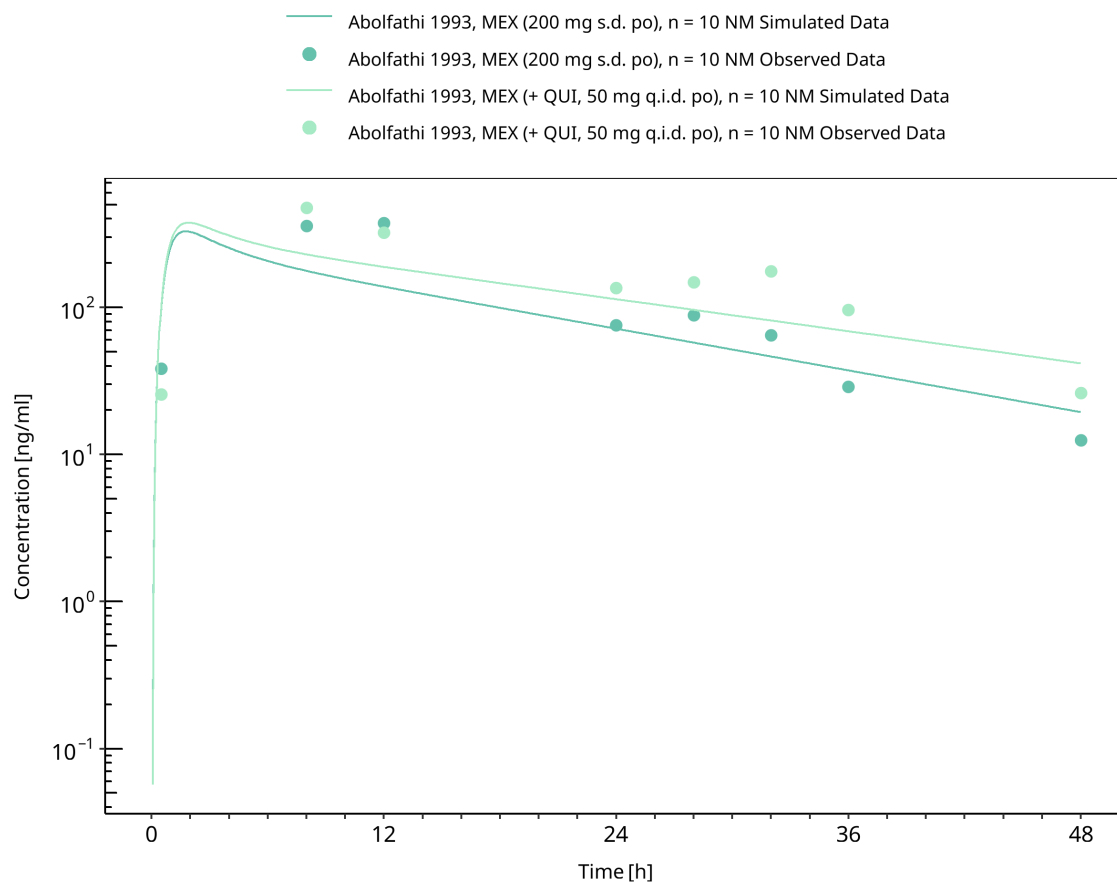


Figure 4-66: Quinidine - Mexiletine DDGI

5 Conclusion

The CYP2D6 DDGI network was qualified against clinical DD(G)I data for AUC and Cmax ratios across the selected perpetrator and victim combinations. In the current qualification run, the aggregate GMFE was 1.40 for AUC ratios and 1.38 for Cmax ratios. Most predictions were within two-fold of the observed values, with 86.32% of AUC ratios and 87.07% of Cmax ratios meeting this criterion.

The stricter limits proposed by Guest et al. were met by 57.26% of AUC ratios and 55.17% of Cmax ratios. The network therefore supports the intended use of the OSP Suite for CYP2D6-mediated DDI and DDGI simulations within the evaluated scenarios, while the remaining deviations indicate that model performance should be interpreted together with the underlying interaction mechanism, genotype group, victim drug, perpetrator drug, and analyte.

This qualification does not replace compound-specific model evaluation. New use cases should remain within the mechanistic and clinical-data domain covered by the qualified network or should be supported by additional verification.

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7 Appendix

7.1 Open Systems Pharmacology Suite (OSPS) Introduction

The Open Systems Pharmacology Suite provides a software environment for whole-body physiologically based pharmacokinetic modeling and simulation. The suite includes PK-Sim for model building and population simulation, MoBi for detailed model extension, and supporting R packages for qualification and reporting workflows.

Models are composed of physiological system information, compound-specific parameters, formulations, administration protocols, and observed data. Simulations can be combined with observed clinical data to evaluate whether model predictions are adequate for a defined context of use.

The qualification workflow uses versioned model snapshots and a machine-readable qualification plan to support reproducibility. The report generated from the plan documents the simulations, observed data comparisons, and summary performance metrics used to evaluate the intended use.

7.2 Mathematical Implementation of Drug-Drug Interactions

Drug-drug interaction simulations in the OSP Suite are represented through mechanistic changes in drug disposition processes. Depending on the compound and interaction scenario, these processes can include reversible enzyme inhibition, time-dependent inactivation, induction, transporter inhibition, transporter induction, or changes in multiple pathways.

For CYP2D6 DDGI scenarios, genotype-dependent activity is represented through phenotype-specific or activity-score-specific model settings where supported by the source model. The resulting simulations compare the victim-drug exposure with and without the perpetrator under the clinical dosing schedule. AUC and Cmax ratios are then calculated from matched control and interaction simulations.

The qualification plots compare predicted and observed exposure ratios. Performance is summarized using predicted versus observed plots, predicted/observed residual plots, geometric mean fold error, two-fold limits, and the limits proposed for ratio endpoints by Guest et al.

7.3 Automatic (re)-qualification workflow

The requalification workflow is driven by the qualification plan. The plan defines the model snapshots, simulations, observed datasets, output paths, comparison windows, figures, tables, and report sections.

The workflow loads the referenced model snapshots, runs the required simulations, calculates exposure ratios and concentration-time profile comparisons, and writes a report-ready output folder. The generated report should be reviewed for scientific consistency, expected model versions, complete figures, correct study metadata, and warnings from the simulation or reporting logs.

Before publication, the final report should be regenerated from a clean environment using the intended OSP Suite and qualification framework versions. The output should be checked against the committed qualification plan and the exact model snapshot releases used for qualification.

8 Glossary

| ALP | alprazolam | | AS | CYP2D6 activity score | | ATO | atomoxetine | | AUC | area under the plasma concentration-time curve | | AUClast | AUC calculated between first and last concentration measurement | | AUCR | AUC ratio | | b.i.d. | twice daily | | BUP | bupropion | | CBZ | carbamazepine | | CIM | cimetidine | | CLA | clarithromycin | | CLO | (E)-clomiphene | | Cmax | maximum plasma concentration | | CmaxR | Cmax ratio | | CYP | cytochrome P450 | | DDI | drug-drug interaction | | DDGI | Drug-drug-gene interaction | | DGI | drug-gene interaction | | DES | desipramine | | DEX | dextromethorphan | | DIG | digoxin | | DTT | total dextrophan | | DXG | dextrophan-O-glucuronide | | DXT | dextrophan | | ERY | erythromycin | | FLV | fluvoxamine | | HDC | (E)-4-hydroxy-N-desethylclomiphene | | IM | CYP2D6 intermediate metabolizer | | ITR | itraconazole | | iv | intravenous | | KET | ketoconazole | | MET | metoprolol | | MEX | mexiletine | | MID | midazolam | | N | number of study participants | | NDC | (E)-N-desethylclomiphene | | NM | CYP2D6 normal metabolizer | | OHC | (E)-4-hydroxyclophene | | OHD | 2-hydroxydesipramine | | OHQ | 3-hydroxyquinidine | | OHR | 9-hydroxyrisperidone | | OME | omeprazole | | OSP | Open Systems Pharmacology | | PAR | paroxetine | | PBPK | physiologically based pharmacokinetic | | P-gp | P-glycoprotein | | PM | CYP2D6 poor metabolizer | | po | oral | | q.d. | once daily | | q.i.d. | four times daily | | QUI | quinidine | | RIF | rifampicin | | RIS | risperidone | | RME | (R)-metoprolol | | s.d. | single dose | | SME | (S)-metoprolol | | t.i.d. | three times daily | | VER | verapamil |